ClaimClarity · Frontend interaction layer & UI states

Field	Value
Owner	*TODO: name / team*
Status	*TODO: Draft / In review / Approved*
Last reviewed	*TODO: YYYY-MM-DD*
Reviewers	*TODO: app team, QA, security, design*
Code	`web/` (React 18 + TypeScript + Vite), contract `web/src/contract/v1.gen.ts` (contract **1.24.0**), API contract `api/CONTRACT.md`
Screenshot source	Local run of this repo (see §2), `images/frontend/*.png`
Related pages	ClaimClarity · Data backend architecture · *TODO: link to API contract page* · *TODO: QA / e2e runbook* · *TODO: ADRs*

> **Note:** Every screenshot on this page was taken from a **local** copy of the web app (demo API with the scripted model, or the offline mock adapter) on synthetic data. The deployed frontend runs elsewhere and was not reachable. Filenames are stable (`images/frontend/<nn>-<slug>.png`); replace the files in place with captures from the deployed app when you can.

> **Note:** Images named `*-annotated.png` carry numbered red callouts. Each has a legend table right under it. The same file without `-annotated` is the clean capture.

> **Note:** Lines like `![TODO screenshot: …](images/frontend/placeholder-….png)` mark states that exist in code but could not be rendered locally. Each one says how to trigger it.

Contents

TODO: insert Confluence Table of Contents macro here (levels 2–3).

---

1. Purpose and audience

This page maps what a ClaimClarity user sees and does in the web app: every page state, every interaction, and how the analyst agent's work shows up on screen. It is the companion to the data backend architecture page, which covers the AWS side (SFTP intake, storage, sign-in, the query path, the analyzer).

- **Purpose:** *TODO: what this page supports (UX review, QA planning, support triage, onboarding, audit evidence that PHI rules hold in the UI).*

- **Audience:** *TODO: frontend and API engineers, QA, product/design, support, security reviewers.*
- **In scope:** *TODO: app shell, sidebar/history, composer, answer rendering, step timeline, artifact panel, errors and recovery, responsive/touch, accessibility, the frontend ↔ API contract as the UI uses it.*
- **Out of scope:** *TODO: AWS infrastructure and data isolation (backend page), analyzer internals beyond what the user sees, deployment (see `docs/frontend_deploy.md`).*

2. How these screenshots were made

TODO: one paragraph on why local captures stand in for the deployed app, and who should replace them.

Mode	How it was started	What it gives
Demo API (HTTP adapter)	`CC_API_MODE=demo CC_API_LLM=scripted CC_API_MODEL_DELAY_SECONDS=0.6 CC_API_STREAM_DELAY_SECONDS=0.03 python -m uvicorn api.app:app --port 8140`, then `CC_API_URL=http://127.0.0.1:8140 VITE_ANALYST_BASE=/ VITE_ANALYST_DEMO_TOKEN=demo:acme_labs:peter npx vite --port 5174`	Real `POST /api/chat` SSE streams from the real `Analyzer` loop with `ScriptedLLM` on synthetic DuckDB data. Demo scenarios (`ChatRequest.scenario`, CONTRACT.md §9) were forced by rewriting the request body in Playwright.
Mock adapter (offline)	`npx vite --port 5173` with `VITE_ANALYST_BASE` unset	`MockAnalystAdapter` replays `web/src/fixtures/*` (charts with an image, PHI-masked table, artifact card, guard rejection, candidates, clarify, abstain). Memory transcript store, so the temporary-chat toggle is available.
Fault injection	Playwright `page.route('**/api/chat', …)`	HTTP 4xx/5xx bodies, 429 + `Retry-After`, refused connections, dropped streams, unknown part types, canned markdown. Same technique as `web/e2e/helpers.ts` (`injectStream`).

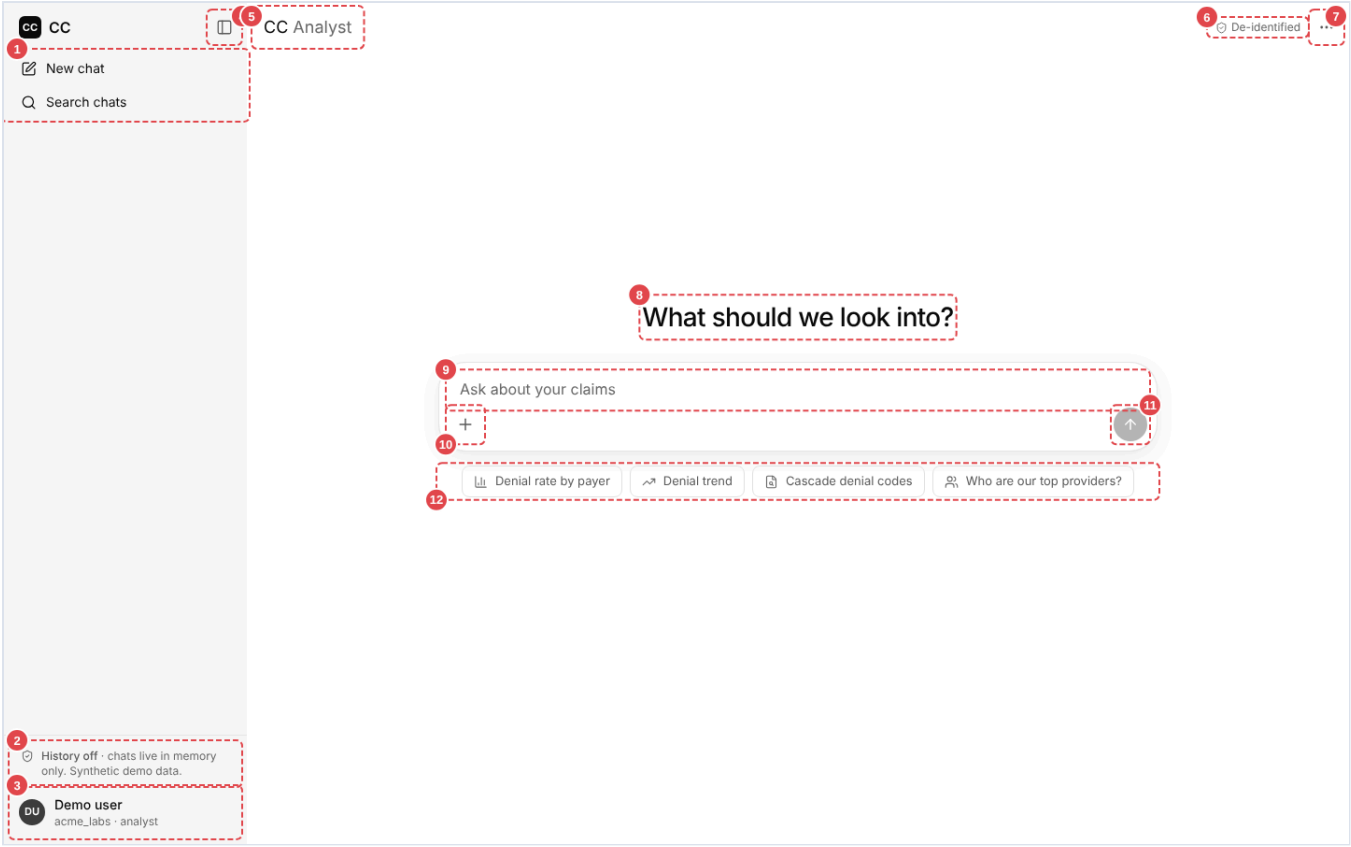
- Browser: Chromium via Playwright, 1440×900 desktop (1000 px tall for some), 390×844 touch (mobile), 820×1000 (tablet). Light theme, Neutral palette unless the name says otherwise.
- Annotations are an SVG overlay drawn over the real element boxes just before the capture.
- Scripts were throwaway and are not in the repo. *TODO: decide whether to add a capture script under `web/scripts/` (there is already `npm run shots` → `scripts/screenshots.mjs`).*

> **Note:** The demo server keeps conversations in memory for the whole run, so later sidebar captures show many threads created by earlier captures ("Stop while planning", "Long query", …). That is expected.

---

3. App shell anatomy

The shell is `web/src/shell/AppShell.tsx`: sidebar (or collapsed rail), header, thread, composer, and the artifact panel on the right when open. Routes are path-based for status pages and hash-based for chats (`#/c/<opaque id>`, never titles or text).



TODO: caption — "Empty chat, first run (desktop)."

#	Element	Component	Behaviour
1	New chat · Search chats	`components/Sidebar.tsx` (`nav[aria-label="Chat actions"]`)	New chat (`Ctrl/⌘ Shift O`); Search chats opens the full-text search dialog (`Ctrl/⌘ K`).
2	History note	`Sidebar` `data-testid="history-note"`	"History off · chats live in memory only" when `TranscriptStore.mode` is `ephemeral`; "History on" otherwise.
3	Account row	`Sidebar` `data-testid="account-button"`	Opens Settings / Keyboard shortcuts / Sign out (Sign out only when a session exists). Identity is a synthetic constant (`ACCOUNT`).
4	Close sidebar	`Sidebar`	Collapses to the 52 px rail (`Ctrl/⌘ Shift S`); preference stored as `cc.pref.sidebarCollapsed`.
5	Product name	`shell/Header.tsx`	Plain text "CC Analyst". No model or data-source picker (contract 1.24.0).
6	PHI mode badge	`Header` `data-testid="phi-badge"`	"De-identified", set by the server. *TODO: confirm which server field drives it; today it is static text.*
7	More options	`Header`	Dark/Light mode, Settings, Keyboard shortcuts.
8	Empty-state heading	`AppShell`	"What should we look into?" Only on an empty chat.
9	Composer	`components/Composer.tsx`	Auto-growing textarea, placeholder "Ask about your claims" (empty chat) / "Ask a follow-up" / "Reply…" (open clarification).
10	Attach text files	`Composer`	`.txt .csv .tsv .md .sql .json .log`, ≤ 256 KB each, read locally into chips; never uploaded as files.

#	Element	Component	Behaviour
11	Send	`Composer`	Disabled until there is text or a chip, and while offline. Turns into Stop while a turn streams.
12	Suggestion chips	`shell/StarterCards.tsx`	Four starters whose prompts match demo fixtures. Phones show a swipeable row of two-line cards above the composer.

3.1 Layout regions and breakpoints

Region	Desktop (≥ 768 px)	Phone (< 768 px)	Notes
Sidebar	260 px column, collapsible to rail	Modal drawer (`role="dialog"`) with scrim, `min(300px, 85vw)`	§6, §14
Header	52 px, title menu when a chat is open	Same, title hidden under 520 px container width	§6.7
Thread	Max ~48 rem column, virtualized rows	Full width	`shell/ThreadView.tsx`
Composer	Pinned at the bottom, footer "CC can make mistakes. Check figures against source before use."	Same	§7
Artifact panel	Right side, resizable 360 px … 75 %, collapses the sidebar under 1400 px	Full-screen sheet	§11

TODO: add a wireframe or diagram of the regions if useful.

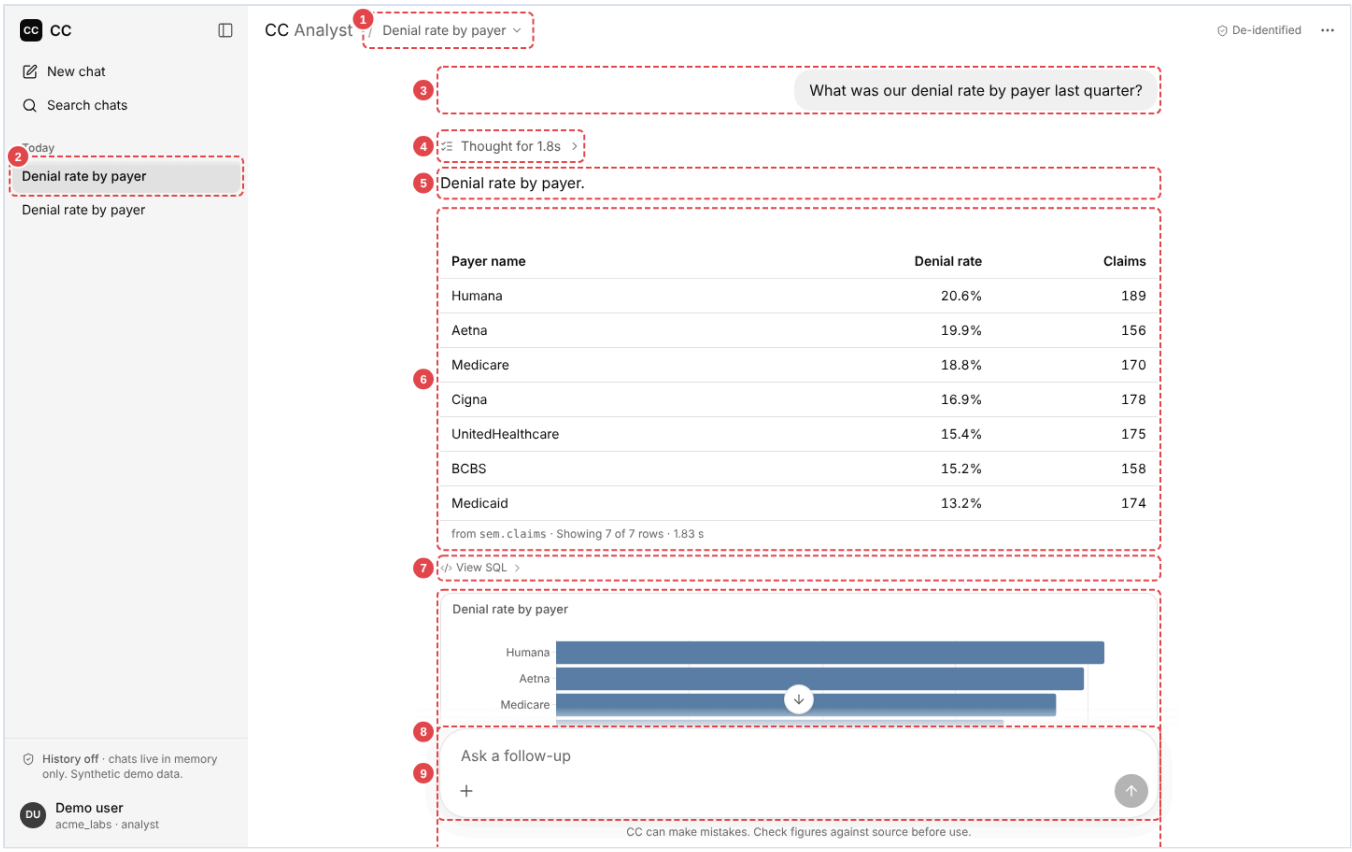
---

4. User journeys end to end

TODO: one-paragraph intro: the journeys below are the paths QA and support should know by heart.

4.1 J1 · First question to finished answer

1. Empty chat → type or pick a starter (`01`, `02`).
2. Send → user bubble appears, the thread gets a placeholder title, a collapsed **Thinking** row and Stop button show at once (`05`).
3. Steps stream: the activity line names the current step ("Checking the claims view", "Running query") (`06`); expanding shows each step (`07`).
4. The answer streams in below the steps with a caret (`08`), then tables, SQL, charts, callouts.
5. `message.end` → action row, Sources chip, usage line, follow-ups; the sidebar title becomes the server's PHI-free title (`10`, `12`).



#	Element	Component	Behaviour
1	Chat title	`Header`	Server title from `message.end.conversationTitle` (derived from column names, never the question). Click → title menu; double-click → rename.
2	Active thread	`Sidebar` thread item	Highlighted; "..." menu on hover/focus.
3	Question bubble	`UserBubble` in `components/Message.tsx`	Hover shows time, Copy, Edit.
4	Step timeline header	`components/answer/StepTimeline.tsx`	Collapsed "Thought for N s"; click to expand (§8).
5	Answer prose	`components/parts/Markdown.tsx` (Streamdown)	Streams with unfinished-markdown repair.
6	Result table	`components/parts/DataTable.tsx`	Sortable, numeric right-aligned, provenance footer "from sem.claims · Showing 7 of 7 rows · 1.83 s".
7	View SQL	`TableSql` in `Message.tsx`	The query that produced this table, linked by `sqlPartId`.
8	Chart	`components/parts/ChartView.tsx`	Vega-Lite, only inline data.
9	Composer	`Composer`	Placeholder switches to "Ask a follow-up".

Full-page capture of the same answer: `images/frontend/11-answer-complete-full.png`.

4.2 J2 · Follow-up, edit, regenerate

- Follow-up: the new question is pinned to the top of the viewport and its answer streams below; "Jump to latest" appears when the user scrolls away (`123`).

- Edit: pencil under the question → inline editor → Send creates a sibling version; `‹ 2/2 ›` switches (`118`, `119`).
- Regenerate: one click re-runs as the next version of the answer; the chevron offers Try again / More concise / More detailed / Ask to change response… (`20`, `120`–`122`).

4.3 J3 · Clarify, then answer

The agent asks before running anything when the question is ambiguous (`finishReason: needs_input`). The composer placeholder becomes "Reply…"; a chip or free text resumes the run (`ChatRequest.resume`) (`38`, `39`, `76`, `77`).

4.4 J4 · The agent can't or won't answer

Abstain (data not available, forecast, unsupported) shows a "Cannot answer" note with suggestions; requests for patient identifiers are refused (`33`, `34`, `44`, `78`).

4.5 J5 · Something fails, user retries

Error line with Retry in the answer column; transport problems (429, offline, reconnecting) show as a status line while the turn is still alive (§13) (`80`–`91`).

4.6 J6 · Come back to an earlier chat

Sidebar (date buckets, pins, filter at ≥ 20 chats) or Search chats (full text, server-side) → thread skeleton while the conversation loads → stored steps are replayed through the same reducer as a live stream (`100`–`117`).

4.7 J7 · Phone

Drawer sidebar, swipeable suggestions, artifacts as a full-screen sheet, action rows always visible (no hover) (`130`–`134`).

TODO: add the journeys support sees most (e.g. "the numbers look wrong" → thumbs down → SQL inspector).

---

5. State inventory

One row per UI state. "Screenshot" points at the image on this page (`P-…` = placeholder). Test files are relative to `web/src/` (unit, Vitest) or `web/e2e/` (Playwright).

ID	State	Trigger	What the user sees	Component(s)	Tests	Screenshot
S01	Empty chat, first run	Load `/`, no chat open	Heading, composer, 4 suggestion chips	`AppShell`, `StarterCards`	`shell/shell.test.tsx`, `e2e/mock.spec.ts`	01
S02	Composer typing	Text in the box	Send enabled (filled), box grows to 240 px	`Composer`	`components/Composer.test.tsx`	02

ID	State	Trigger	What the user sees	Component(s)	Tests	Screenshot
S03	Slash command menu	Type `/`	Listbox of 12 commands, arrow keys / Tab / Enter	`Composer`, `shell/slash.ts`	`Composer.test.tsx`	03
S04	Attachment chips	Paste ≥ 1200 chars or > 15 lines; attach text files	Chips "PASTED" / file extension, remove ×, Backspace removes last	`Composer`	`Composer.test.tsx`	04
S05	Attachment rejected	Non-text file or > 256 KB	Red notice "x isn't a text file · y is over 256 KB"	`Composer` (`attach-notice`)	`Composer.test.tsx`	P-attach-rejected
S06	Drag-over	Drag files onto the composer	Focus ring around the composer	`Composer` (`data-dragging`)	`Composer.test.tsx`	P-attach-drag
S07	Sending / Thinking	Send, before the first event	Question bubble, collapsed "Thinking" row, Stop button	`StepTimeline`, `Composer`	`components/answer/thinkingRow.test.tsx`, `e2e/visual-states.spec.ts`	05
S08	Live step line	Step events arriving	One muted line naming the current step, no spinner/timer	`StepTimeline`	`components/answer/thinkingDisclosure.test.tsx`, `quietStream.test.tsx`	06
S09	Live steps expanded	Click the live header	Rows turning running → done, running dot, escalation note, table skeleton	`StepTimeline`, `TableSkeleton`	`answer/answer.test.tsx`, `e2e/answer-edge.spec.ts`	07
S10	Text streaming	`text.delta`	Paced prose with a caret	`Markdown`, `lib/pacedText.ts`	`lib/pacedText.test.tsx`, `answer/liveTurnPacing.test.tsx`, `parts/streaming.test.tsx`	08
S11	Answer complete	`message.end` `stop`	Prose, table, SQL, chart, callouts, Sources, action row, usage	`AssistantTurn`	`v1.test.tsx`, `e2e/contract-outcomes.spec.ts`	10, 12

ID	State	Trigger	What the user sees	Component(s)	Tests	Screenshot
S12	Steps expanded (done)	Click "Thought for…"	Counts, reasoning summary, each step with detail	`StepTimeline`	`answer/thinkingDisclosure.test.tsx`	13, 14
S13	Answer details popover	Click usage line	Tokens, cost (est.), queries, first token, total time, stage bar	`answer/AnswerMeta.tsx`	`answer/answer.test.tsx`	16
S14	SQL inspector	"SQL" in action row	Tabs per query, SQL, row count, columns	`answer/SqlInspector.tsx`	`answer/sqlMasking.test.tsx`	17
S15	Sources popover	"Looked at …" chip	Views used, columns, "No row values are shown"	`answer/Sources.tsx`	`answer/sources.test.tsx`, `sourcesConsulted.test.tsx`	18
S16	More actions menu	"…" in action row	Branch in new chat, Print this answer, Copy as Markdown, Copy SQL	`AssistantTurn`	`answer/messageActions.test.tsx`, `shell/branch.test.tsx`	19
S17	Regenerate menu	Chevron next to Regenerate	Try again, More concise, More detailed, Ask to change response…	`AssistantTurn`	`contract/regenerate.test.ts`, `messageActions.test.tsx`	20
S18	Feedback panel	Thumbs down	"Tell us more" reason chips + comment ≤ 1000 chars	`FeedbackPanel`	`lib/feedbackWire.test.ts`, `contract/feedback.test.ts`	21
S19	Repair	Query fails (binder error)	Failed step ✕ red, "Repaired the query", re-run attempt 2	`StepTimeline`	`answer/edgeScenarios.test.tsx`	30, 43
S20	Escalation	Small model gives up	"Handed off to Sonnet 5.5", "Escalated to Sonnet 5.5 · repairs ran out"	`EscalationNote`, `StepTimeline`	`answer/answer.test.tsx`	31, 09
S21	Candidates	Multi-candidate run	"Compared 3 candidates · execution agreement"	`StepTimeline`	`answer/comparisonPass.test.tsx`	32, 75
S22	Abstain	`finishReason: abstain`	"The data needed for this isn't available", Try asking chips	`answer/States.tsx` `AbstainCard`	`answer/answer.test.tsx`	33, 78

ID	State	Trigger	What the user sees	Component(s)	Tests	Screenshot
S23	PHI refusal	Ask for patient identifiers	"I can't share patient-identifying details" + alternatives	`AbstainCard`	`security/phi.test.tsx`	34
S24	Analysis error	Run fails server-side	"Something went wrong / The analysis failed. Try again." + Retry	`ErrorView` (`parts/misc.tsx`)	`answer/malformedParts.test.tsx`	35
S25	Chart answer	Time series / ≤ 50-row categorical	Line or bar chart under the table	`ChartView`	`parts/chartActions.test.tsx` , `render/chartEdge.test.ts`	36, 66
S26	Report artifact	"report" questions	Artifact card + panel opens on its own	`ArtifactCard` , `ArtifactPanel`	`components/artifacts.test.tsx` , `parts/artifactKind.test.tsx`	37
S27	Clarify (single)	`finishReason: needs_input`	Question, choice chips, "Or type your answer below", composer "Reply…"	`ClarifyCard`	`contract/clarify.test.ts` , `shell/clarifyReply.test.tsx`	38, 76
S28	Clarify (multi-question form)	`ClarificationPart.questions`	Tabs (Metric / Period / Group by), options, Other…, Skip	`answer/ClarifyForm.tsx`	`answer/clarifyForm.test.tsx`	39
S29	Clarify answered + resumed	Pick a chip	Chip filled, "You answered … Continuing…", new answer	`ClarifyCard`	`e2e/shell-clarify-reply.spec.ts`	77
S30	Nested clarify	Second question after resume	Another clarification card	`ClarifyCard`	`contract/edgeCases.test.ts`	45
S31	Empty result	Query returns 0 rows	Explanation, table header with "Query returned 0 rows", caveat	`DataTable` (`table-empty`)	`parts/tableEdge.test.tsx`	40
S32	Multiple tables	Two result sets	Two titled tables, each with its own SQL and chart	`AssistantTurn`	`contract/edgeCases.test.ts`	41
S33	Long SQL / wide table	100+ line query, many columns	Horizontal scroll inside the table, edge fades	`CodeBlock` , `DataTable`	`parts/codeChrome.test.tsx` , `parts/tableEdge.test.tsx`	42, 142

ID	State	Trigger	What the user sees	Component(s)	Tests	Screenshot
S34	Stopped (planning)	Stop before any SQL	"Stopped · Retry"	`StoppedNote`	`lib/useChat.races.test.tsx`	46
S35	Stopped (mid-query / repair / candidates / escalation / render)	Stop at that stage	Steps closed as "stopped", "Stopped" note, action row	`StepTimeline`, `StoppedNote`	`contract/edgeCases.test.ts`, `e2e/chaos.spec.ts`	47–52
S36	Table hover toolbar	Hover a table	Copy table, Download CSV, Expand	`DataTable`	`parts/tableEdge.test.tsx`	60
S37	Table sorted	Click a header	Arrow on the sorted column	`DataTable`	`parts/polish.test.tsx`	61
S38	Column resized	Drag the column edge	Wider column	`DataTable` (`col-resize`)	`parts/tableResize.test.tsx`	62
S39	File card	Result > 50 rows or CSV attached	CSV chip with rows and size; download via opaque id	`FileCard`	`lib/csvFilename.test.ts`, `security/csvInjection.test.ts`	59
S40	Artifact panel · table	Expand table	Table/SQL tabs, Copy, Download, Full screen, Close, Filter rows	`ArtifactPanel`	`components/ArtifactPanel.test.tsx`	63, 64
S41	Artifact panel resized	Drag or arrow keys on the separator	Narrower/wider panel	`ArtifactPanel`	`ArtifactPanel.test.tsx`	65
S42	Chart point picked	Click a mark	"Ask about Northwind Health Plan" chip	`ChartView` (`chart-ask`)	`parts/chartActions.test.tsx`	67
S43	Artifact panel · chart	Expand chart	Chart / Spec / Data tabs	`ArtifactPanel`	`ArtifactPanel.test.tsx`	68, 69
S44	Image part + caveat	Analyst-rendered figure	Image with caption, Caveat callout	`ImageView`, `Callout`	`render/render.test.tsx`	70
S45	PHI-masked column	`Column.phi: true`	"•••• 56", shield icon, PHI masked callout	`DataTable`, `Callout`	`security/phi.test.tsx`, `answer/dataWithheld.test.tsx`	71, 72
S46	Artifact panel · code	Artifact card (code)	SQL in the panel	`ArtifactPanel`	`artifacts.test.tsx`	73

ID	State	Trigger	What the user sees	Component(s)	Tests	Screenshot
S47	Guard rejection	Query names a relation outside `sem.*`	Failed step + "Access denied … I can answer this from the sem.* semantic views instead" + Retry/Details	`ErrorView`, `StepTimeline`	`adapters/http.guard.test.ts` *(TODO: confirm; that file tests the stream guard)*, `e2e/mock.spec.ts`	74
S48	HTTP 5xx	`POST /api/chat` 500	"Analyst unavailable · The analyst returned 500" + Retry	`httpError` → `ErrorView`	`adapters/http.httpError.test.ts`, `e2e/chaos.spec.ts`	80
S49	413 too large	Body > 1 MiB	"Message too long · Shorten the message to under 1024 KB" (no Retry)	`httpError`	`adapters/http.httpError.test.ts`	81
S50	422 validation	Bad body	"Couldn't send that" + server's fixed message	`httpError`	`adapters/http.errors.test.ts`	82
S51	409 clarification expired	Stale resume	"Ask again"	`httpError`	`adapters/http.errors.test.ts`	83
S52	Network failure	Connection refused / DNS / CORS	"Can't reach the analyst" + Retry	`networkError()`	`adapters/http.errors.test.ts`	84
S53	Retry recovered	Retry after S52	Normal answer	`useChat`	`lib/useChat.resilience.test.tsx`	85
S54	429 countdown	429 + Retry-After	"Busy right now, retrying in 8s…", auto-retry	`useChat`, `rateLimitLabel`	`adapters/http.ratelimit.test.ts`, `e2e/edge.spec.ts`	86
S55	Transport announcement	429 / offline / reconnecting	Screen-reader text (polite, throttled)	`TransportAnnouncer`	`components/transportAnnouncer.test.tsx`, `lib/transportAnnounce.test.ts`	87

ID	State	Trigger	What the user sees	Component(s)	Tests	Screenshot
S56	Reconnecting	Stream drops mid-answer	"Reconnecting…" after 1 s quiet	`useChat` (`RECONNECT_QUIET_MS`)	`adapters/http.midstream.test.ts` , `lib/useChat.connection.test.tsx`	88
S57	Connection lost	Reconnect budget used / replay gone	"Connection lost · Response interrupted" + Retry	`useChat` , `ErrorView`	`adapters/http.streamgone.test.ts`	89
S58	Offline	Browser offline ≥ 1 s	Pill "You're offline. Sending resumes when you reconnect.", Send disabled, draft kept	`OfflineBanner` , `useSettledOnline`	`adapters/http.flapping.test.ts` , `e2e/edge.spec.ts`	90
S59	Session expired (in chat)	401 with a demo token	"Session expired · Sign in again to continue."	`httpError`	`auth/session.test.ts`	91
S60	Signed out page	`/signed-out`	"You're signed out" + Sign in again	`StatusPage`	`shell/routes.test.tsx` , `e2e/shell-routes.spec.ts`	92
S61	Session expired page	`/signed-out?reason=expired` (cookie session 401)	"Your session expired" + Sign in again	`StatusPage`	`auth/session.test.ts` , `auth/crossTabSignOut.test.ts`	93
S62	Not found	Any other path	"Page not found" + Go to CC	`NotFoundPage`	`shell/routes.test.tsx`	94
S63	Unsupported part	Unknown `part.type`	Muted chip "Unsupported content (sankey)", rest renders	`Unsupported`	`answer/malformedParts.test.tsx` , `contract/fuzz.test.ts`	95
S64	Chart failed to load	Vega chunk fails	"Chart failed to render: …" inside the chart frame	`ChartView` , `lib/onceOk.ts`	`lib/idlePrefetch.test.ts` , `parts/chartSplit.test.tsx`	96

ID	State	Trigger	What the user sees	Component(s)	Tests	Screenshot
S65	App crash	Render error escapes every section	"Something went wrong", Reload, Start a new chat, Reference id + diagnostics	`ErrorBoundary` (app scope)	`components/ErrorBoundary.test.tsx`, `lib/errorReporting.test.tsx`	97
S66	Section crash	Render error inside a part / panel	"This part couldn't be displayed. Ref ..." + Try again / Close	`ErrorBoundary` (section scope)	`components/sectionBoundaries.test.tsx`	P-section-crash
S67	Stalled stream	No bytes for 45 s	Before start: "The analyst stopped responding" (retryable); mid-answer: reconnect	`guardBody`	`adapters/http.guard.test.ts`	P-stalled
S68	Answer too large	One SSE line > 32 MiB	What arrived + "Answer too large to show"	`guardBody`	`adapters/http.guard.test.ts`	P-too-large
S69	Sidebar list + filter	≥ 20 chats	"Filter chats" box, Pinned / Today / Yesterday / Previous 7 days / Previous 30 days / month / year buckets	`Sidebar`	`components/sidebarFilter.test.tsx`, `e2e/shell-sidebar-filter.spec.ts`	100, 101
S70	Filter, no match	Filter text matches nothing	"No chat titles match "..."." + "Search messages" link	`Sidebar` (`filter-empty`)	`sidebarFilter.test.tsx`	102
S71	Thread options	"..." or right-click on a thread	Export as Markdown, Rename, Pin, Archive, Delete	`Sidebar`	`shell/archive.test.tsx`, `adapters/httpStore.pinArchive.test.ts`	103
S72	Pinned	Pin	"Pinned" group at the top	`Sidebar`	`contract/pinArchive.test.ts`	104
S73	Rename inline	Rename / double-click	Input in the row; Enter saves, Esc cancels	`Sidebar`	`shell/title.test.ts`	105
S74	Delete / archive toast	Delete or Archive	Toast "Deleted "..."" with Undo (5 s)	`ToastProvider`	`shell/toast.test.tsx`	107

ID	State	Trigger	What the user sees	Component(s)	Tests	Screenshot
S75	Search dialog	`Ctrl/⌘ K`	Search box, New chat, results with snippets (titles and message text)	`shell/SearchDialog.tsx`	`shell/SearchDialog.test.tsx`, `e2e/shell-find.spec.ts`	108
S76	Sidebar rail	Collapse	52 px rail: open, new chat, search, account	`SidebarRail`	`components/sidebarRail.test.tsx`	109
S77	Account menu	Account row	Settings, Keyboard shortcuts, Sign out	`Sidebar`	`shell/shell.test.tsx`	110
S78	Settings · General	`Ctrl/⌘ ,`	Theme (System/Light/Dark), Style (Neutral/Paper), Send with Enter	`shell/SettingsDialog.tsx`	`shell/settings.test.tsx`	111
S79	Settings · Data controls	Tab	Chat history mode, Unsent drafts (memory only), Archived chats (manage/unarchive), Export chats	`SettingsDialog`	`shell/settings.test.tsx`, `shell/archive.test.tsx`	112
S80	Keyboard shortcuts	`Ctrl/⌘ /`	Shortcut table (Chats / Navigation / Composer)	`ShortcutsDialog`, `shell/shortcuts.ts`	`e2e/keyboard-sr.spec.ts`	113
S81	Header title menu	Click chat title	Rename, Copy as Markdown, Export as Markdown, Export as JSON (live API), Delete	`Header`	`shell/fidelity.test.ts`	114
S82	Header more menu	"..." top right	Dark/Light mode, Settings, Keyboard shortcuts	`Header`	`shell/shell.test.tsx`	115
S83	Opening a chat	Click a thread (HTTP store)	Thread skeleton after a short grace period	`AppShell` (`thread-skeleton`)	`shell/routes.test.tsx`	116
S84	Response interrupted	Stored chat ends on a question (reload mid-answer)	"Response interrupted · This answer stopped before it finished." + Retry	`withInterrupted` in `lib/useChat.ts`	`lib/useChat.interrupted.test.tsx`, `e2e/answer-edge.spec.ts`	117
S85	Edit question	Pencil under a question	Inline editor, Cancel / Send	`UserBubble`	`components/userBubble.test.tsx`	118

ID	State	Trigger	What the user sees	Component(s)	Tests	Screenshot
S86	Question versions	Edited question	`‹ 2/2 ›` under the question	`VersionNav`	`lib/useChat.branches.test.tsx`, `lib/threadTree.test.ts`	119
S87	Answer versions	Regenerate	`‹ 2/2 ›` in the action row	`VersionNav`	`useChat.branches.test.tsx`	120
S88	Styled regeneration	More concise / More detailed	Version labelled "Concise" / "Detailed"	`AssistantTurn` (`regen-style`)	`contract/regenerate.test.ts`	121
S89	Ask to change response	Menu item	Inline input "Ask to change response…"	`answer/ChangeResponse.tsx`	`answer/messageActions.test.tsx`	122
S90	Jump to latest	Scroll up during/after an answer	Round ↓ button above the composer	`AppShell` (`jump-latest`)	`shell/turnSpacing.test.tsx`	123
S91	Long question	> 700 chars or > 9 lines	Bubble clipped with fade + "Show more"	`UserText`	`components/userBubble.test.tsx`	124
S92	Keyboard focus	Tab / Shift+Tab	Visible focus ring	all	`shell/focus.test.tsx`, `e2e/shell-focus.spec.ts`, `e2e/a11y.spec.ts`	125
S93	Phone empty	< 768 px	Sidebar button, swipeable cards, composer	`AppShell`, `StarterCards`	`e2e/visual.spec.ts` (mobile)	130
S94	Phone drawer	Open sidebar / edge swipe	Drawer + scrim	`Sidebar` (drawer), `shell/swipe.ts`	`components/sidebarTouch.test.tsx`, `e2e/shell-swipe.spec.ts`	131
S95	Phone answer / sheet	Answer, expand table	Answer full width; panel as full-screen sheet	`ArtifactPanel`	`e2e/visual.spec.ts`	132–134
S96	Tablet	820 px	Sidebar + thread, chart below table	—	—	135
S97	Themes	Settings / `/dark` / OS	Neutral or Paper × light or dark	`lib/theme.tsx`	`lib/themeSync.test.tsx`	136–138

ID	State	Trigger	What the user sees	Component(s)	Tests	Screenshot
S98	Temporary chat	Toggle on an empty chat (memory store)	"Temporary chat" heading and note; not listed, searched or titled	`Header`, `shell/temporaryStore.ts`	`shell/temporary.test.tsx`	139, 140
S99	Markdown / code	Prose with quote, table, list, link, fenced code	Rendered GFM, Shiki code block with language label and Copy	`Markdown`, `CodeBlock`	`render/render.test.tsx`, `parts/codeChrome.test.tsx`	141
S100	Math (KaTeX)	`$…$` or `$$…$$` in prose	**Locally: whole-app crash** (see §21)	`Markdown` + `@streamdown/math`	`parts/math.test.tsx`, `parts/displayMath.test.tsx` (jsdom)	P-math
S101	Large table	> 200 rows	Virtualized rows, sticky header, "Showing 60 of 600 rows · Open full table"	`DataTable` (TanStack Virtual)	`parts/tableEdge.test.tsx`	142
S102	Image lightbox	Expand image	Full-screen dialog, Close	`ImageView`	`render/render.test.tsx`	P-lightbox
S103	Branch in new chat	More actions → Branch	New chat with the thread up to that answer, toast "Branched into a new chat"	`AppShell`	`shell/branch.test.tsx`	P-branch
S104	Print one answer	More actions → Print	Print view with question + answer only	`printOne`, `index.css` print rules	`printCss.test.ts`, `e2e/print.spec.ts`	P-print
S105	Signed out in another tab	Sign out elsewhere	This tab leaves to the signed-out page	`auth/session.ts` (BroadcastChannel)	`auth/crossTabSignOut.test.ts`	P-cross-tab
S106	Thumbs up / copied	Click thumbs up / Copy	Filled thumb; toast "Copied as Markdown" / "SQL copied"	`AssistantTurn`	`answer/actionRowFidelity.test.tsx`	*TODO*
S107	History gone (410)	Customer key destroyed	"Chat unavailable · This chat is no longer available."	`httpError`	`adapters/httpStore.test.ts`	P-410

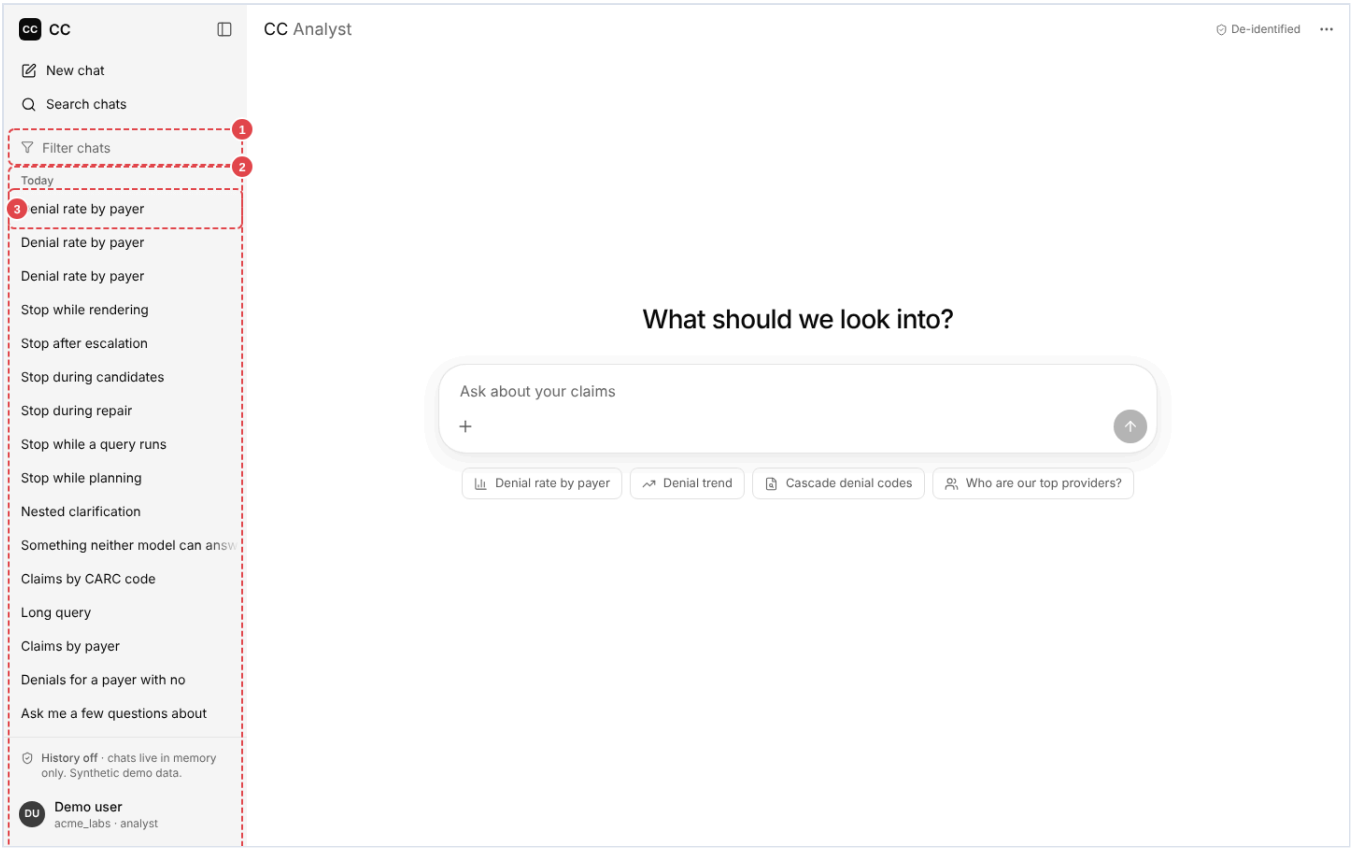
TODO: review the inventory with QA; add any state missing from the e2e suites.

---

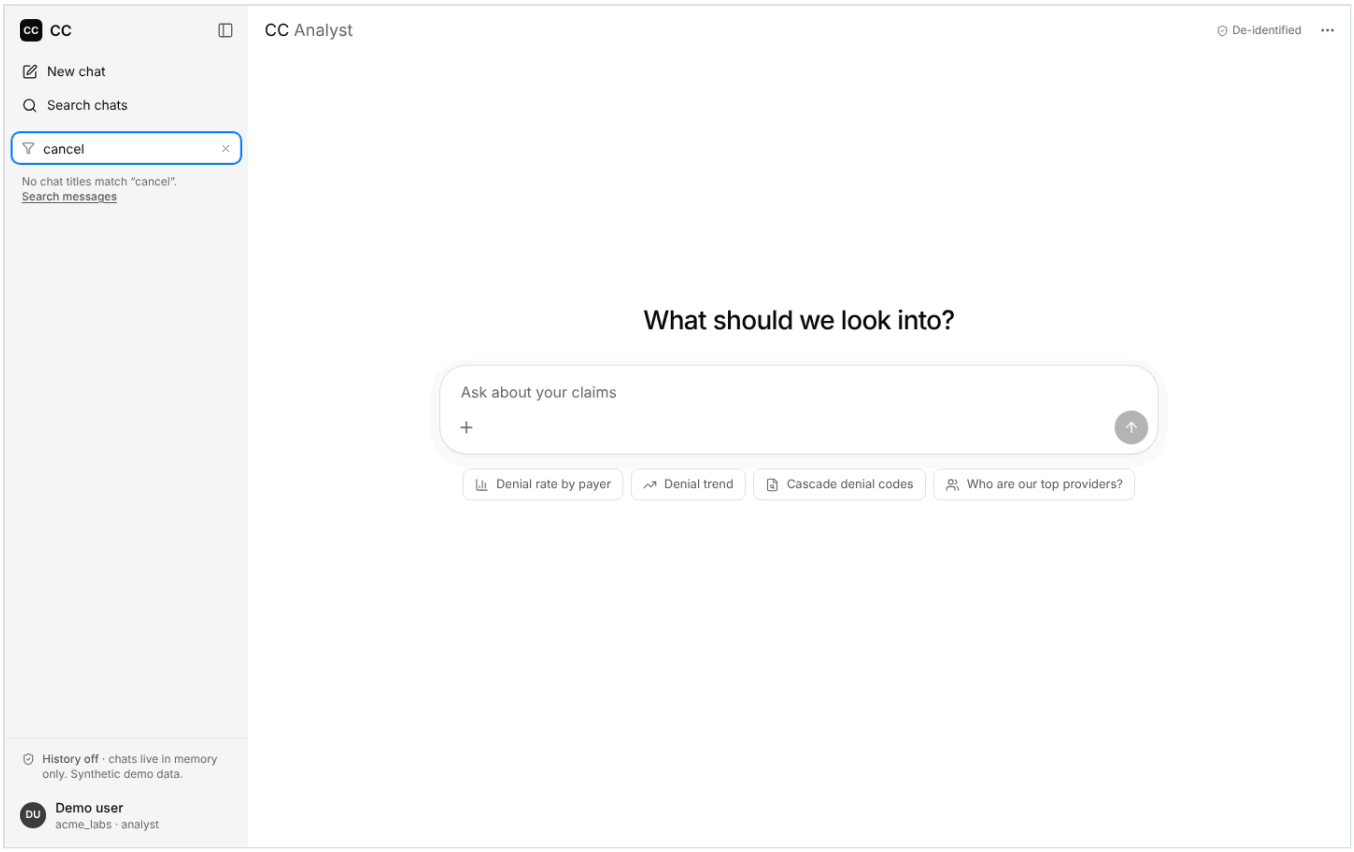
6. Sidebar & history

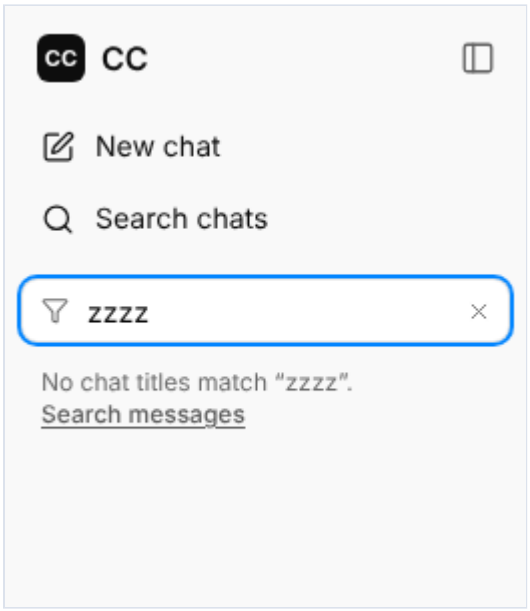
TODO: one paragraph: what the sidebar is for, where history lives (server `HttpTranscriptStore` with the live API, memory `TemporaryAwareStore` with the mock), and that titles never come from the question.

6.1 Thread list, buckets and filter

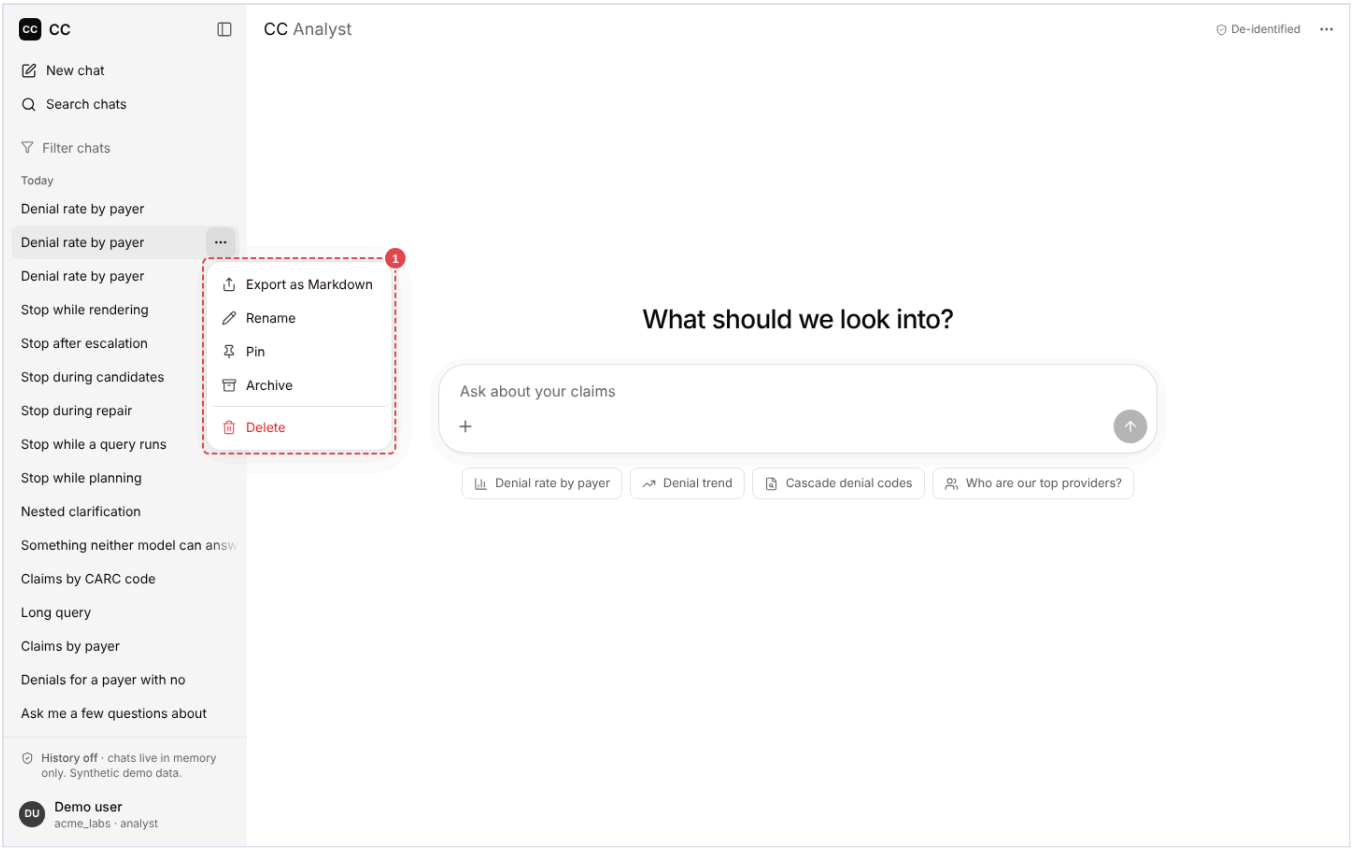


#	Element	Behaviour
1	Filter chats	Appears only at ≥ 20 chats (`FILTER_AT`). Case- and accent-insensitive title match; Esc clears, ↓ moves into the list, Enter opens the first hit. Titles never leave the page.
2	Date bucket	Pinned, Today, Yesterday, Previous 7 days, Previous 30 days, then month (this year) or year. Calendar days, DST-safe (`bucket()`).
3	Thread item	One tab stop for the list; ↑ ↓ Home End move (roving focus).

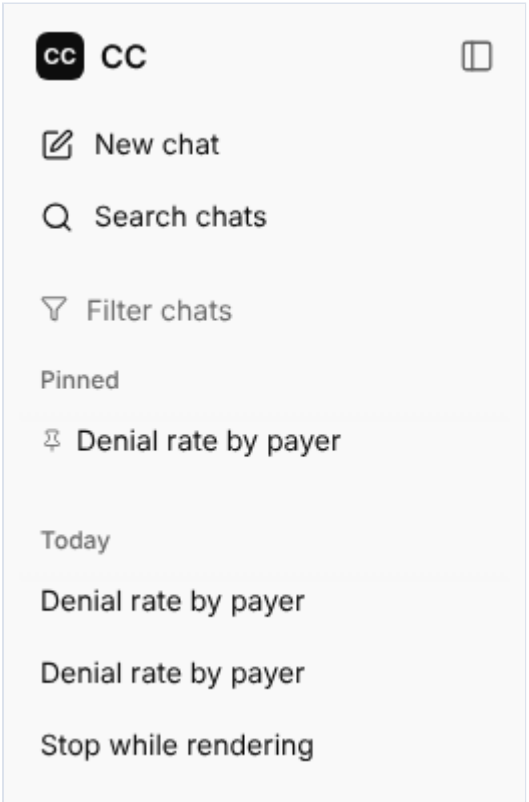

CC
New chat
Search chats
cancel
No chat titles match "cancel".
Search messages
History off · chats live in memory only. Synthetic demo data.
Demo user
acme_labs · analyst
CC Analyst
De-identified
What should we look into?
Ask about your claims
Denial rate by payer
Denial trend
Cascade denial codes
Who are our top providers?

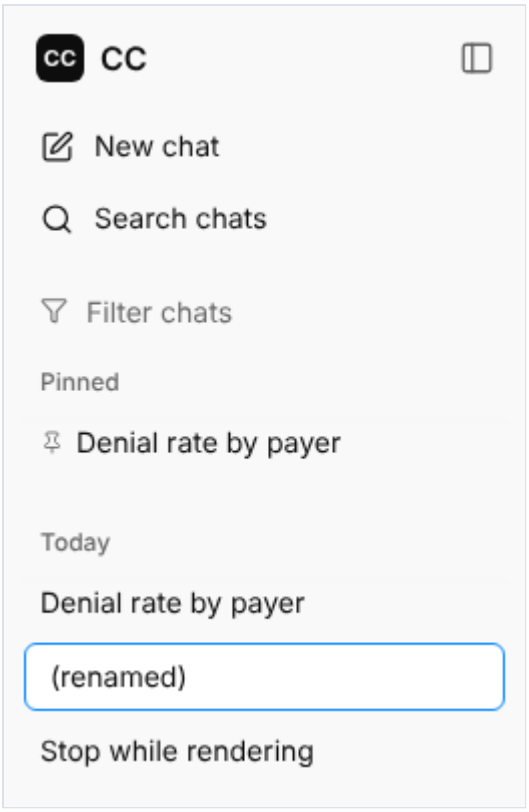

CC
New chat
Search chats
zzzz
No chat titles match "zzzz".
Search messages

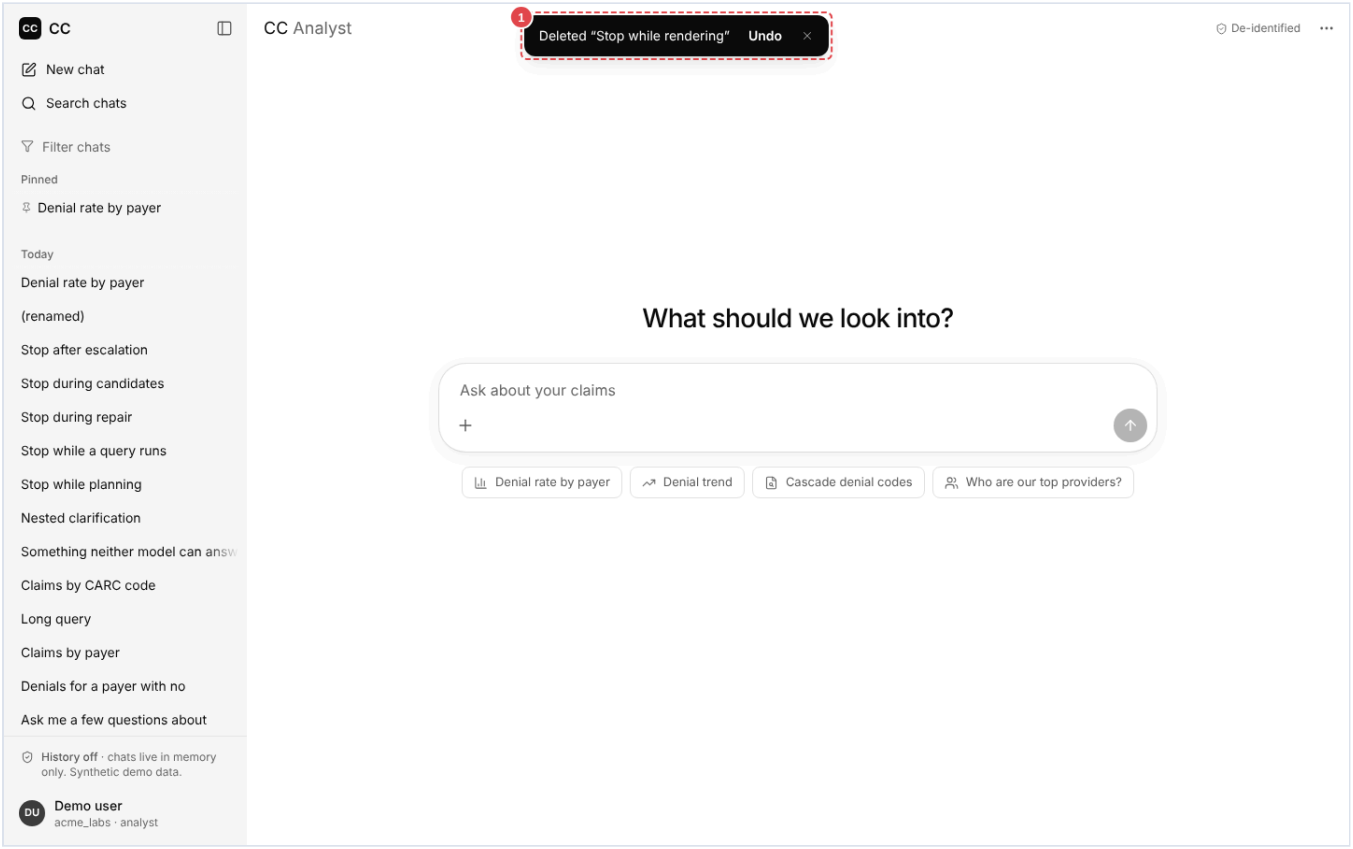
6.2 Thread actions



#	Element	Behaviour
1	Thread menu	Opens from "…" (hover/focus) or right-click. Pin/Archive appear only when the store supports them (`Capabilities.pinArchive` , contract 1.17.0).



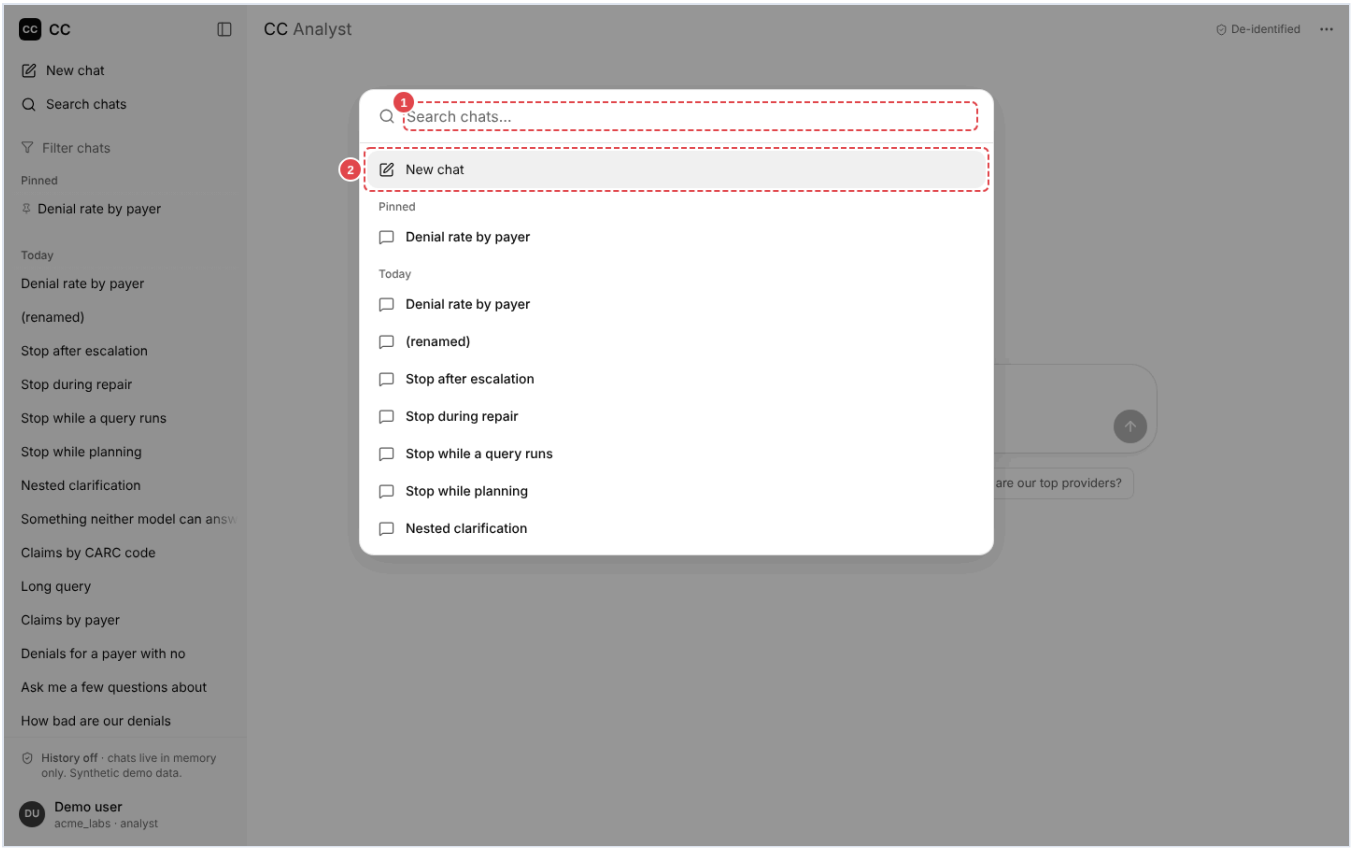




#	Element	Behaviour
1	Toast	"Deleted "<title>"" + Undo for 5 s (`UNDO_MS`). Archive shows the same pattern; a server refusal shows "Couldn't archive that chat. Try again."

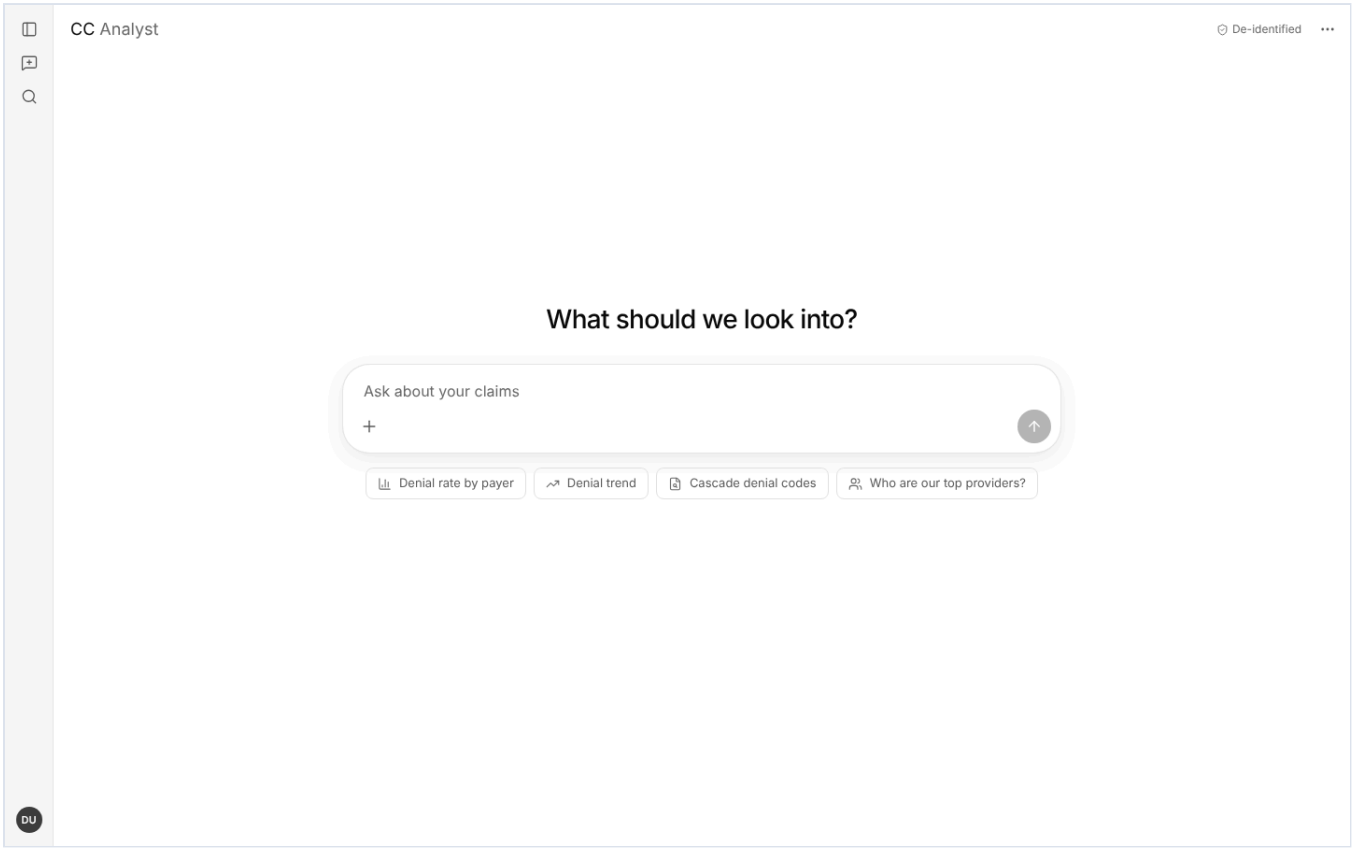
TODO: confirm whether delete is optimistic on the live API and what happens if the DELETE fails after Undo expires.

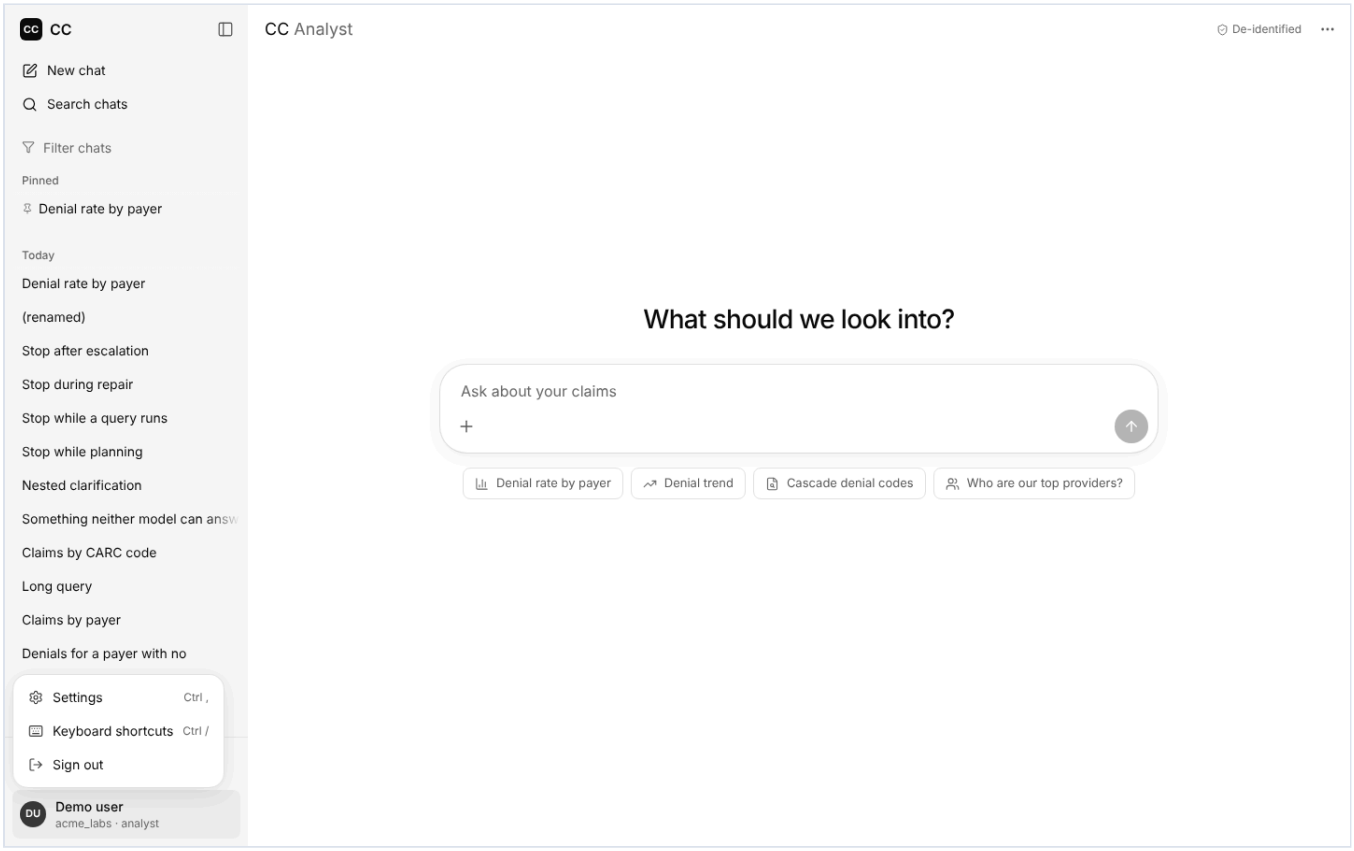
6.3 Search chats



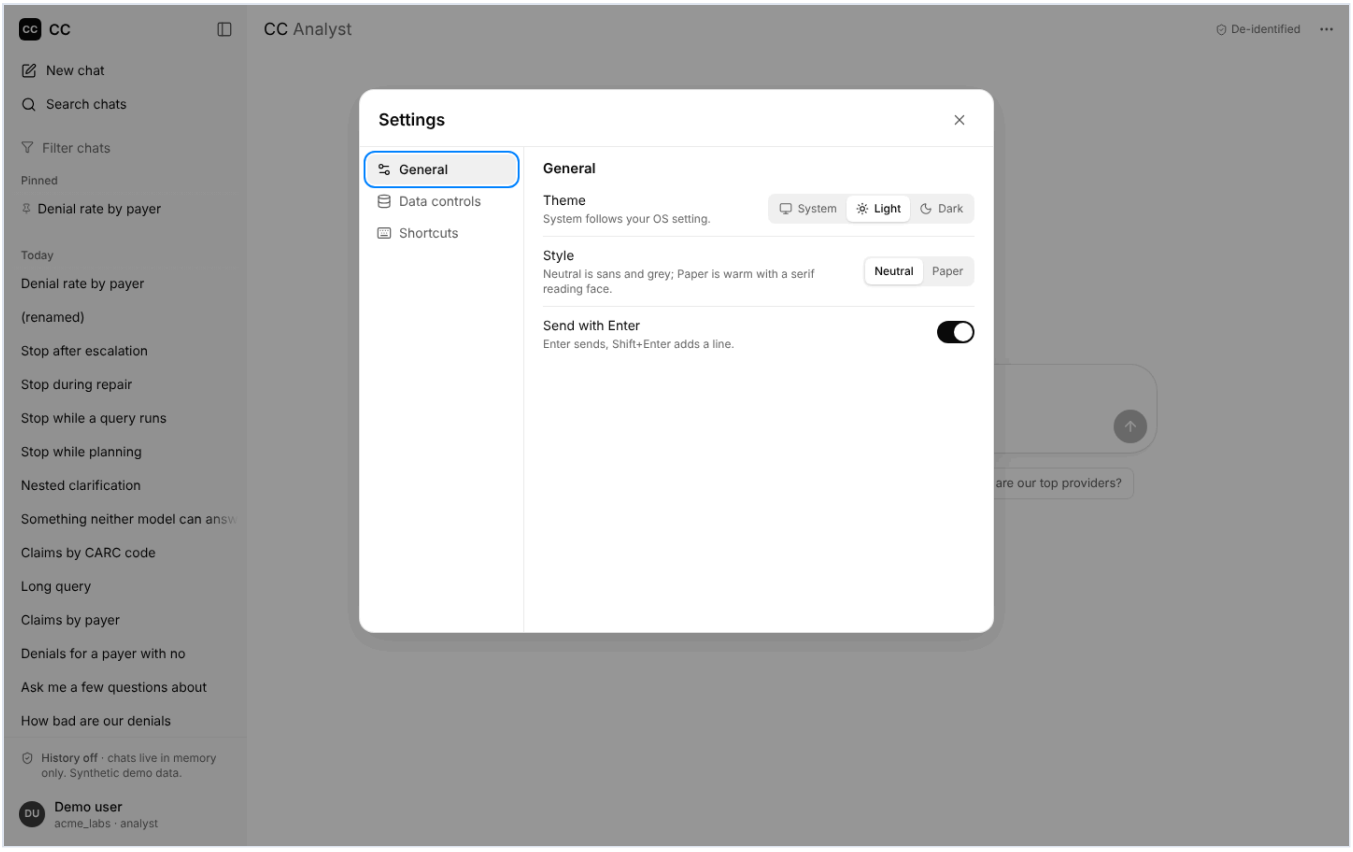
#	Element	Behaviour
1	Search box	Full-text over titles **and** message text. On the live API it is `POST /api/conversations/search` (POST so the text never sits in a URL).
2	Result	Opens the chat (and the matched message). Archived chats are included.
3	Match highlight	`search-mark` around the hit in the snippet. *(Not captured: the demo store returned title-only hits.)*

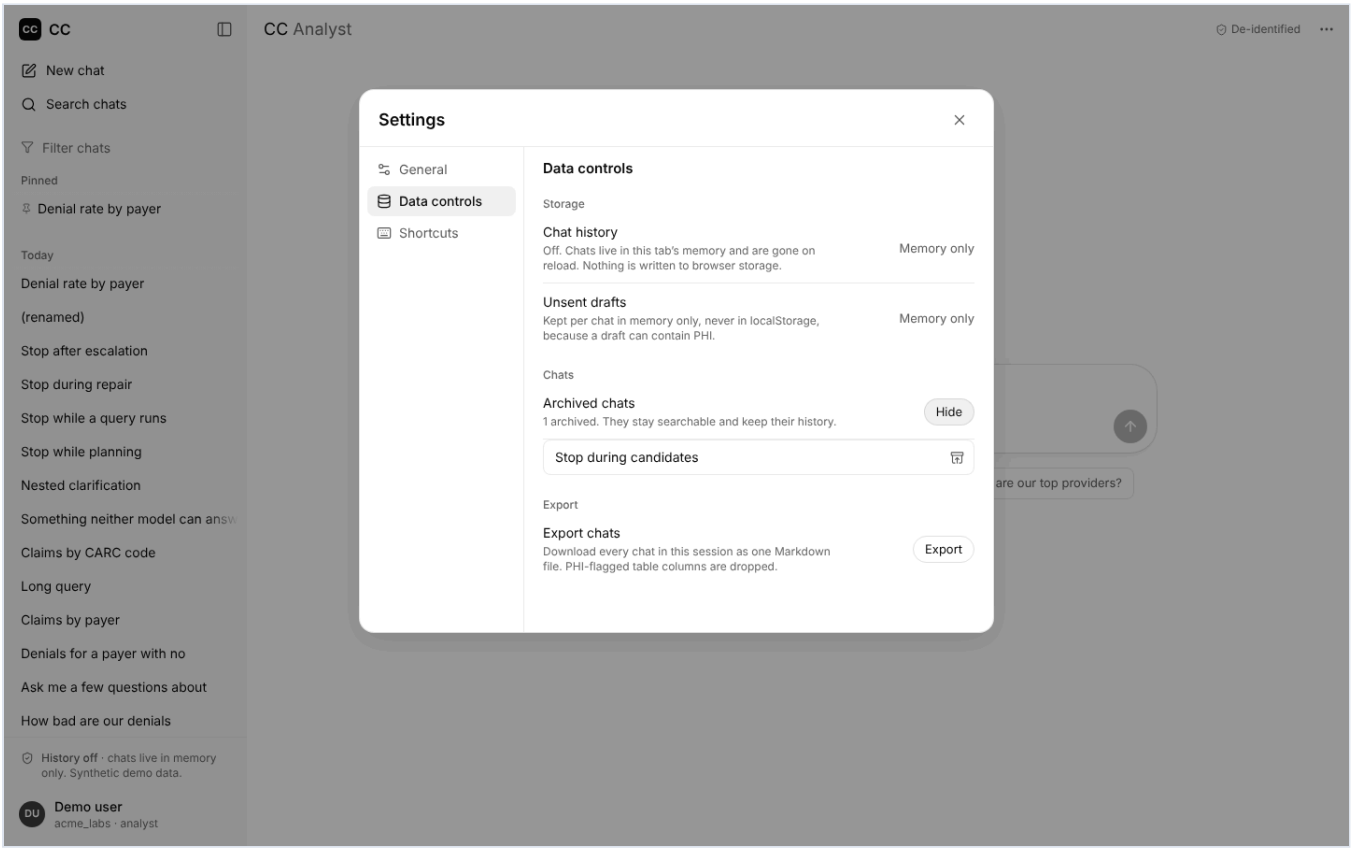
6.4 Collapsed rail and account

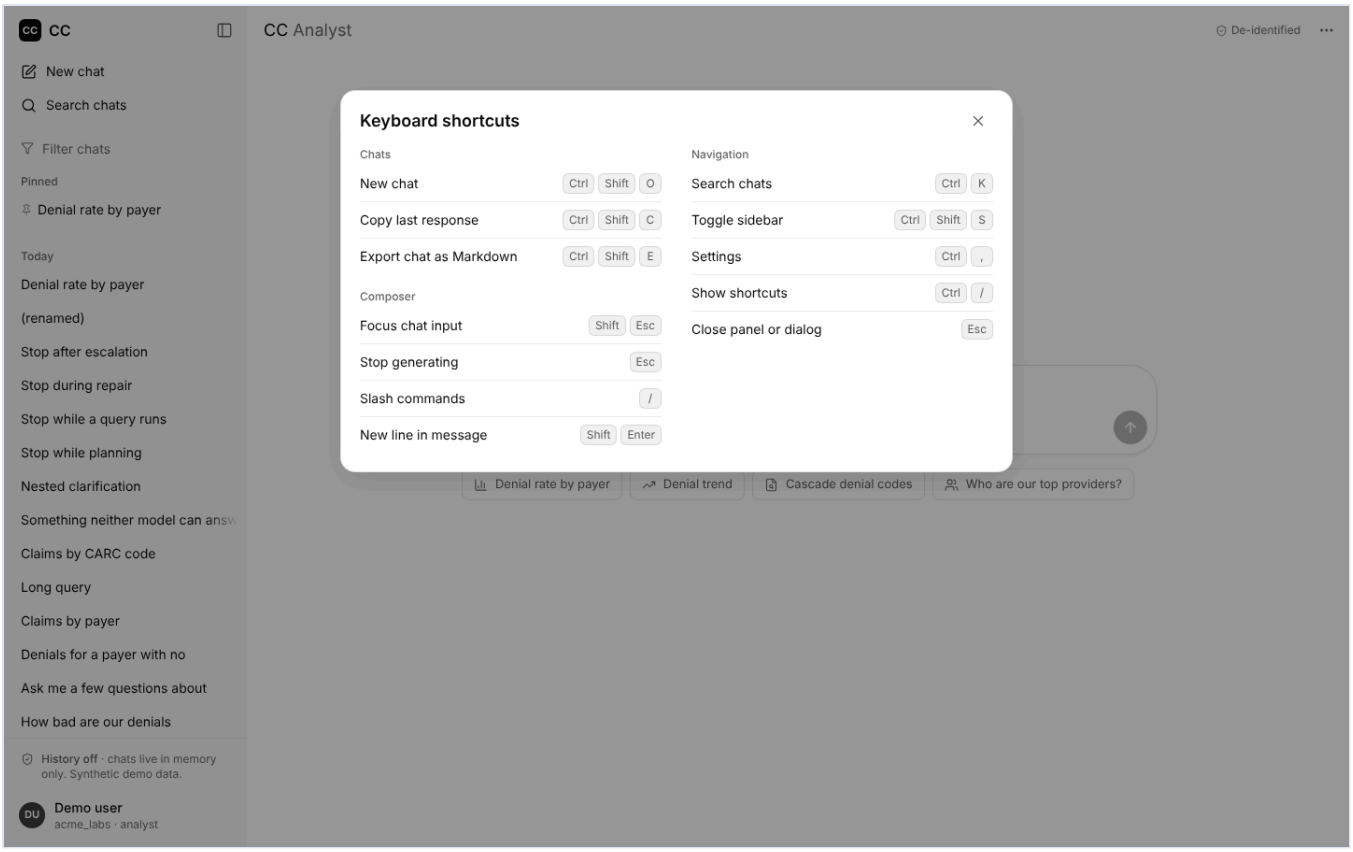




6.5 Settings, archived chats, shortcuts

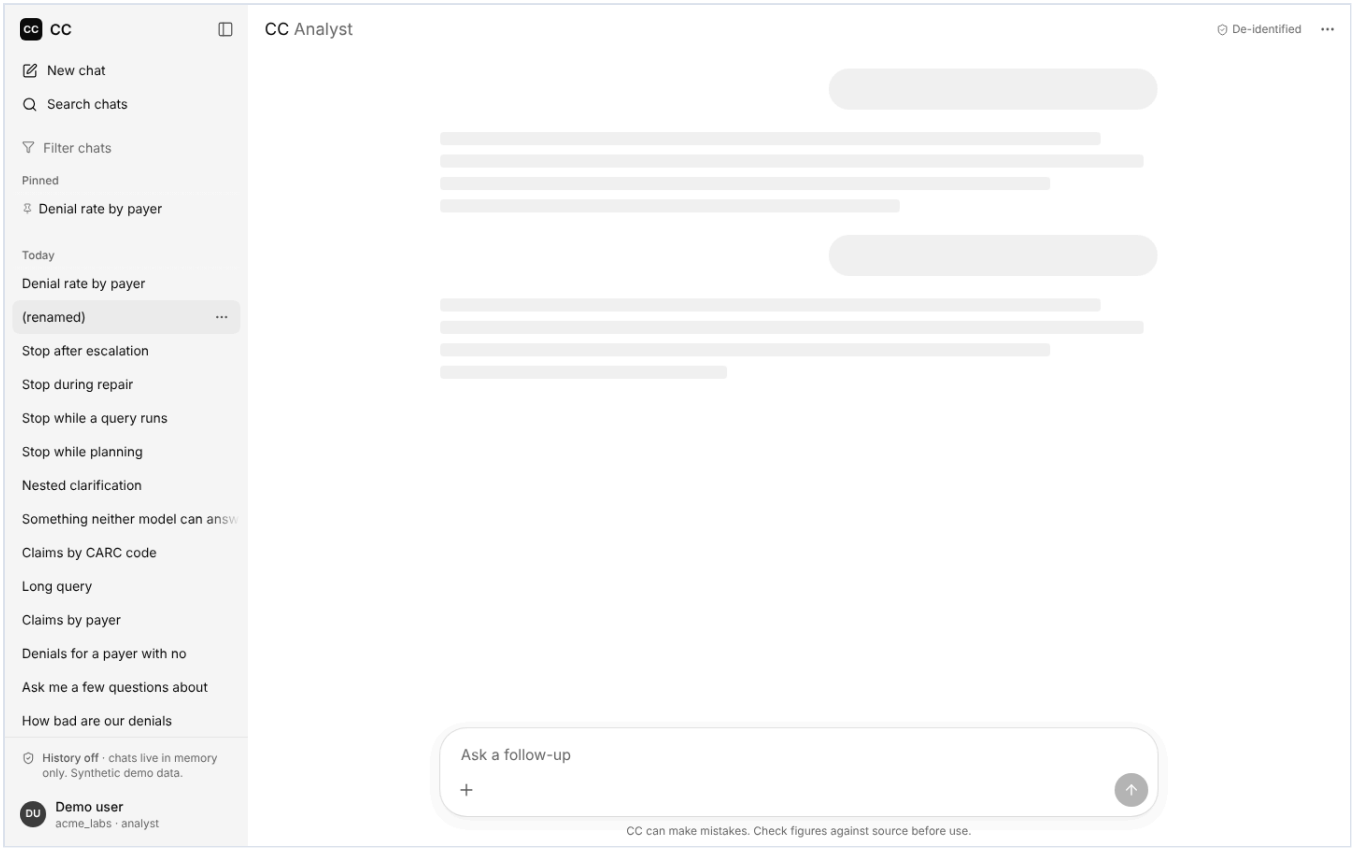


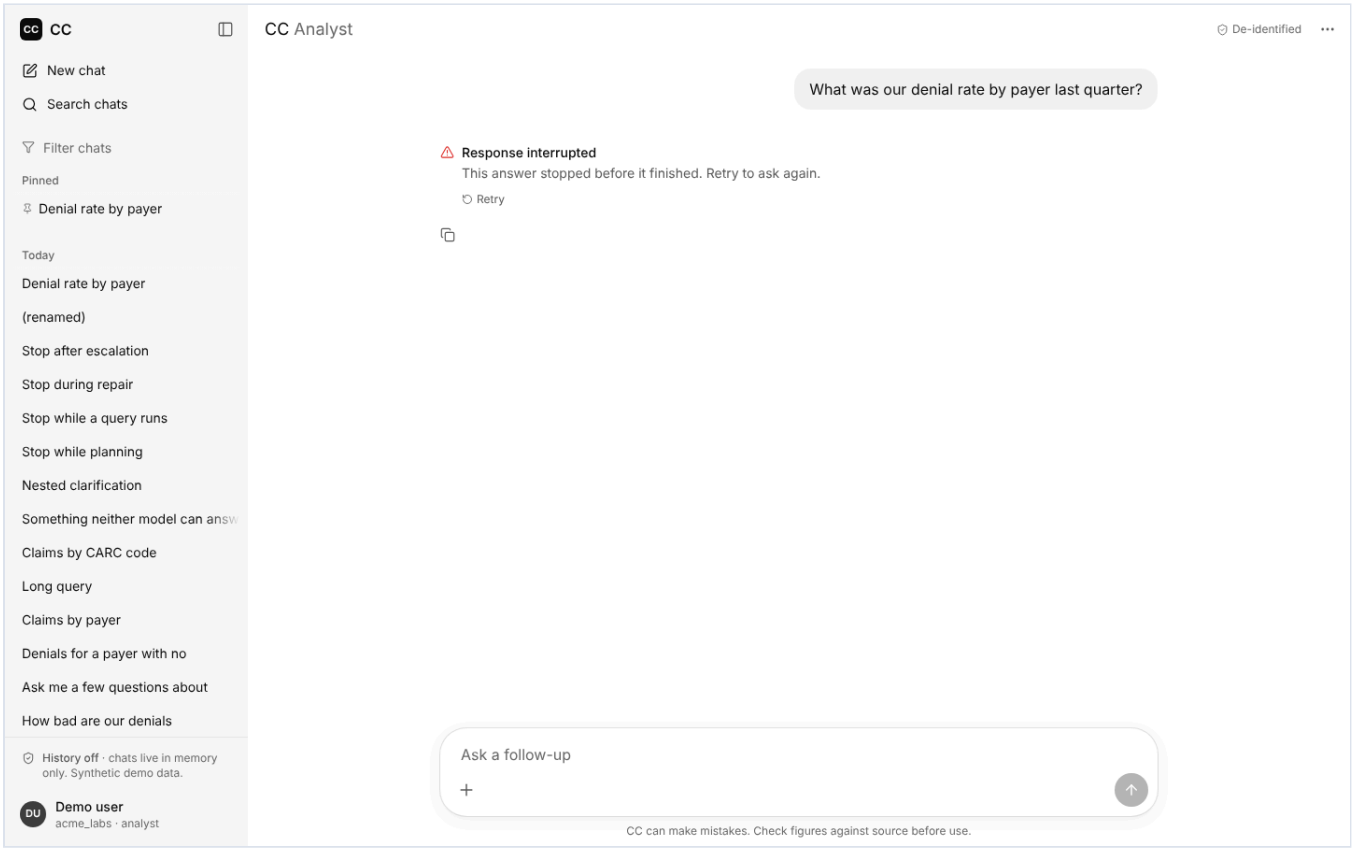




Shortcut	Action (from `shell/shortcuts.ts`)
`Ctrl/⌘ Shift O`	New chat
`Ctrl/⌘ K`	Search chats
`Ctrl/⌘ Shift S`	Toggle sidebar
`Shift Esc`	Focus chat input
`Ctrl/⌘ Shift C`	Copy last response
`Ctrl/⌘ Shift E`	Export chat as Markdown
`Ctrl/⌘ ,`	Settings
`Ctrl/⌘ /`	Show shortcuts
`Esc`	Stop generating · close panel or dialog
`/`	Slash commands
`Shift Enter`	New line in message
`↑` in an empty composer	Edit the last question

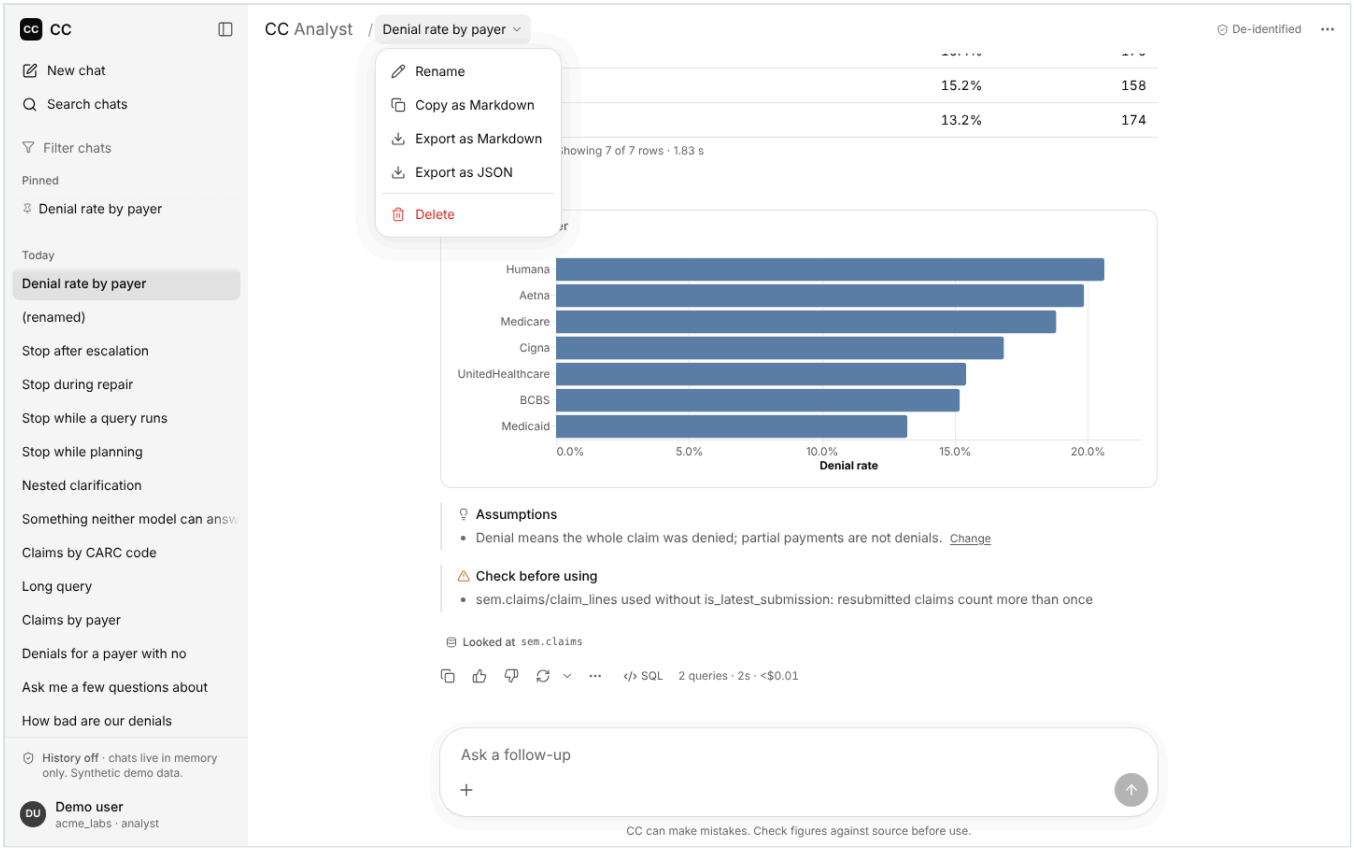
6.6 Opening a stored chat

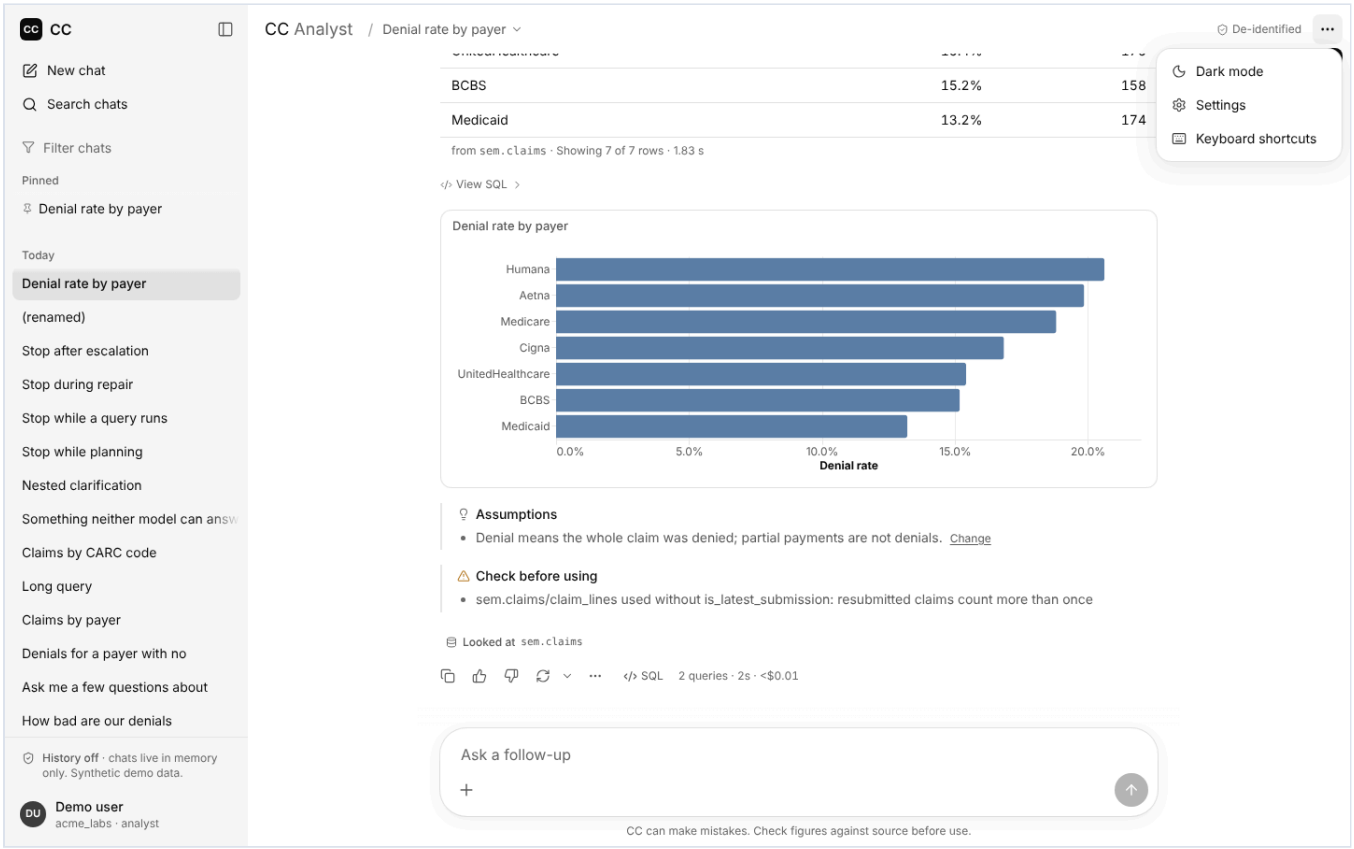




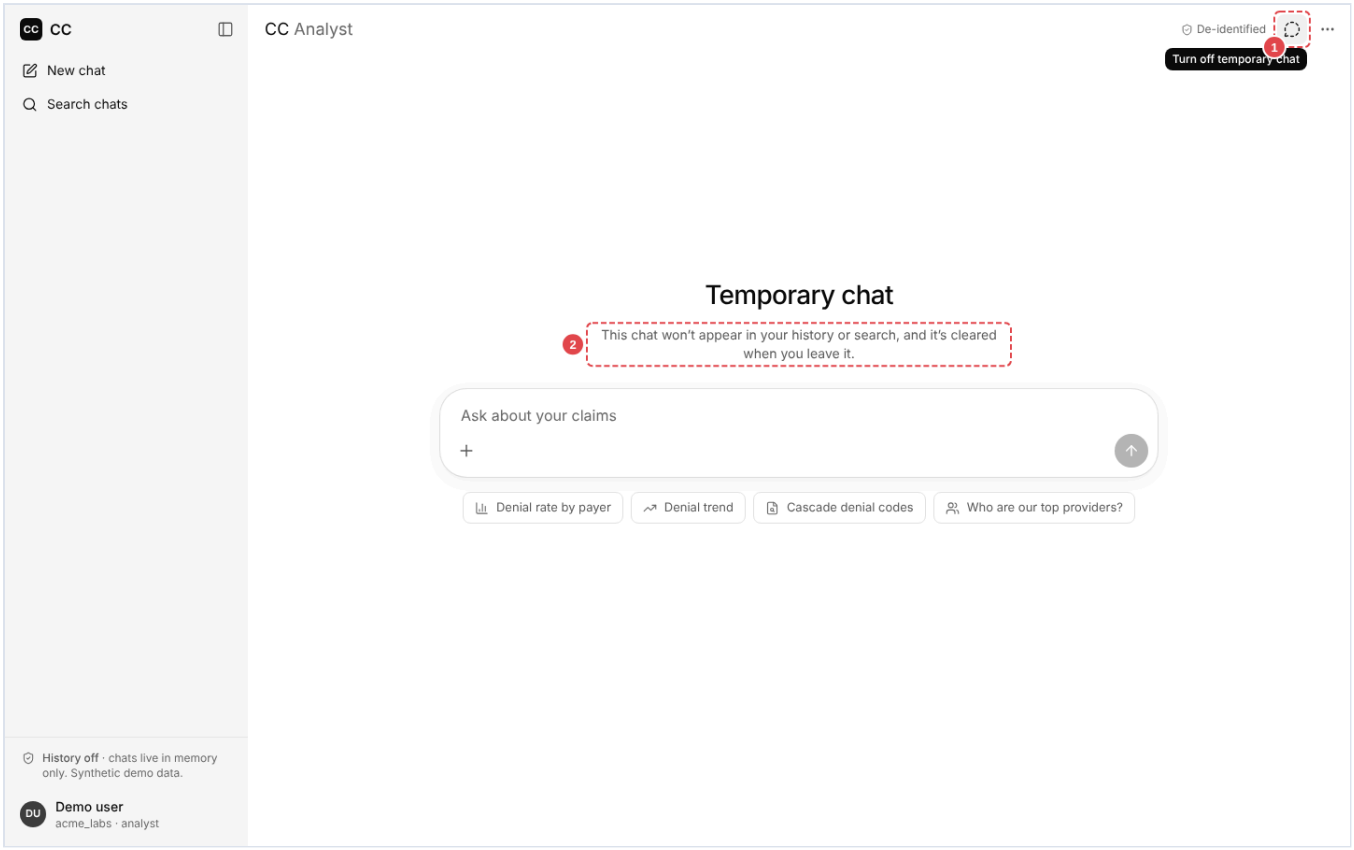
TODO: explain the disconnect grace (`CC_API_DISCONNECT_GRACE_SECONDS`, 10 s) and why a reload mid-answer usually reattaches instead of showing S84.

6.7 Header title menu and more menu

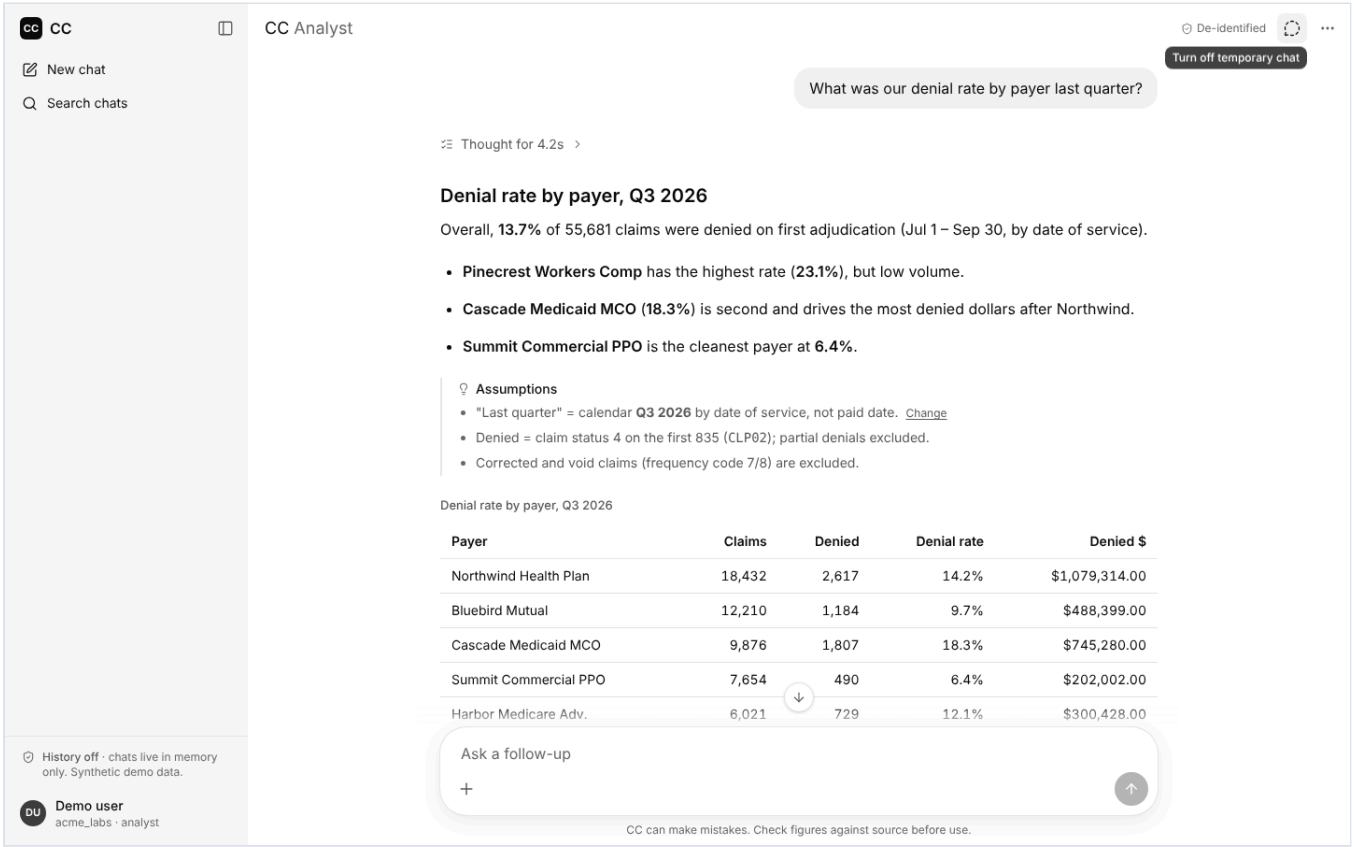




6.8 Temporary chat (memory store only)



#	Element	Behaviour
1	Temporary toggle	Shown only with the memory store, and only on an empty chat or while temporary is on.
2	Note	"This chat won't appear in your history or search, and it's cleared when you leave it."

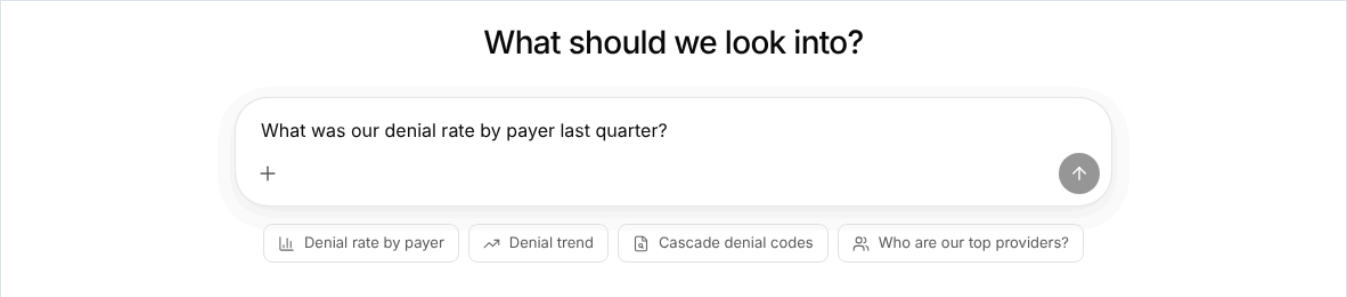


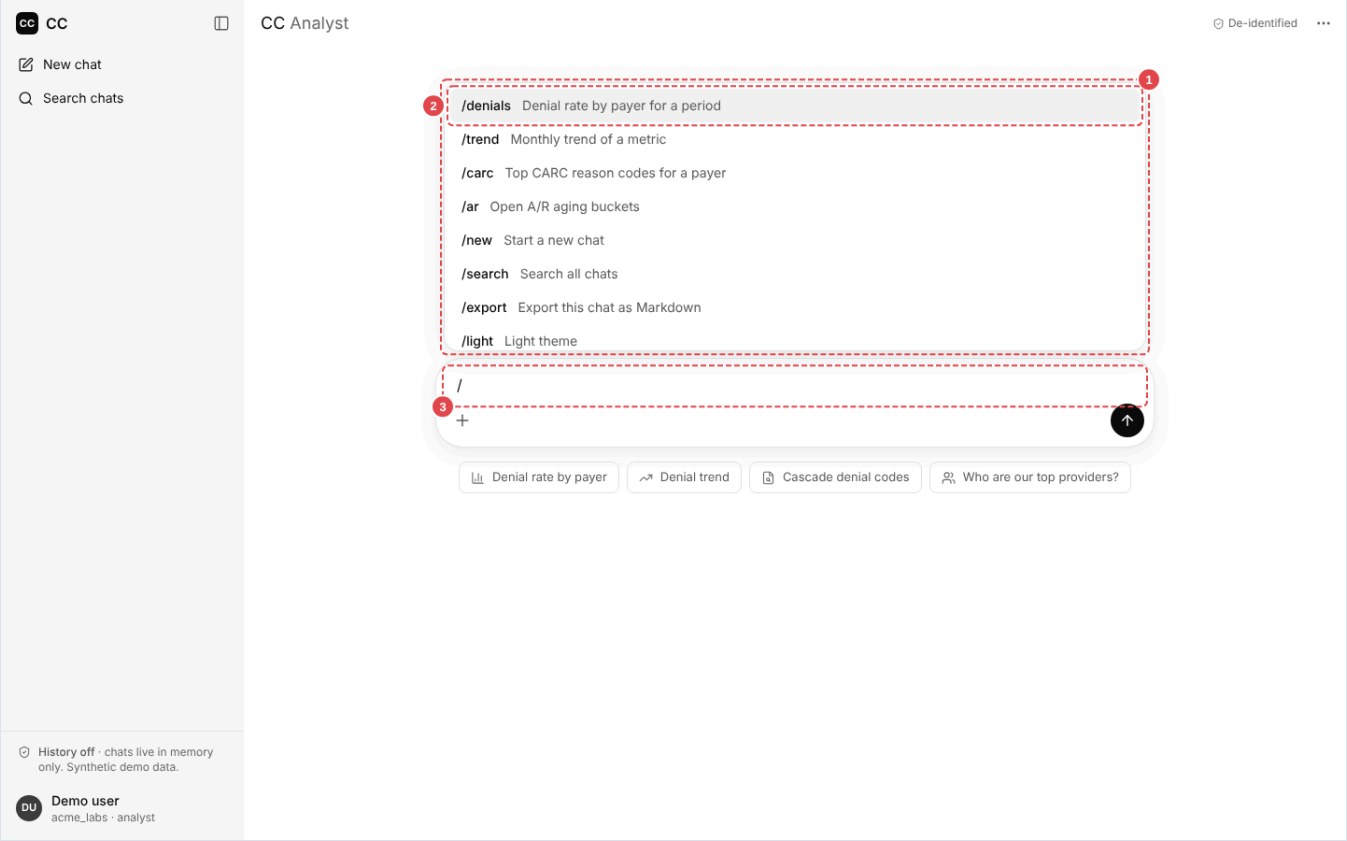
> **Note:** With the live API the toggle is hidden today (`tempStore` is null for `HttpTranscriptStore`) even though the server announces `Capabilities.temporaryChat` . *TODO: confirm whether that is intended.*

---

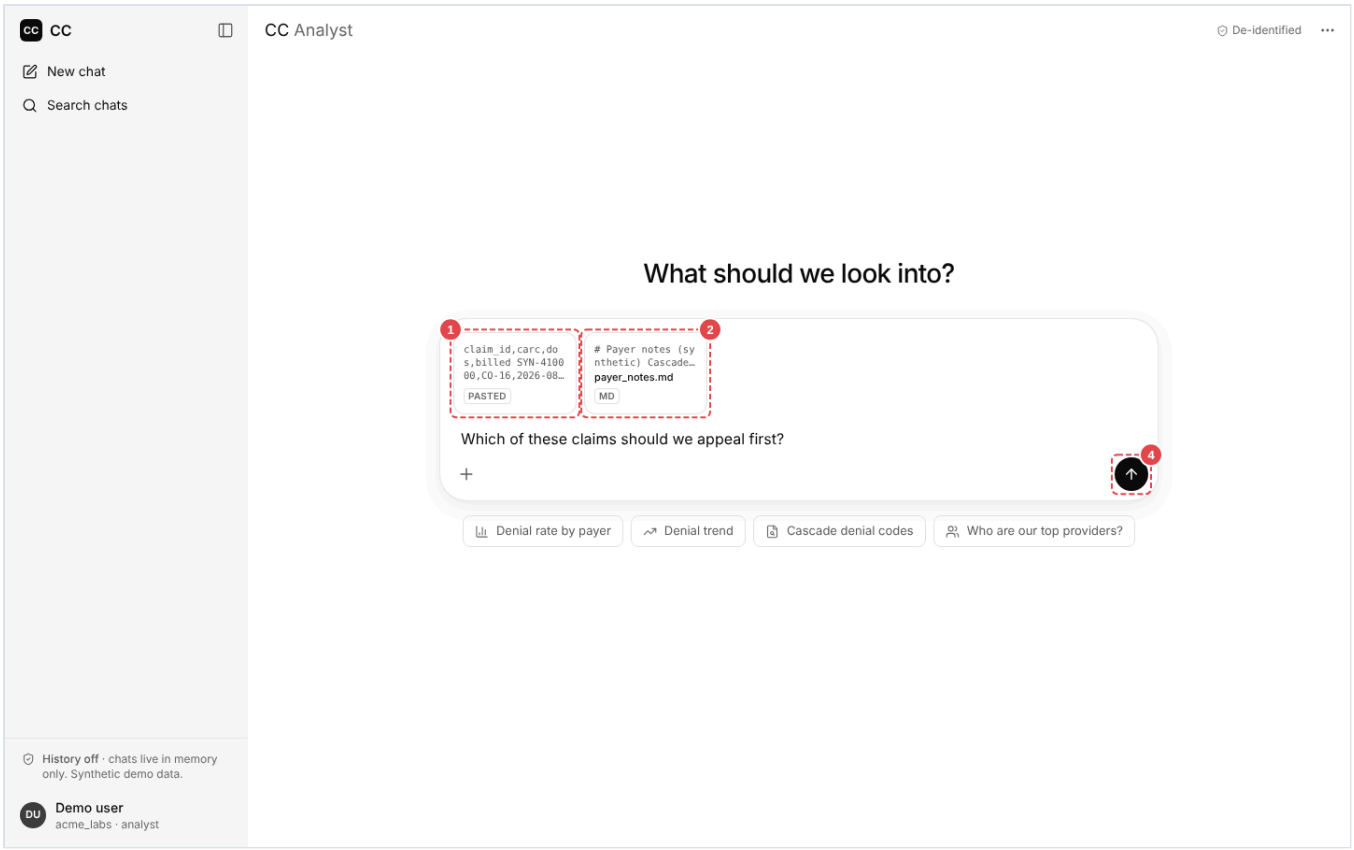
7. Composer

TODO: one paragraph: the composer is the only input; drafts are kept per chat in memory only (never `localStorage` , a draft can hold PHI).





#	Element	Behaviour
1	Command list	`/denials` , `/trend` , `/carc` , `/ar` insert a prompt; `/new` , `/search` , `/export` , `/light` , `/dark` , `/system` , `/settings` , `/shortcuts` run an action.
2	Active option	↑ ↓ move, Tab or Enter runs, Esc dismisses (`aria-activedescendant`).
3	Textarea	Combobox; the menu shows only while the text is `/word` .



#	Element	Behaviour
1	Pasted chip	A paste of ≥ 1200 characters or > 15 lines becomes a chip instead of flooding the box.
2	File chip	Text file read locally; name and extension shown.
4	Send	Sends text + chips joined as one user message (`name:\ncontent`).

TODO screenshot: attachment rejected — attach a .png or a file over 256 KB; the red notice "scan.png isn't a text file" appears above the textarea

TODO screenshot: drag-over — drag a file onto the composer; it gets a focus-colour ring (data-dragging)

State	Send button	Placeholder	Notes
Idle, empty	Disabled (30 % opacity)	"Ask about your claims" / "Ask a follow-up"	
Typing	Enabled		Enter sends, Shift+Enter new line (or Ctrl/⌘+Enter when "Send with Enter" is off); IME composition never sends
Clarification open	Enabled	"Reply…"	The text resumes the run instead of starting a new turn
Streaming	Becomes **Stop** (Esc)		

State	Send button	Placeholder	Notes
Offline	Disabled, tooltip "You're offline"		Draft stays editable (§13)

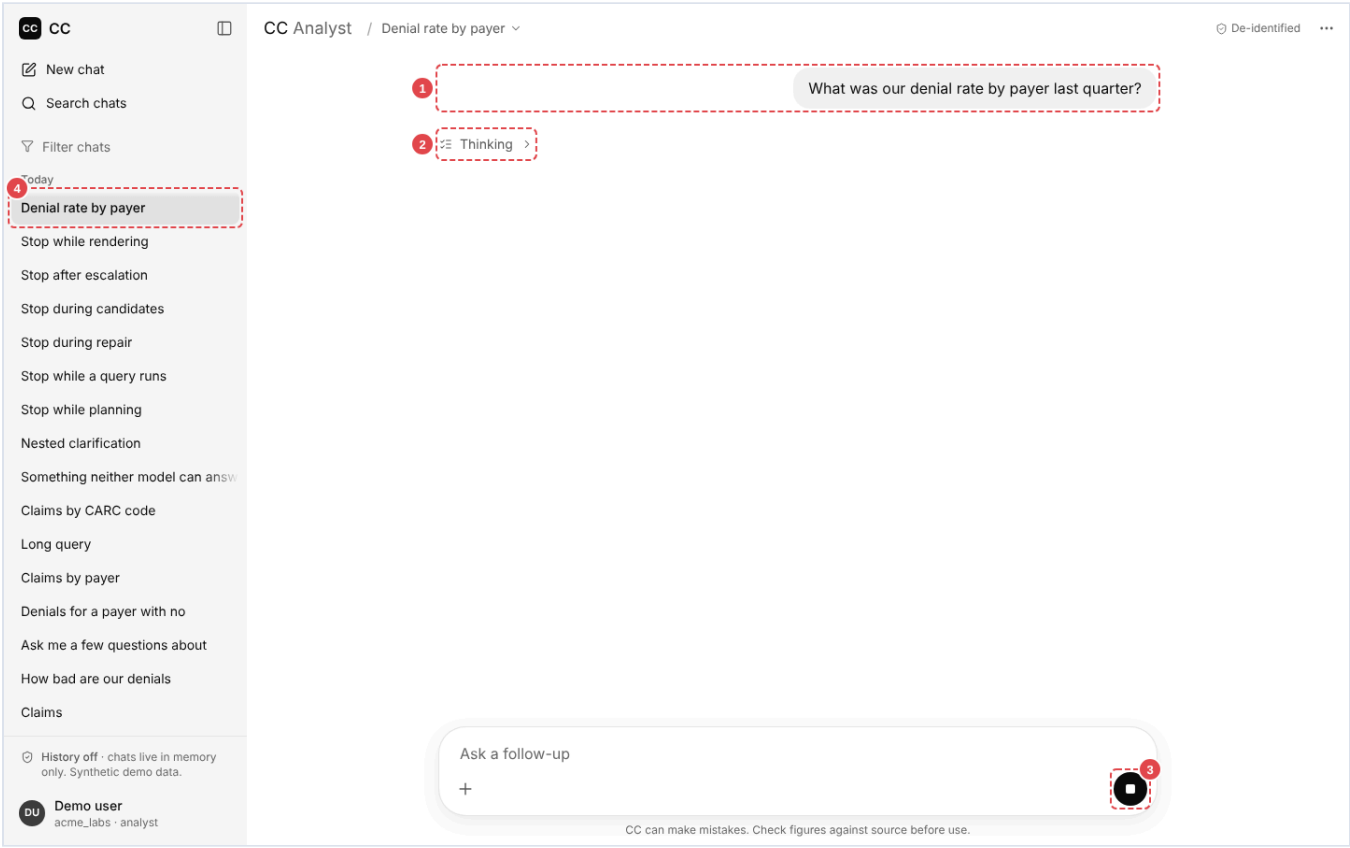
TODO: document the boot shell (`lib/bootShell.ts`): text typed before JS loads is adopted as the new-chat draft.

---

8. Agent turn: thinking, streaming and steps

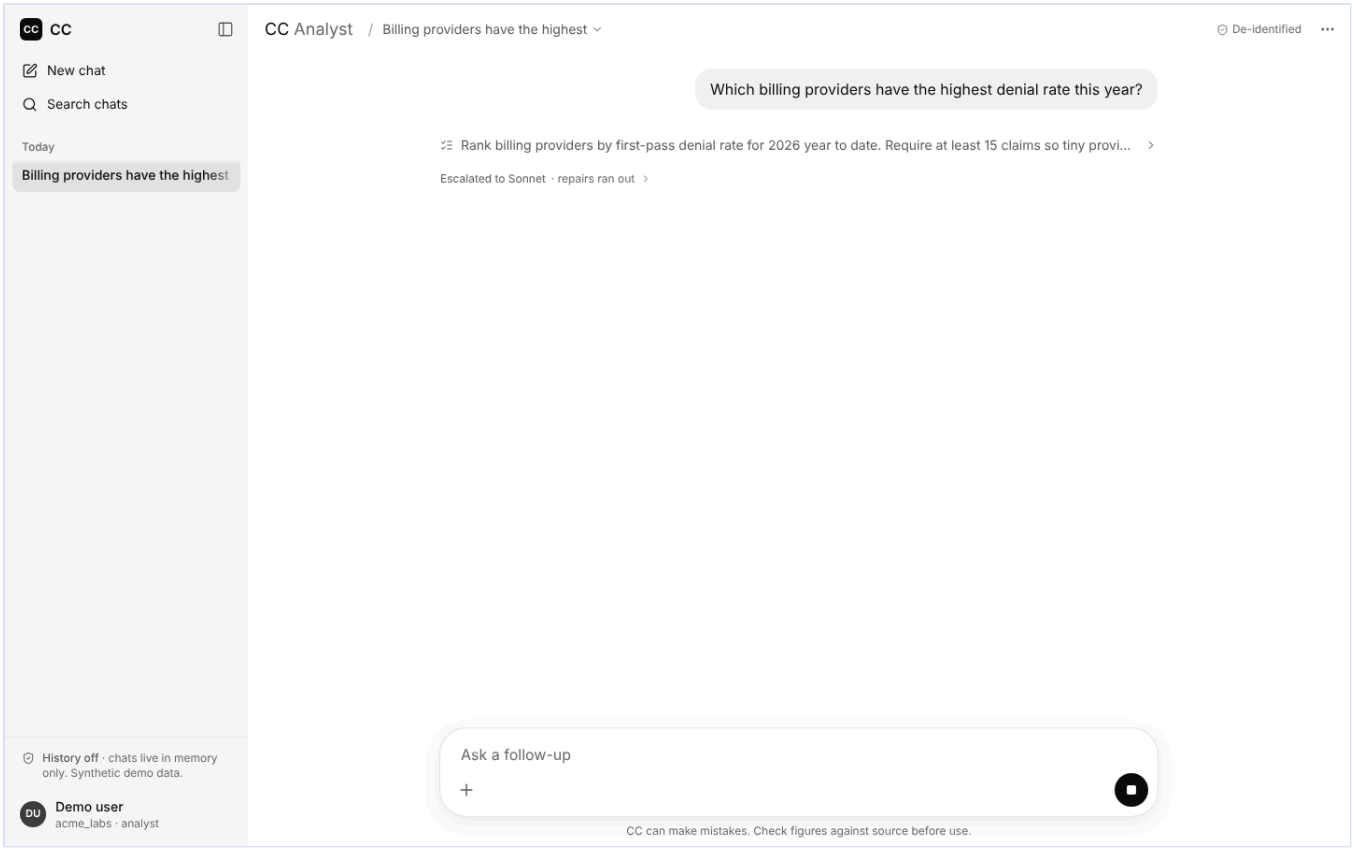
TODO: intro: one collapsed row covers the whole think-and-query phase (ChatGPT/Claude style), the answer appears below it.

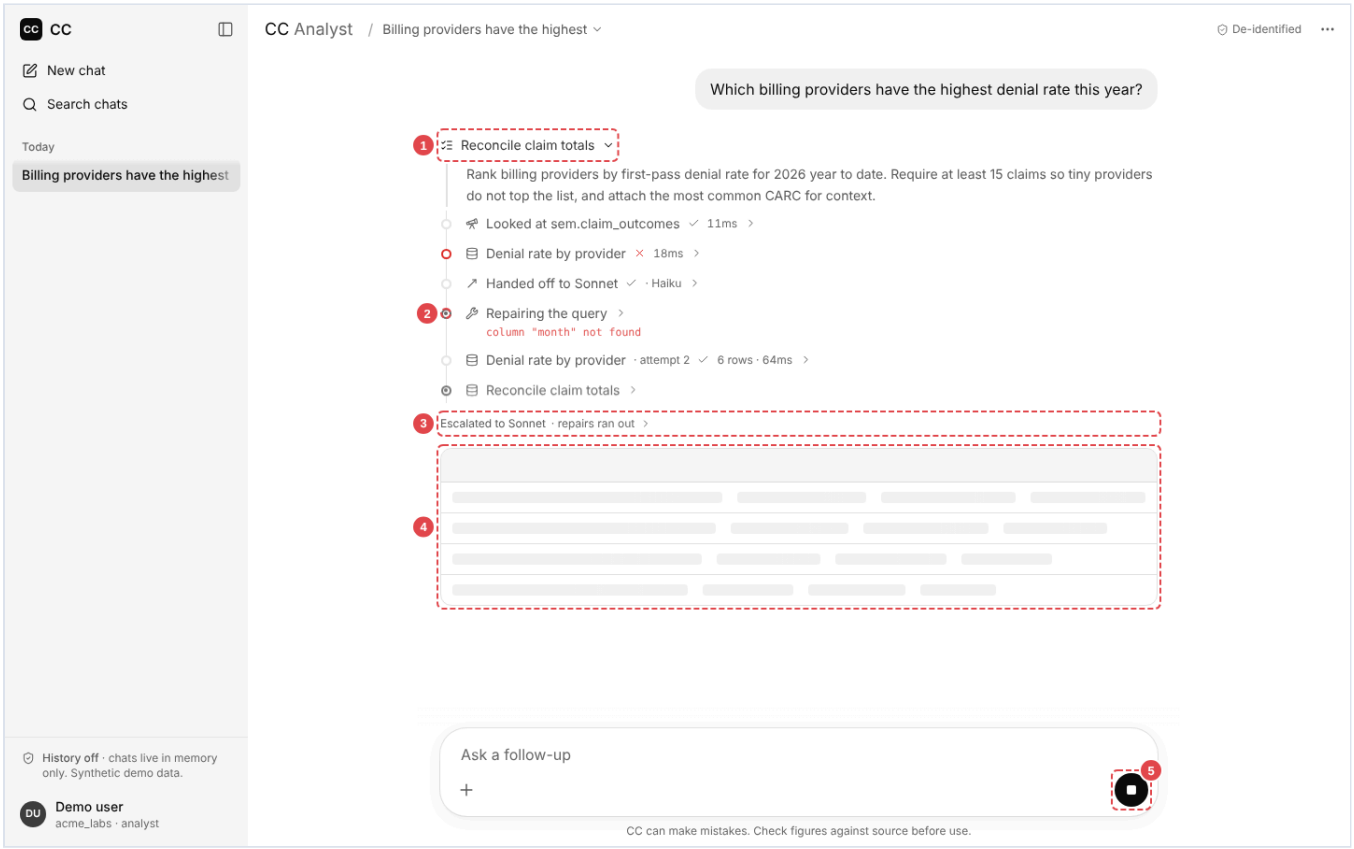
8.1 Sending / Thinking



#	Element	Behaviour
1	Question	Added at once; the follow-up is pinned to the top of the viewport.
2	Thinking row	Shown from the moment of sending, before `message.start` (QA-A15). Same element later carries the steps, so nothing jumps.
3	Stop	Aborts the fetch and calls `POST /api/chat/{messageId}/stop`.
4	Sidebar entry	New thread appears right away.

8.2 Live step line and expanded live steps

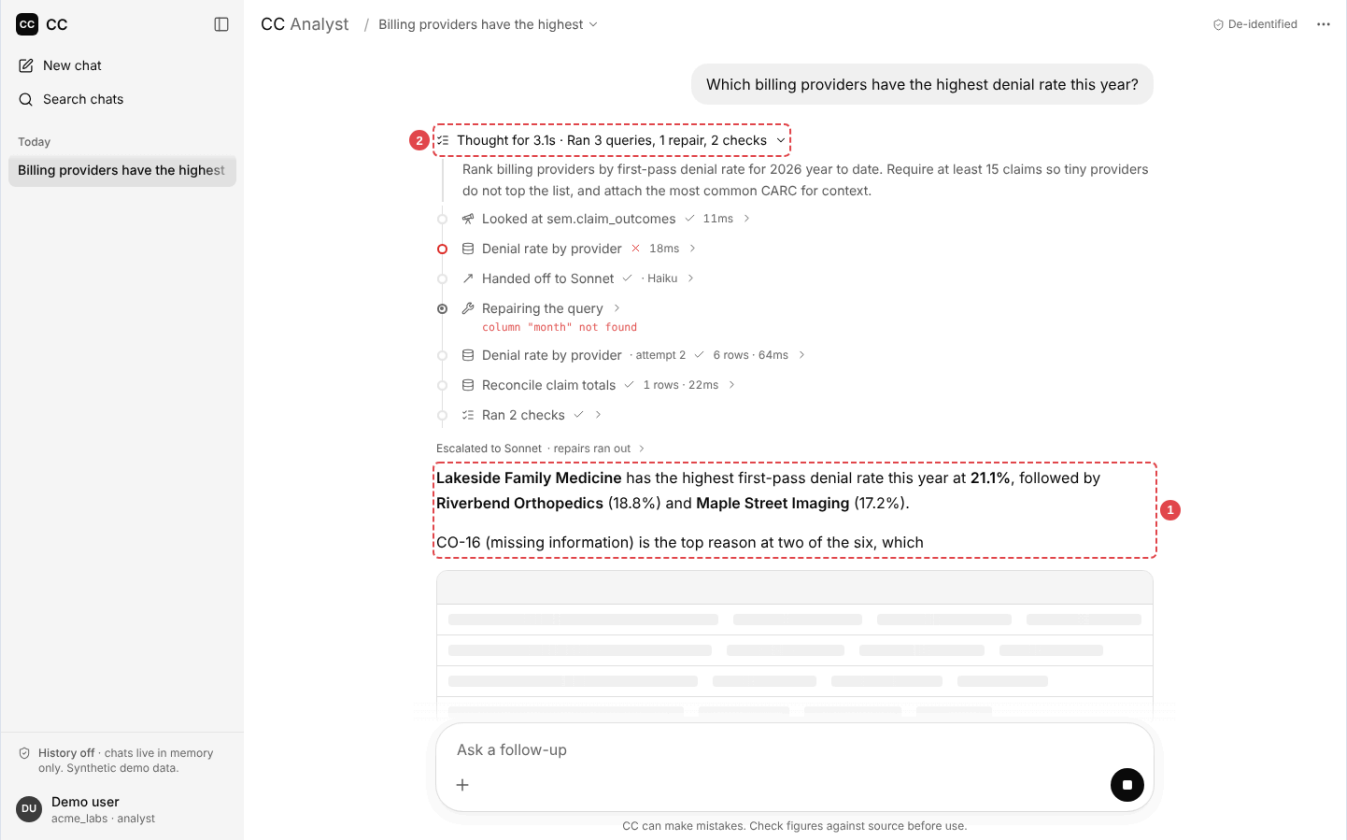




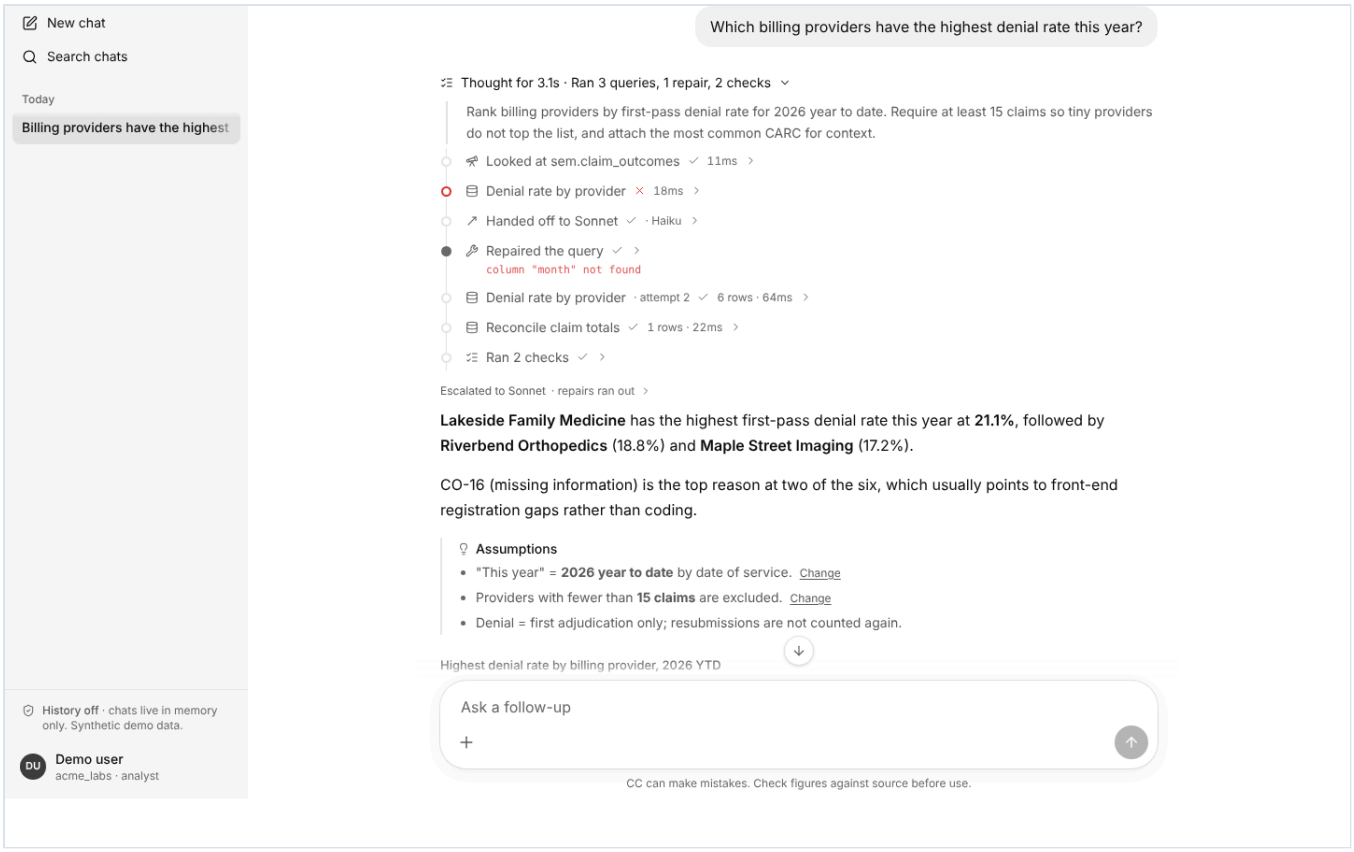
#	Element	Behaviour
1	Live header	Current step's label ("Reconcile claim totals"), no spinner or ticking timer; fades when it changes (off under reduced motion).

#	Element	Behaviour
2	Running step	Dot marks the step still running; it turns ✓ or ✕ in place (rows keyed by `stepId`).
3	Escalation note	"Escalated to Sonnet · repairs ran out" (`escalation` event).
4	Table skeleton	While a SQL/repair step runs and its table has not arrived.
5	Stop	Available throughout.

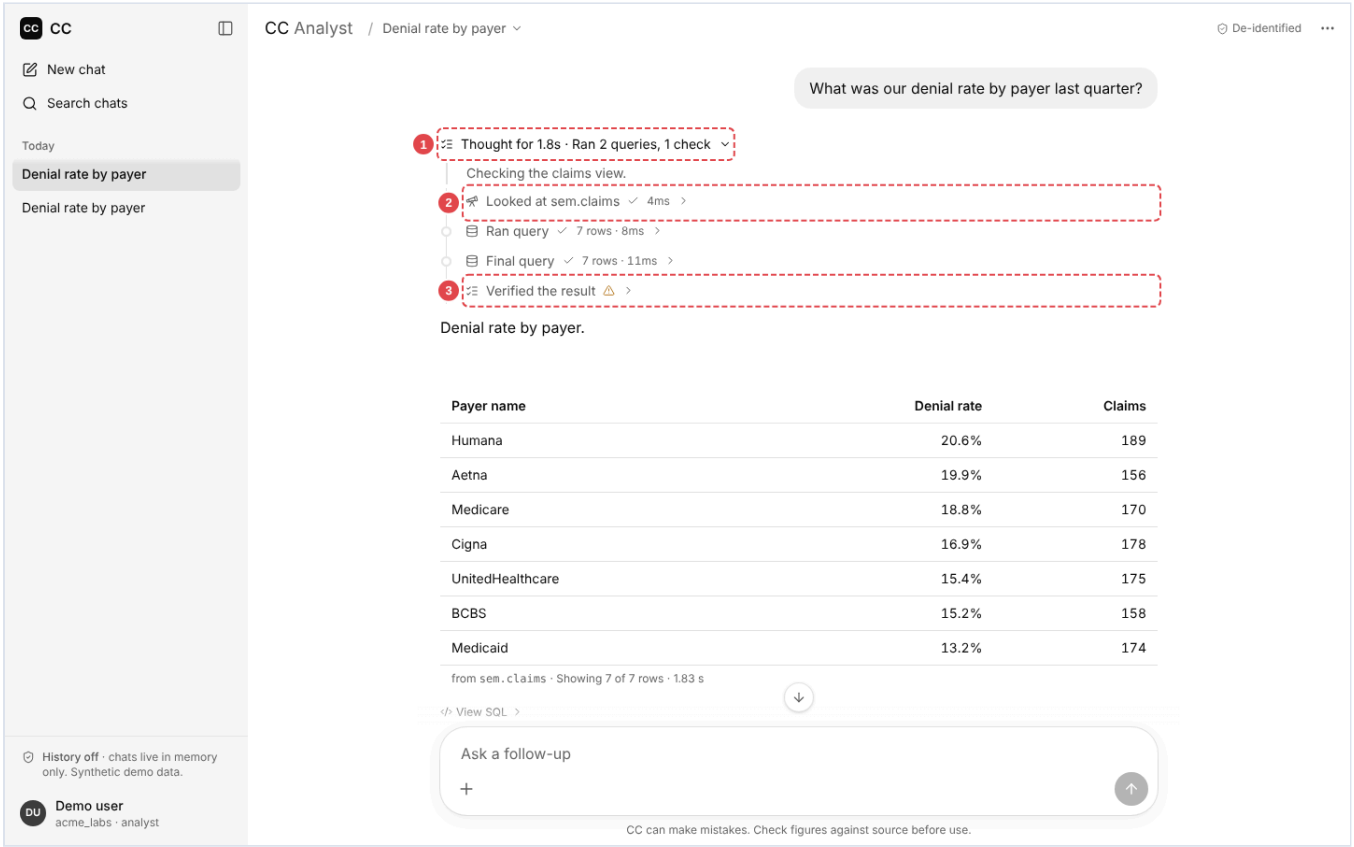
8.3 Text streaming



#	Element	Behaviour
1	Caret	Paced text (`lib/pacedText.ts`) flows in over frames even when it arrives in one chunk; at once under reduced motion.
2	Timeline header	Collapses to "Thought for N s · Ran 3 queries, 1 repair, 2 checks" as soon as the answer starts after the last step.



8.4 Steps after the answer

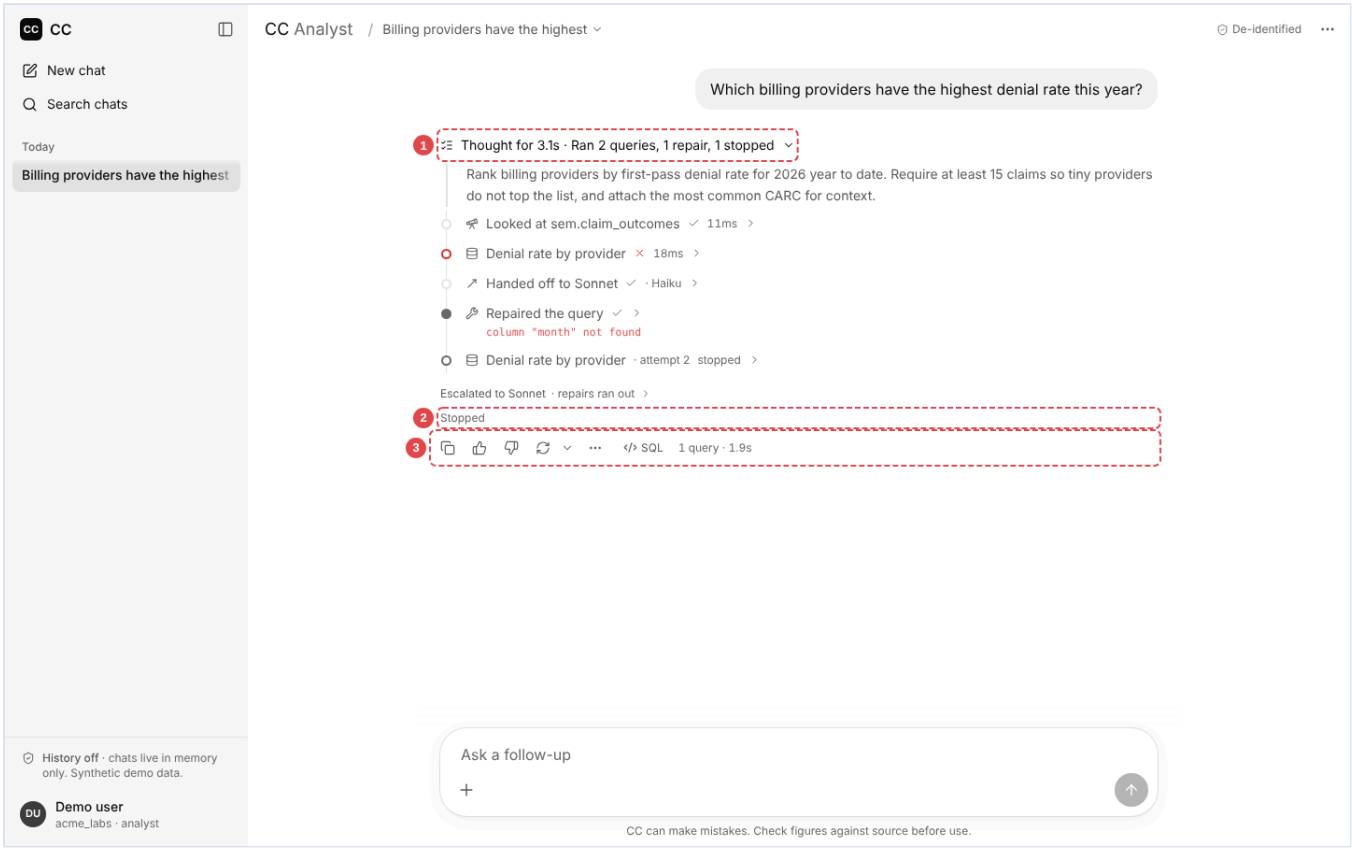


#	Element	Behaviour
1	Header	"Thought for 1.8s · Ran 2 queries, 1 check" (never shorter than its longest step, QA-A16).

#	Element	Behaviour
2	First step	"Looked at sem.claims ✓ 4ms".
3	Last step	"Verified the result ⚠" (a check with a warning).



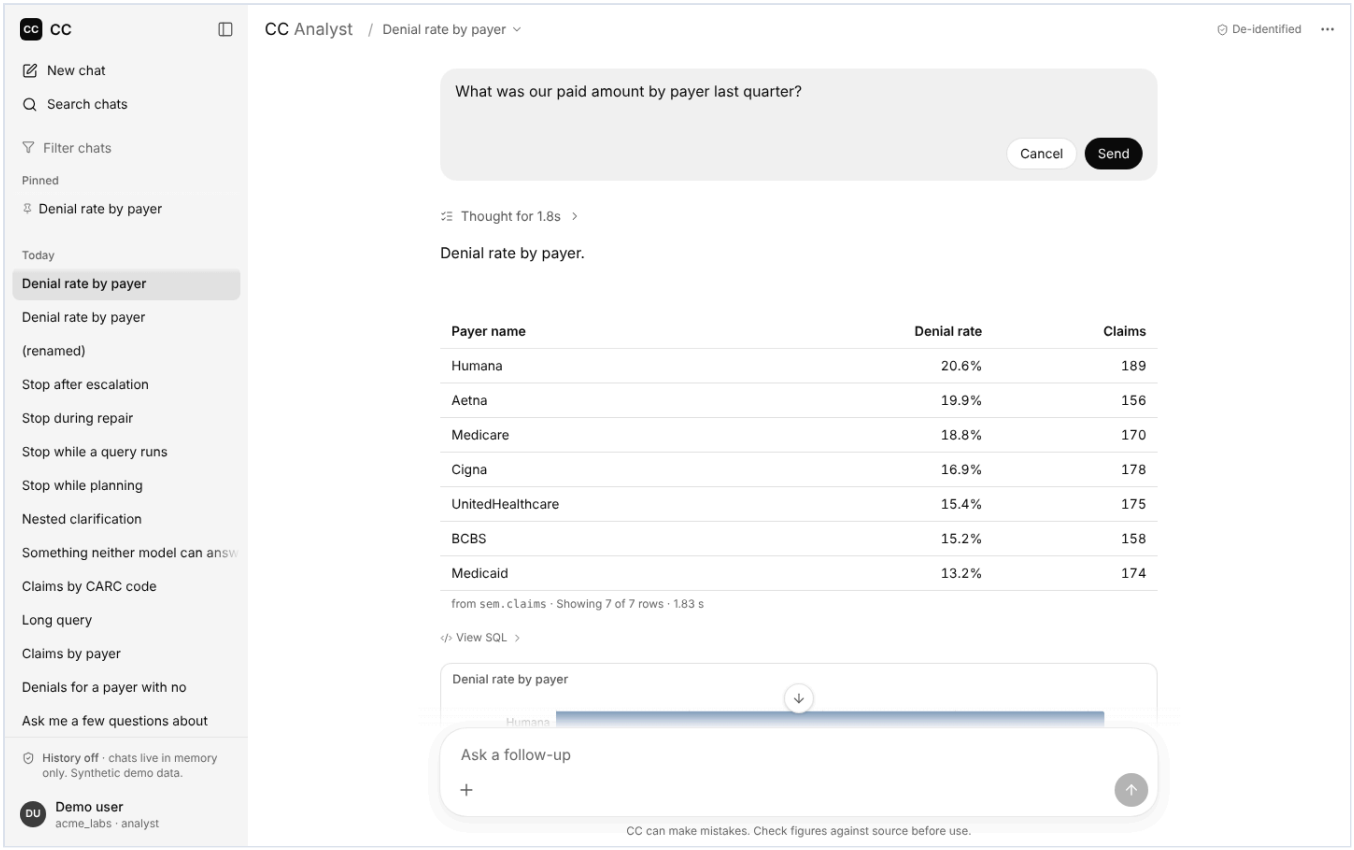
8.5 Stopped by the user

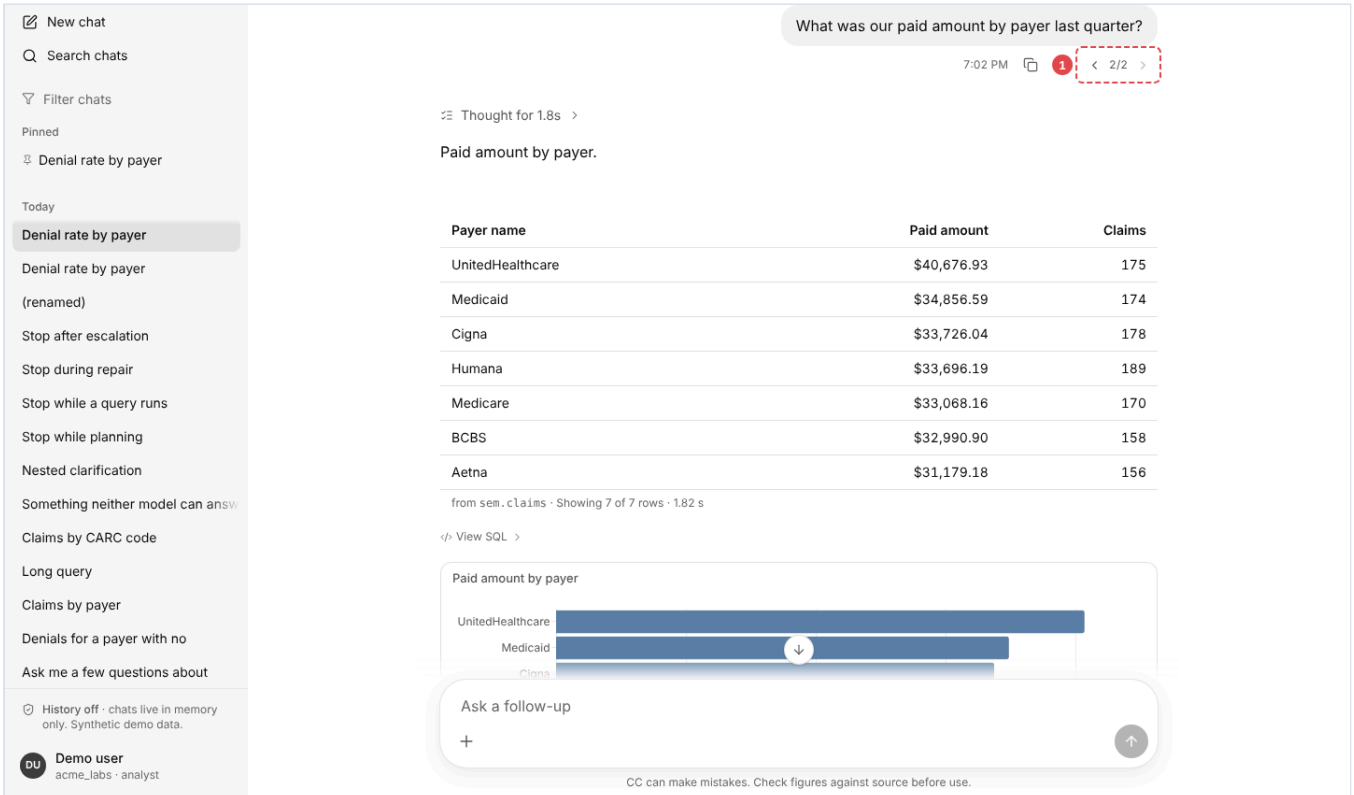


#	Element	Behaviour
1	Header	"Thought for 3.1s · Ran 2 queries, 1 repair, 1 stopped".
2	Stopped note	`finishReason: aborted`; Retry only when nothing came back.
3	Action row	Copy, thumbs, Regenerate on what did arrive.

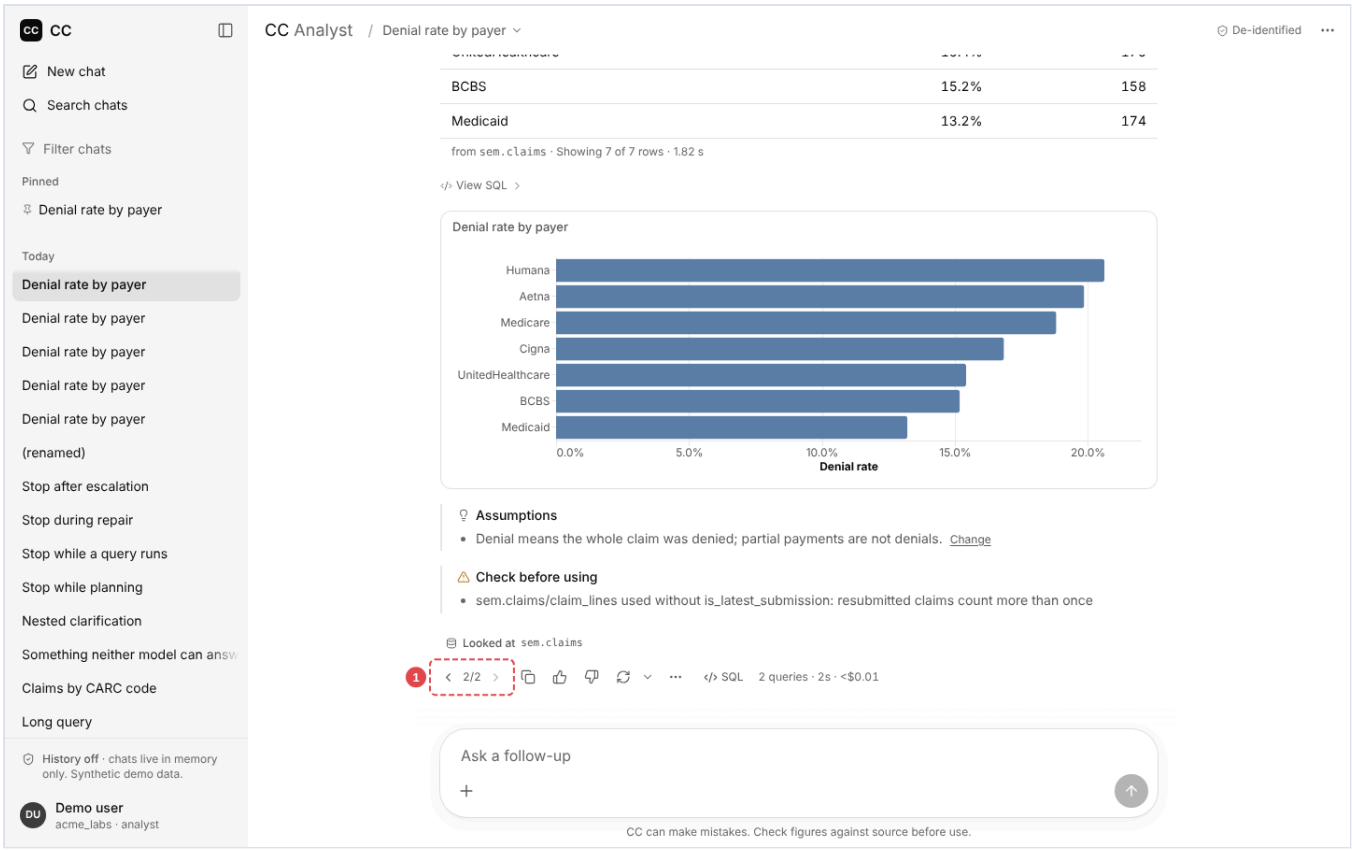
TODO: does the UI show a distinct "Stopping…" between `/stop` and `message.end` (CONTRACT.md §8 recommends it)? Not observed locally; the stop landed within one frame.

8.6 Follow-ups, edits and versions

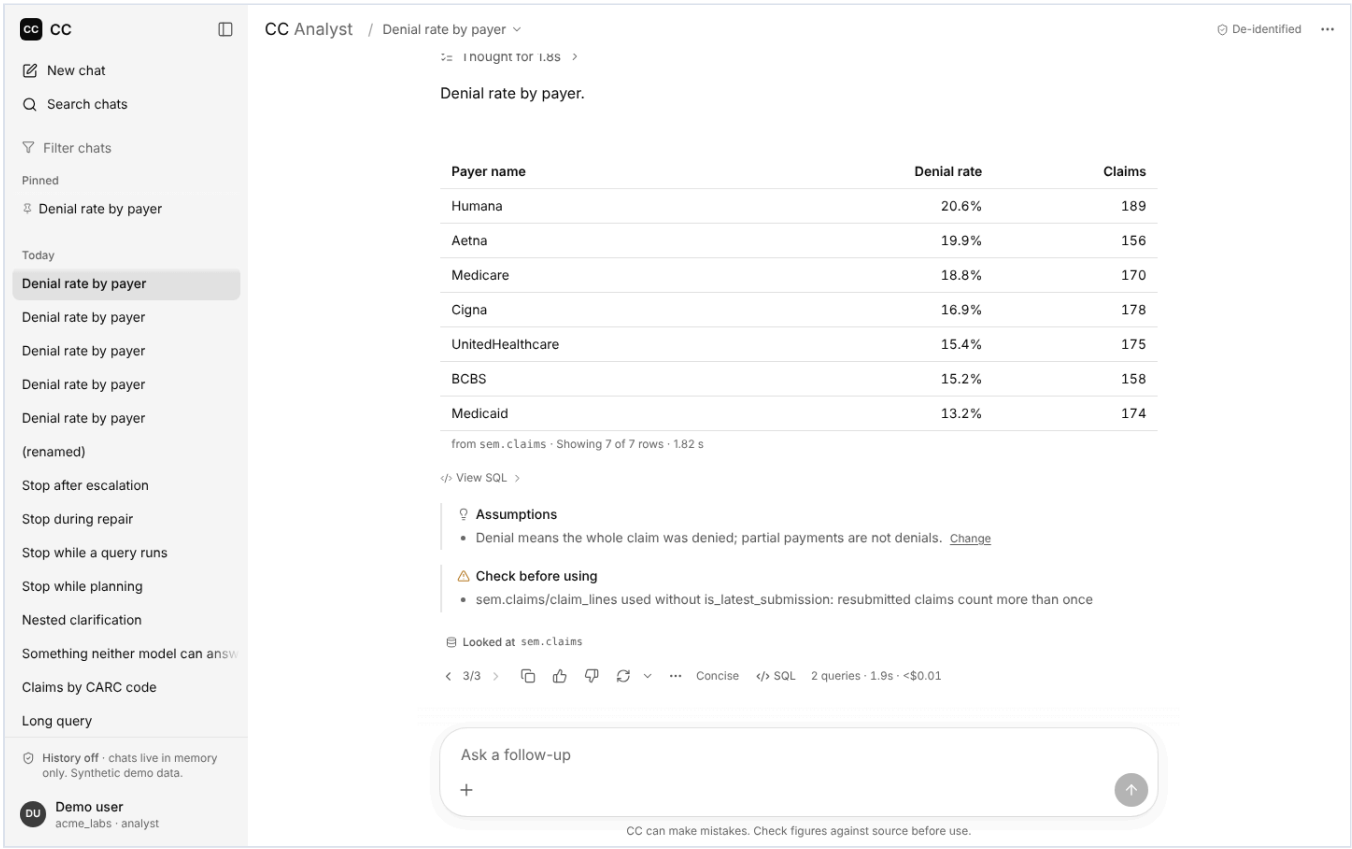


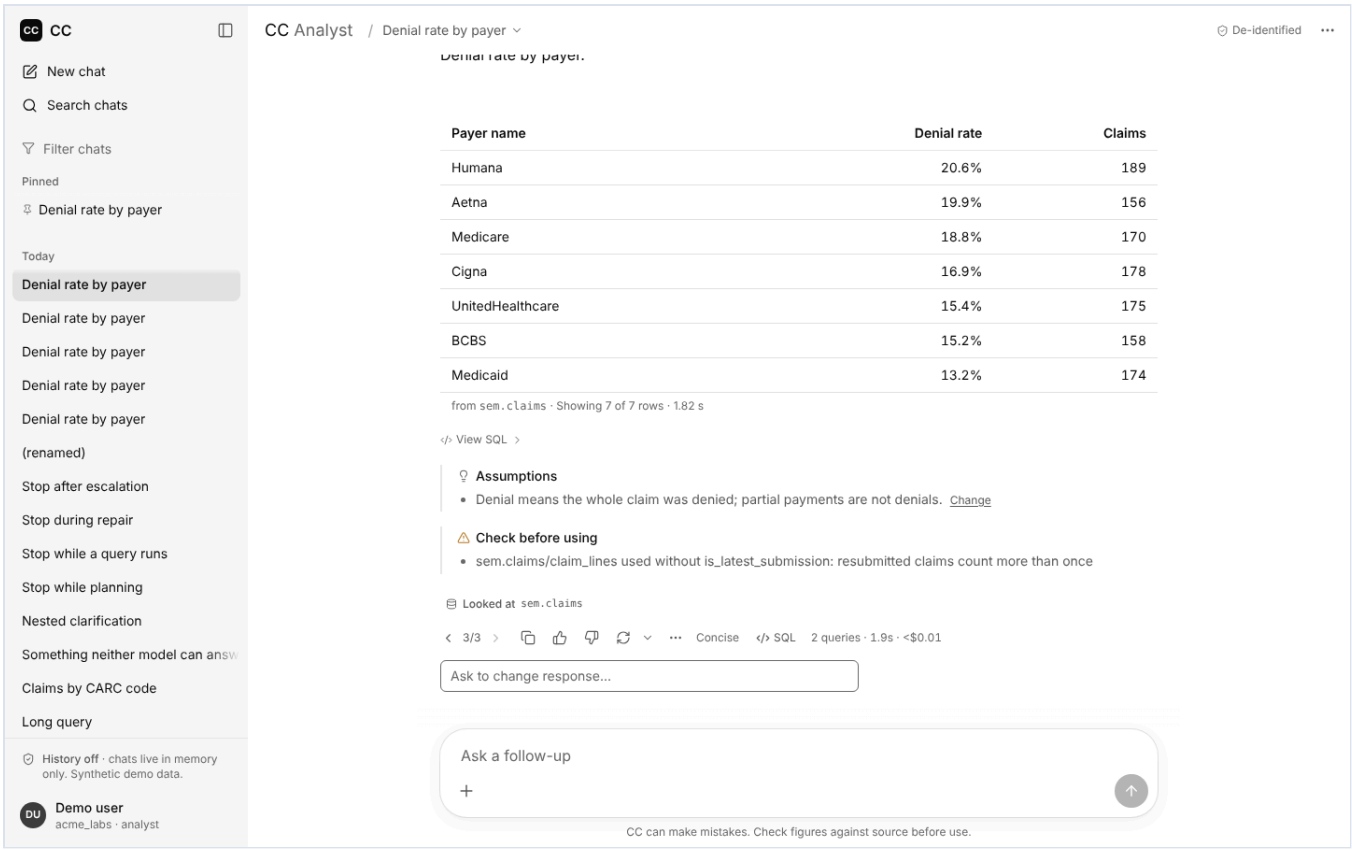


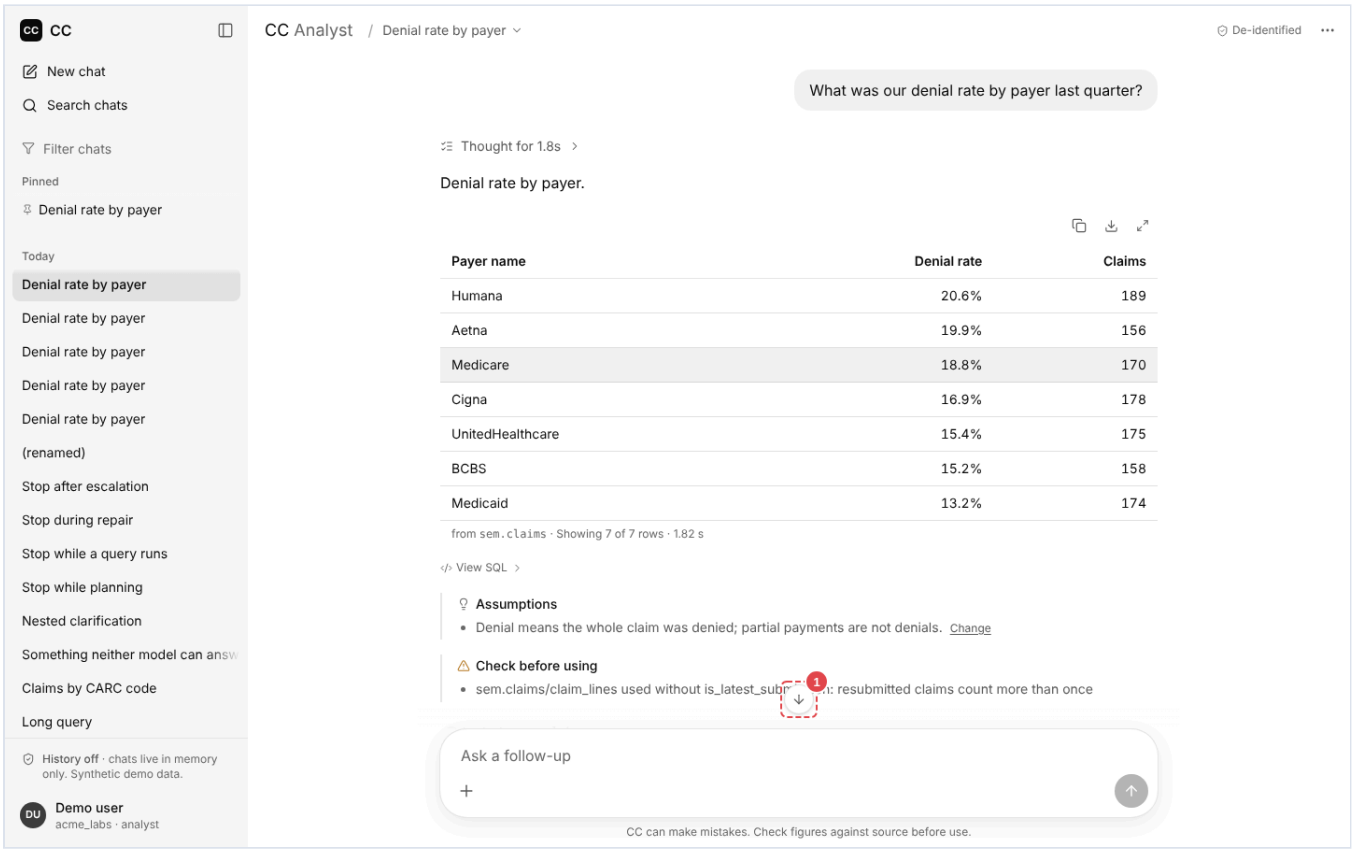
#	Element	Behaviour
1	Question versions	`‹ 2/2 ›` (`role="group"` "Versions", read as "2 of 2"). The edit is sent with `editOf` and stored as a sibling; the old tail is kept as version 1.



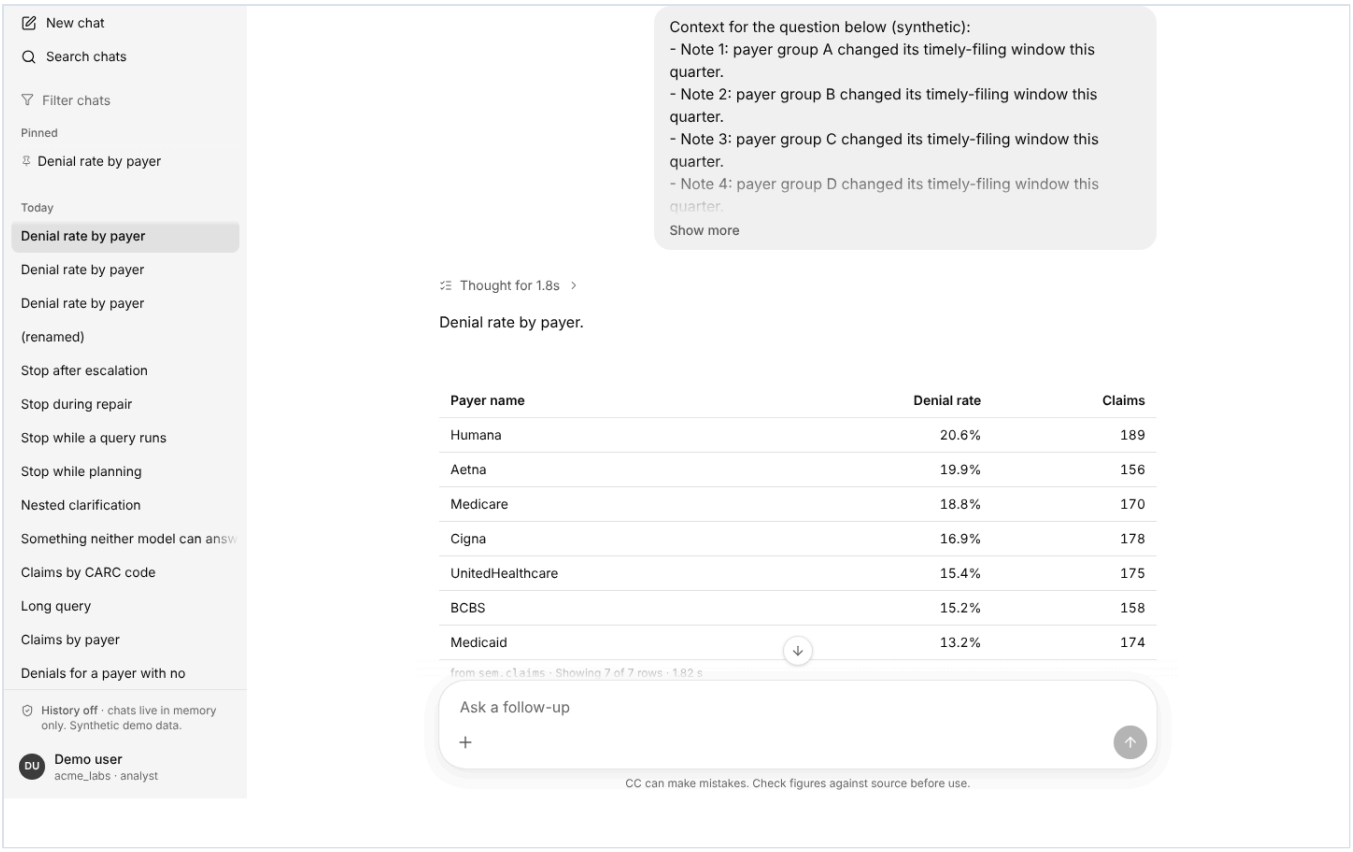
#	Element	Behaviour
1	Answer versions	Regenerate sends `regenerate {messageId}`, `messages: []`; the reply is a sibling with `regenerateOf`.







#	Element	Behaviour
1	Jump to latest	Appears when the user scrolls away from the bottom; releases the pinned follow-up.



TODO screenshot: branch in new chat — More actions → Branch in new chat on an earlier answer; a new chat opens with the thread up to that answer and the toast "Branched into a new chat"

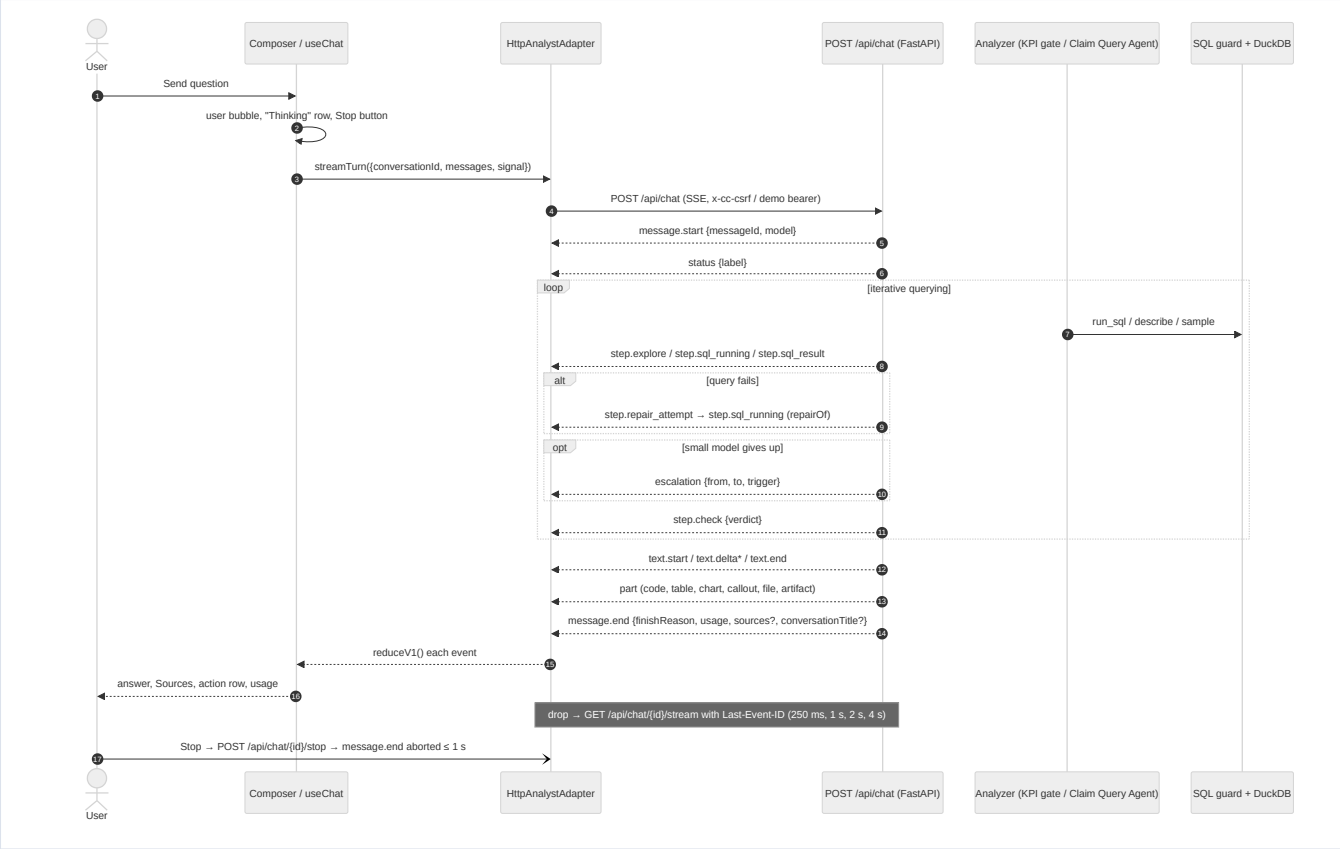
TODO screenshot: print one answer — More actions → Print this answer; capture the print preview (only that question and answer)

---

9. How the agent behaves and operates (user's view)

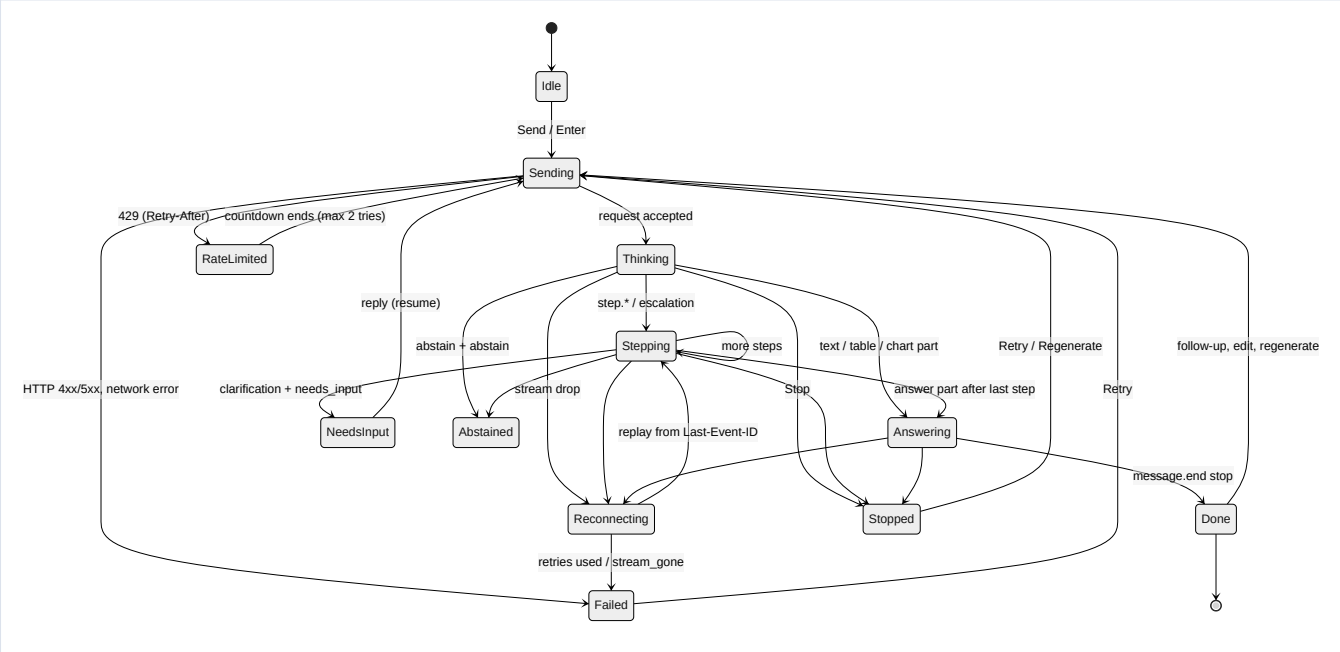
TODO: one paragraph on the agent: one analyzer configuration (bundle B1: Haiku, low effort, single agent; contract 1.24.0), no model or strategy choice in the UI, model names are labels only. Link to backend page §7 (Analyzer).

9.1 Request lifecycle



TODO: confirm the KPI gate path (`via=template:<id>@v1` vs `b1`) shows up in step labels on the deployed build; LOCAL_STATE.md says the UI step labels were not checked in a browser.

9.2 Turn state machine



9.3 Stream events → what the user sees

Event (contract 1.24.0)	Payload highlights	UI effect	Screenshot
`message.start`	`messageId`, `conversationId`, `contractVersion`, `model?`, `regenerateStyle?`, `temporary?`	Swaps in the server id (row keeps its `uiKey`); Stop now also calls `/stop`	05
`status`	`label` or `null`	Transient line; hidden while the timeline is live unless it is a transport status	86, 88
`step.explore`	`probe`, `target`, `summary`, `elapsedMs`	"Looked at sem.claims", "Read schema", "Used the full EDI data" (`request_escalation`)	13
`step.sql_running`	`sql`, `purpose` (explore/answer/candidate/repair/check), `attempt`, `label`, `maskedLiterals?`	Running row "Running query" / server label; table skeleton	07
`step.sql_result`	`ok`, `elapsedMs`, `rowCount`, `columns`, `error {code}`	Row turns ✓ "7 rows · 8ms" or ✕ with the error; `cancelled` → "stopped"	30, 47
`step.repair_attempt`	`failedStepId`, `attempt/maxAttempts`, `error`, `revisedSql`, `explanation`	"Repairing the query" → "Repaired the query", error text in red	30, 43
`step.check`	`check`, `verdict` pass/warn/fail, `message`, `action`	"Verified the result ⚠", "Ran 2 checks"	13, 75
`step.candidates`	`candidates[]`, `chosenId`, `method`, `rationale`	"Compared 3 candidates · execution agreement"	32
`escalation`	`from`, `to` (model, tier), `reason`, `trigger`	"Handed off to Sonnet 5.5" step + note under the timeline	31
`reasoning.delta` / `reasoning.end`	summary text, `durationMs`	Folded into the timeline (one "Thought for" row)	13
`text.start` / `text.delta` / `text.end`	markdown	Paced prose with caret	08
`code.delta`		Streaming code part	—
`table.start` / `table.rows`	columns, then rows with `offset`	Header first, rows fill in; dropped if the turn aborts	*TODO*
`chart.start` / `chart.points`	strict `PlotHeader`, points	Chart draws from the first points	*TODO*
`part` / `part.patch`	complete part / shallow patch	Part rendered or replaced by id	10
`message.end`	`finishReason` stop/abstain/needs_input/error/aborted, `usage`, `sources?`, `conversationTitle?`	Action row, Sources, usage line, title in sidebar	12

Event (contract 1.24.0)	Payload highlights	UI effect	Screenshot
`heartbeat`	`at?` (no `seq`)	Nothing visible; resets the 45 s idle guard	—

Unknown events and fields are ignored; unknown part types render as "Unsupported content" (S63).

9.4 Step types and labels

Step kind	Icon	Live label → done label	Source
explore	Telescope	"Probing the data" → "Looked at <relation>" / "Read schema" / "Sampled values" / "Looked up glossary" / "Used the full EDI data"	`answer/normalize.ts` `PROBE`
sql	Database	"Running query" → "Ran query"; check purpose → "Ran check query"; final → "Final query"	`PURPOSE` / `PRESENT`
repair	Wrench	"Repairing the query" → "Repaired the query" ("Re-ran repaired query · attempt 2")	`normalize.ts`
check	ListChecks	"Verified the result" / "Checked row count" / "Ran N checks"	`CHECK`
candidates	GitCompare	"Compared N candidates · <method>"	
escalation	MoveUpRight	"Handed off to <model>"	
chart / schema / search / other	BarChart3 / List / Search / Wrench	legacy `tool_call` parts	

9.5 Outcomes the user can get

Outcome	finishReason	What it looks like	Screenshot
Answer	`stop`	Prose + table(s) + SQL + chart + callouts	10, 36, 41
Answer after repair	`stop`	Red failed steps, then repaired query	30, 43
Answer after escalation	`stop`	"Handed off to Sonnet 5.5", note "repairs ran out"	31
Answer from candidates	`stop`	"Compared 3 candidates"	32, 75
Empty result	`stop`	"Query returned 0 rows" + caveat	40
Clarification	`needs_input`	Question + choices; multi-question form; nested	38, 39, 45
Abstain	`abstain`	"The data needed for this isn't available" + Try asking	33, 44
Refusal (PHI)	`stop` *(TODO: confirm)*	"I can't share patient-identifying details"	34
Error	`error`	"Something went wrong" + Retry	35
Stopped	`aborted`	"Stopped" + closed steps	46–52

New chat

Search chats

Today

Claims by CARC code

Denial rate by payer

Denial rate by payer

History off · chats live in memory only. Synthetic demo data.

Demo user
acme_labs · analyst

Which CARC reason codes drive our denials?

Thought for 2.5s · Ran 3 queries, 1 repair, 1 check

Checking the claims view. The denial reason column is carc_code; retrying.

Looked at sem.claims ✓ 4ms

Ran query ✕ 226ms

Repaired the query ✓

Binder Error: Referenced column "denial_code" not found in FROM clause! Hint: check column names with …

Re-ran repaired query · attempt 2 ✓ 6 rows · 10ms

Final query ✓ 6 rows · 8ms

Verified the result ⚠

Denied and partially paid claims by CARC code.

CARC code	Claims	Charge amount
CO-97	84	$37,320.70
CO-45	72	$31,350.38
CO-50	67	$26,242.09
PR-1	64	$29,558.41
CO-16	57	$24,311.14
PR-2	52	$20,740.77

from sem.claims · Showing 6 of 6 rows · 2.65 s

View SQL

Claims by CARC code

Ask a follow-up

CC can make mistakes. Check figures against source before use.

New chat

Search chats

Today

Net collection rate by payer

Claims by CARC code

Denial rate by payer

Denial rate by payer

History off · chats live in memory only. Synthetic demo data.

Demo user
acme_labs · analyst

What is our net collection rate?

Thought for 3.5s · Ran 4 queries, 1 repair, 1 check

Checking the claims view. I could not find a working definition of net collection rate. Net collection rate is paid over charged; this view has no allowed amount.

Looked at sem.claims ✓ 3ms

Ran query ✕ 3ms

Repaired the query ✕

Binder Error: Referenced column "net_paid" not found in FROM clause! Hint: check column names with `co…

Re-ran repaired query · attempt 2 ✕ 3ms

Handed off to Sonnet 5.5 ✓ · Haiku 5.5

Ran query ✓ 7 rows · 9ms

Final query ✓ 7 rows · 9ms

Verified the result ⚠

Escalated to Sonnet 5.5 · repairs ran out

Net collection rate by payer (paid ÷ charged).

Payer name	Net collection rate	Claims
UnitedHealthcare	49.4%	175
BCBS	44.8%	158
Medicaid	43.5%	174
Medicare	42.9%	170
Aetna	42.1%	156
Cigna	41.5%	178
Humana	38.8%	189

Ask a follow-up

CC can make mistakes. Check figures against source before use.

New chat

Search chats

Today

Avg paid amount by payer

Net collection rate by payer

Claims by CARC code

Denial rate by payer

Denial rate by payer

History off · chats live in memory only. Synthetic demo data.

Demo user
acme_labs · analyst

What is the average paid amount per claim?

Thought for 3.7s · Ran 1 check, compared candidates

Verified the result

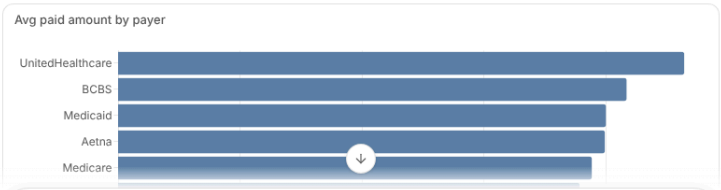
Compared 3 candidates · execution agreement · 3.7s

Average paid amount per claim, by payer.

Payer name	Avg paid amount	Claims
UnitedHealthcare	$232.44	175
BCBS	$208.80	158
Medicaid	$200.33	174
Aetna	$199.87	156
Medicare	$194.52	170
Cigna	$189.47	178
Humana	$178.29	189

from sem.claims · Showing 7 of 7 rows · 3.66 s

View SQL



Ask a follow-up

CC can make mistakes. Check figures against source before use.

CC

New chat

Search chats

Today

Forecast our denial rate

Avg paid amount by payer

Net collection rate by payer

Claims by CARC code

Denial rate by payer

Denial rate by payer

History off · chats live in memory only. Synthetic demo data.

Demo user
acme_labs · analyst

CC Analyst / Forecast our denial rate

De-identified

Forecast our denial rate for next year

Thought for 1.3s · Looked at sem.claims

Checking the claims view.

Looked at sem.claims · 4ms

The data needed for this isn't available

This data doesn't include **appeals, prior authorizations or forecasts**, so I can't answer that reliably. I can work with claims, payers, service dates, charges, payments and denial codes.

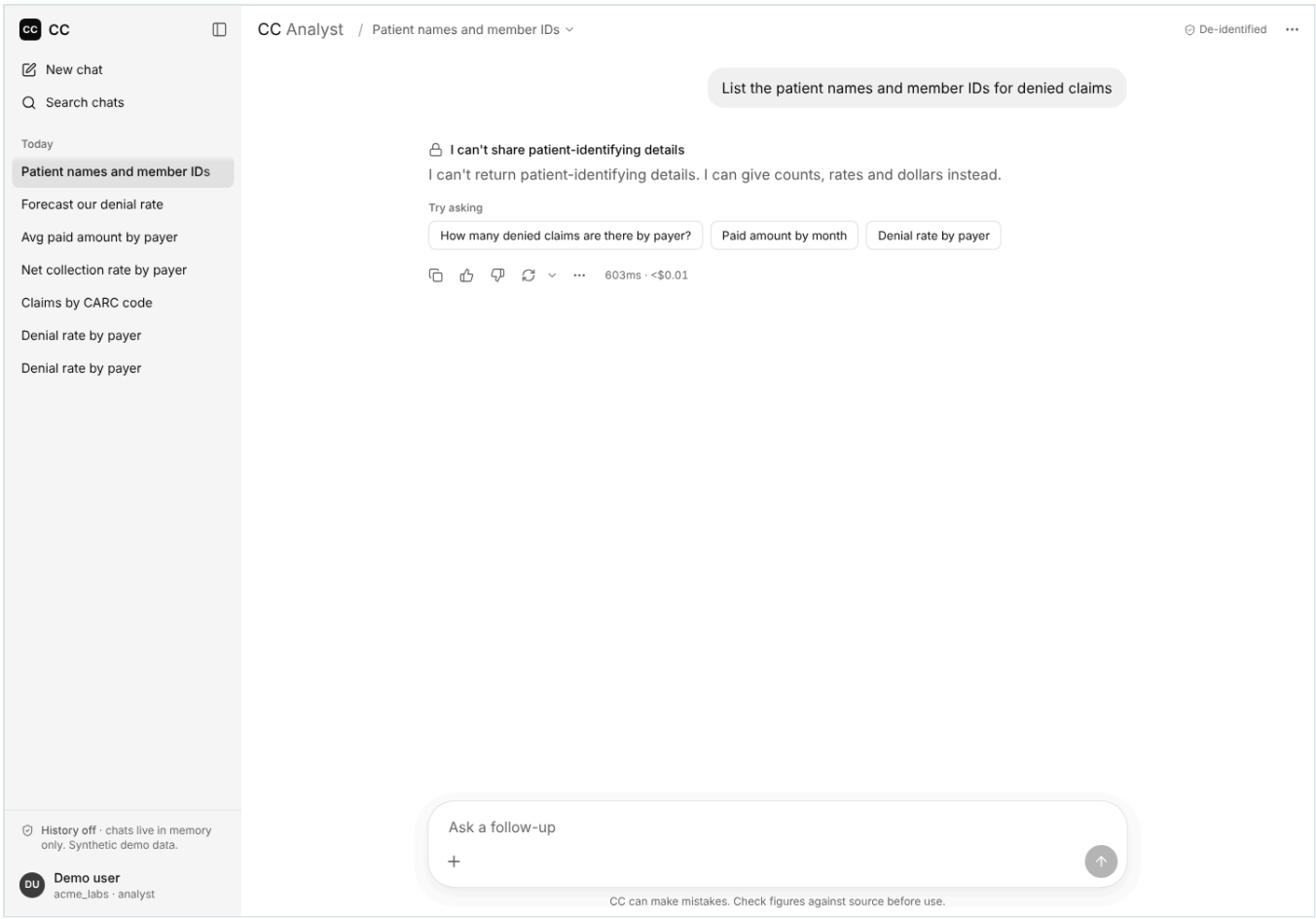
Try asking

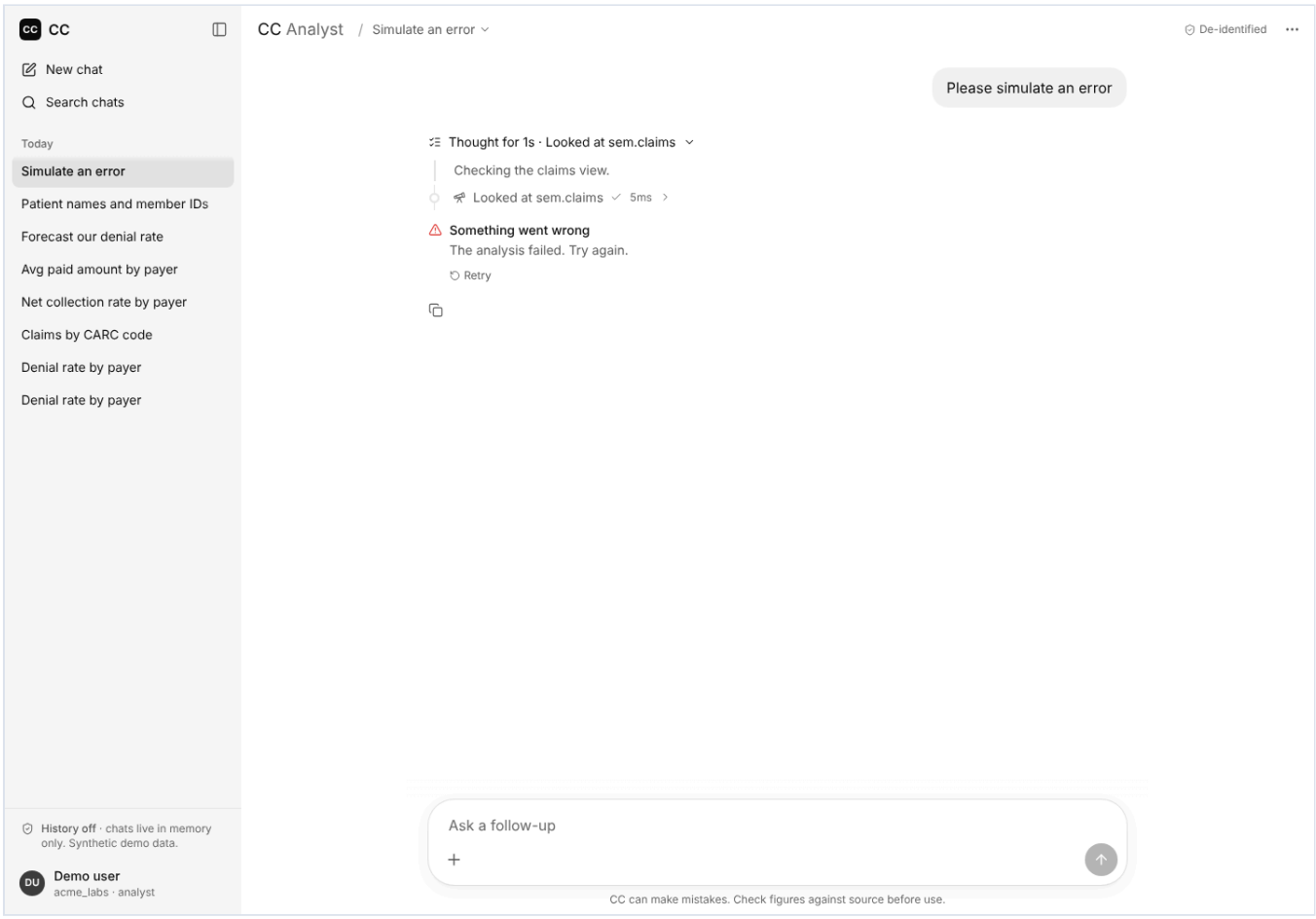
Denial rate by payer | Paid amount trend by month | Top denial reason codes (CARC)

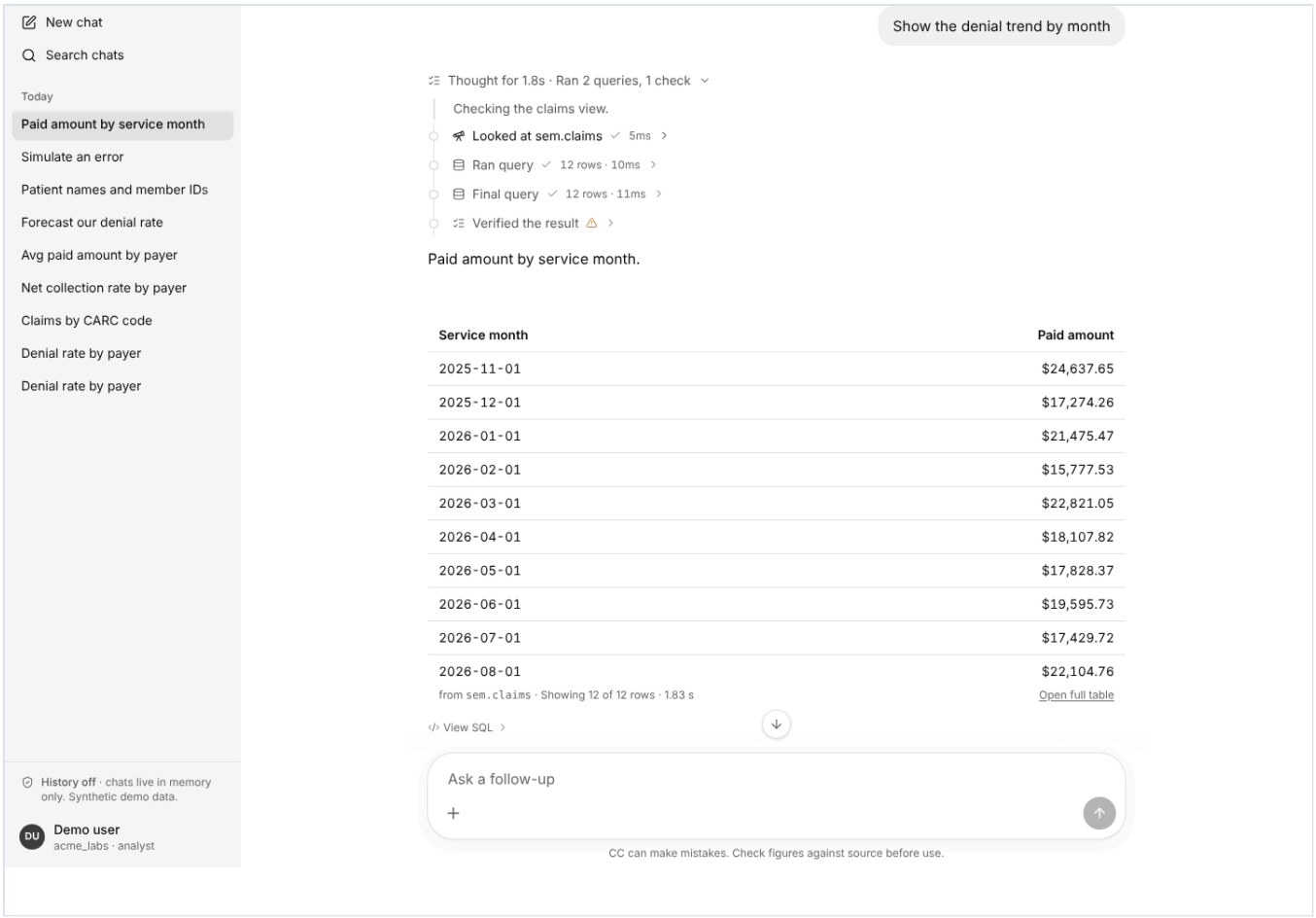
1.3s · <$0.01

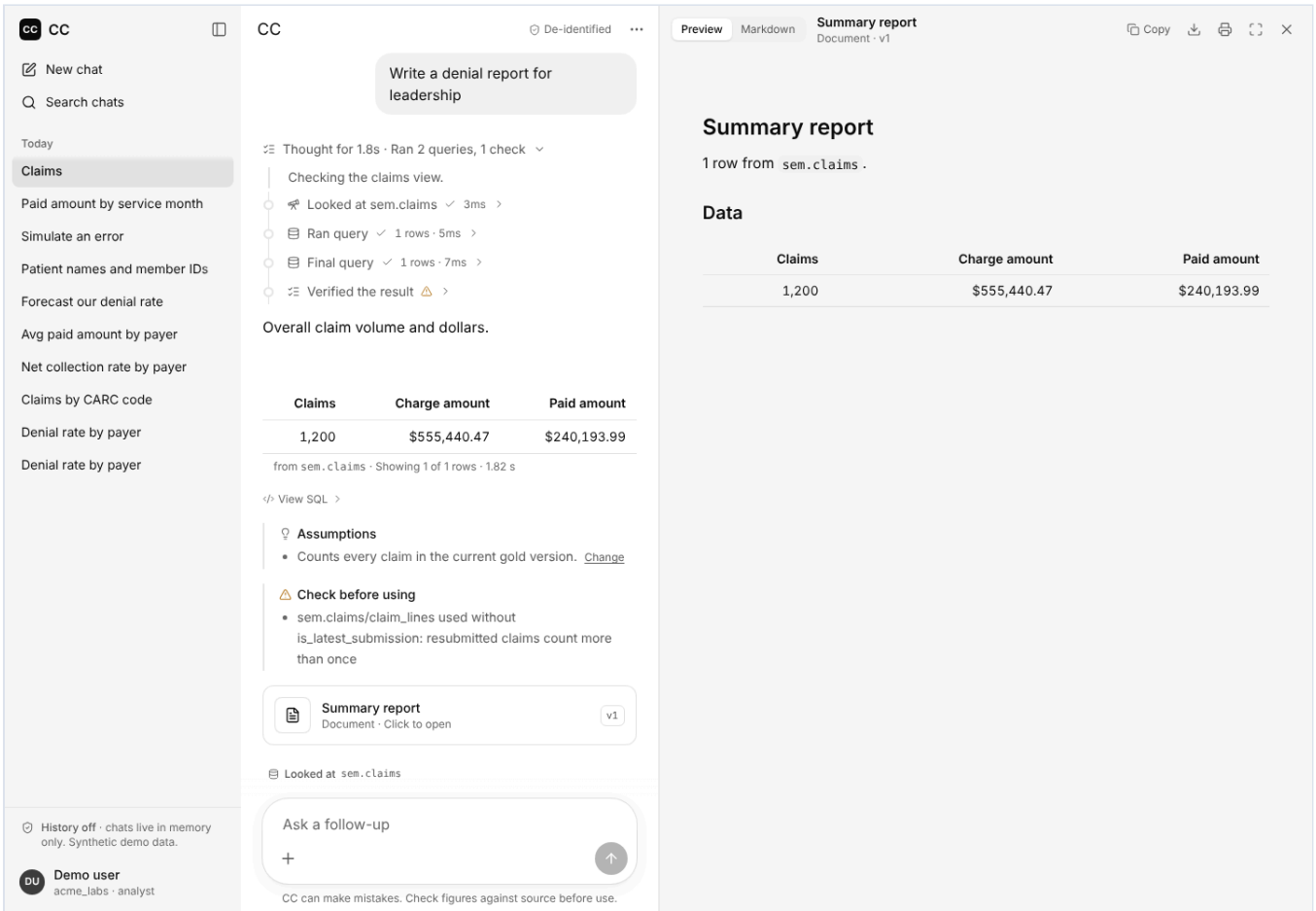
Ask a follow-up

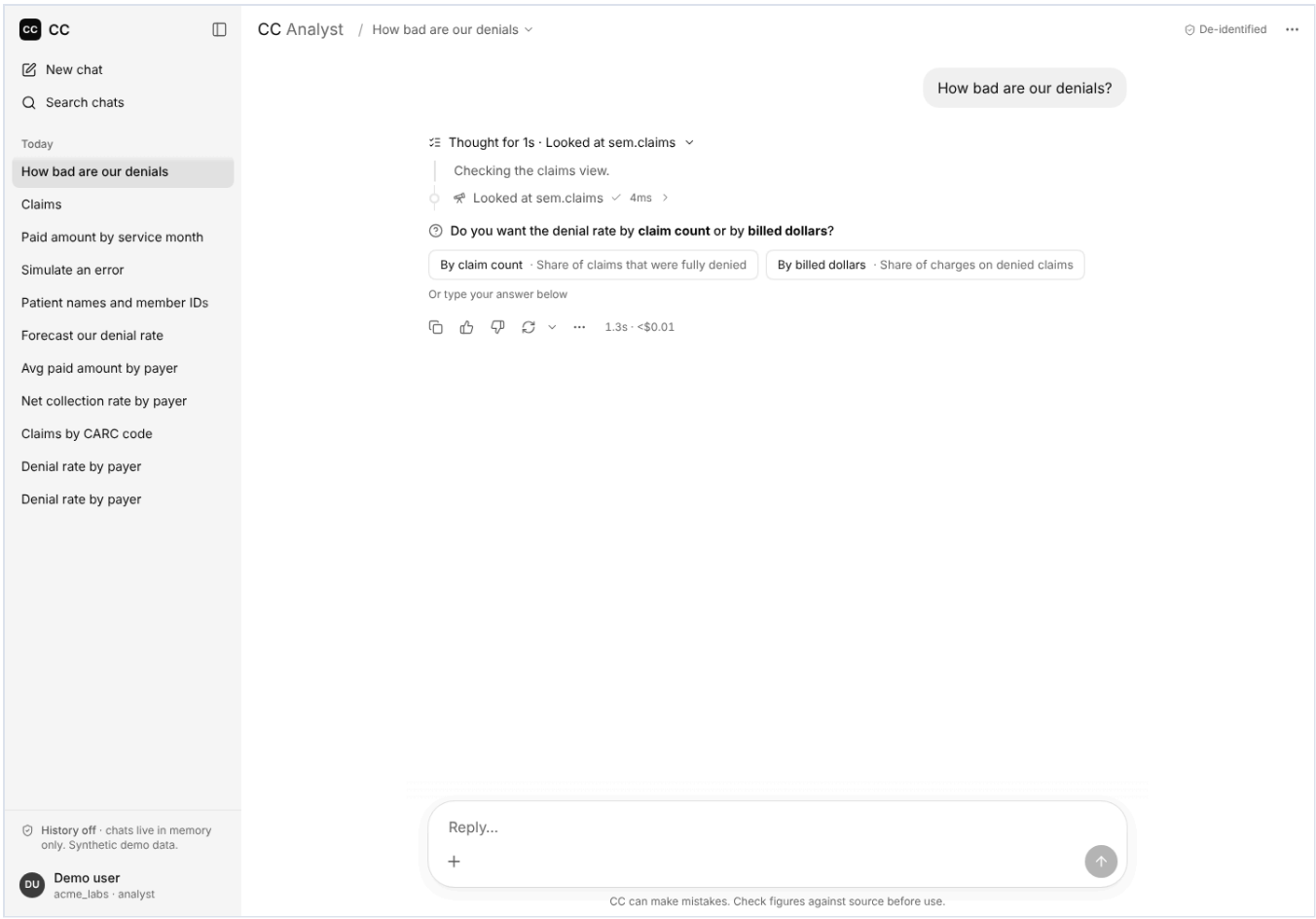
CC can make mistakes. Check figures against source before use.

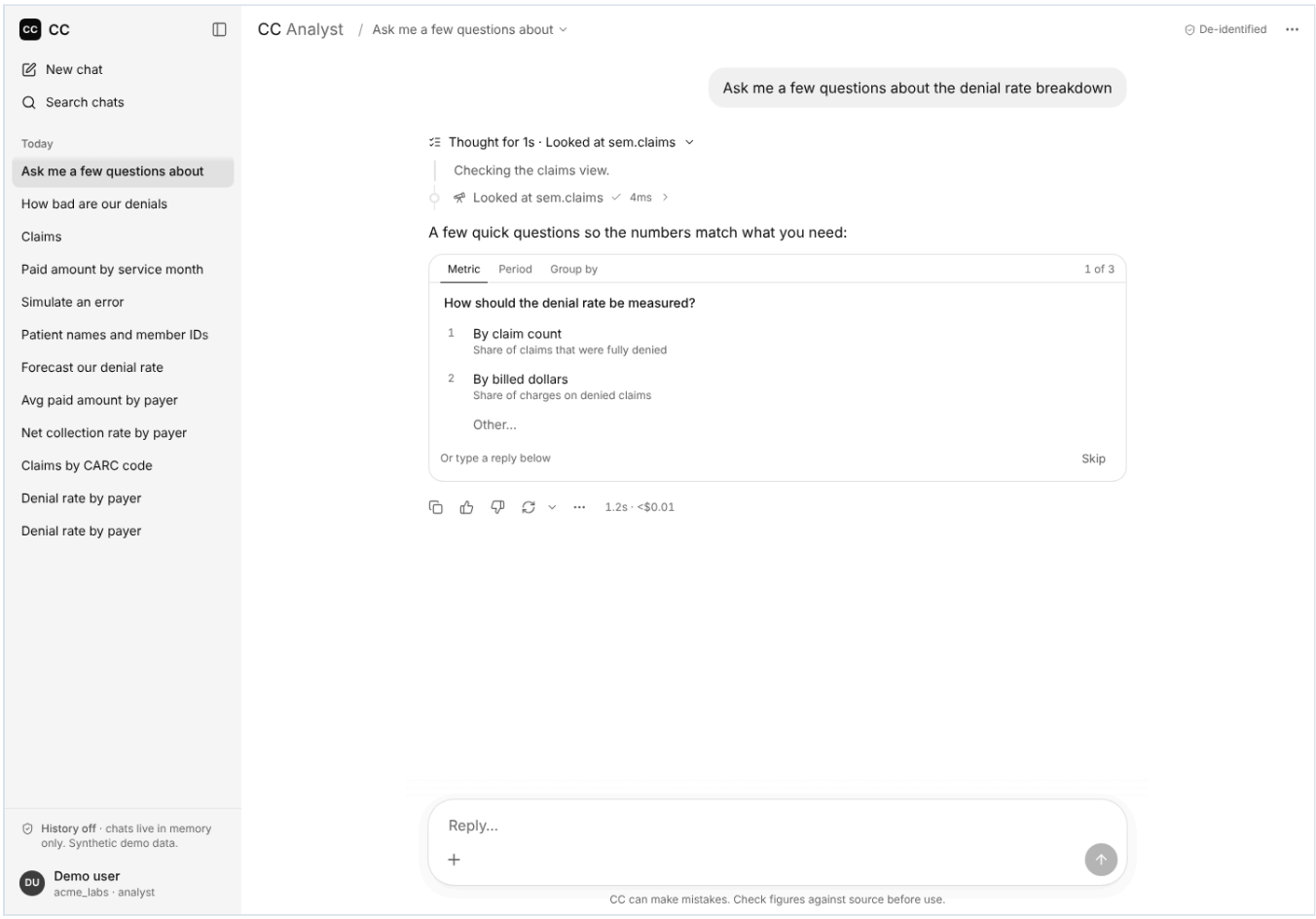

CC
New chat
Search chats
Today
Patient names and member IDs
Forecast our denial rate
Avg paid amount by payer
Net collection rate by payer
Claims by CARC code
Denial rate by payer
Denial rate by payer
History off · chats live in memory only. Synthetic demo data.
Demo user
acme_labs · analyst
CC Analyst / Patient names and member IDs
De-identified
List the patient names and member IDs for denied claims
I can't share patient-identifying details
I can't return patient-identifying details. I can give counts, rates and dollars instead.
Try asking
How many denied claims are there by payer?
Paid amount by month
Denial rate by payer
603ms · <$0.01
Ask a follow-up
CC can make mistakes. Check figures against source before use.


CC
New chat
Search chats
Today
Simulate an error
Patient names and member IDs
Forecast our denial rate
Avg paid amount by payer
Net collection rate by payer
Claims by CARC code
Denial rate by payer
Denial rate by payer
History off · chats live in memory only. Synthetic demo data.
Demo user
acme_labs · analyst
CC Analyst / Simulate an error
De-identified
Please simulate an error
Thought for 1s · Looked at sem.claims
Checking the claims view.
Looked at sem.claims
5ms
Something went wrong
The analysis failed. Try again.
Retry
Ask a follow-up
CC can make mistakes. Check figures against source before use.


New chat
Search chats
Today
Paid amount by service month
Simulate an error
Patient names and member IDs
Forecast our denial rate
Avg paid amount by payer
Net collection rate by payer
Claims by CARC code
Denial rate by payer
Denial rate by payer
History off · chats live in memory only. Synthetic demo data.
Demo user
acme_labs · analyst
Show the denial trend by month
Thought for 1.8s · Ran 2 queries, 1 check
Checking the claims view.
Looked at sem.claims 5ms
Ran query 12 rows · 10ms
Final query 12 rows · 11ms
Verified the result
Paid amount by service month.
Service month
Paid amount
2025-11-01
$24,637.65
2025-12-01
$17,274.26
2026-01-01
$21,475.47
2026-02-01
$15,777.53
2026-03-01
$22,821.05
2026-04-01
$18,107.82
2026-05-01
$17,828.37
2026-06-01
$19,595.73
2026-07-01
$17,429.72
2026-08-01
$22,104.76
from sem.claims · Showing 12 of 12 rows · 1.83 s
Open full table
View SQL
Ask a follow-up
CC can make mistakes. Check figures against source before use.


CC
New chat
Search chats
Today
Claims
Paid amount by service month
Simulate an error
Patient names and member IDs
Forecast our denial rate
Avg paid amount by payer
Net collection rate by payer
Claims by CARC code
Denial rate by payer
Denial rate by payer
History off · chats live in memory only. Synthetic demo data.
Demo user
acme_labs · analyst
CC
De-identified
Write a denial report for leadership
Thought for 1.8s · Ran 2 queries, 1 check
Checking the claims view.
Looked at sem.claims 3ms
Ran query 1 rows · 5ms
Final query 1 rows · 7ms
Verified the result
Overall claim volume and dollars.
Claims
Charge amount
Paid amount
1,200
$555,440.47
$240,193.99
from sem.claims · Showing 1 of 1 rows · 1.82 s
View SQL
Assumptions
Counts every claim in the current gold version. Change
Check before using
sem.claims/claim_lines used without is_latest_submission: resubmitted claims count more than once
Summary report
Document · Click to open
v1
Looked at sem.claims
Ask a follow-up
CC can make mistakes. Check figures against source before use.
Preview
Markdown
Summary report
Document · v1
Copy
Summary report
1 row from sem.claims .
Data
Claims
Charge amount
Paid amount
1,200
$555,440.47
$240,193.99


CC
New chat
Search chats
Today
How bad are our denials
Claims
Paid amount by service month
Simulate an error
Patient names and member IDs
Forecast our denial rate
Avg paid amount by payer
Net collection rate by payer
Claims by CARC code
Denial rate by payer
Denial rate by payer
History off · chats live in memory only. Synthetic demo data.
Demo user
acme_labs · analyst
CC Analyst / How bad are our denials
De-identified
How bad are our denials?
Thought for 1s · Looked at sem.claims
Checking the claims view.
Looked at sem.claims 4ms
Do you want the denial rate by claim count or by billed dollars?
By claim count · Share of claims that were fully denied
By billed dollars · Share of charges on denied claims
Or type your answer below
1.3s · <$0.01
Reply...
CC can make mistakes. Check figures against source before use.


CC
New chat
Search chats
Today
Ask me a few questions about
How bad are our denials
Claims
Paid amount by service month
Simulate an error
Patient names and member IDs
Forecast our denial rate
Avg paid amount by payer
Net collection rate by payer
Claims by CARC code
Denial rate by payer
Denial rate by payer
History off · chats live in memory only. Synthetic demo data.
Demo user
acme_labs · analyst
CC Analyst / Ask me a few questions about
De-identified
Ask me a few questions about the denial rate breakdown
Thought for 1s · Looked at sem.claims
Checking the claims view.
Looked at sem.claims 4ms
A few quick questions so the numbers match what you need:
Metric
Period
Group by
1 of 3
How should the denial rate be measured?
1 By claim count
Share of claims that were fully denied
2 By billed dollars
Share of charges on denied claims
Other...
Or type a reply below
Skip
1.2s · <$0.01
Reply...
CC can make mistakes. Check figures against source before use.

CC

New chat

Search chats

Today

Denials for a payer with no

Ask me a few questions about

How bad are our denials

Claims

Paid amount by service month

Simulate an error

Patient names and member IDs

Forecast our denial rate

Avg paid amount by payer

Net collection rate by payer

Claims by CARC code

Denial rate by payer

Denial rate by payer

History off · chats live in memory only. Synthetic demo data.

Demo user
acme_labs · analyst

CC Analyst / Denials for a payer with no

De-identified

Denials for a payer with no claims

Thought for 1.8s · Ran 2 queries, 1 check

Checking the claims view.

Looked at sem.claims 4ms

Ran query 0 rows · 6ms

Final query 0 rows · 8ms

Verified the result

No denied claims from Tricare in this data, so there is nothing to break down.

Payer name	Claims	Paid amount

Query returned 0 rows

from sem.claims · 1.82 s

View SQL

Assumptions

- Tricare is matched by exact payer name. Change

Check before using

- the query returned no rows: check filters and value spellings
- sem.claims/claim_lines used without is_latest_submission: resubmitted claims count more than once

Looked at sem.claims

SQL 2 queries · 2s · <$0.01

Ask a follow-up

CC can make mistakes. Check figures against source before use.

New chat

Search chats

Today

Claims by payer

Denials for a payer with no

Ask me a few questions about

How bad are our denials

Claims

Paid amount by service month

Simulate an error

Patient names and member IDs

Forecast our denial rate

Avg paid amount by payer

Net collection rate by payer

Claims by CARC code

Denial rate by payer

Denial rate by payer

History off · chats live in memory only. Synthetic demo data.

Demo user
acme_labs · analyst

Denials by payer and by claim status

Thought for 2.3s · Ran 3 queries, 1 check

Checking the claims view.

Looked at sem.claims 3ms

Ran query 7 rows · 6ms

Ran query 4 rows · 8ms

Final query 7 rows · 7ms

Verified the result

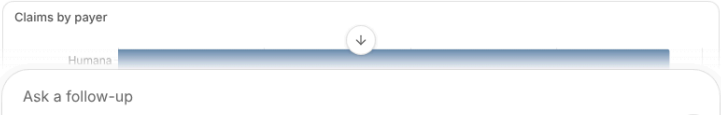
Claims and paid amount by payer, and claims and charges by status.

By payer

Payer name	Claims	Paid amount
UnitedHealthcare	175	$40,676.93
Medicaid	174	$34,856.59
Cigna	178	$33,726.04
Humana	189	$33,696.19
Medicare	170	$33,068.16
BCBS	158	$32,990.90
Aetna	156	$31,179.18

from sem.claims · Showing 7 of 7 rows · 2.43 s

View SQL



Ask a follow-up

CC can make mistakes. Check figures against source before use.

New chat

Search chats

Today

Long query

Claims by payer

Denials for a payer with no

Ask me a few questions about

How bad are our denials

Claims

Paid amount by service month

Simulate an error

Patient names and member IDs

Forecast our denial rate

Avg paid amount by payer

Net collection rate by payer

Claims by CARC code

Denial rate by payer

Denial rate by payer

History off · chats live in memory only. Synthetic demo data.

Demo user
acme_labs · analyst

A long query please

Thought for 1.8s · Ran 2 queries, 1 check

Checking the claims view. One monthly series per payer, joined on service month.

Looked at sem.claims 4ms

Ran query 12 rows · 51ms

Final query 12 rows · 47ms

Verified the result

Paid amount and denied claims by month, one column pair per payer.

Service month	Paid aetna	Denied aetna	Paid BCBS	Denied BCBS	Paid cigna	Denied cigna
2025-11-01	$3,474.24	2	$3,233.98	2	$3,744.42	0
2025-12-01	$2,516.02	3	$3,100.57	3	$2,515.24	1
2026-01-01	$3,366.40	3	$3,726.30	3	$1,810.29	3
2026-02-01	$1,185.70	3	$1,100.09	3	$2,502.19	2
2026-03-01	$4,669.66	2	$4,586.42	0	$3,331.18	3
2026-04-01	$3,956.80	4	$1,324.86	3	$1,626.25	2
2026-05-01	$2,256.91	1	$2,384.47	5	$2,866.80	2
2026-06-01	$2,081.64	6	$1,542.33	1	$3,758.56	4
2026-07-01	$1,427.04	2	$2,748.69	1	$3,786.37	3
2026-08-01	$1,425.20	2	$3,165.58	2	$2,203.95	3

from sem.claims · Showing 12 of 12 rows · 1.91 s

Open full table

View SQL

Ask a follow-up

CC can make mistakes. Check figures against source before use.

New chat

Search chats

Today

Claims by CARC code

Long query

Claims by payer

Denials for a payer with no

Ask me a few questions about

How bad are our denials

Claims

Paid amount by service month

Simulate an error

Patient names and member IDs

Forecast our denial rate

Avg paid amount by payer

Net collection rate by payer

Claims by CARC code

Denial rate by payer

Denial rate by payer

History off · chats live in memory only. Synthetic demo data.

Demo user
acme_labs · analyst

Hard query with many repairs

Thought for 4.3s · Ran 6 queries, 4 repairs, 1 check

Checking the claims view. Retrying with a corrected query. Retrying with a corrected query. Retrying with a corrected query. The denial reason column is carc_code on sem.claims.

Looked at sem.claims 7ms

Ran query 3ms

Repaired the query

Binder Error: Referenced column "denial_reason" not found in FROM clause! Hint: check column names wit…

Re-ran repaired query · attempt 2 2ms

Repaired the query · attempt 2

Binder Error: Referenced column "carc" not found in FROM clause! Hint: check column names with `contex…

Re-ran repaired query · attempt 3 6ms

Repaired the query · attempt 3

Catalog Error: Table with name claim does not exist!

Re-ran repaired query · attempt 4 3ms

Repaired the query · attempt 4

Binder Error: Referenced column "claimz" not found in FROM clause! Hint: check column names with `cont…

Re-ran repaired query · attempt 5 6 rows · 8ms

Final query 6 rows · 7ms

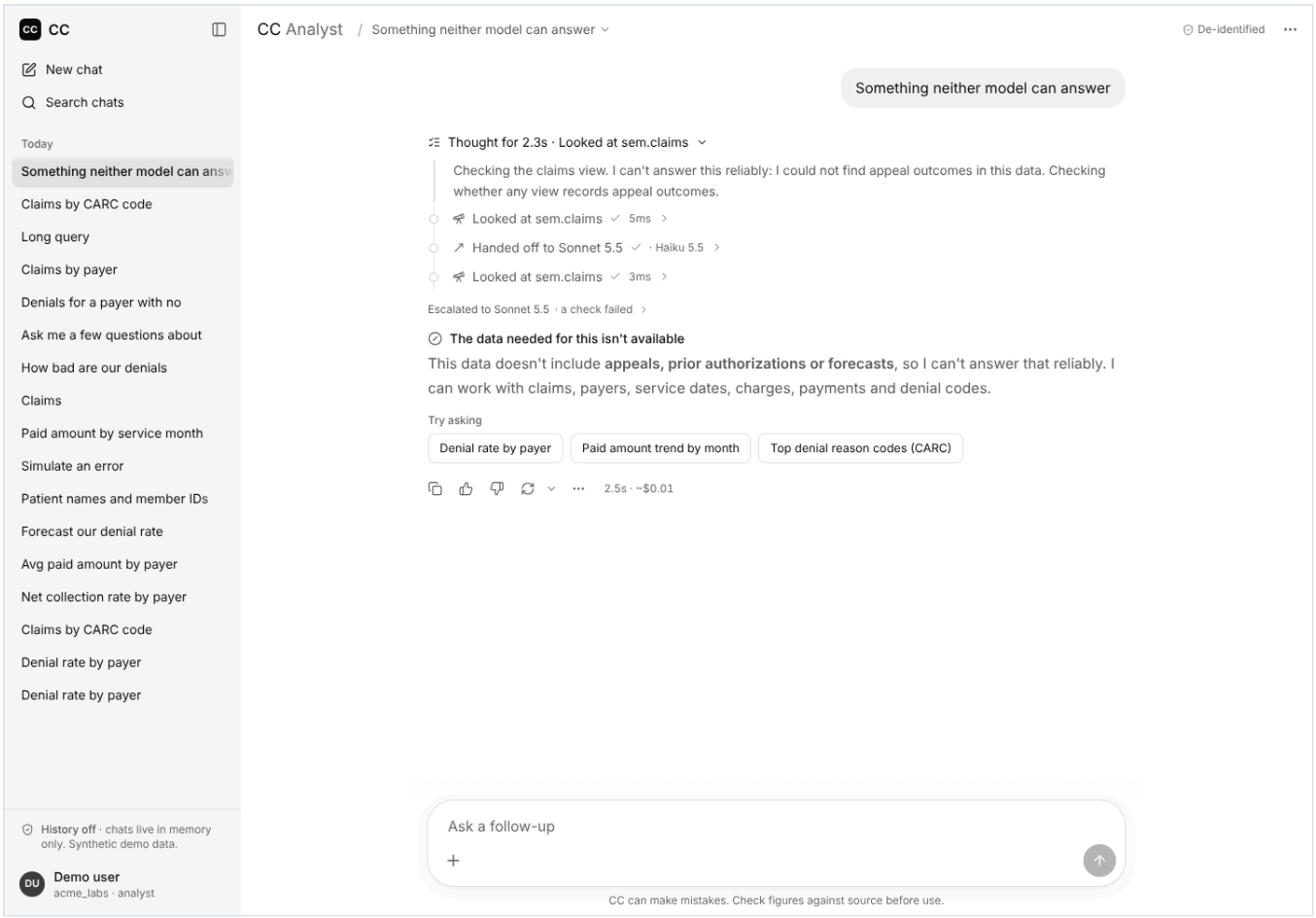
Verified the result

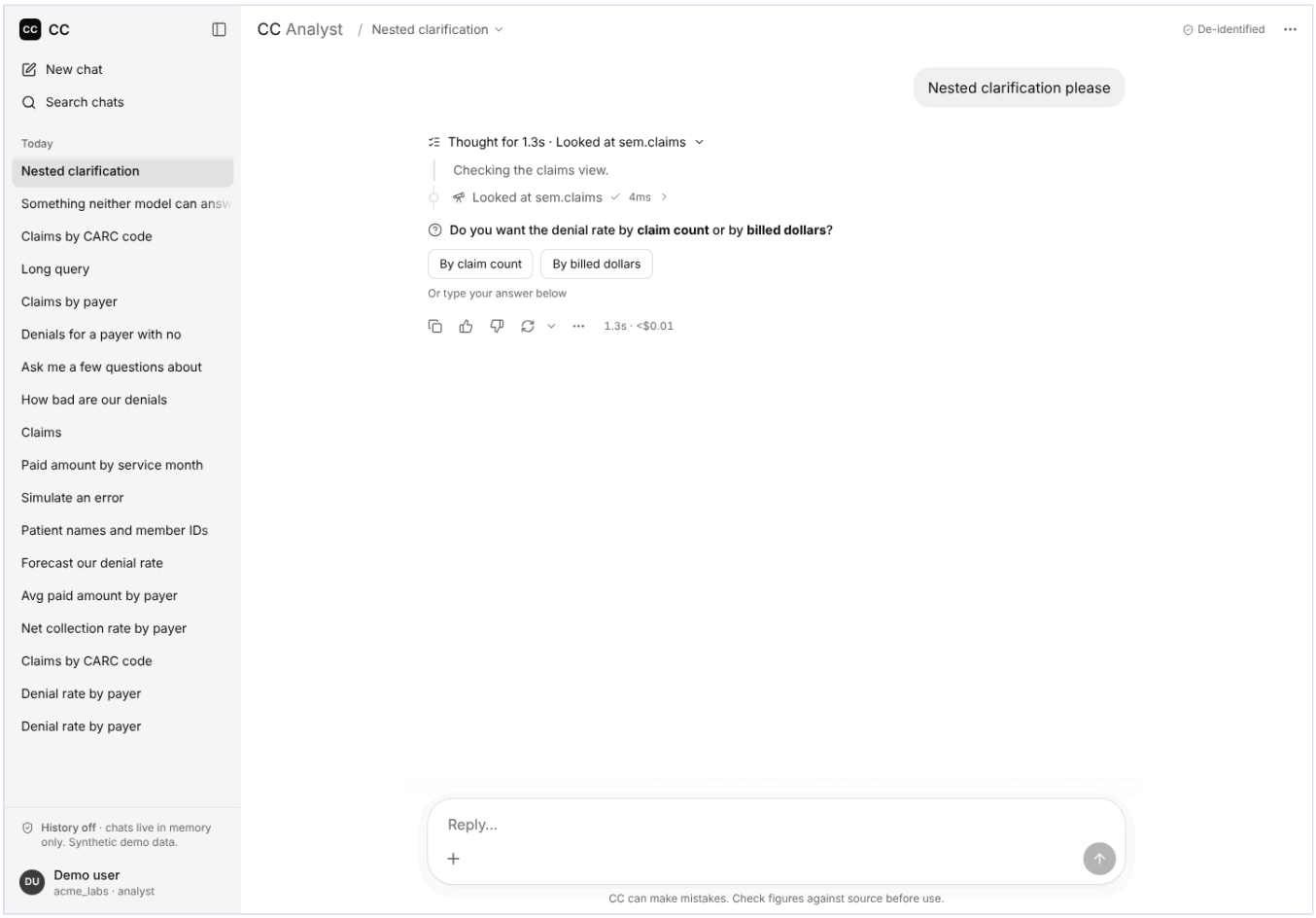
Denied and partially paid claims by CARC code (after four corrections).

CARC code	Claims	Charge amount
CO-97	84	$37,320.70
CO-45	72	$31,350.38

Ask a follow-up

CC can make mistakes. Check figures against source before use.


CC
New chat
Search chats
Today
Something neither model can answ
Claims by CARC code
Long query
Claims by payer
Denials for a payer with no
Ask me a few questions about
How bad are our denials
Claims
Paid amount by service month
Simulate an error
Patient names and member IDs
Forecast our denial rate
Avg paid amount by payer
Net collection rate by payer
Claims by CARC code
Denial rate by payer
Denial rate by payer
History off · chats live in memory only. Synthetic demo data.
DU
Demo user
acme_labs · analyst
CC Analyst / Something neither model can answer
De-identified
Something neither model can answer
Thought for 2.3s · Looked at sem.claims
Checking the claims view. I can't answer this reliably: I could not find appeal outcomes in this data. Checking whether any view records appeal outcomes.
Looked at sem.claims 5ms
Handed off to Sonnet 5.5 · Haiku 5.5
Looked at sem.claims 3ms
Escalated to Sonnet 5.5 · a check failed
The data needed for this isn't available
This data doesn't include appeals, prior authorizations or forecasts, so I can't answer that reliably. I can work with claims, payers, service dates, charges, payments and denial codes.
Try asking
Denial rate by payer
Paid amount trend by month
Top denial reason codes (CARC)
2.5s · ~$0.01
Ask a follow-up
CC can make mistakes. Check figures against source before use.


CC
New chat
Search chats
Today
Nested clarification
Something neither model can answ
Claims by CARC code
Long query
Claims by payer
Denials for a payer with no
Ask me a few questions about
How bad are our denials
Claims
Paid amount by service month
Simulate an error
Patient names and member IDs
Forecast our denial rate
Avg paid amount by payer
Net collection rate by payer
Claims by CARC code
Denial rate by payer
Denial rate by payer
History off · chats live in memory only. Synthetic demo data.
DU
Demo user
acme_labs · analyst
CC Analyst / Nested clarification
De-identified
Nested clarification please
Thought for 1.3s · Looked at sem.claims
Checking the claims view.
Looked at sem.claims 4ms
Do you want the denial rate by claim count or by billed dollars?
By claim count
By billed dollars
Or type your answer below
1.3s · <$0.01
Reply...
CC can make mistakes. Check figures against source before use.

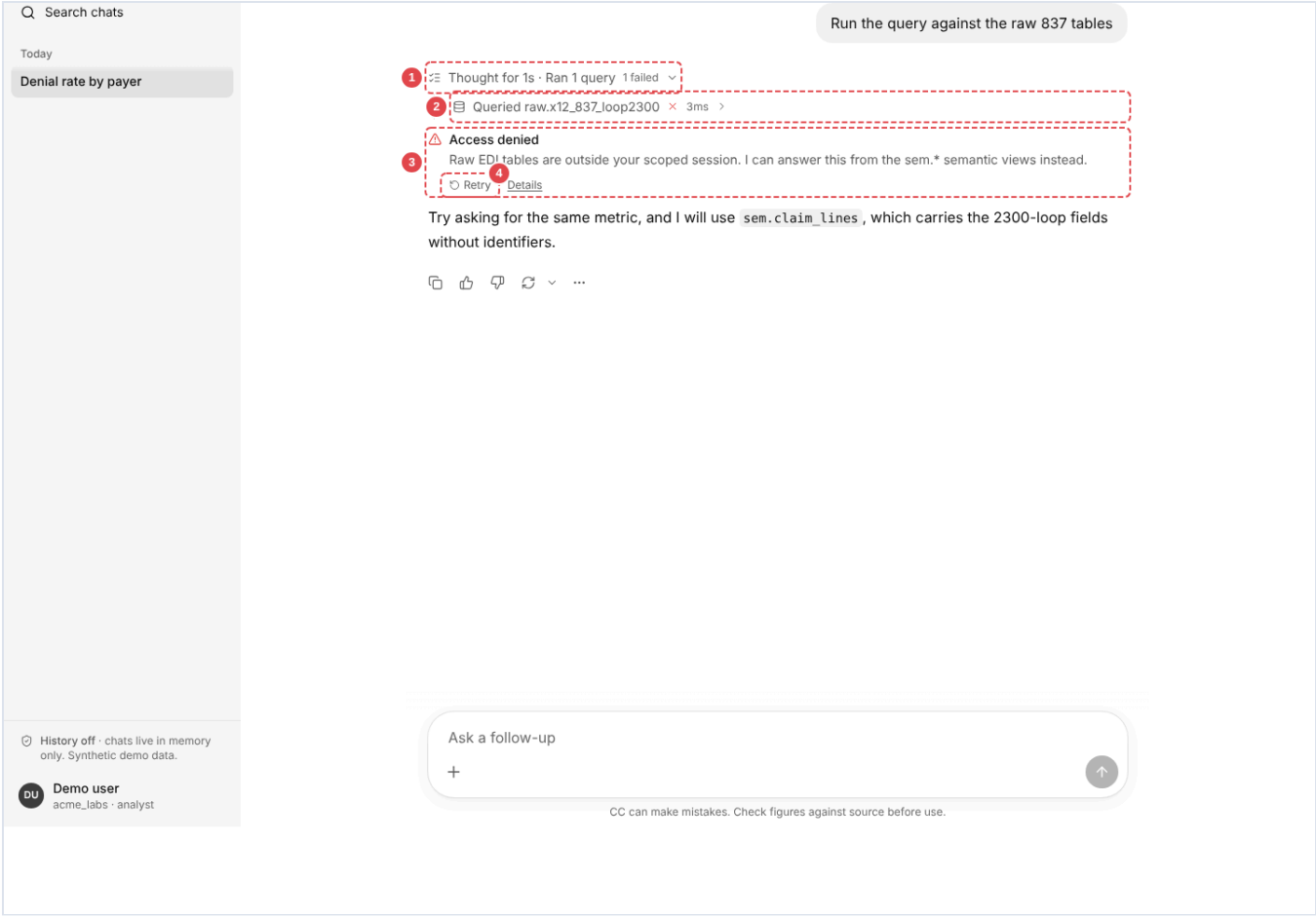
9.6 Limits and escalations shown to the user

Limit	Value / source	What the user sees
Repair attempts	`step.repair_attempt.maxAttempts` (demo: up to attempt 5)	Each attempt as its own step; "repairs ran out" on escalation
Escalation	`escalation.trigger`: verifier_fail, low_confidence, repair_exhausted, complexity, disagreement, user	"Handed off to ..."; usage popover lists both models
Data escalation (lossless)	`request_escalation(reason)` → `step.explore` probe `request_escalation`	"Used the full EDI data" / "Asked for the full EDI data" when refused
Concurrent runs	3 unfinished runs per user (`CC_API_MAX_CONCURRENT_RUNS`) → 429	"Busy right now, retrying in Ns..."
Request body	1 MiB (`CC_API_MAX_BODY_BYTES`) → 413	"Message too long"
Preview rows	≤ 50 rows in the table part; full CSV via `GET /api/download/{id}` (15 min, principal-bound)	"Showing 7 of 7 rows" / file card
Stream idle	45 s client guard; server heartbeat every 15 s	S67
Reconnect	4 retries (250 ms, 1 s, 2 s, 4 s); replay kept 60 s after the run	S56, S57
Disconnect grace	10 s server-side, then the run is cancelled	S84
Literal masking	`Capabilities.sqlLiteralMasking` (phi / all / off), `maskedLiterals`	`?` in shown SQL, "N values masked" (`sql-masked`)

TODO: add the analyzer's own budgets (`max_turns`, `max_tool_calls`, `row_limit` 5000, `time_limit_seconds`) once the deployed values are confirmed (backend page §7).

9.7 Guard rejection

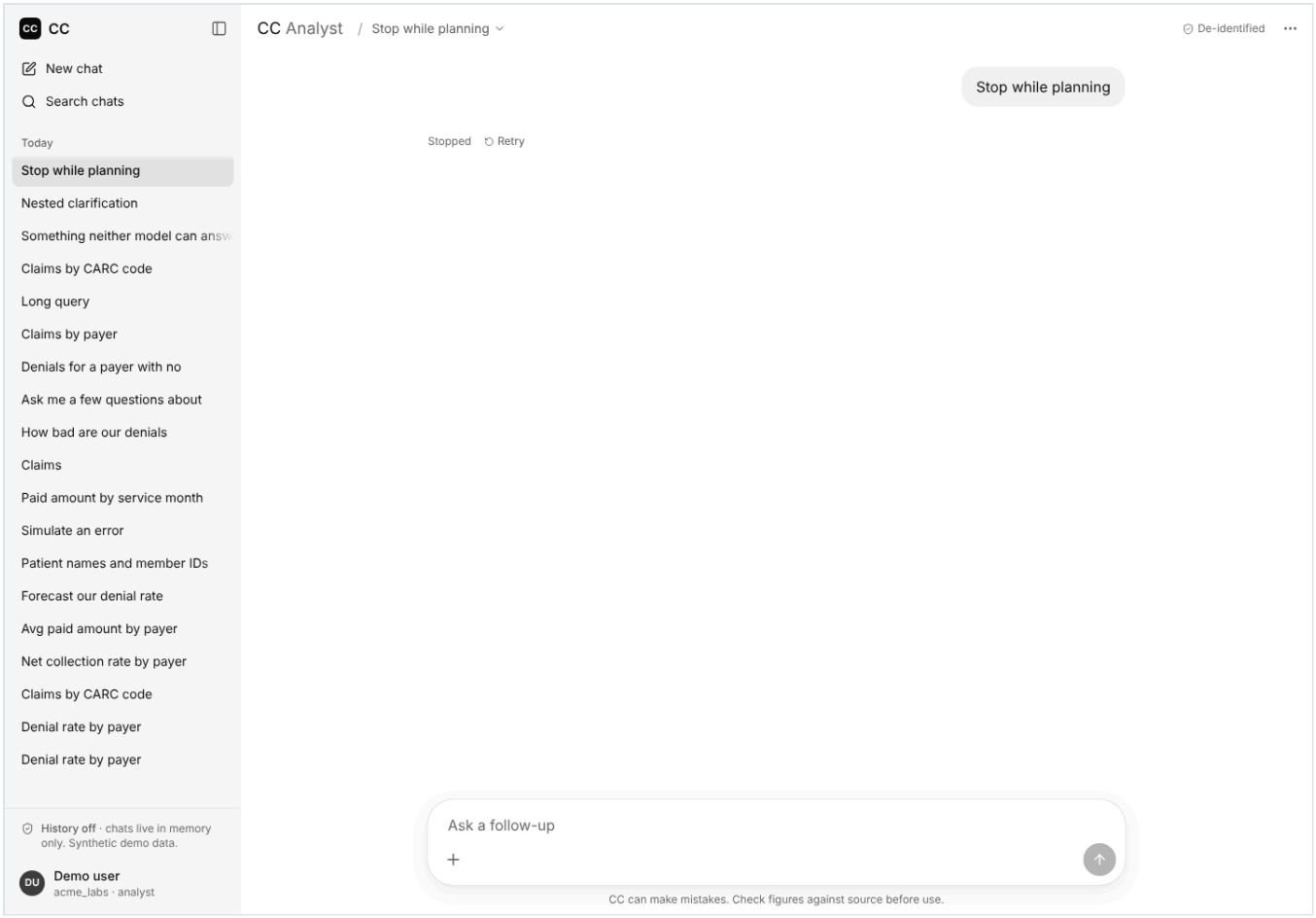
When the agent's SQL names something outside the allowed set (only `sem.*`; `lossless.*` only after an audited escalation), the guard blocks it before DuckDB runs. The user sees a failed step and, in the mock, an `access_denied` error part that offers the semantic views instead.

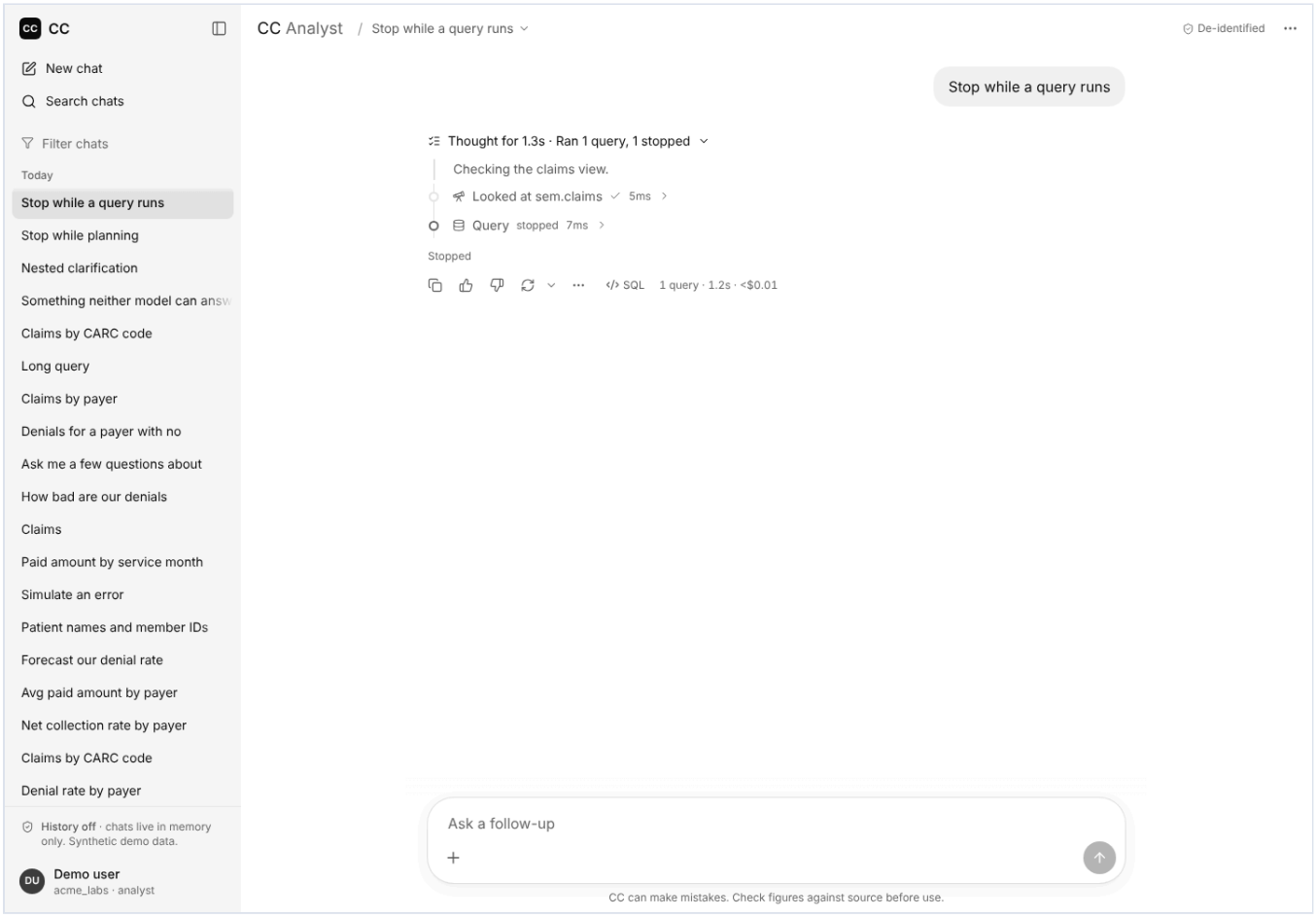


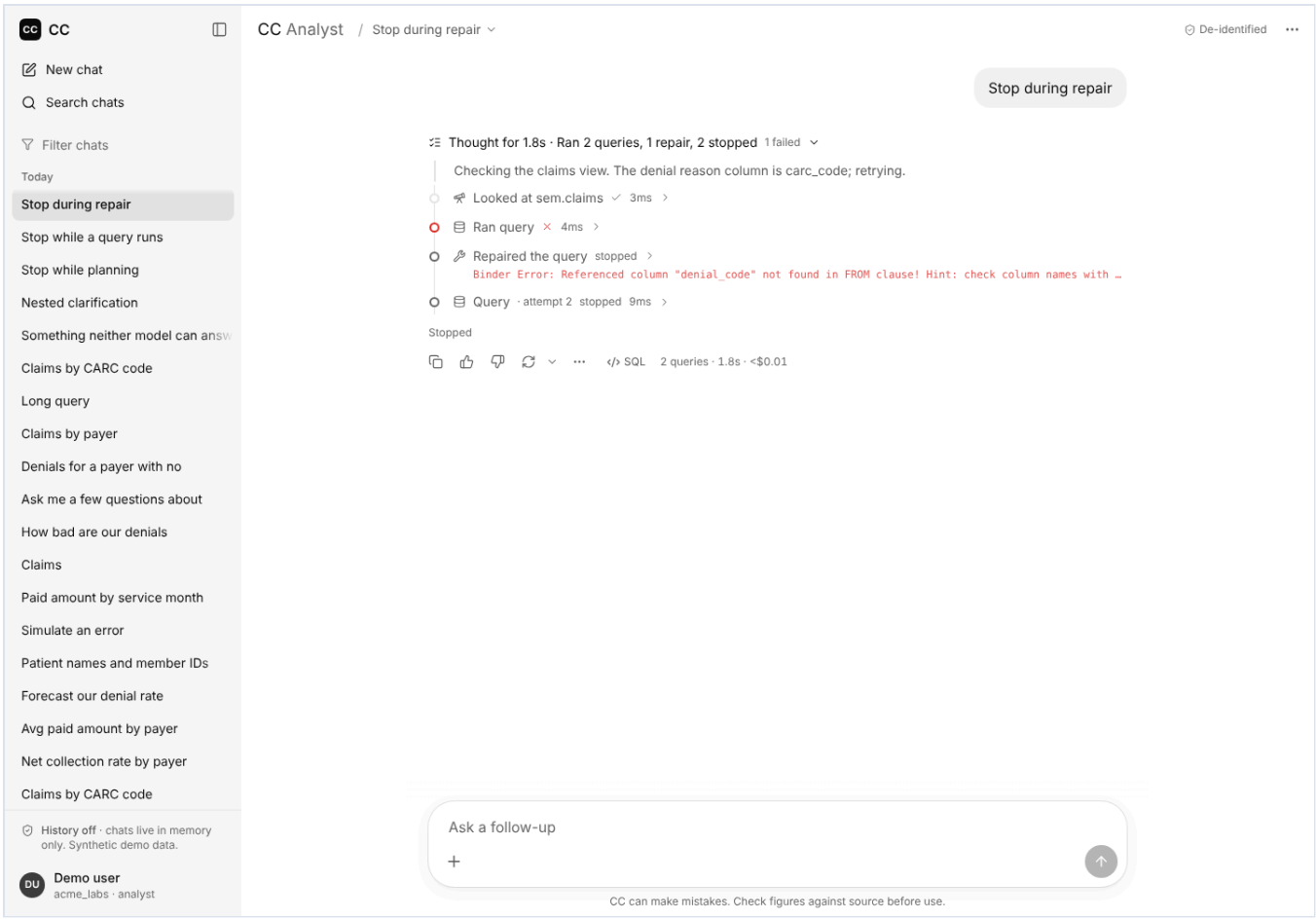
#	Element	Behaviour
1	Timeline header	"Thought for 1s · Ran 1 query · 1 failed".
2	Failed step	"Blocked by guard: only sem.* views are allowed."
3	Error part	`code: access_denied`, "Raw EDI tables are outside your scoped session..."; Details shows `guard=run_semantic_sql; relation=…; policy=sem_only`.
4	Retry	Re-runs the same question (regenerate).

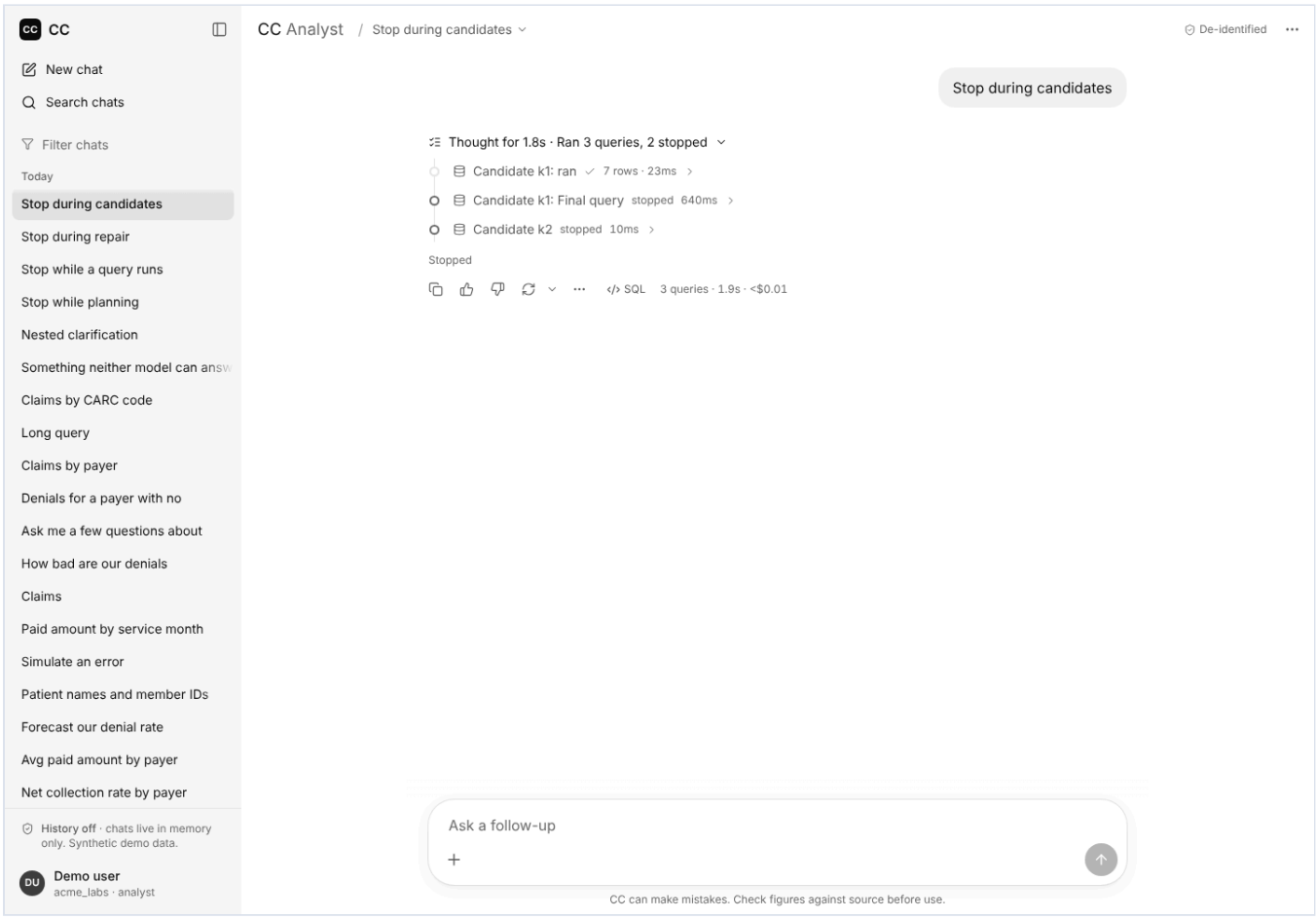
> **Note:** This capture is the **mock** fixture. On the real analyzer a guard error usually goes back to the model as a `step.sql_result {ok:false, error.code: "guard_blocked"}` and is repaired rather than surfaced. *TODO: capture a real* `guard_blocked` *step from the deployed app.*

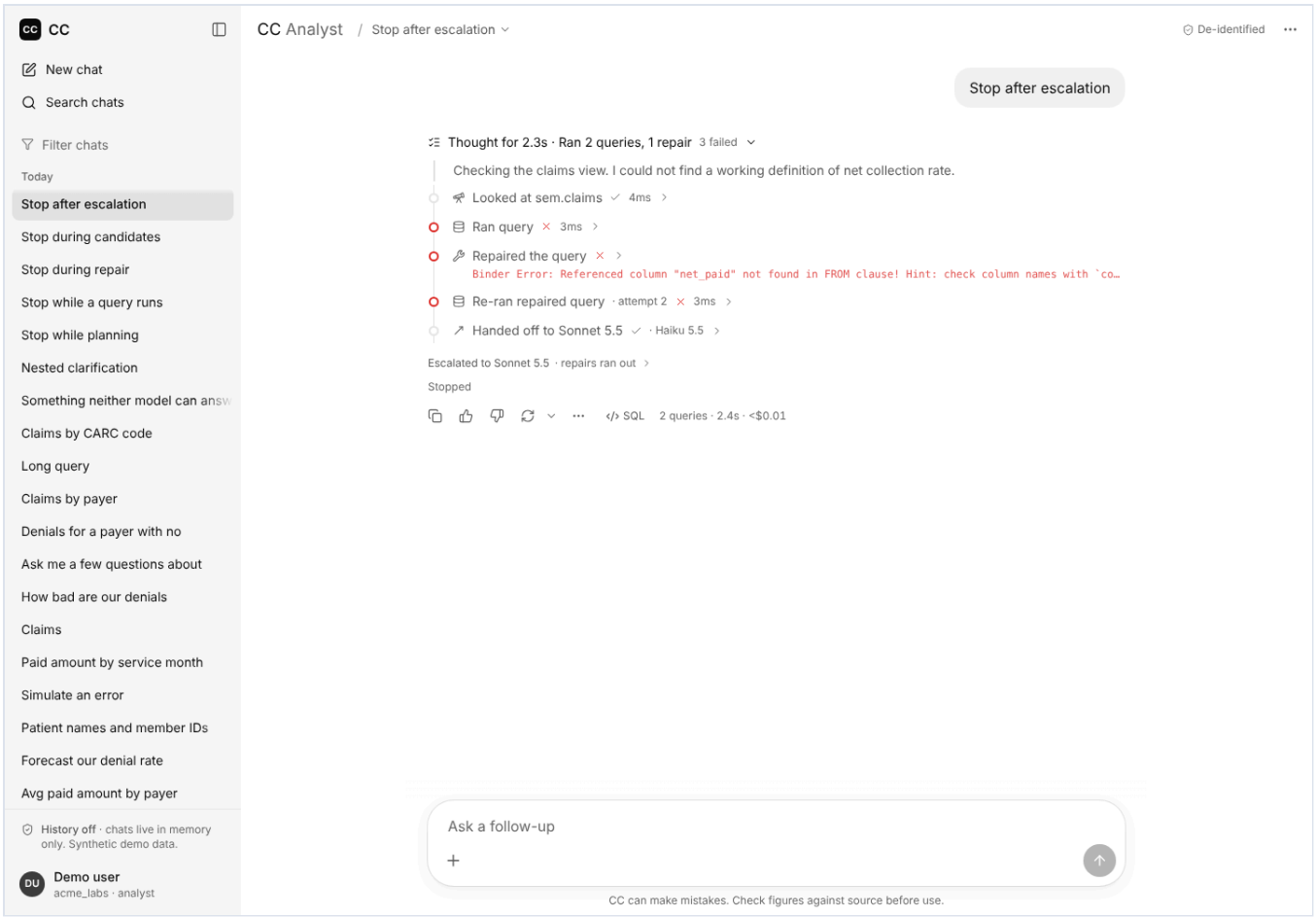
9.8 Cancellation at each stage

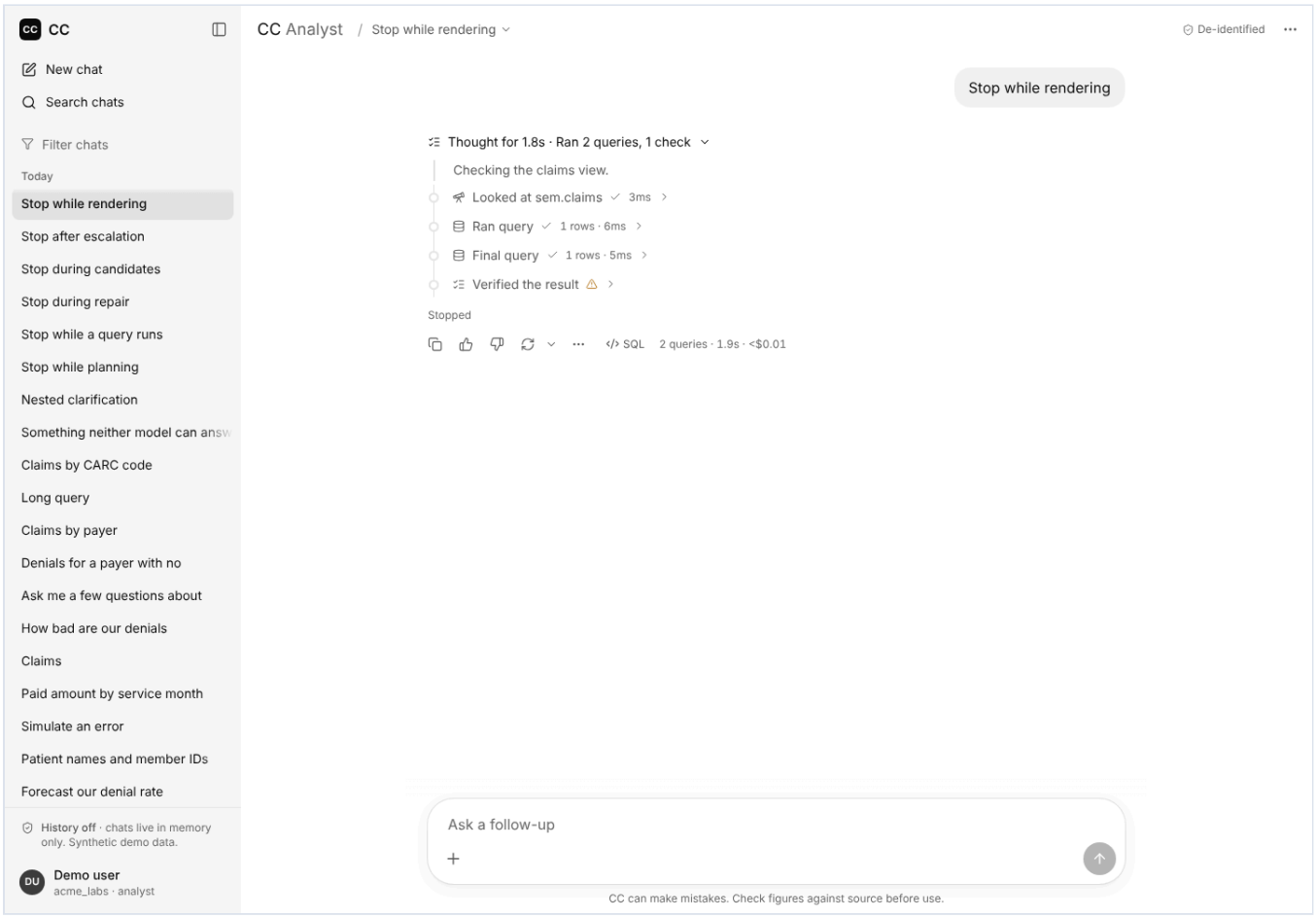





CC
New chat
Search chats
Filter chats
Today
Stop during repair
Stop while a query runs
Stop while planning
Nested clarification
Something neither model can answ
Claims by CARC code
Long query
Claims by payer
Denials for a payer with no
Ask me a few questions about
How bad are our denials
Claims
Paid amount by service month
Simulate an error
Patient names and member IDs
Forecast our denial rate
Avg paid amount by payer
Net collection rate by payer
Claims by CARC code
History off · chats live in memory only. Synthetic demo data.
Demo user
acme_labs · analyst
CC Analyst / Stop during repair
De-identified
Stop during repair
Thought for 1.8s · Ran 2 queries, 1 repair, 2 stopped 1 failed
Checking the claims view. The denial reason column is carc_code; retrying.
Looked at sem.claims 3ms
Ran query 4ms
Repaired the query stopped
Binder Error: Referenced column "denial_code" not found in FROM clause! Hint: check column names with …
Query · attempt 2 stopped 9ms
Stopped
SQL 2 queries · 1.8s · <$0.01
Ask a follow-up
CC can make mistakes. Check figures against source before use.


CC
New chat
Search chats
Filter chats
Today
Stop during candidates
Stop during repair
Stop while a query runs
Stop while planning
Nested clarification
Something neither model can answ
Claims by CARC code
Long query
Claims by payer
Denials for a payer with no
Ask me a few questions about
How bad are our denials
Claims
Paid amount by service month
Simulate an error
Patient names and member IDs
Forecast our denial rate
Avg paid amount by payer
Net collection rate by payer
History off · chats live in memory only. Synthetic demo data.
Demo user
acme_labs · analyst
CC Analyst / Stop during candidates
De-identified
Stop during candidates
Thought for 1.8s · Ran 3 queries, 2 stopped
Candidate k1: ran 7 rows · 23ms
Candidate k1: Final query stopped 640ms
Candidate k2 stopped 10ms
Stopped
SQL 3 queries · 1.9s · <$0.01
Ask a follow-up
CC can make mistakes. Check figures against source before use.


CC
New chat
Search chats
Filter chats
Today
Stop after escalation
Stop during candidates
Stop during repair
Stop while a query runs
Stop while planning
Nested clarification
Something neither model can answ
Claims by CARC code
Long query
Claims by payer
Denials for a payer with no
Ask me a few questions about
How bad are our denials
Claims
Paid amount by service month
Simulate an error
Patient names and member IDs
Forecast our denial rate
Avg paid amount by payer
History off · chats live in memory only. Synthetic demo data.
DU
Demo user
acme_labs · analyst
CC Analyst / Stop after escalation
De-identified
Stop after escalation
Thought for 2.3s · Ran 2 queries, 1 repair 3 failed
Checking the claims view. I could not find a working definition of net collection rate.
Looked at sem.claims 4ms
Ran query 3ms
Repaired the query
Binder Error: Referenced column "net_paid" not found in FROM clause! Hint: check column names with `co…
Re-ran repaired query · attempt 2 3ms
Handed off to Sonnet 5.5 · Haiku 5.5
Escalated to Sonnet 5.5 · repairs ran out
Stopped
SQL 2 queries · 2.4s · <$0.01
Ask a follow-up
CC can make mistakes. Check figures against source before use.


CC
New chat
Search chats
Filter chats
Today
Stop while rendering
Stop after escalation
Stop during candidates
Stop during repair
Stop while a query runs
Stop while planning
Nested clarification
Something neither model can answ
Claims by CARC code
Long query
Claims by payer
Denials for a payer with no
Ask me a few questions about
How bad are our denials
Claims
Paid amount by service month
Simulate an error
Patient names and member IDs
Forecast our denial rate
History off · chats live in memory only. Synthetic demo data.
DU
Demo user
acme_labs · analyst
CC Analyst / Stop while rendering
De-identified
Stop while rendering
Thought for 1.8s · Ran 2 queries, 1 check
Checking the claims view.
Looked at sem.claims 3ms
Ran query 1 rows · 6ms
Final query 1 rows · 5ms
Verified the result
Stopped
SQL 2 queries · 1.9s · <$0.01
Ask a follow-up
CC can make mistakes. Check figures against source before use.

TODO: one line per stage on what is kept: steps up to the stop, no partial table, `usage.queries` counts the interrupted statement.

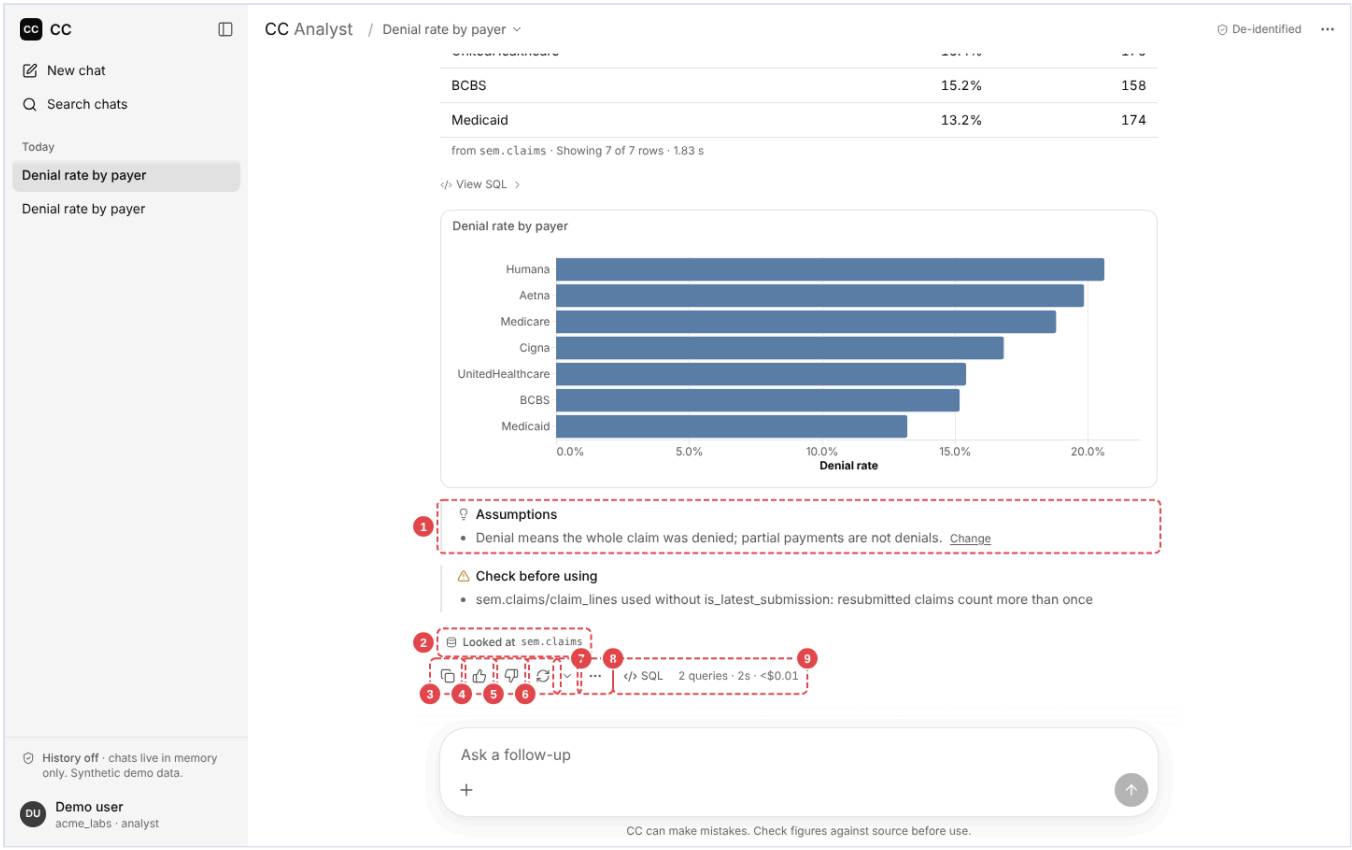
---

10. Message rendering parts

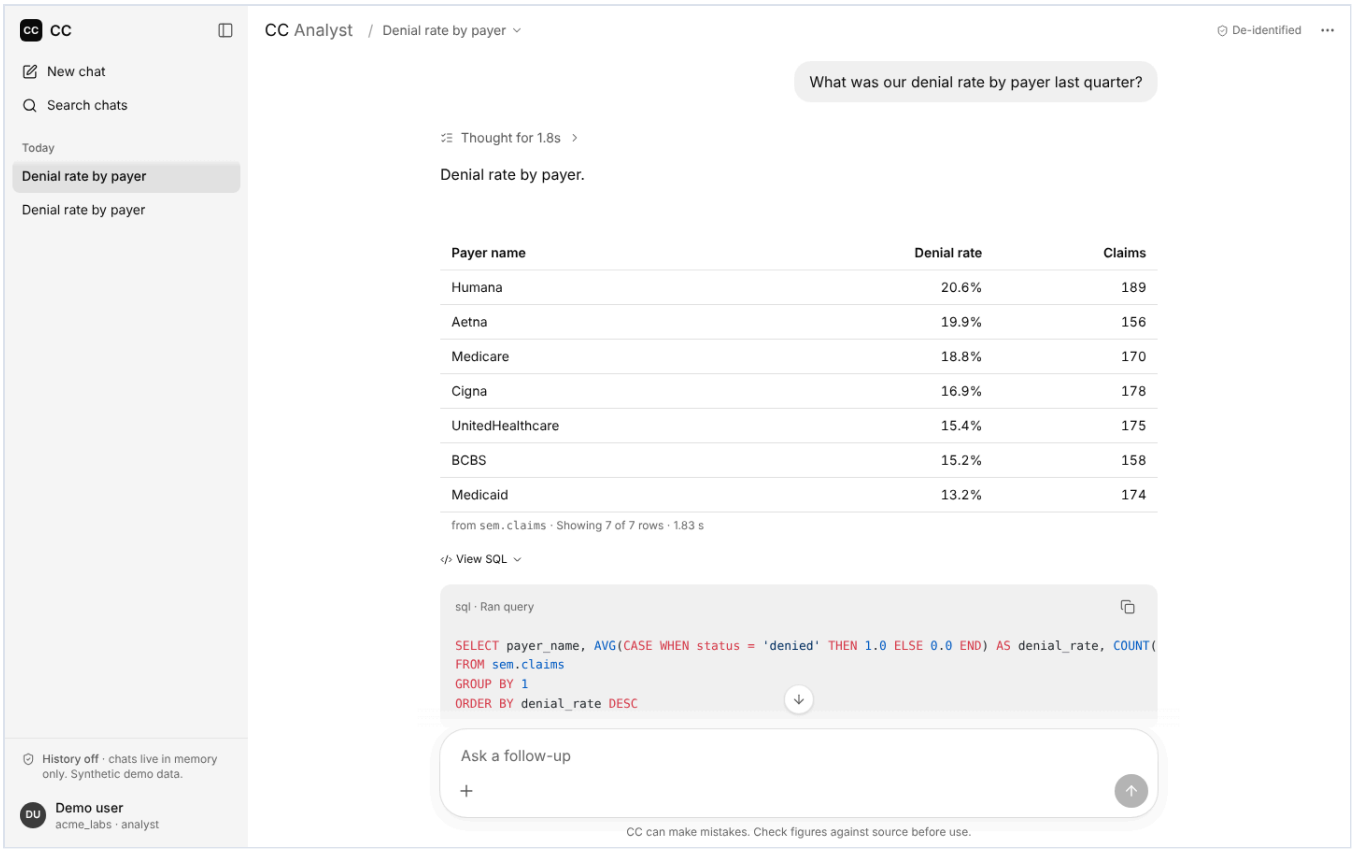
TODO: intro: the answer is a list of typed parts (`Part` union in `contract/v1.gen.ts`); each has a renderer in `components/parts/` or `components/answer/`.

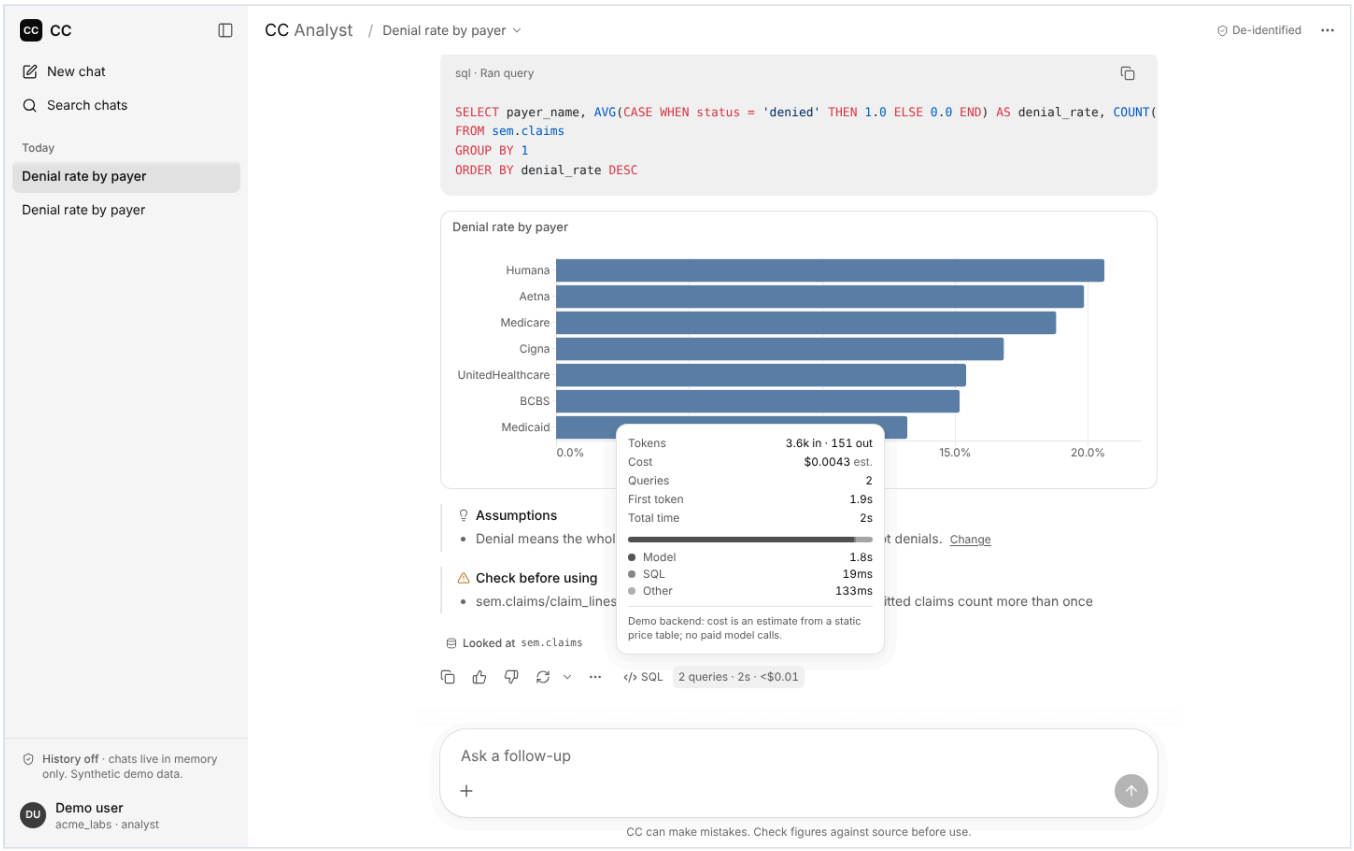
Part	Renderer	Notes
`text`	`Markdown` (Streamdown, GFM)	No raw HTML; remote images replaced by alt text; links `noopener noreferrer nofollow`
`reasoning`	folded into `StepTimeline`	Real backends send label/count summaries only
`tool_call` (legacy)	`StepTimeline` rows	Input SQL/args only, never result rows
`table`	`DataTable`	PHI columns masked, sortable, virtualized > 200 rows
`chart`	`ChartView` (Vega-Lite, SVG)	Strict `PlotSpec` (1.18.0); `data.url` stripped
`code`	`CodePartView` / `CodeBlock` (Shiki)	`role: query` collapses under "View SQL" when linked to a table
`image`	`ImageView`	Fetched by opaque `blobId` into a revoked object URL
`file`	`FileCard`	Download via adapter blob / `GET /api/download/{id}`
`callout`	`Callout`	`assumption` (editable → "Change"), `caveat`, `data_quality`, `phi_redaction`, `info`
`artifact`	`ArtifactCard` → panel	kinds: code, report, … (open enum, fallback `report`)
`error`	`ErrorView`	Plain line, small red icon, Retry, Details
`abstain`	`AbstainCard`	category (open, fallback `unsupported`), suggestions
`clarification`	`ClarifyCard` / `ClarifyForm`	single or 1–4 questions
`status`	—	not rendered as a part
unknown	`Unsupported`	"Unsupported content (<type>)"

10.1 Answer chrome: action row, sources, usage



#	Element	Behaviour
1	Assumption callout	"Change" pre-fills the composer with "Instead, assume ".
2	Sources chip	"Looked at sem.claims"; only when something was consulted (QA-A31).
3	Copy	Markdown of the answer; PHI columns dropped.
4	Good response	`PUT …/feedback {rating: up}` on server-known answers.
5	Bad response	Opens "Tell us more" (S18).
6	Regenerate	Re-runs as the next version (last answer only).
7	Answer style	Try again / More concise / More detailed / Ask to change response…
8	More actions	Branch in new chat, Print, Copy as Markdown, Copy SQL.
9	Usage / SQL	"SQL · 2 queries · 2s · <$0.01"; opens SQL inspector and the details popover.
10	Follow-ups	Suggested follow-up questions when the answer has them (not in this capture).


CC
New chat
Search chats
Today
Denial rate by payer
Denial rate by payer
CC Analyst / Denial rate by payer
De-identified
What was our denial rate by payer last quarter?
Thought for 1.8s
Denial rate by payer.
Payer name | Denial rate | Claims
Humana | 20.6% | 189
Aetna | 19.9% | 156
Medicare | 18.8% | 170
Cigna | 16.9% | 178
UnitedHealthcare | 15.4% | 175
BCBS | 15.2% | 158
Medicaid | 13.2% | 174
from sem.claims · Showing 7 of 7 rows · 1.83 s
View SQL
sql · Ran query
SELECT payer_name, AVG(CASE WHEN status = 'denied' THEN 1.0 ELSE 0.0 END) AS denial_rate, COUNT(
FROM sem.claims
GROUP BY 1
ORDER BY denial_rate DESC
Ask a follow-up
History off · chats live in memory only. Synthetic demo data.
Demo user
acme_labs · analyst
CC can make mistakes. Check figures against source before use.


CC
New chat
Search chats
Today
Denial rate by payer
Denial rate by payer
CC Analyst / Denial rate by payer
De-identified
sql · Ran query
SELECT payer_name, AVG(CASE WHEN status = 'denied' THEN 1.0 ELSE 0.0 END) AS denial_rate, COUNT(
FROM sem.claims
GROUP BY 1
ORDER BY denial_rate DESC
Denial rate by payer
Humana
Aetna
Medicare
Cigna
UnitedHealthcare
BCBS
Medicaid
0.0%
15.0%
20.0%
Tokens 3.6k in · 151 out
Cost $0.0043 est.
Queries 2
First token 1.9s
Total time 2s
Model 1.8s
SQL 19ms
Other 133ms
Demo backend: cost is an estimate from a static price table; no paid model calls.
Assumptions
Change
Check before using
Looked at sem.claims
SQL 2 queries · 2s · <$0.01
Ask a follow-up
History off · chats live in memory only. Synthetic demo data.
Demo user
acme_labs · analyst
CC can make mistakes. Check figures against source before use.

CC

New chat

Search chats

Today

Denial rate by payer

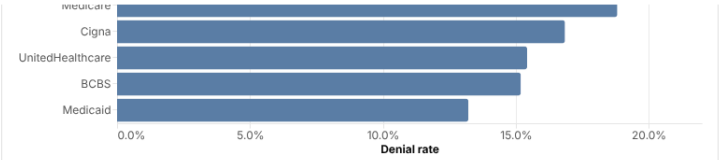
Denial rate by payer

History off · chats live in memory only. Synthetic demo data.

Demo user
acme_labs · analyst

CC Analyst / Denial rate by payer

De-identified



Assumptions

- Denial means the whole claim was denied; partial payments are not denials. Change

Check before using

- sem.claims/claim_lines used without is_latest_submission: resubmitted claims count more than once

Looked at `sem.claims`

SQL 2 queries · 2s · <$0.01

1. Result

Close

Ran query 7 rows 8ms produced "Result"

sql · query 1 of 1

```
SELECT payer_name, AVG(CASE WHEN status = 'denied' THEN 1.0 ELSE 0.0 END) AS denial_rate, COU
FROM sem.claims
GROUP BY 1
ORDER BY denial_rate DESC
```

Columns: payer_name denial_rate claims

Ask a follow-up

CC can make mistakes. Check figures against source before use.

CC

New chat

Search chats

Today

Denial rate by payer

Denial rate by payer

History off · chats live in memory only. Synthetic demo data.

Demo user
acme_labs · analyst

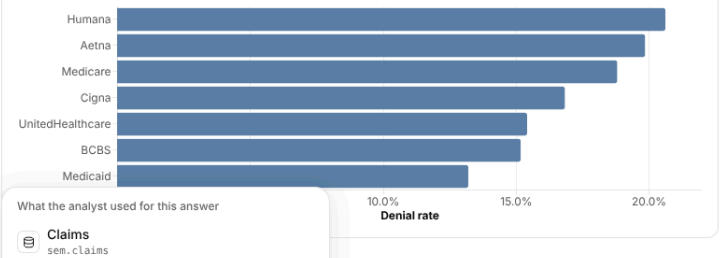
CC Analyst / Denial rate by payer

De-identified

sql · Ran query

```
SELECT payer_name, AVG(CASE WHEN status = 'denied' THEN 1.0 ELSE 0.0 END) AS denial_rate, COUNT(
FROM sem.claims
GROUP BY 1
ORDER BY denial_rate DESC
```

Denial rate by payer



What the analyst used for this answer

Claims
`sem.claims`

One row per claim submission with billed, allowed and paid amounts, payer and status.

Payer name · Denial rate · Claims

Table and column names only. No row values are shown here.

al payments are not denials. Change

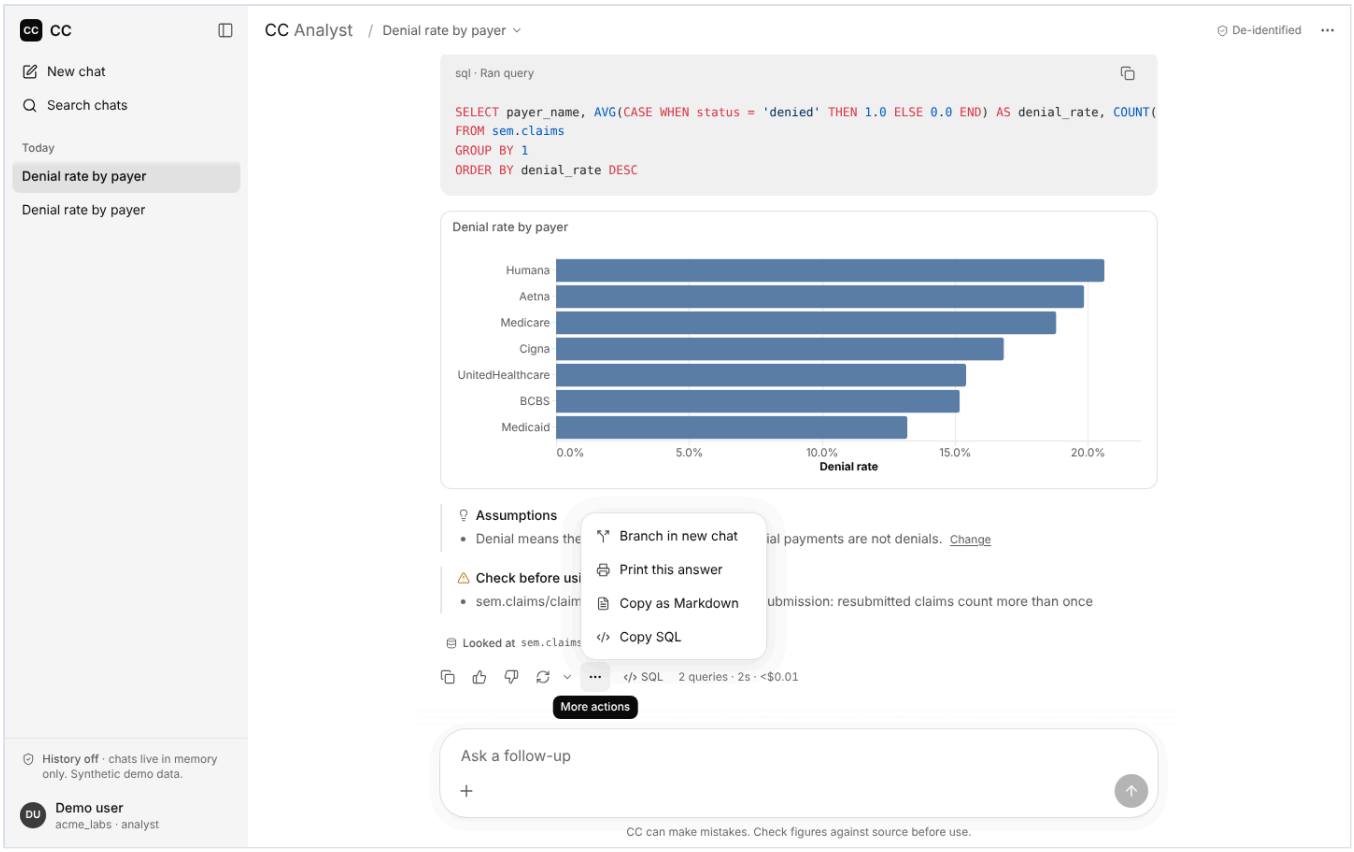
ubmission: resubmitted claims count more than once

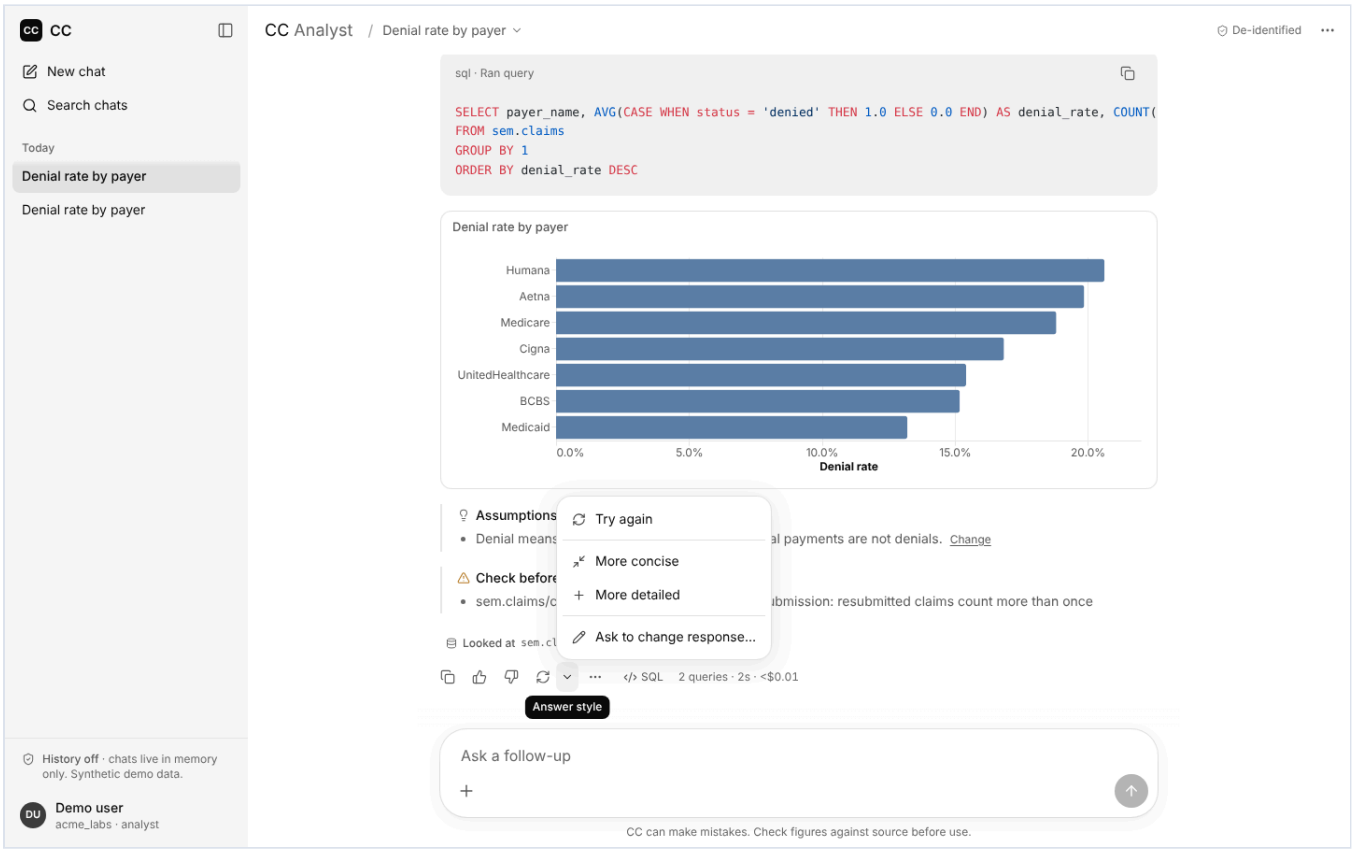
Looked at `sem.claims`

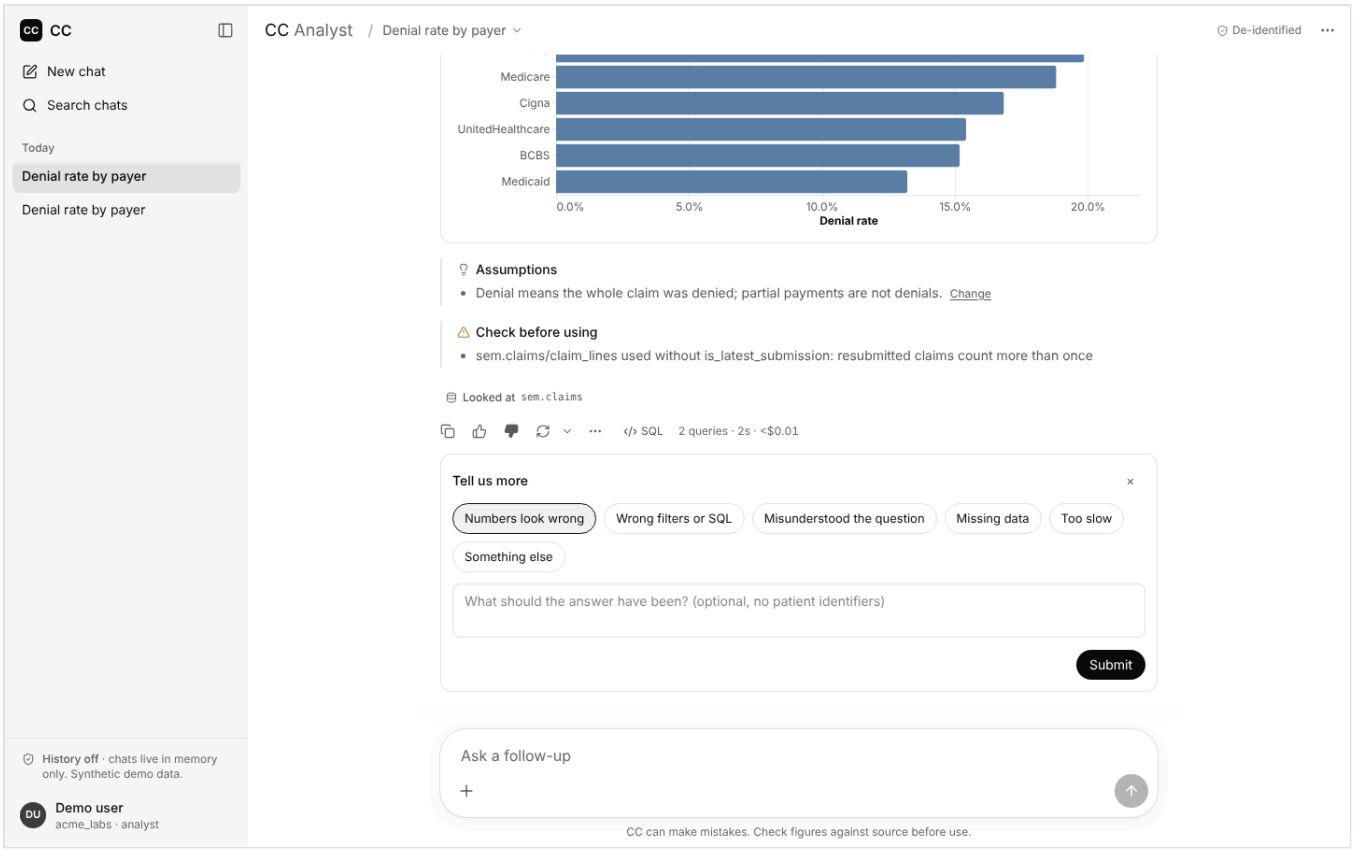
SQL 2 queries · 2s · <$0.01

Ask a follow-up

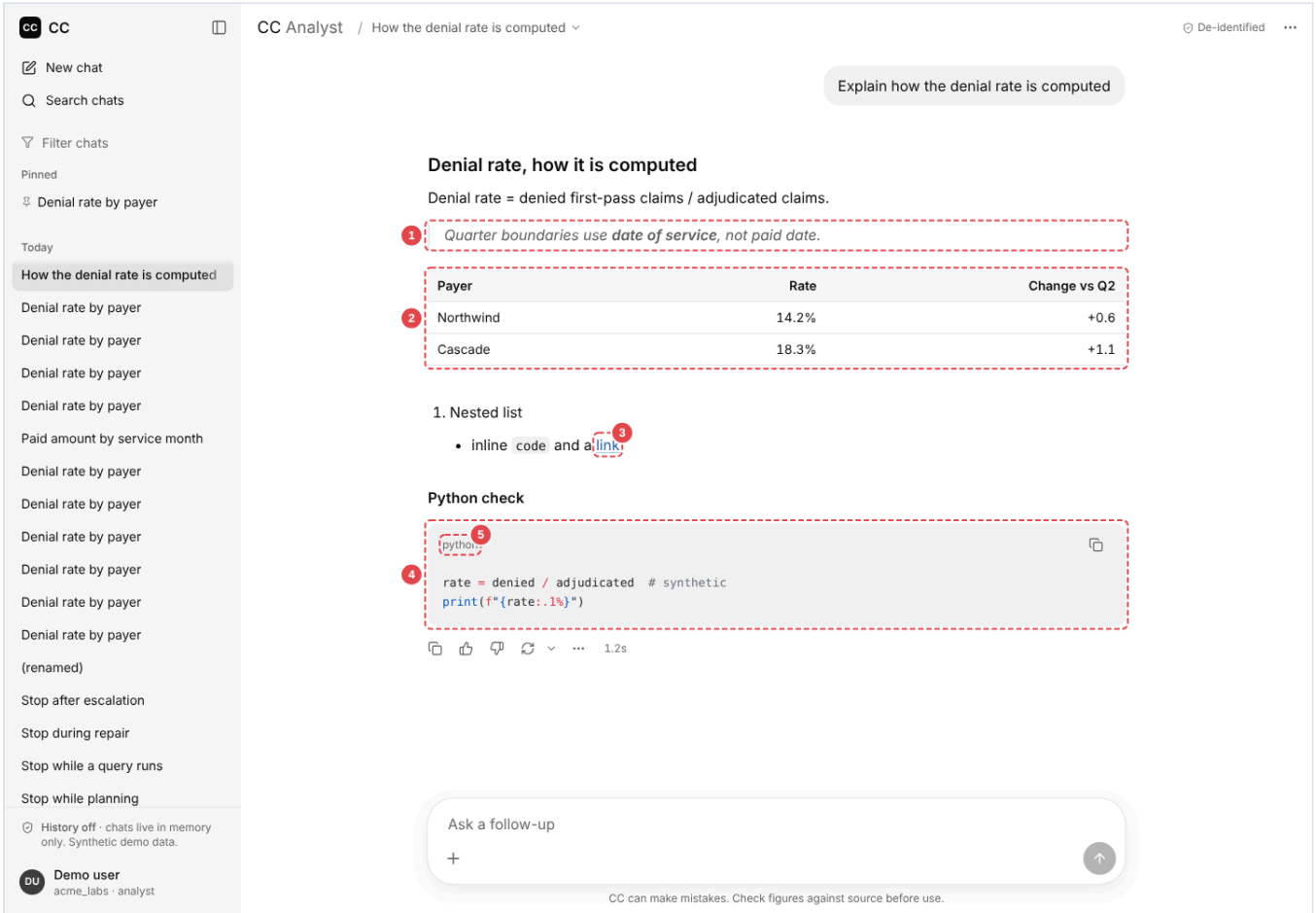
CC can make mistakes. Check figures against source before use.


CC
New chat
Search chats
Today
Denial rate by payer
Denial rate by payer
CC Analyst / Denial rate by payer
De-identified
sql · Ran query
SELECT payer_name, AVG(CASE WHEN status = 'denied' THEN 1.0 ELSE 0.0 END) AS denial_rate, COUNT(
FROM sem.claims
GROUP BY 1
ORDER BY denial_rate DESC
Denial rate by payer
Humana
Aetna
Medicare
Cigna
UnitedHealthcare
BCBS
Medicaid
0.0%
5.0%
10.0%
15.0%
20.0%
Denial rate
Assumptions
Denial means the
ial payments are not denials. Change
Check before usi
sem.claims/claim
ubmission: resubmitted claims count more than once
Looked at sem.claims
Branch in new chat
Print this answer
Copy as Markdown
Copy SQL
SQL 2 queries · 2s · <$0.01
More actions
Ask a follow-up
History off · chats live in memory only. Synthetic demo data.
Demo user
acme_labs · analyst
CC can make mistakes. Check figures against source before use.


CC
New chat
Search chats
Today
Denial rate by payer
Denial rate by payer
CC Analyst / Denial rate by payer
De-identified
sql · Ran query
SELECT payer_name, AVG(CASE WHEN status = 'denied' THEN 1.0 ELSE 0.0 END) AS denial_rate, COUNT(
FROM sem.claims
GROUP BY 1
ORDER BY denial_rate DESC
Denial rate by payer
Humana
Aetna
Medicare
Cigna
UnitedHealthcare
BCBS
Medicaid
0.0%
5.0%
10.0%
15.0%
20.0%
Denial rate
Assumptions
Denial means
ial payments are not denials. Change
Check before
sem.claims/c
bmission: resubmitted claims count more than once
Looked at sem.cl
Try again
More concise
More detailed
Ask to change response...
SQL 2 queries · 2s · <$0.01
Answer style
Ask a follow-up
History off · chats live in memory only. Synthetic demo data.
Demo user
acme_labs · analyst
CC can make mistakes. Check figures against source before use.



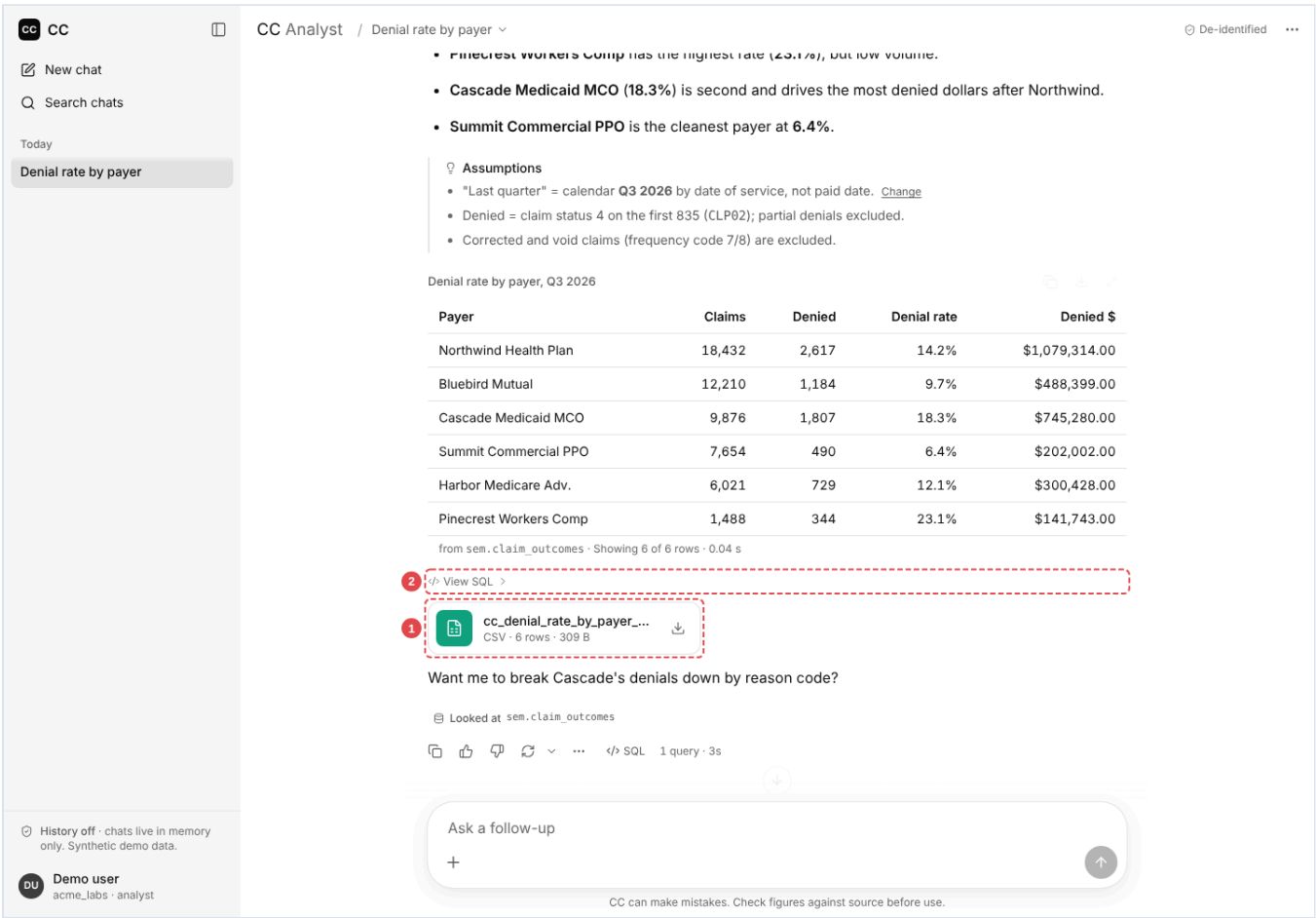
10.2 Prose, code and math



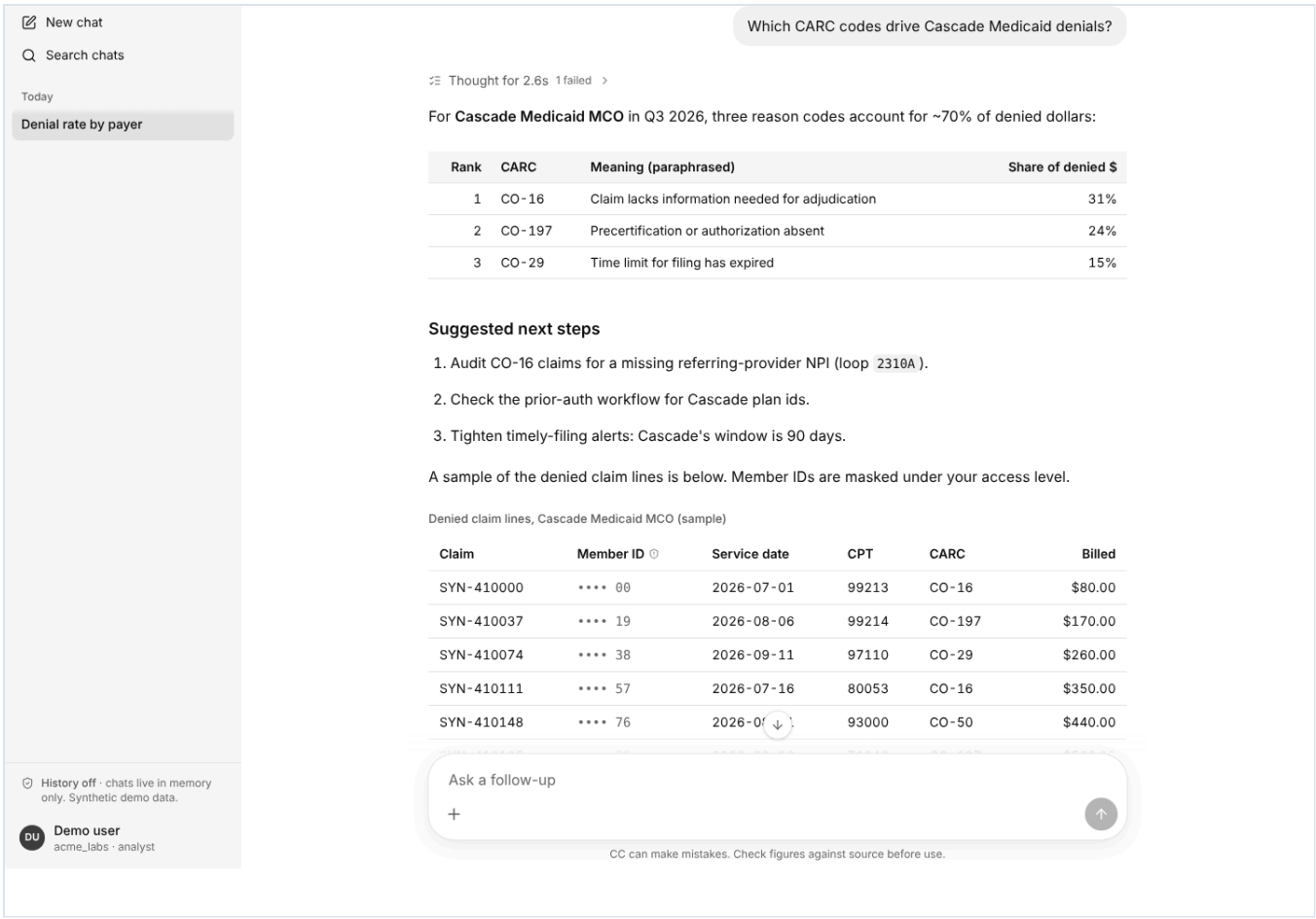
#	Element	Behaviour
1	Blockquote	Styled quote.
2	GFM table	Markdown table (not a `table` part: no toolbar).
3	Link	Quiet underline; `rel="noopener noreferrer nofollow"`.
4	Code block	Shiki highlighting, Copy, Download, Open in panel (on hover), Wrap lines.
5	Language label	`code-lang`.

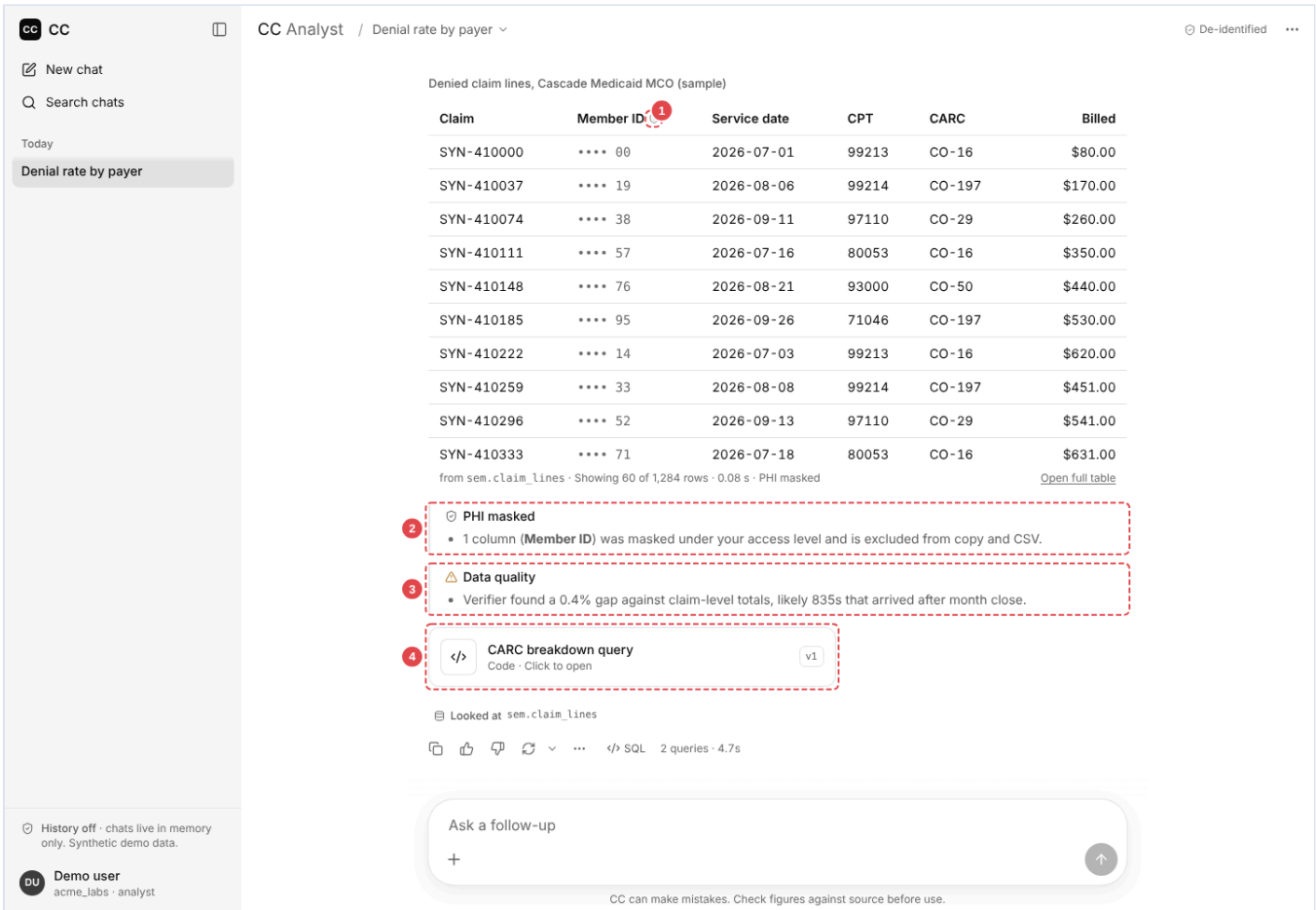
TODO screenshot: KaTeX math — send a turn whose text contains $r = \frac{d}{n}$ and $$...$$; locally this crashed the whole app instead (see §21, open question 1)

10.3 Files, PHI masking, artifacts

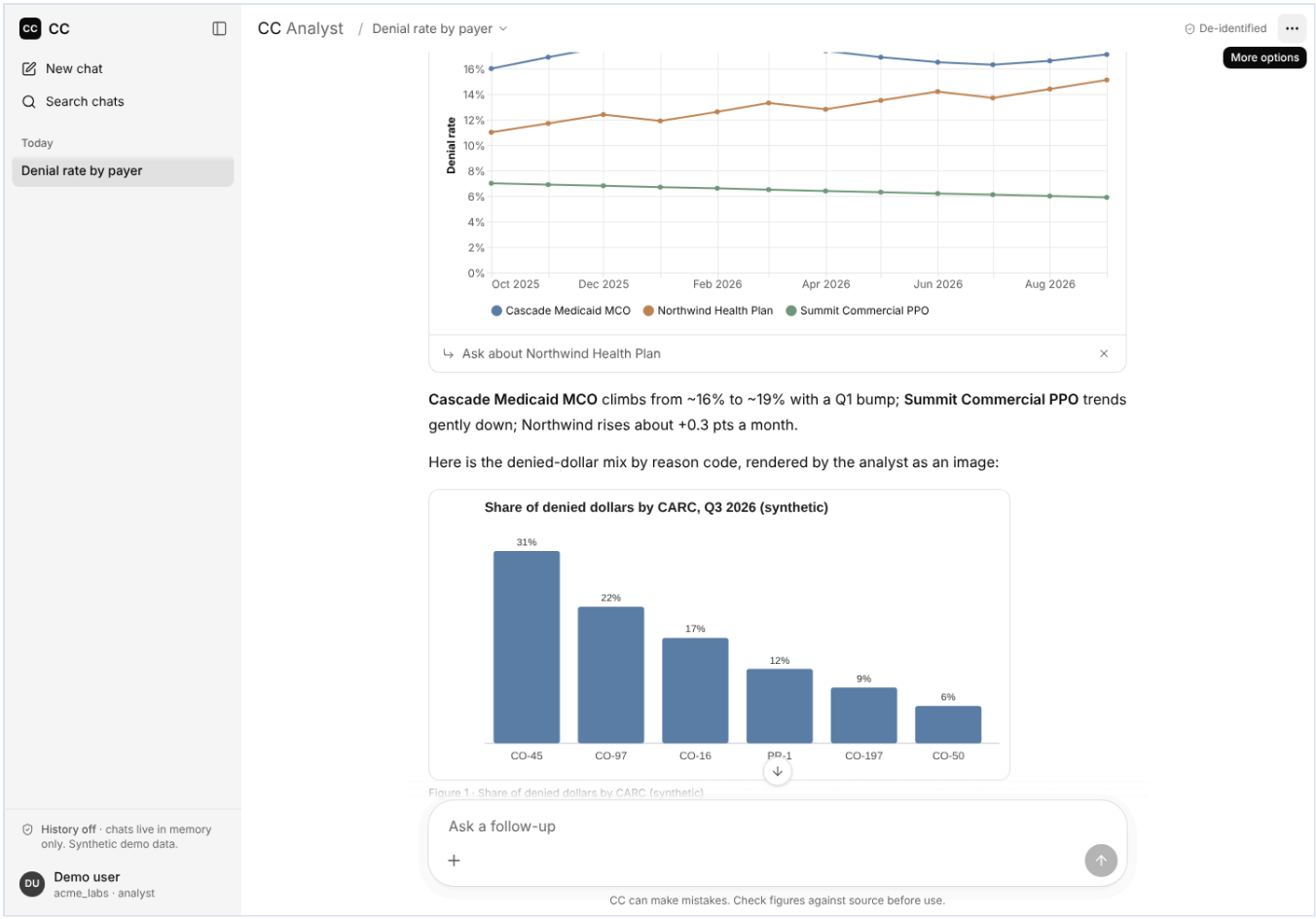


#	Element	Behaviour
1	File card	`cc_denial_rate_by_payer_2026-10-25.csv · 6 rows · 309 B`; download; preview opens the matching table in the panel.
2	View SQL	Collapsed query linked to the table.


New chat
Search chats
Today
Denial rate by payer
History off · chats live in memory only. Synthetic demo data.
Demo user
acme_labs · analyst
Which CARC codes drive Cascade Medicaid denials?
Thought for 2.6s 1 failed
For Cascade Medicaid MCO in Q3 2026, three reason codes account for ~70% of denied dollars:
Rank CARC Meaning (paraphrased) Share of denied $
1 CO-16 Claim lacks information needed for adjudication 31%
2 CO-197 Precertification or authorization absent 24%
3 CO-29 Time limit for filing has expired 15%
Suggested next steps
1. Audit CO-16 claims for a missing referring-provider NPI (loop 2310A).
2. Check the prior-auth workflow for Cascade plan ids.
3. Tighten timely-filing alerts: Cascade's window is 90 days.
A sample of the denied claim lines is below. Member IDs are masked under your access level.
Denied claim lines, Cascade Medicaid MCO (sample)
Claim Member ID Service date CPT CARC Billed
SYN-410000 •••• 00 2026-07-01 99213 CO-16 $80.00
SYN-410037 •••• 19 2026-08-06 99214 CO-197 $170.00
SYN-410074 •••• 38 2026-09-11 97110 CO-29 $260.00
SYN-410111 •••• 57 2026-07-16 80053 CO-16 $350.00
SYN-410148 •••• 76 2026-0 93000 CO-50 $440.00
Ask a follow-up
CC can make mistakes. Check figures against source before use.

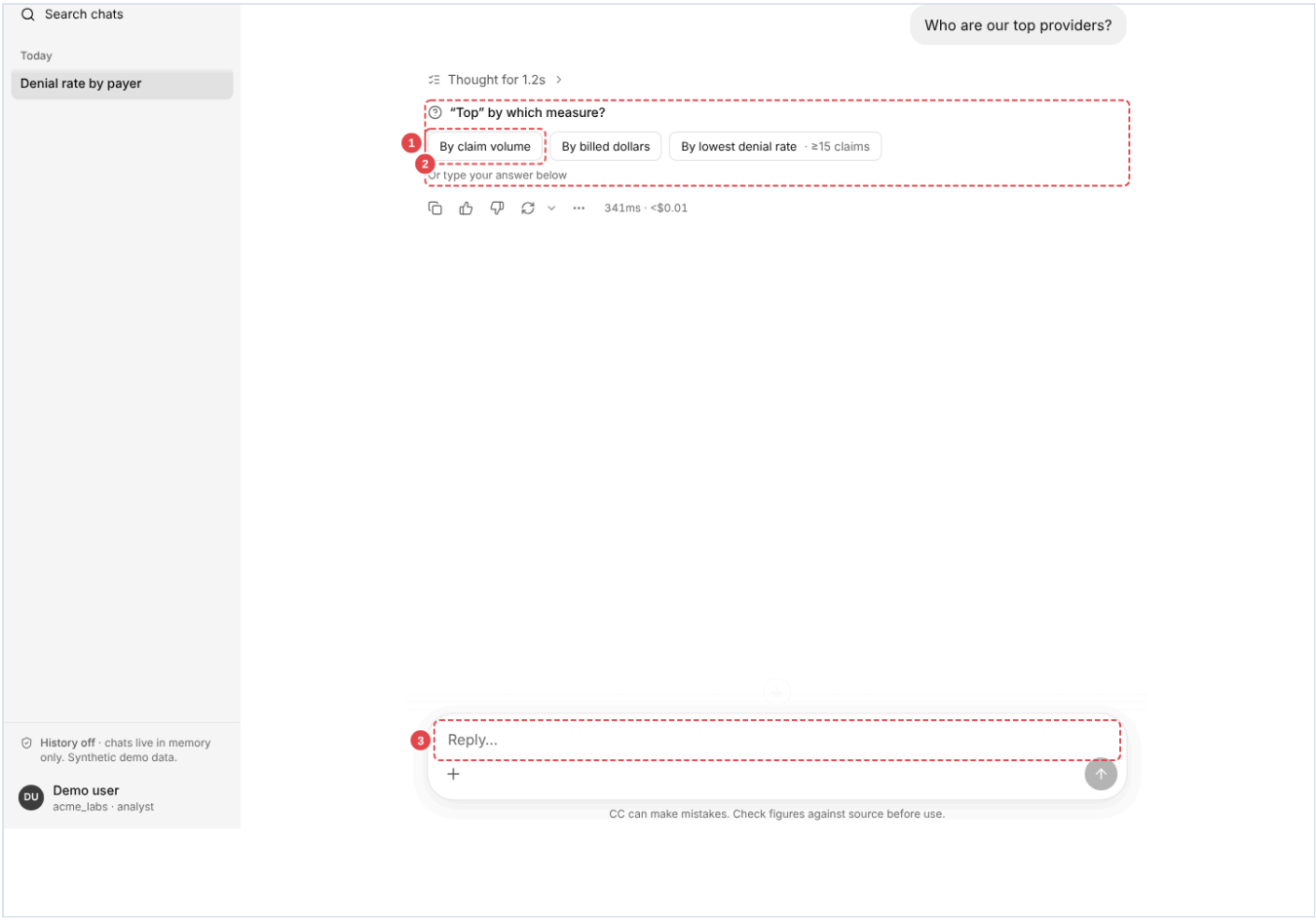

CC
New chat
Search chats
Today
Denial rate by payer
History off · chats live in memory only. Synthetic demo data.
Demo user
acme_labs · analyst
CC Analyst / Denial rate by payer
De-identified
Denied claim lines, Cascade Medicaid MCO (sample)
Claim Member ID Service date CPT CARC Billed
SYN-410000 •••• 00 2026-07-01 99213 CO-16 $80.00
SYN-410037 •••• 19 2026-08-06 99214 CO-197 $170.00
SYN-410074 •••• 38 2026-09-11 97110 CO-29 $260.00
SYN-410111 •••• 57 2026-07-16 80053 CO-16 $350.00
SYN-410148 •••• 76 2026-08-21 93000 CO-50 $440.00
SYN-410185 •••• 95 2026-09-26 71046 CO-197 $530.00
SYN-410222 •••• 14 2026-07-03 99213 CO-16 $620.00
SYN-410259 •••• 33 2026-08-08 99214 CO-197 $451.00
SYN-410296 •••• 52 2026-09-13 97110 CO-29 $541.00
SYN-410333 •••• 71 2026-07-18 80053 CO-16 $631.00
from sem.claim_lines · Showing 60 of 1,284 rows · 0.08 s · PHI masked
Open full table
1
2
PHI masked
1 column (Member ID) was masked under your access level and is excluded from copy and CSV.
3
Data quality
Verifier found a 0.4% gap against claim-level totals, likely 835s that arrived after month close.
4
CARC breakdown query
Code · Click to open
v1
Looked at sem.claim_lines
SQL 2 queries · 4.7s
Ask a follow-up
CC can make mistakes. Check figures against source before use.

#	Element	Behaviour
1	PHI column	"Member ID" shown as `• • • • 56`, shield icon, not sortable, excluded from copy and CSV.
2	PHI masked callout	"1 column (Member ID) was masked under your access level and is excluded from copy and CSV."
3	Data quality callout	Verifier gap note.
4	Artifact card	"CARC breakdown query · Code · Click to open".

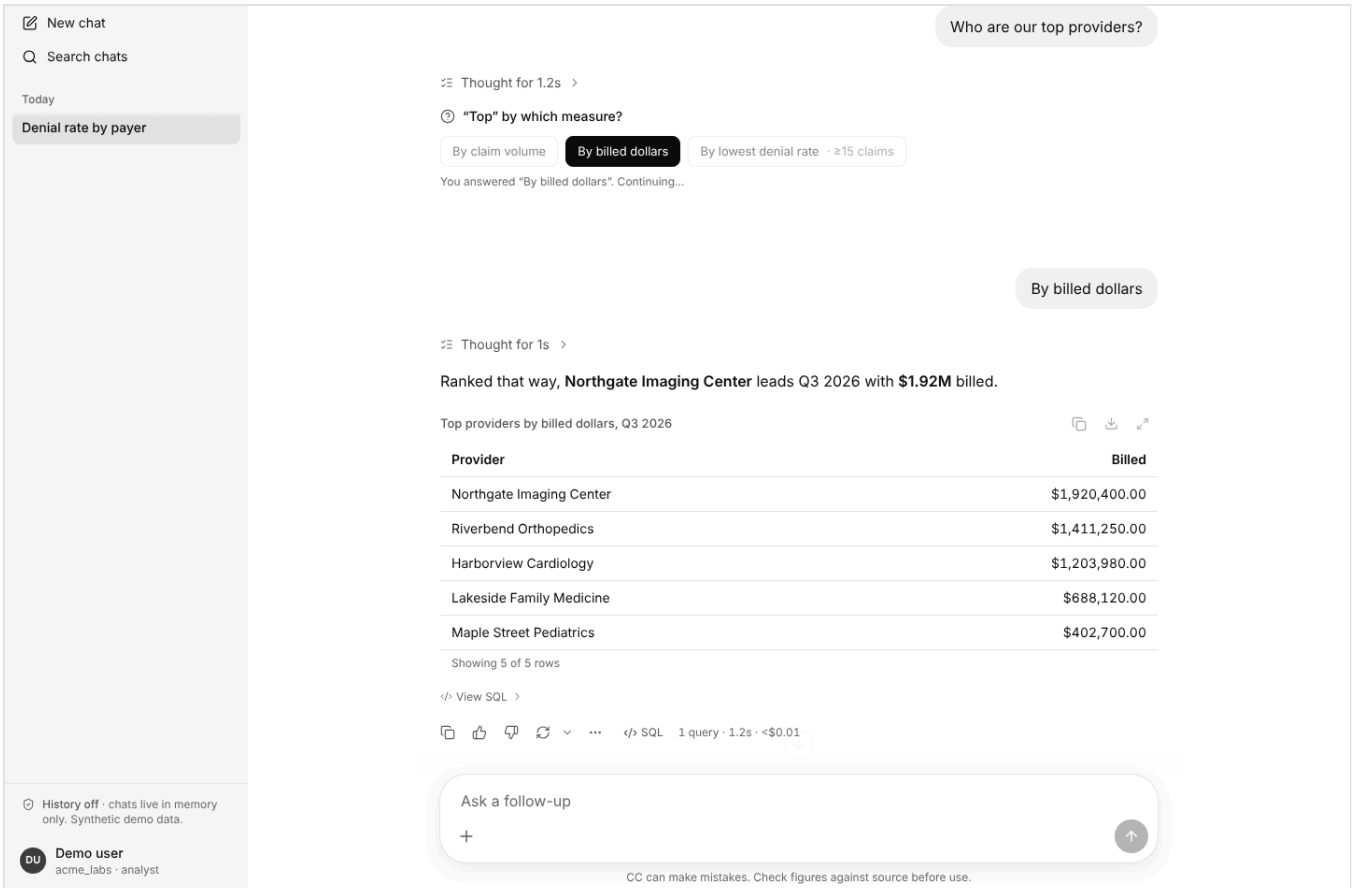


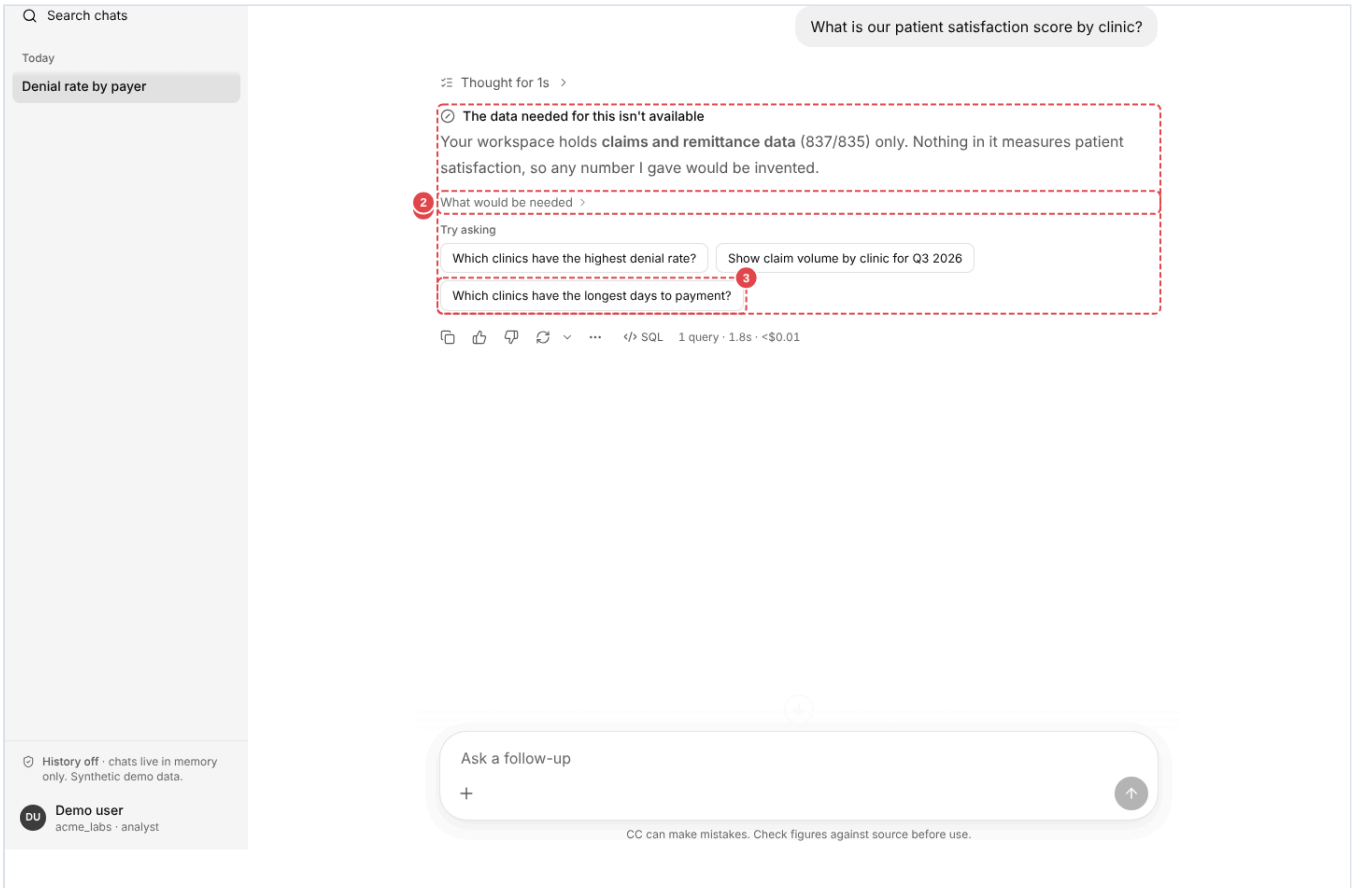
TODO screenshot: image lightbox — mock mode, ask "Plot the monthly denial rate trend for our top 3 payers", hover the figure, press Expand image

10.4 Clarify and abstain cards



#	Element	Behaviour
1	Clarifying question	`role="group"` "Clarifying question"; only the last turn is active.
2	Choice chip	Sends `resume {clarificationId, choiceIds}`.
3	Composer	Placeholder "Reply..."; free text resumes too (`allowFreeText`).

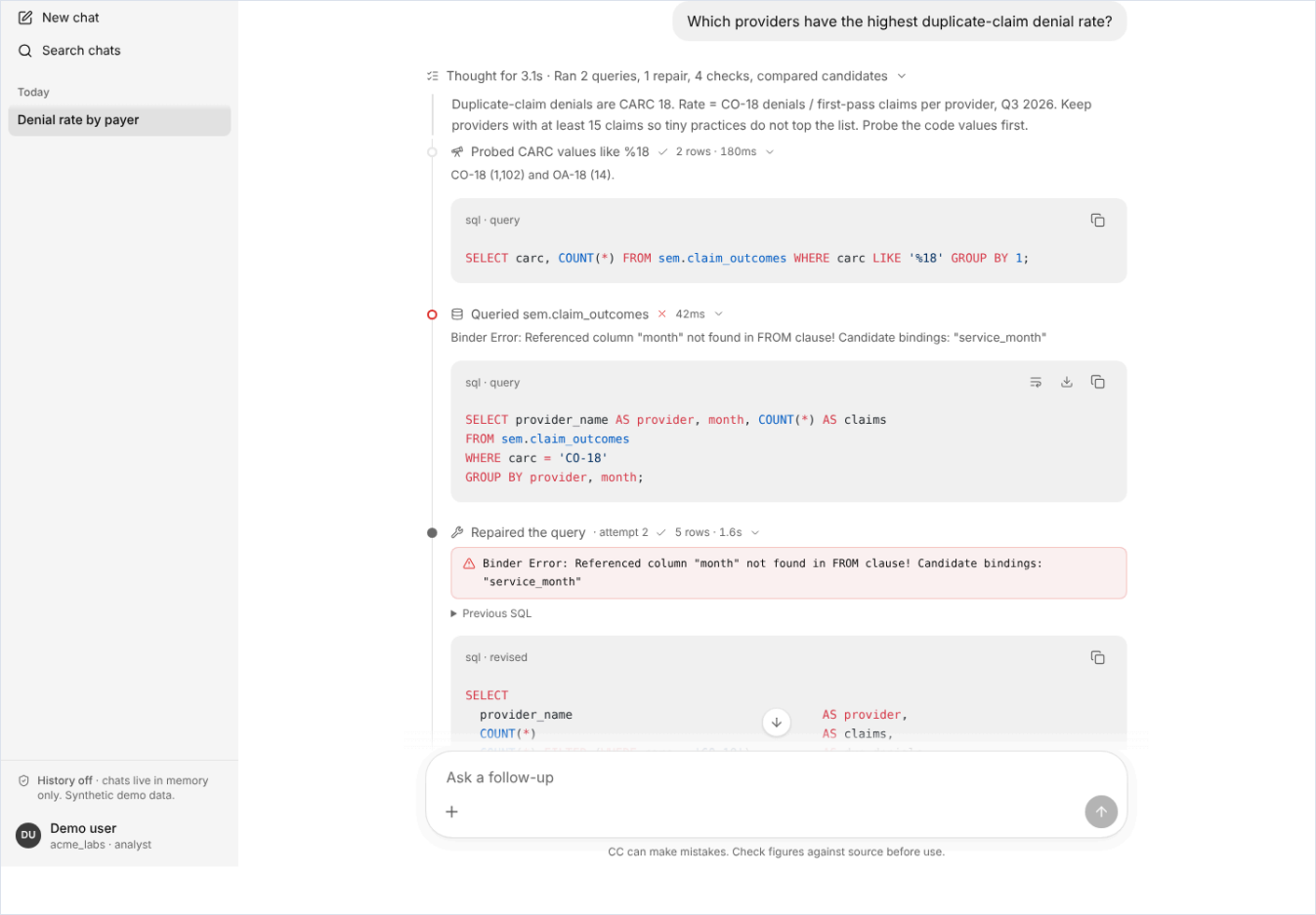

New chat
Search chats
Today
Denial rate by payer
History off · chats live in memory only. Synthetic demo data.
Demo user
acme_labs · analyst
Who are our top providers?
Thought for 1.2s
"Top" by which measure?
By claim volume
By billed dollars
By lowest denial rate · ≥15 claims
You answered "By billed dollars". Continuing...
By billed dollars
Thought for 1s
Ranked that way, Northgate Imaging Center leads Q3 2026 with $1.92M billed.
Top providers by billed dollars, Q3 2026
Provider
Billed
Northgate Imaging Center
$1,920,400.00
Riverbend Orthopedics
$1,411,250.00
Harborview Cardiology
$1,203,980.00
Lakeside Family Medicine
$688,120.00
Maple Street Pediatrics
$402,700.00
Showing 5 of 5 rows
View SQL
SQL
1 query · 1.2s · <$0.01
Ask a follow-up
CC can make mistakes. Check figures against source before use.


Search chats
Today
Denial rate by payer
History off · chats live in memory only. Synthetic demo data.
Demo user
acme_labs · analyst
What is our patient satisfaction score by clinic?
Thought for 1s
The data needed for this isn't available
Your workspace holds claims and remittance data (837/835) only. Nothing in it measures patient satisfaction, so any number I gave would be invented.
2
What would be needed
Try asking
Which clinics have the highest denial rate?
Show claim volume by clinic for Q3 2026
3
Which clinics have the longest days to payment?
SQL
1 query · 1.8s · <$0.01
Ask a follow-up
CC can make mistakes. Check figures against source before use.

#	Element	Behaviour
1	Reason	Markdown reason, `data-category` (data_unavailable, unsupported, ...).
2	What would be needed	`details`.
3	Try asking	Suggestion chips send a new question.

TODO screenshot: multi-question clarification answered — demo API, ask "Ask me a few questions about the denial rate breakdown", answer Metric and Period, Skip Group by, Submit

10.5 Candidates and checks (mock)

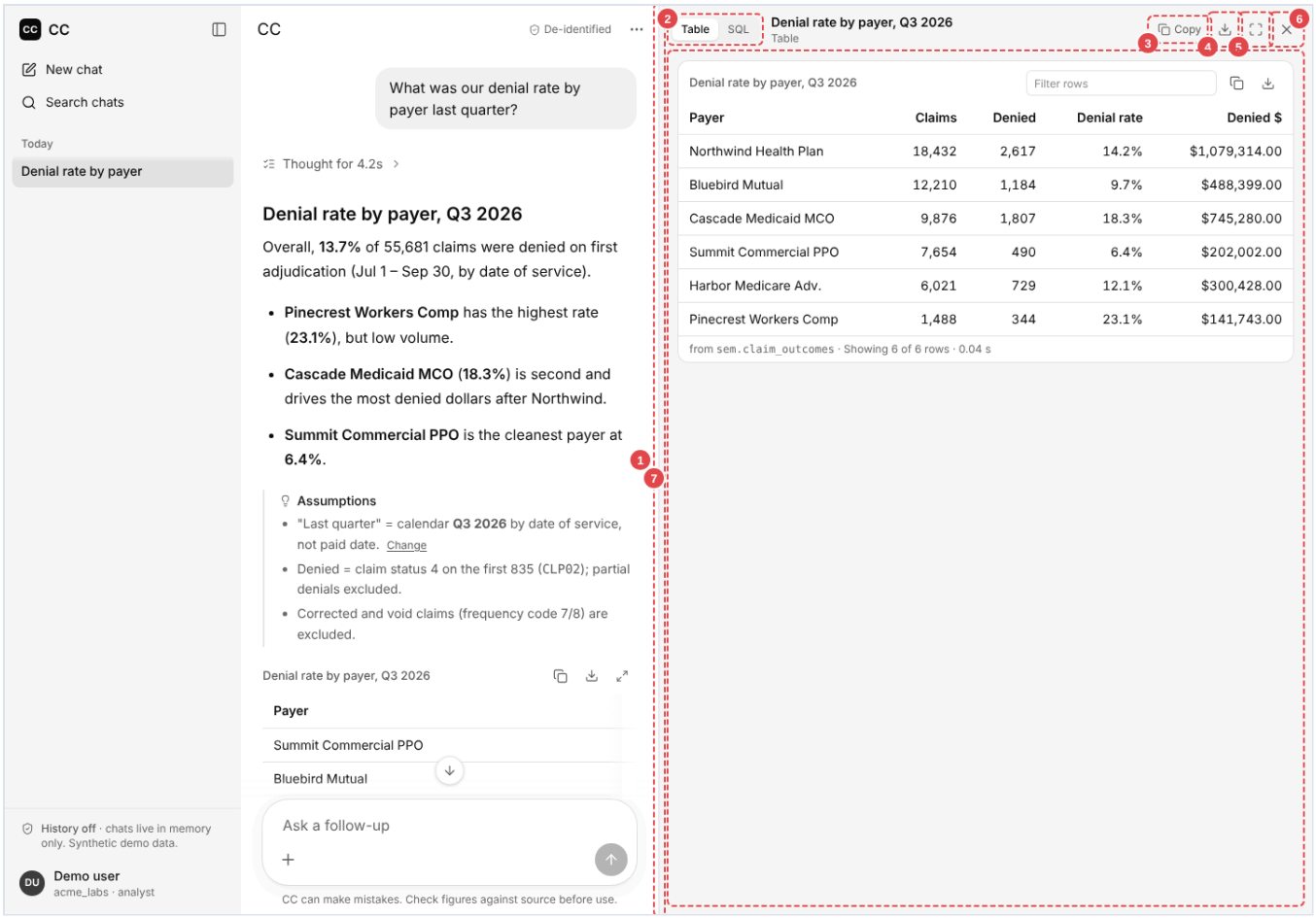


Full-page version: `images/frontend/75-candidates-and-checks-full.png`.

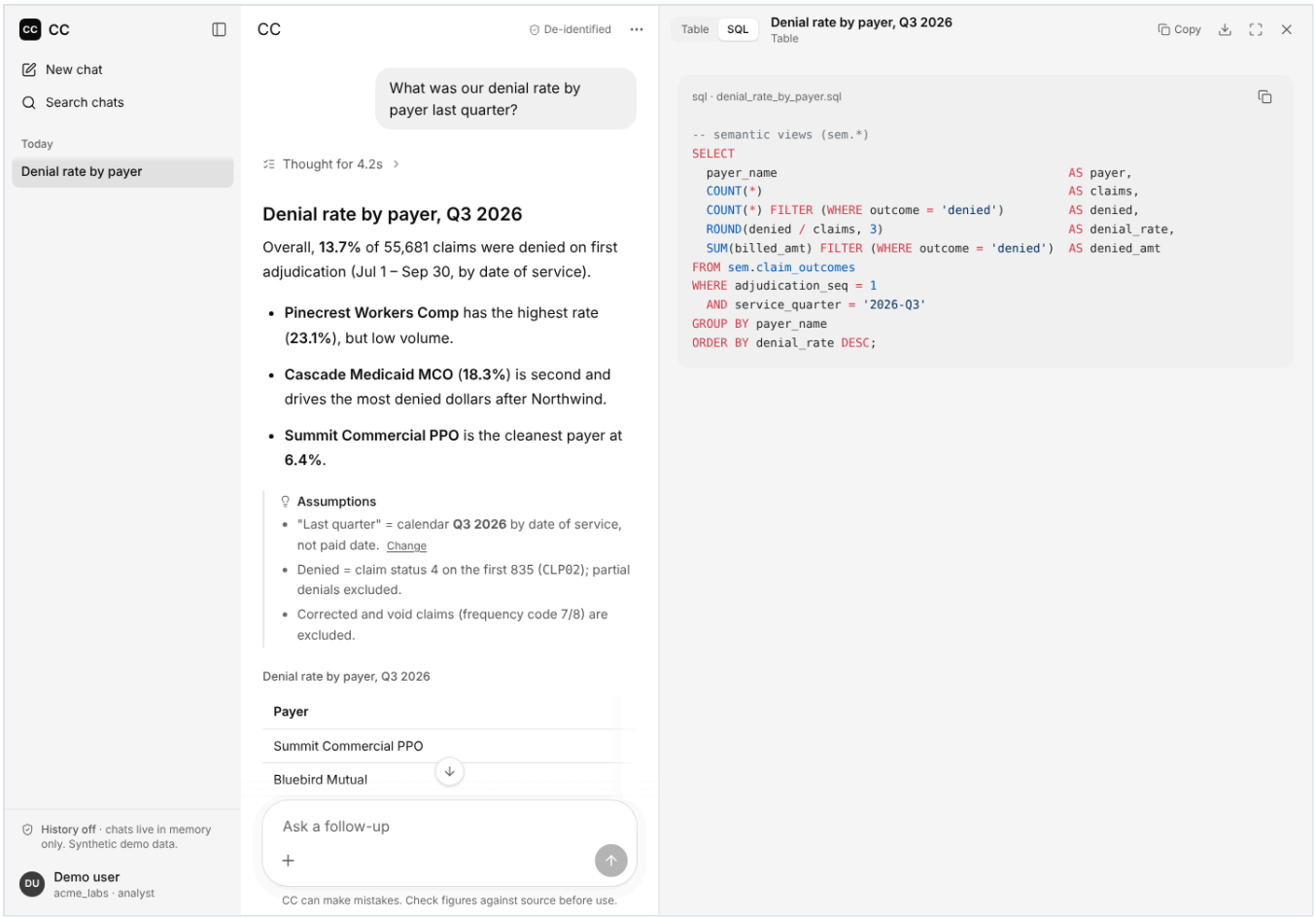
---

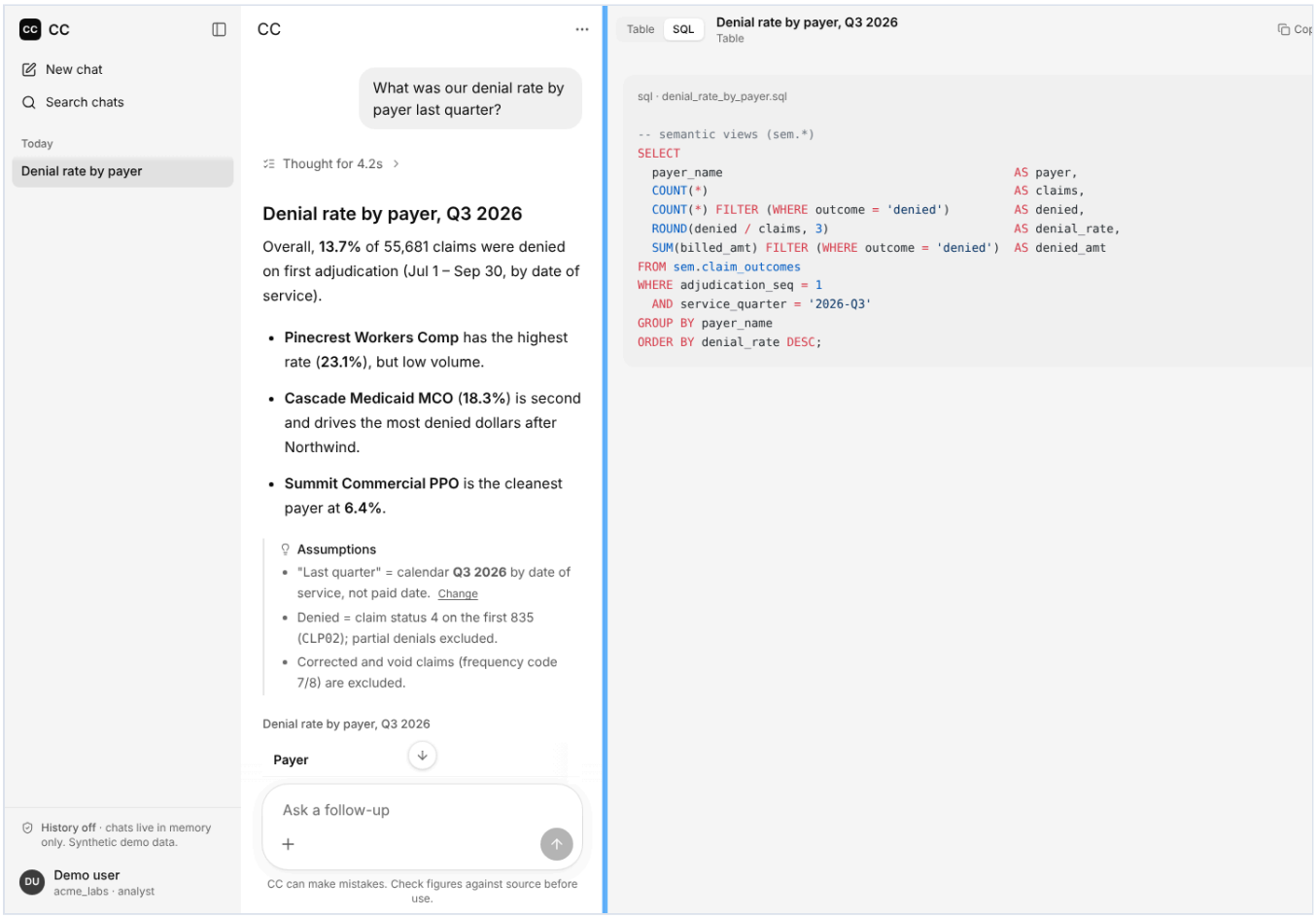
11. Artifact panel

TODO: intro: the right-hand panel for tables, charts, code, images and reports; read-only result views, versioned.



#	Element	Behaviour
1	Resize handle	`role="separator"`; drag or ←/→; double-click resets to 50 %; min 360 px, max 75 %; width stored as `cc.pref.artifactWidth`.
2	Tabs	Table
3	Copy	Copies the current tab.
4	Download	CSV / PNG / SVG / file.
5	Full screen	Hidden on phones.
6	Close	Esc.
7	Body	Table with "Filter rows" (panel only).


CC
New chat
Search chats
Today
Denial rate by payer
History off · chats live in memory only. Synthetic demo data.
Demo user
acme_labs · analyst
CC
De-identified
What was our denial rate by payer last quarter?
Thought for 4.2s
Denial rate by payer, Q3 2026
Overall, 13.7% of 55,681 claims were denied on first adjudication (Jul 1 – Sep 30, by date of service).
Pinecrest Workers Comp has the highest rate (23.1%), but low volume.
Cascade Medicaid MCO (18.3%) is second and drives the most denied dollars after Northwind.
Summit Commercial PPO is the cleanest payer at 6.4%.
Assumptions
"Last quarter" = calendar Q3 2026 by date of service, not paid date. Change
Denied = claim status 4 on the first 835 (CLP02); partial denials excluded.
Corrected and void claims (frequency code 7/8) are excluded.
Denial rate by payer, Q3 2026
Payer
Summit Commercial PPO
Bluebird Mutual
Ask a follow-up
CC can make mistakes. Check figures against source before use.
Table
SQL
Denial rate by payer, Q3 2026
Table
Copy
sql · denial_rate_by_payer.sql
-- semantic views (sem.*)
SELECT
payer_name AS payer,
COUNT(*) AS claims,
COUNT(*) FILTER (WHERE outcome = 'denied') AS denied,
ROUND(denied / claims, 3) AS denial_rate,
SUM(billed_amt) FILTER (WHERE outcome = 'denied') AS denied_amt
FROM sem.claim_outcomes
WHERE adjudication_seq = 1
AND service_quarter = '2026-Q3'
GROUP BY payer_name
ORDER BY denial_rate DESC;


CC
New chat
Search chats
Today
Denial rate by payer
History off · chats live in memory only. Synthetic demo data.
Demo user
acme_labs · analyst
CC
What was our denial rate by payer last quarter?
Thought for 4.2s
Denial rate by payer, Q3 2026
Overall, 13.7% of 55,681 claims were denied on first adjudication (Jul 1 – Sep 30, by date of service).
Pinecrest Workers Comp has the highest rate (23.1%), but low volume.
Cascade Medicaid MCO (18.3%) is second and drives the most denied dollars after Northwind.
Summit Commercial PPO is the cleanest payer at 6.4%.
Assumptions
"Last quarter" = calendar Q3 2026 by date of service, not paid date. Change
Denied = claim status 4 on the first 835 (CLP02); partial denials excluded.
Corrected and void claims (frequency code 7/8) are excluded.
Denial rate by payer, Q3 2026
Payer
Ask a follow-up
CC can make mistakes. Check figures against source before use.
Table
SQL
Denial rate by payer, Q3 2026
Table
sql · denial_rate_by_payer.sql
-- semantic views (sem.*)
SELECT
payer_name AS payer,
COUNT(*) AS claims,
COUNT(*) FILTER (WHERE outcome = 'denied') AS denied,
ROUND(denied / claims, 3) AS denial_rate,
SUM(billed_amt) FILTER (WHERE outcome = 'denied') AS denied_amt
FROM sem.claim_outcomes
WHERE adjudication_seq = 1
AND service_quarter = '2026-Q3'
GROUP BY payer_name
ORDER BY denial_rate DESC;

CC

New chat

Search chats

Today

Denial rate by payer

History off · chats live in memory only. Synthetic demo data.

Demo user
acme_labs · analyst

CC

from sem.claim_outcomes · Showing 6 of 6 rows · 0.04 s

View SQL

cc_denial_rate_by_payer_...
CSV · 6 rows · 309 B

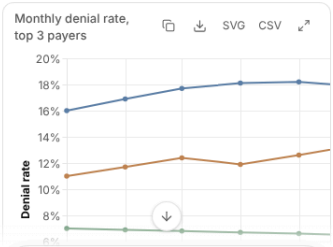
Want me to break Cascade's denials down by reason code?

Looked at sem.claim_outcomes

Plot the monthly denial rate trend for our top 3 payers

Thought for 1.3s

Monthly first-pass denial rate, **Oct 2025 – Sep 2026**:

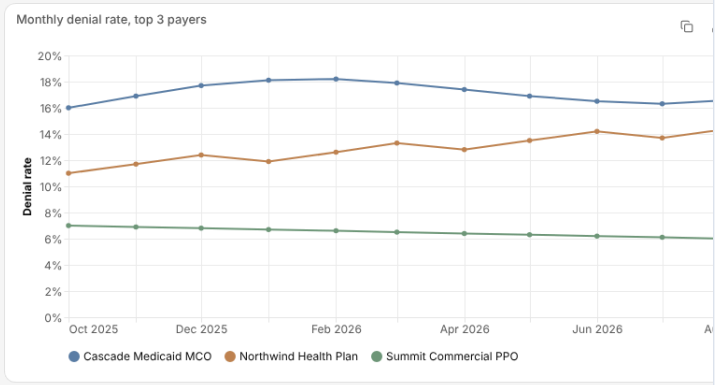


Ask a follow-up

CC can make mistakes. Check figures against source before use.

Chart | Spec | Data

Monthly denial rate, top 3 payers
Chart



CC

New chat

Search chats

Today

Denial rate by payer

History off · chats live in memory only. Synthetic demo data.

Demo user
acme_labs · analyst

CC

from sem.claim_outcomes · Showing 6 of 6 rows · 0.04 s

View SQL

cc_denial_rate_by_payer_...
CSV · 6 rows · 309 B

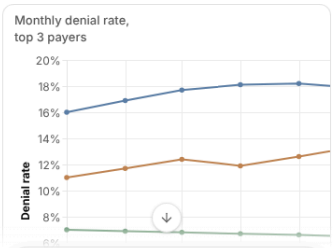
Want me to break Cascade's denials down by reason code?

Looked at sem.claim_outcomes

Plot the monthly denial rate trend for our top 3 payers

Thought for 1.3s

Monthly first-pass denial rate, **Oct 2025 – Sep 2026**:



Ask a follow-up

CC can make mistakes. Check figures against source before use.

Chart | Spec | Data

Monthly denial rate, top 3 payers
Chart

json

```
{
  "width": "container",
  "height": 280,
  "mark": {
    "type": "line",
    "point": true,
    "strokeWidth": 2
  },
  "encoding": {
    "x": {
      "field": "month",
      "type": "temporal",
      "timeUnit": "yearmonth",
      "title": null
    },
    "y": {
      "field": "denial_rate",
      "type": "quantitative",
      "title": "Denial rate",
      "axis": {
        "format": ".0%"
      }
    },
    "color": {
      "field": "payer",
      "type": "nominal",
      "title": null,
      "legend": {
        "orient": "bottom"
      }
    },
    "tooltip": [
      {
        "field": "payer",
        "type": "nominal"
      },
      {
        "field": "month",
        "type": "temporal",
        "timeUnit": "yearmonth"
      },
      {
```

CC

New chat

Search chats

Today

Denial rate by payer

History off · chats live in memory only. Synthetic demo data.

Demo user
acme_labs · analyst

CC

from sem.claim_outcomes · Showing 6 of 6 rows · 0.04 s

View SQL

cc_denial_rate_by_payer_...
CSV · 6 rows · 309 B

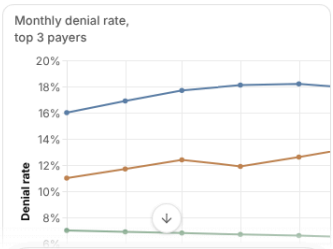
Want me to break Cascade's denials down by reason code?

Looked at sem.claim_outcomes

Plot the monthly denial rate trend for our top 3 payers

Thought for 1.3s

Monthly first-pass denial rate, **Oct 2025 – Sep 2026**:



Ask a follow-up

CC can make mistakes. Check figures against source before use.

Chart | Spec | Data — Monthly denial rate, top 3 payers — Chart

Monthly denial rate, top 3 payers data

Filter rows

month	payer
2025-10	Northwind Health Plan
2025-10	Cascade Medicaid MCO
2025-10	Summit Commercial PPO
2025-11	Northwind Health Plan
2025-11	Cascade Medicaid MCO
2025-11	Summit Commercial PPO
2025-12	Northwind Health Plan
2025-12	Cascade Medicaid MCO
2025-12	Summit Commercial PPO
2026-01	Northwind Health Plan
2026-01	Cascade Medicaid MCO
2026-01	Summit Commercial PPO
2026-02	Northwind Health Plan
2026-02	Cascade Medicaid MCO
2026-02	Summit Commercial PPO
2026-03	Northwind Health Plan
2026-03	Cascade Medicaid MCO
2026-03	Summit Commercial PPO
2026-04	Northwind Health Plan
2026-04	Cascade Medicaid MCO
2026-04	Summit Commercial PPO

Showing 36 of 36 rows

New chat

Search chats

Today

Denial rate by payer

History off · chats live in memory only. Synthetic demo data.

Demo user
acme_labs · analyst

Rank	CARC	Meaning (paraphrased)	Share
1	CO-16	Claim lacks information needed for adjudication	
2	CO-197	Precertification or authorization absent	
3	CO-29	Time limit for filing has expired	

Suggested next steps

1. Audit CO-16 claims for a missing referring-provider NPI (loop `2310A`).
2. Check the prior-auth workflow for Cascade plan ids.
3. Tighten timely-filing alerts: Cascade's window is 90 days.

A sample of the denied claim lines is below. Member IDs are masked under your access level.

Denied claim lines, Cascade Medicai...

Claim	Member ID	Service date
SYN-410000	•••• 00	2026-07-01
SYN-410037	•••• 19	2026-08-06
SYN-410074	••••	2026-09-11

Ask a follow-up

CC can make mistakes. Check figures against source before use.

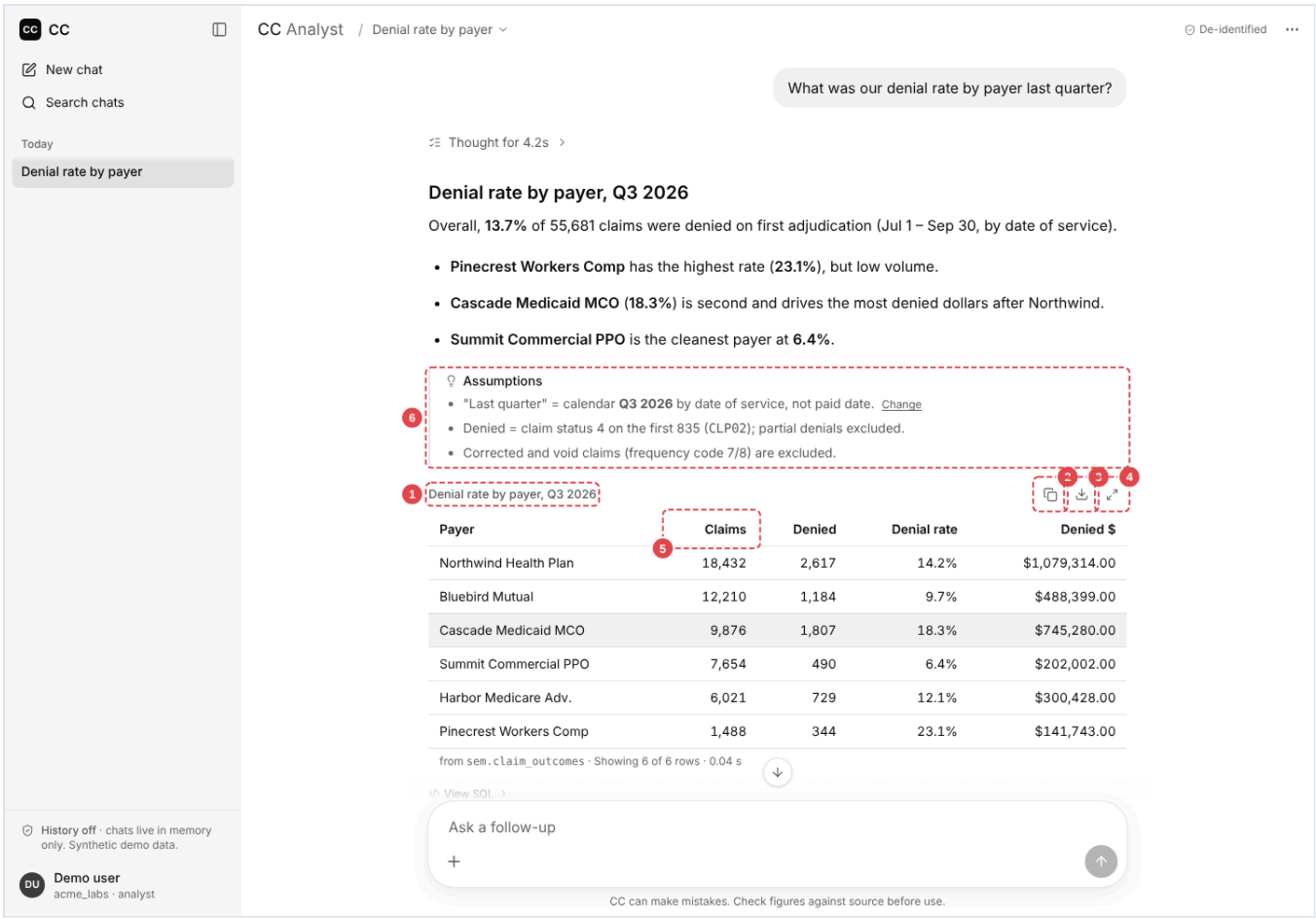
sql · carc_breakdown.sql

```sql
SELECT group_code || '-' || reason_code AS carc,
       SUM(adj_amt)                     AS denied_amt,
       SUM(adj_amt) / SUM(SUM(adj_amt)) OVER () AS share
FROM sem.claim_adjustments
WHERE payer_name = 'Cascade Medicaid MCO'
  AND service_quarter = '2026-Q3'
GROUP BY 1
ORDER BY denied_amt DESC;
```

TODO: versions (`artifact-versions`, "v1/v2"), auto-open rules (`autoOpenIds`, `userIsTyping`), and print report.

12. Tables and charts

12.1 Tables



#	Element	Behaviour
1	Title	Real titles only; a generic "Result" is hidden.
2	Copy table	TSV, PHI columns dropped.
3	Download CSV	Server `csvFileId` first (`GET /api/download/{id}`), client CSV of the preview as fallback; CSV injection guarded.
4	Expand	Opens the panel.
5	Column header	Click to sort; numeric columns right-aligned.
6	Callout above	Assumptions for this answer.

Denial rate by payer, Q3 2026

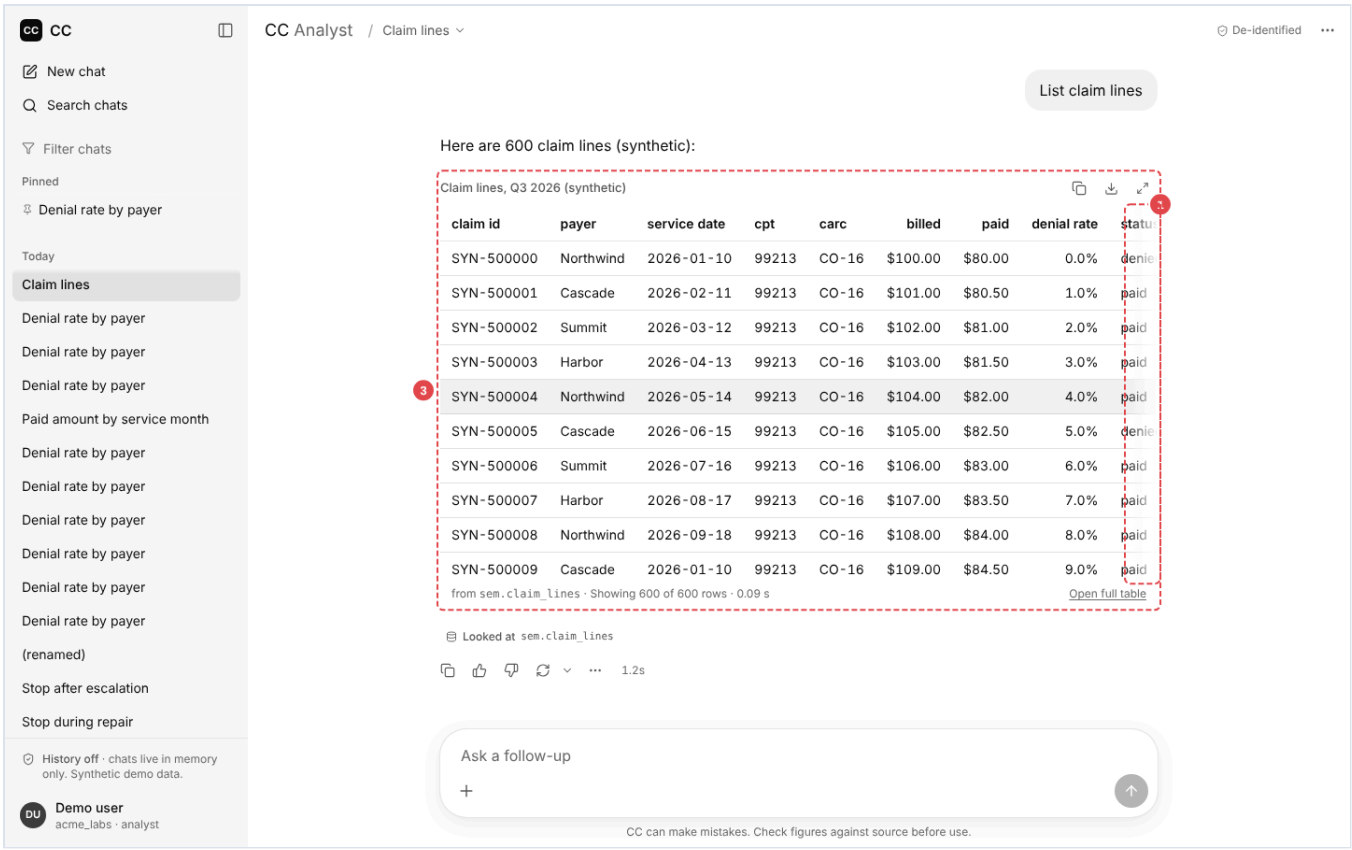
Payer	Claims	Denied	↑ Denial rate	Denied $
Summit Commercial PPO	7,654	490	6.4%	$202,002.00
Bluebird Mutual	12,210	1,184	9.7%	$488,399.00
Harbor Medicare Adv.	6,021	729	12.1%	$300,428.00
Northwind Health Plan	18,432	2,617	14.2%	$1,079,314.00
Cascade Medicaid MCO	9,876	1,807	18.3%	$745,280.00
Pinecrest Workers Comp	1,488	344	23.1%	$141,743.00

from sem.claim_outcomes · Showing 6 of 6 rows · 0.04 s

Denial rate by payer, Q3 2026

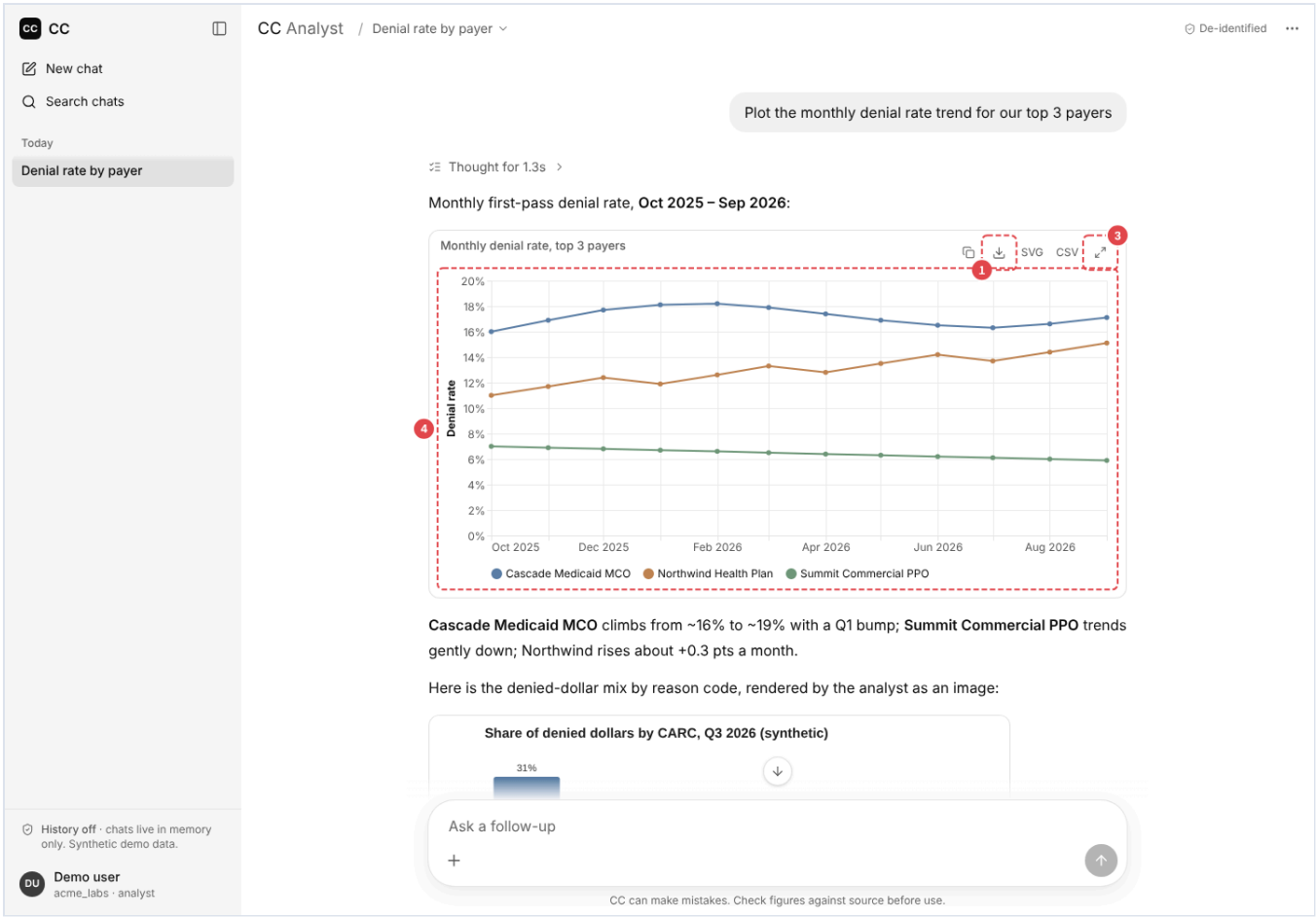
Payer	Claims	Denied	↑ Denial rate	Denied $
Summit Commercial PPO	7,654	490	6.4%	$202,002.00
Bluebird Mutual	12,210	1,184	9.7%	$488,399.00
Harbor Medicare Adv.	6,021	729	12.1%	$300,428.00
Northwind Health Plan	18,432	2,617	14.2%	$1,079,314.00
Cascade Medicaid MCO	9,876	1,807	18.3%	$745,280.00
Pinecrest Workers Comp	1,488	344	23.1%	$141,743.00

from sem.claim_outcomes · Showing 6 of 6 rows · 0.04 s

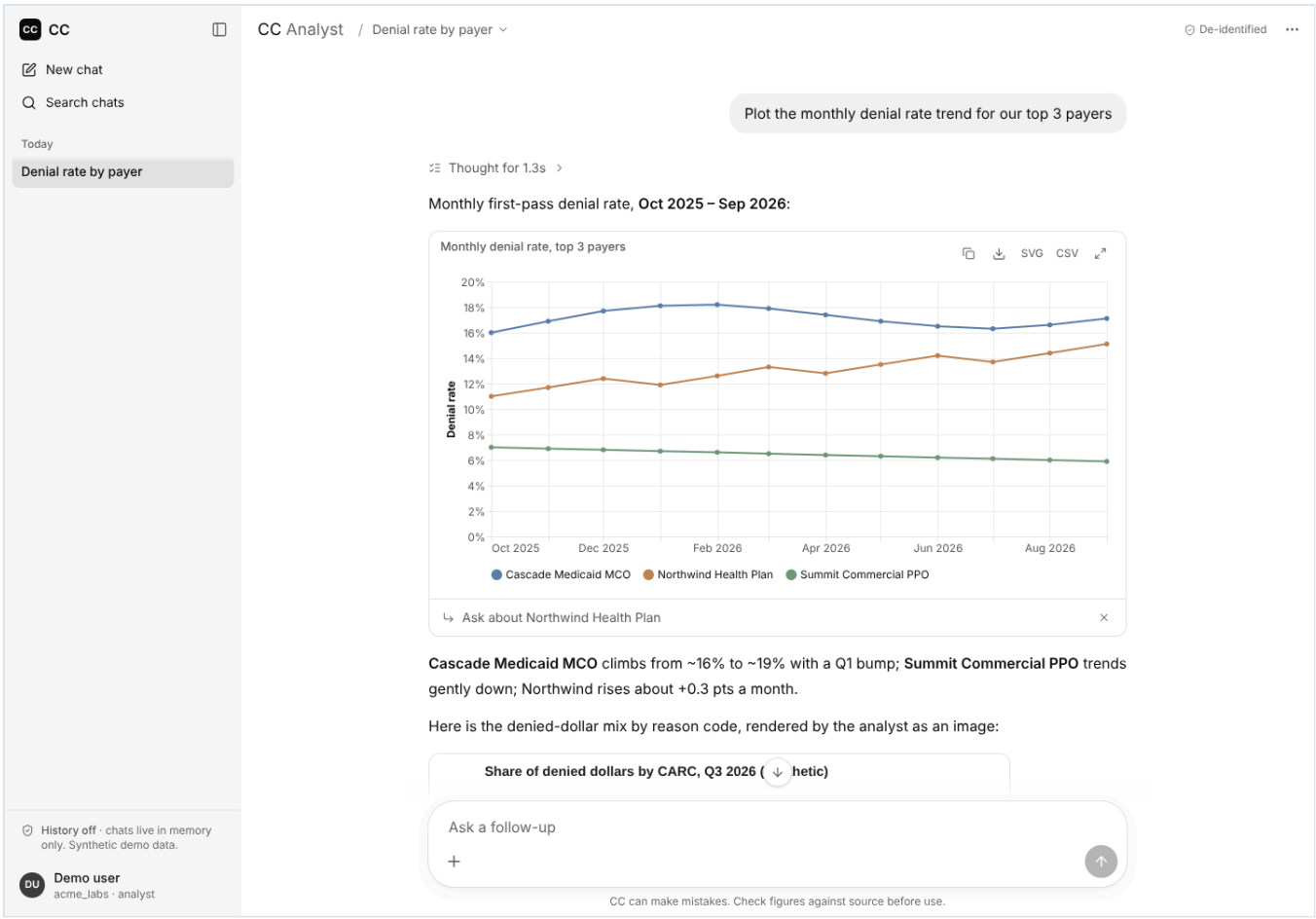


#	Element	Behaviour
1	Right fade	`table-fade-right` : more columns off to the right; the table scrolls in its own frame, never the page.
3	Table	> 200 rows virtualized (TanStack Virtual) with spacer rows; "Open full table".

12.2 Charts



#	Element	Behaviour
1	Download	PNG / SVG / CSV of the data.
3	Expand chart	Opens the panel (Chart / Spec / Data).
4	Plot	`role="img"` with the part's `alt`; theme-aware; tooltips escaped through the Trusted Types policy.



TODO: progressive charts (`chart.start` / `chart.points`, contract 1.18.0) and the client plot gate (`contract/plot.ts`).

---

13. Errors and recovery

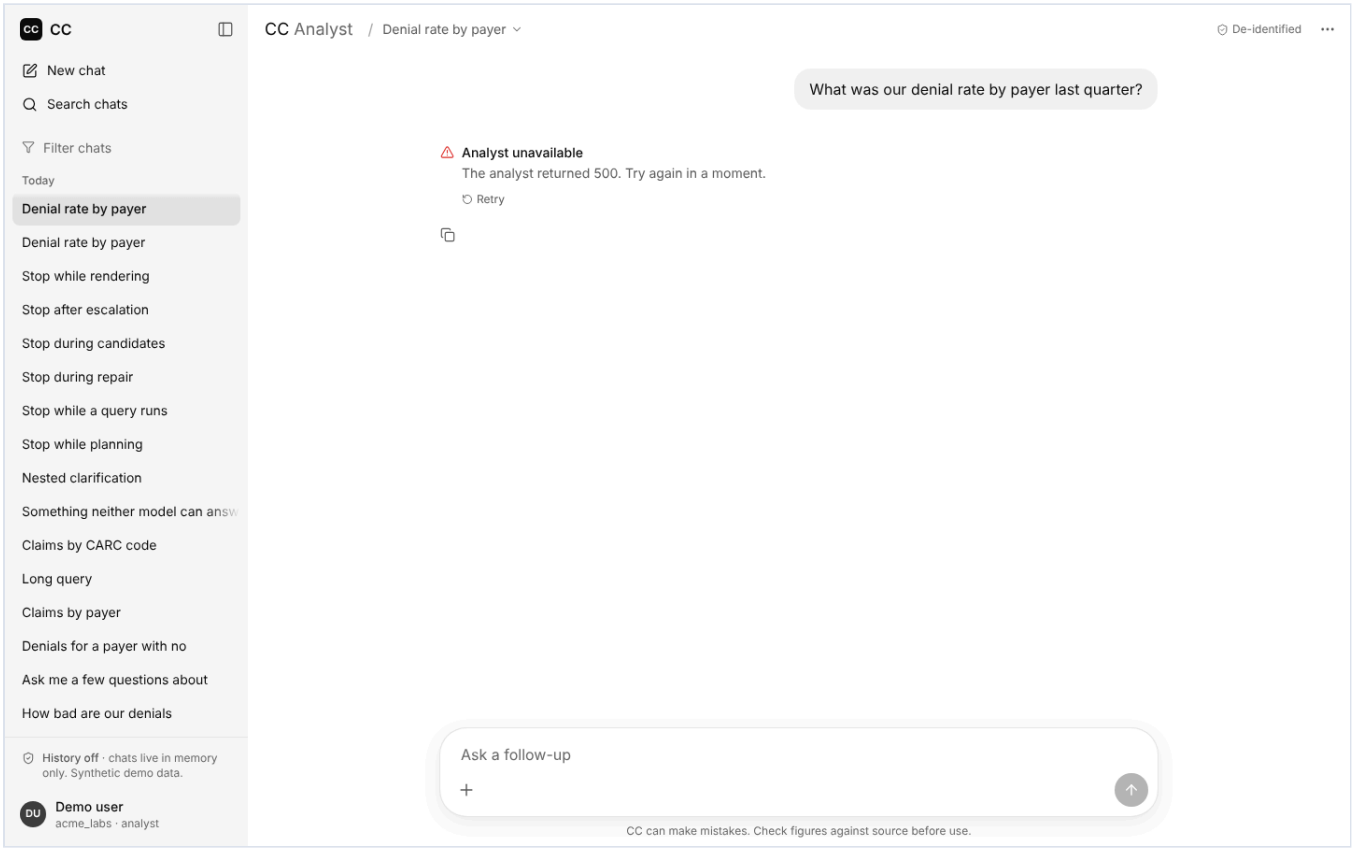
TODO: intro: errors are a plain line in the answer column (no tinted box) with one Retry; transport states are a status line while the turn is still alive; nothing spins forever.

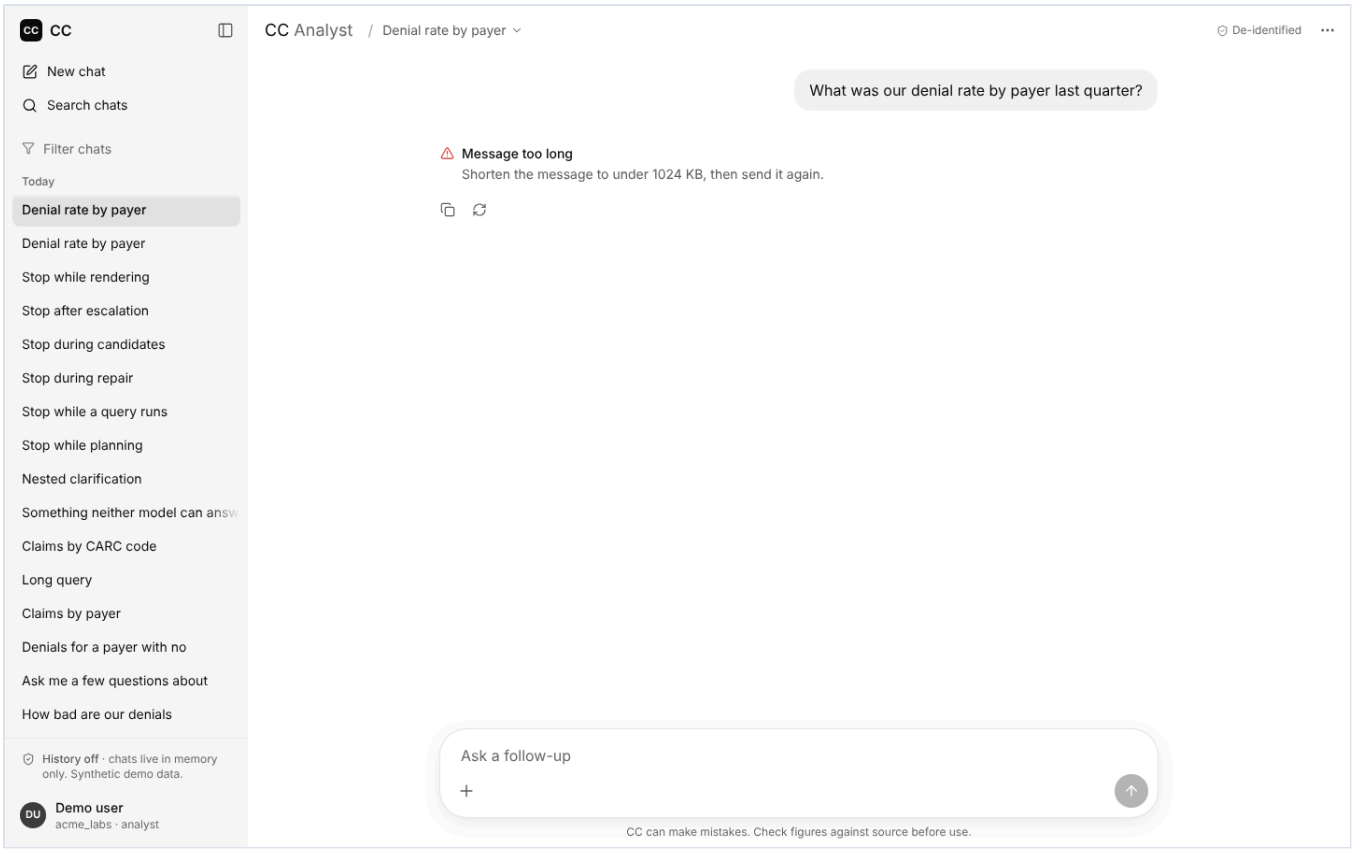
13.1 HTTP errors before the stream starts

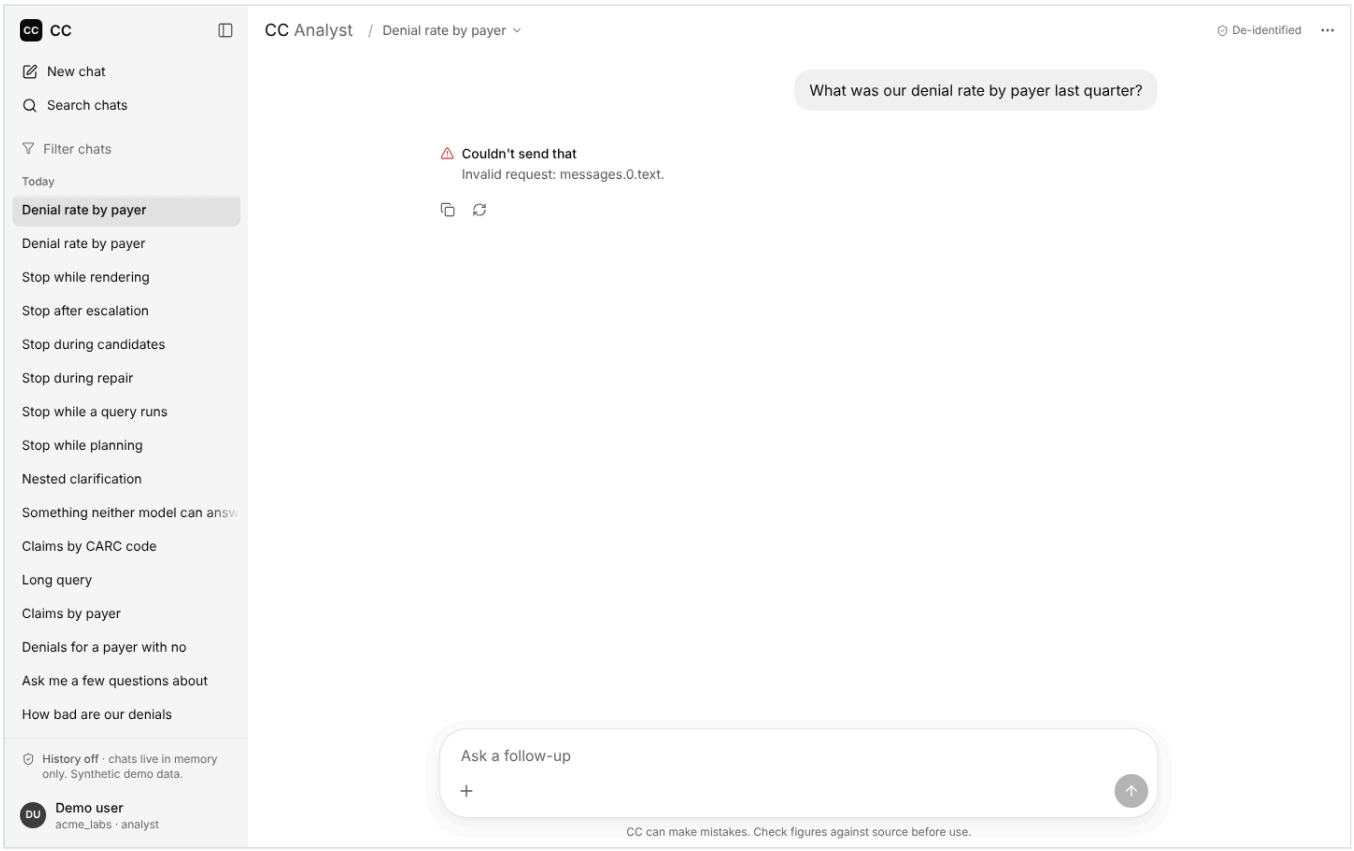
`httpError()` in `adapters/http.ts` resolves `error.code` → `error.legacyCode` → HTTP status.

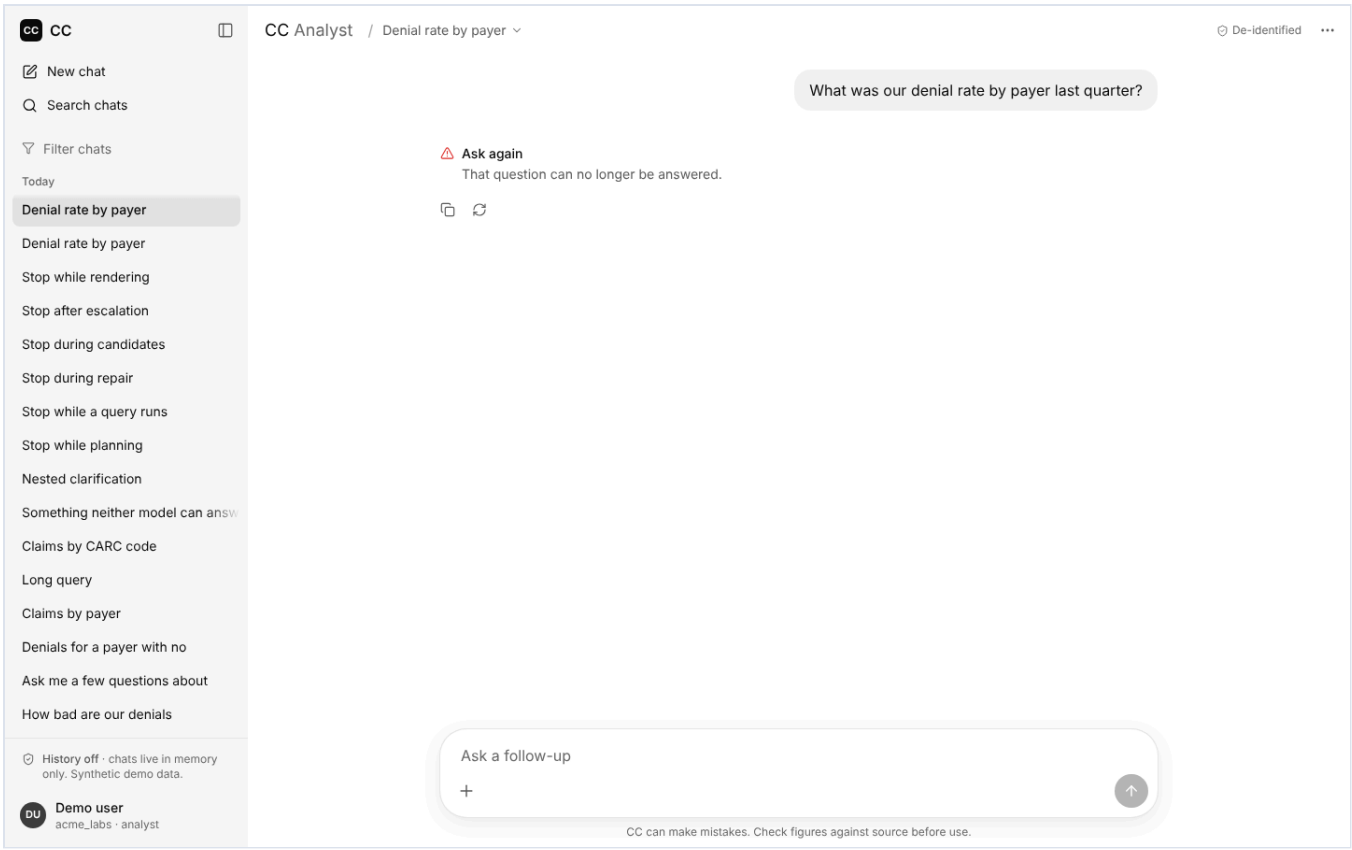
Status	code	Title	Retry	Screenshot
400 / 422	`validation`	Couldn't send that	no	82
401	`auth_expired`	Session expired (cookie session: redirect to sign-in or `/signed-out?reason=expired`)	no	91, 93
403	`access_denied`	Not available	no	—
404	`not_found`	Not found	no	—
404	`stream_gone`	Answer no longer streaming	no	—
405	`method_not_allowed`	Not supported	no	—

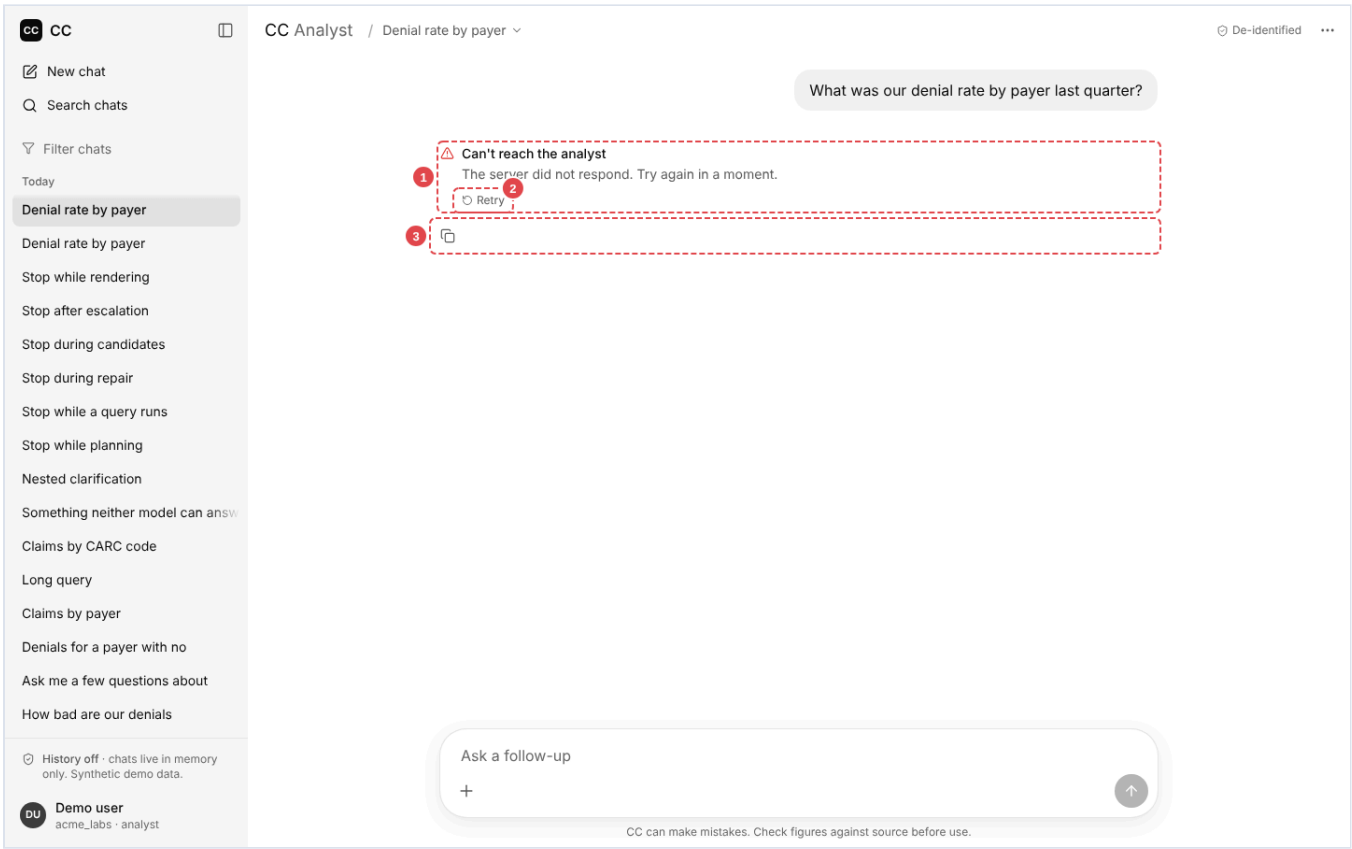
Status	code	Title	Retry	Screenshot
409	`clarification_expired` / `conflict`	Ask again	no	83
410	`transcripts_unavailable`	Chat unavailable	no	P-410
413	`payload_too_large`	Message too long (limit from `maxBytes`)	no	81
422	`invalid_title`	Couldn't rename	no	—
429	`rate_limited`	Too many requests (auto-retry first, §13.2)	yes	86
5xx	`internal`	Analyst unavailable (generic copy, never the server message)	yes	80
other 4xx	raw code	Request not accepted	no	—
no response	`internal` / `offline`	Can't reach the analyst / You're offline	yes	84



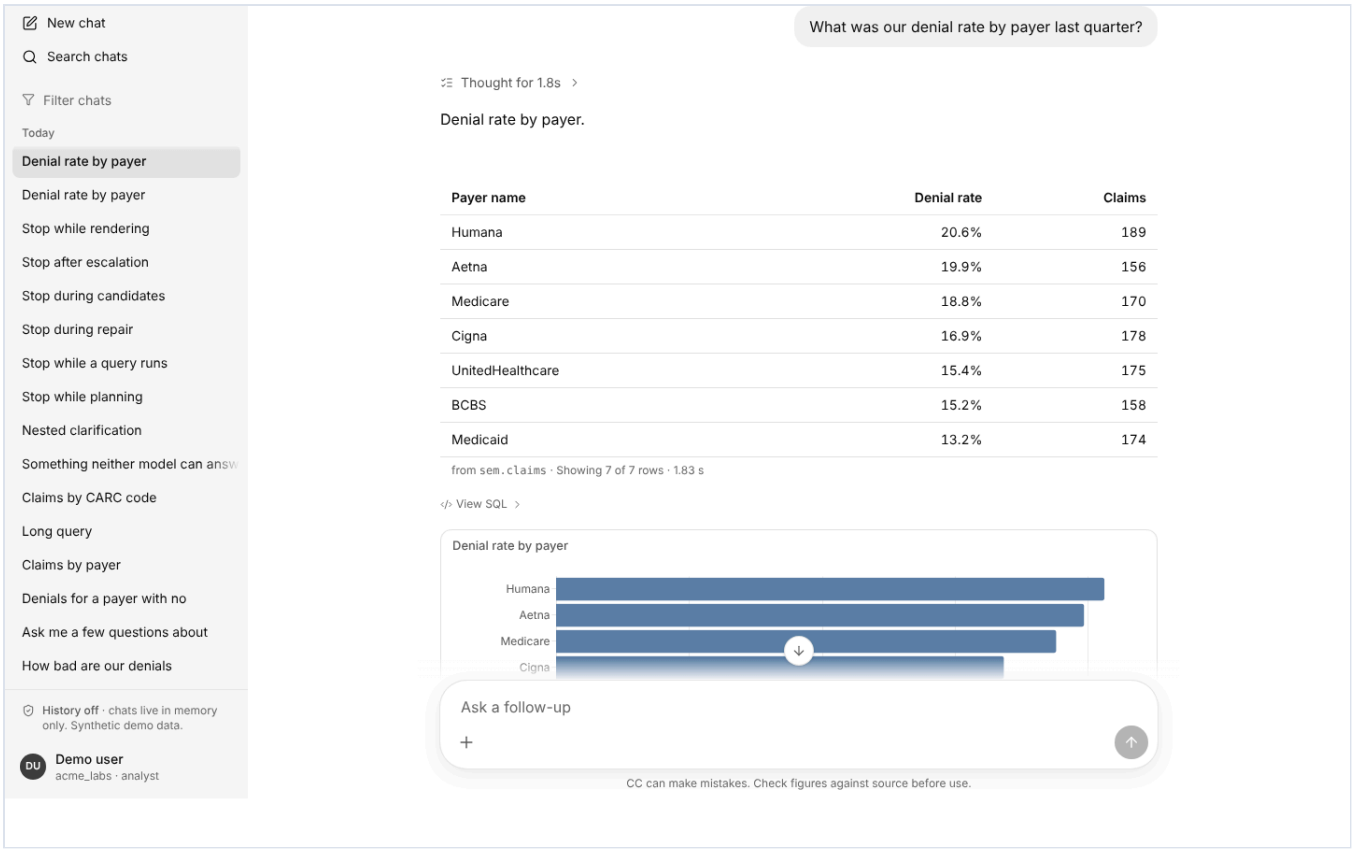

CC
New chat
Search chats
Filter chats
Today
Denial rate by payer
Denial rate by payer
Stop while rendering
Stop after escalation
Stop during candidates
Stop during repair
Stop while a query runs
Stop while planning
Nested clarification
Something neither model can answ
Claims by CARC code
Long query
Claims by payer
Denials for a payer with no
Ask me a few questions about
How bad are our denials
History off · chats live in memory only. Synthetic demo data.
Demo user
acme_labs · analyst
CC Analyst / Denial rate by payer
De-identified
What was our denial rate by payer last quarter?
Message too long
Shorten the message to under 1024 KB, then send it again.
Ask a follow-up
CC can make mistakes. Check figures against source before use.

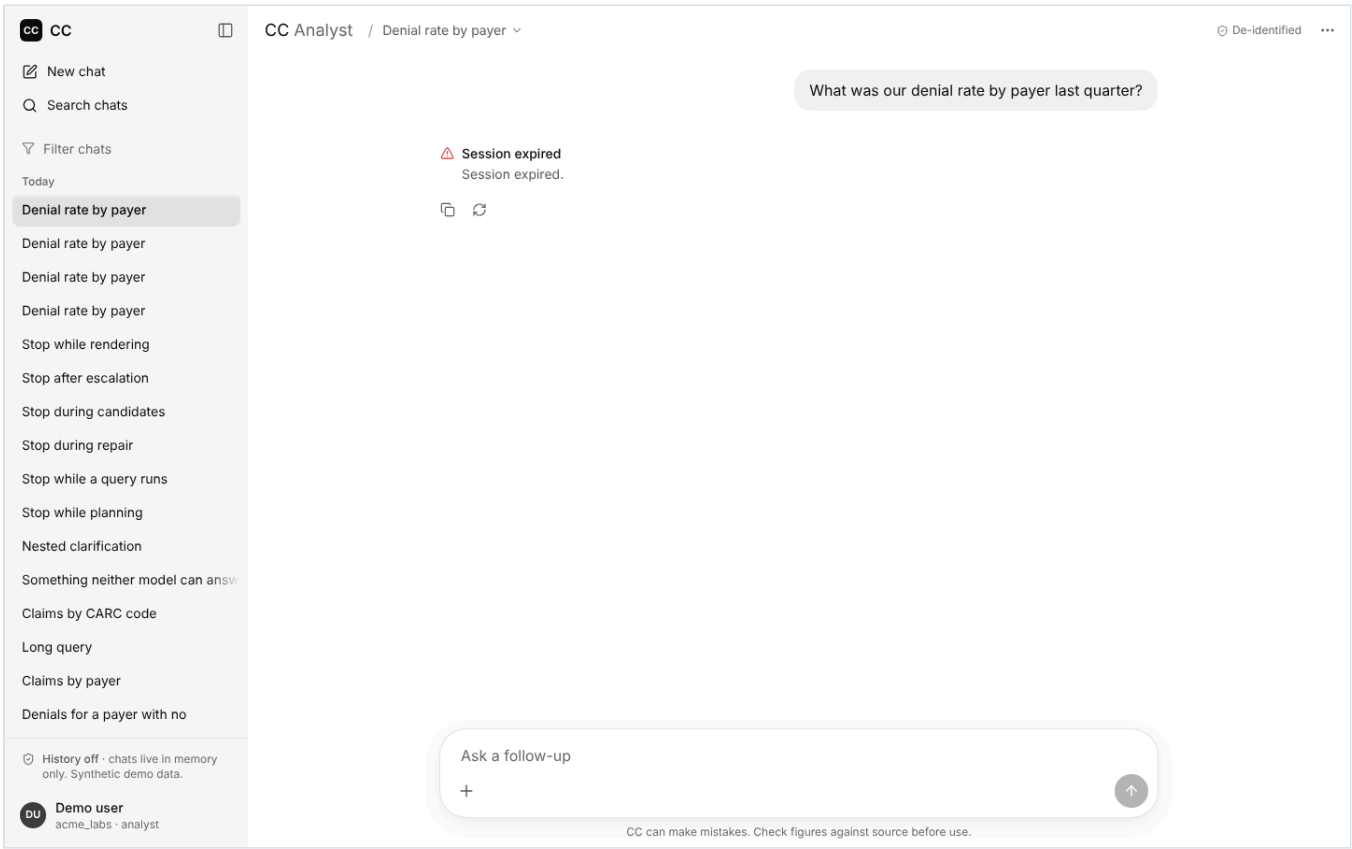

CC
New chat
Search chats
Filter chats
Today
Denial rate by payer
Denial rate by payer
Stop while rendering
Stop after escalation
Stop during candidates
Stop during repair
Stop while a query runs
Stop while planning
Nested clarification
Something neither model can answ
Claims by CARC code
Long query
Claims by payer
Denials for a payer with no
Ask me a few questions about
How bad are our denials
History off · chats live in memory only. Synthetic demo data.
Demo user
acme_labs · analyst
CC Analyst / Denial rate by payer
De-identified
What was our denial rate by payer last quarter?
Couldn't send that
Invalid request: messages.0.text.
Ask a follow-up
CC can make mistakes. Check figures against source before use.





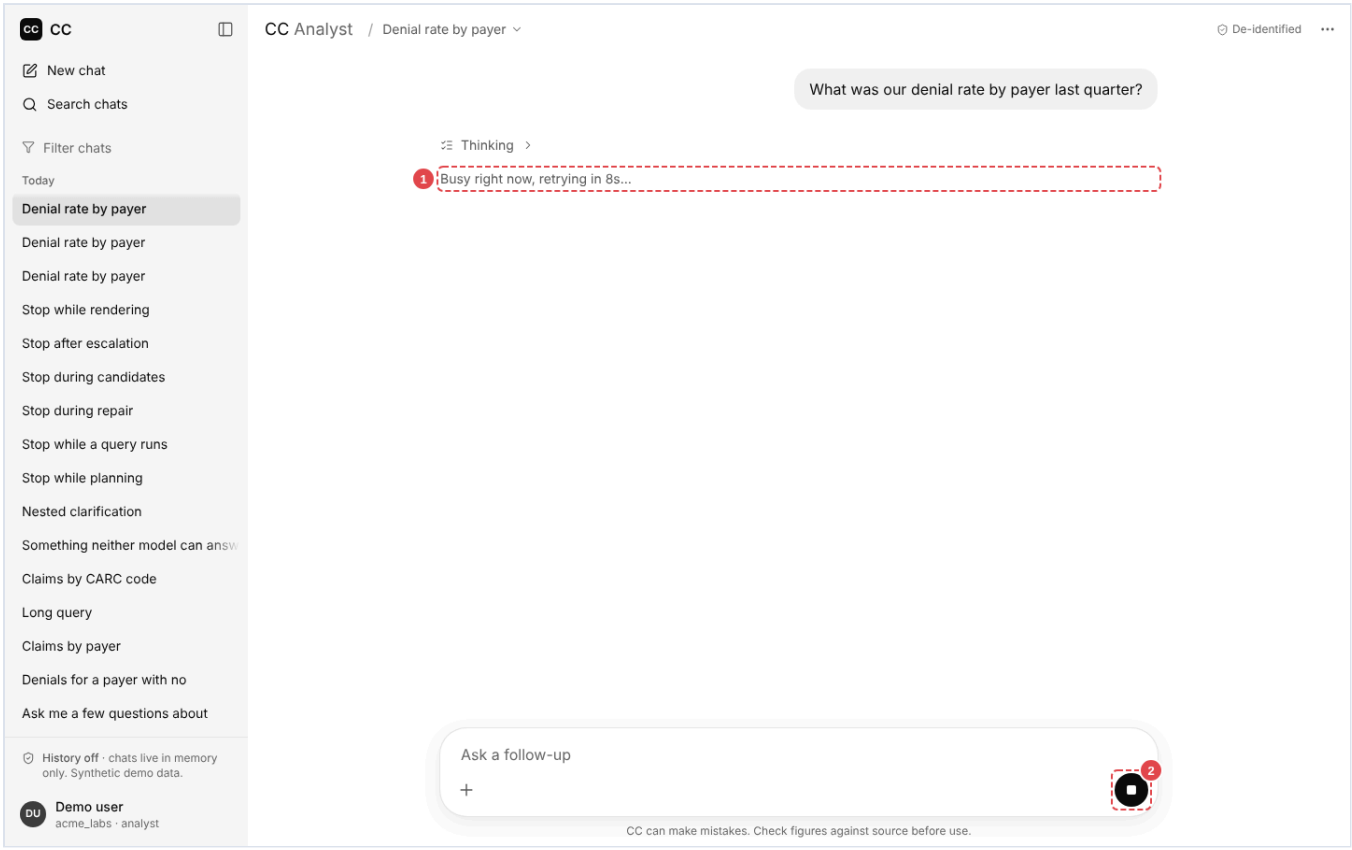
#	Element	Behaviour
1	Error line	"Can't reach the analyst · The server did not respond. Try again in a moment."
2	Retry	Re-sends; the composer kept the question.
3	Action row	Copy only on a failed turn (QA-A22).





TODO screenshot: 410 transcripts_unavailable — route GET /api/conversations/{id} to 410 {error: {code:"transcripts_unavailable"}} and open the chat

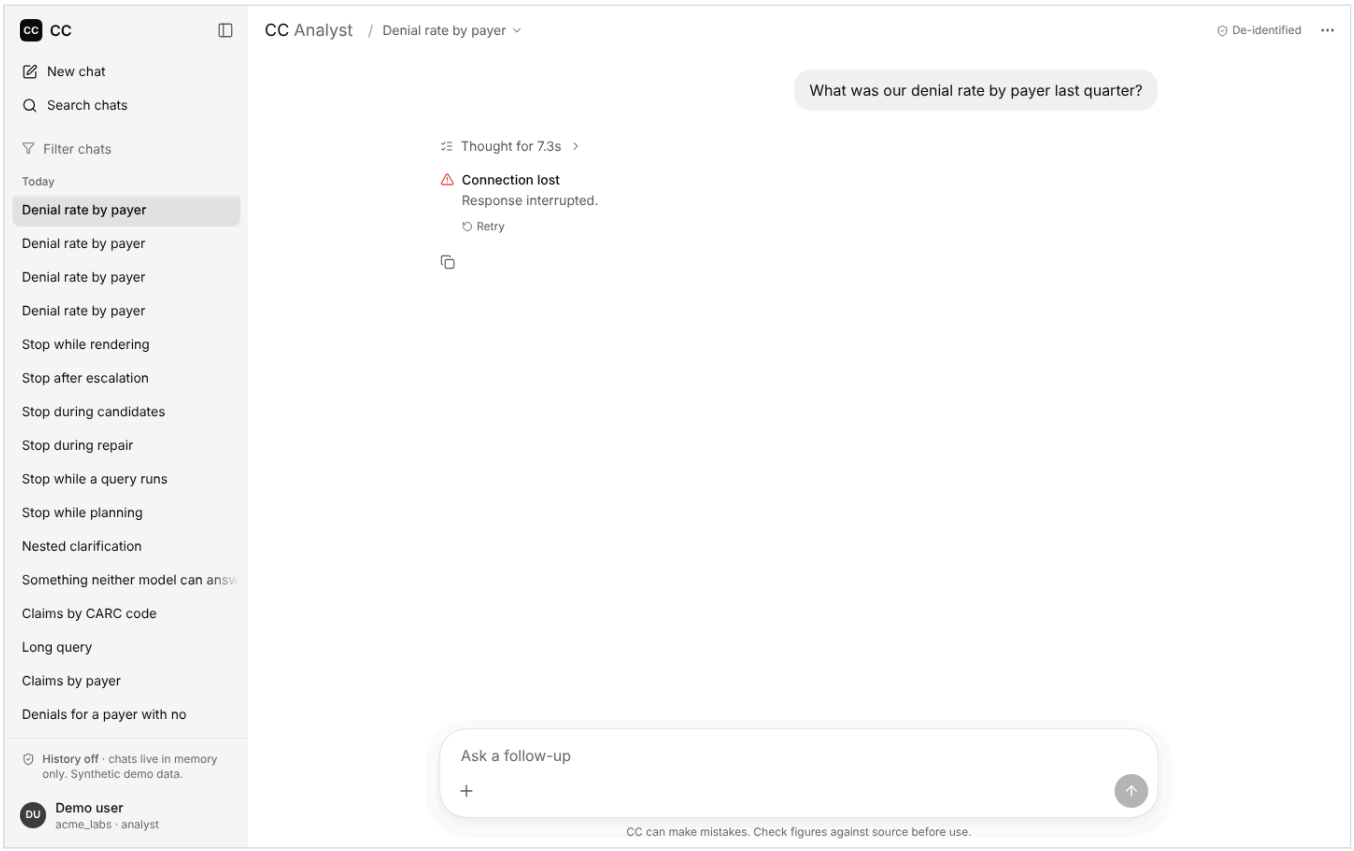
13.2 Transport: rate limit, reconnecting, offline

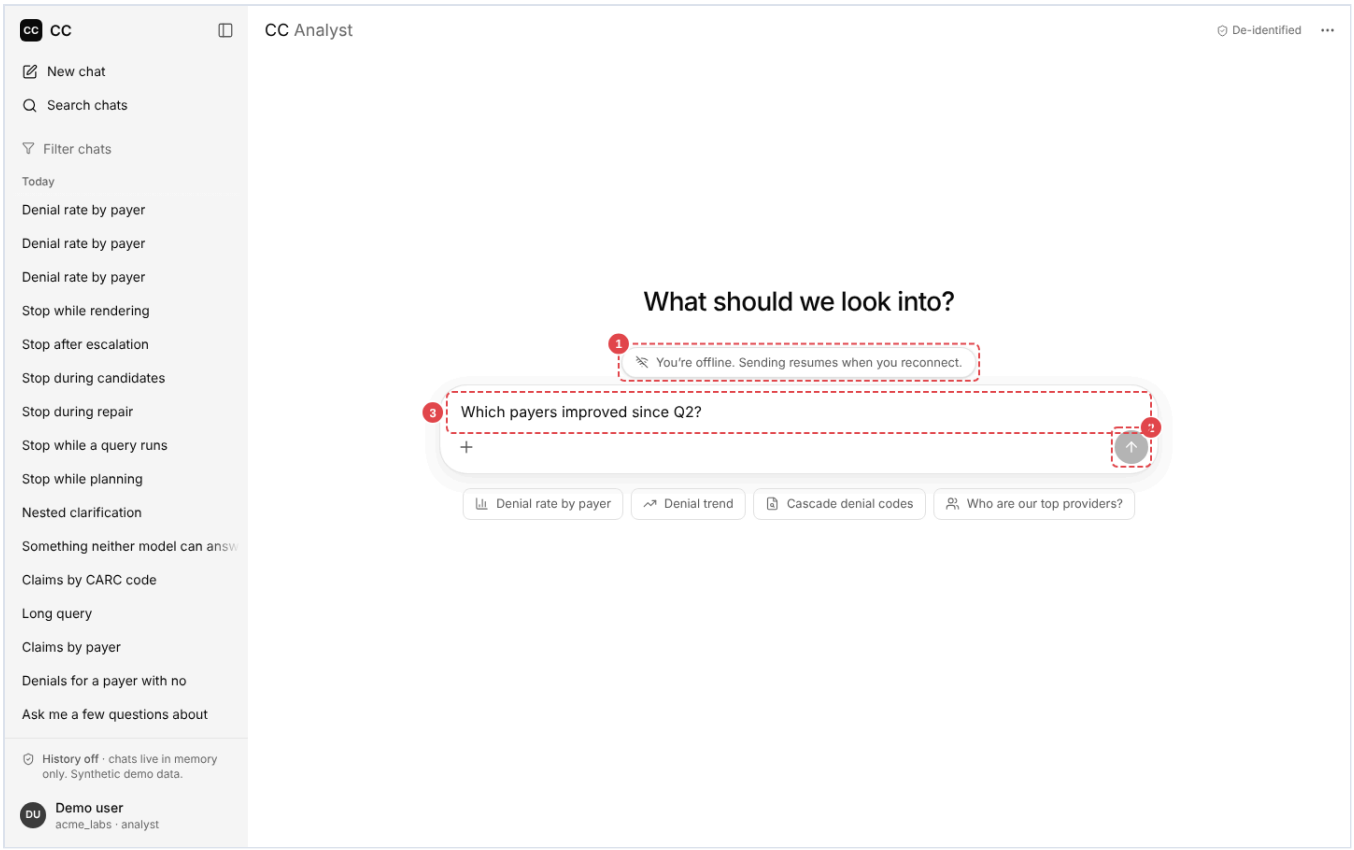


#	Element	Behaviour
1	Status line	Countdown from `Retry-After` (seconds or HTTP date against the response `Date`), up to 2 automatic retries (`maxRateRetries`), 30 s max wait.
2	Stop	Cancels the wait.

Dev shortcut: `?fault=429` on a `vite dev` build makes the first try of each answer 429 (`demoFaultOption`).


CC
New chat
Search chats
Filter chats
Today
Denial rate by payer
Denial rate by payer
Denial rate by payer
Stop while rendering
Stop after escalation
Stop during candidates
Stop during repair
Stop while a query runs
Stop while planning
Nested clarification
Something neither model can answ
Claims by CARC code
Long query
Claims by payer
Denials for a payer with no
Ask me a few questions about
History off · chats live in memory only. Synthetic demo data.
Demo user
acme_labs · analyst
CC Analyst / Denial rate by payer
De-identified
What was our denial rate by payer last quarter?
Running query
Reconnecting...
Ask a follow-up
CC can make mistakes. Check figures against source before use.


CC
New chat
Search chats
Filter chats
Today
Denial rate by payer
Denial rate by payer
Denial rate by payer
Denial rate by payer
Stop while rendering
Stop after escalation
Stop during candidates
Stop during repair
Stop while a query runs
Stop while planning
Nested clarification
Something neither model can answ
Claims by CARC code
Long query
Claims by payer
Denials for a payer with no
History off · chats live in memory only. Synthetic demo data.
Demo user
acme_labs · analyst
CC Analyst / Denial rate by payer
De-identified
What was our denial rate by payer last quarter?
Thought for 7.3s
Connection lost
Response interrupted.
Retry
Ask a follow-up
CC can make mistakes. Check figures against source before use.

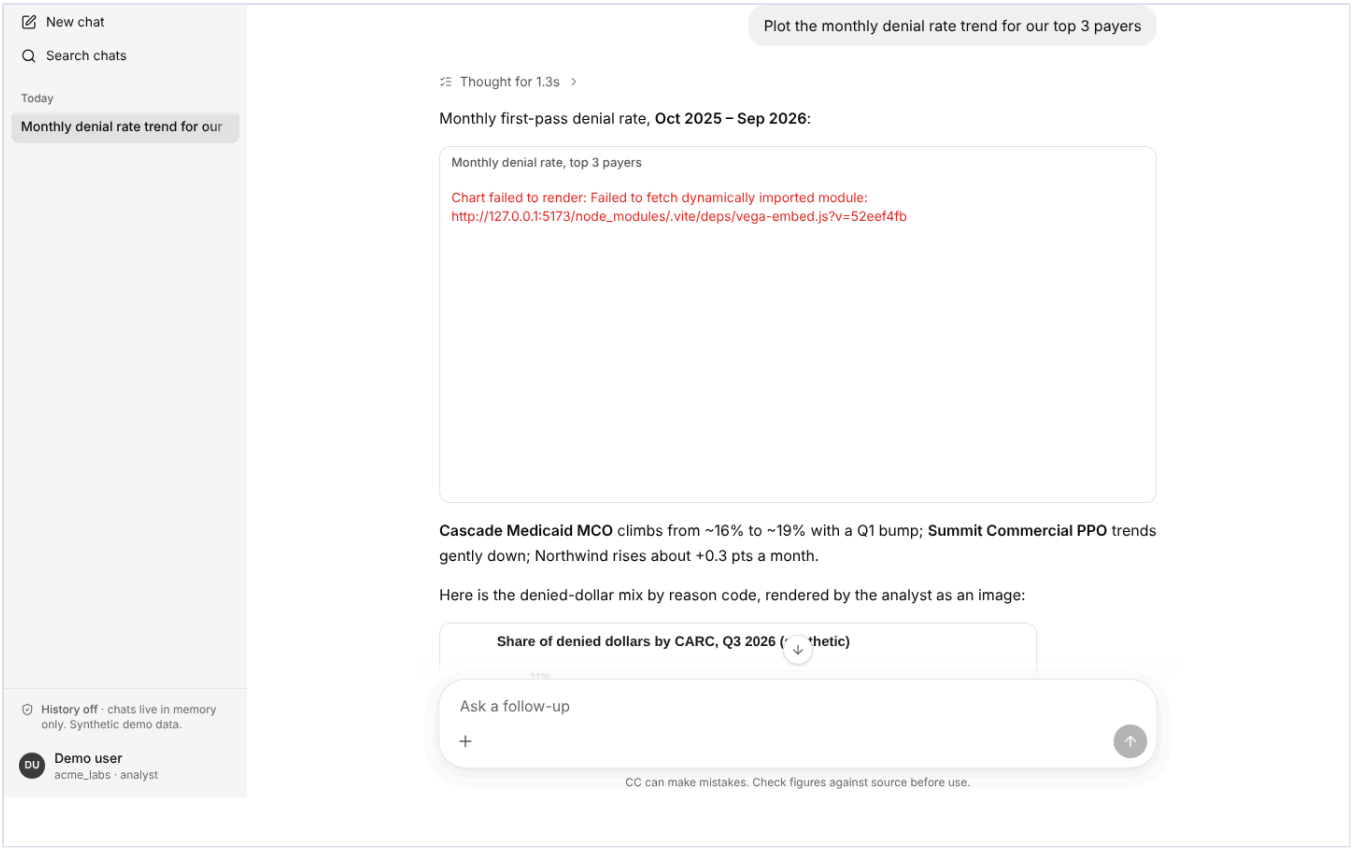


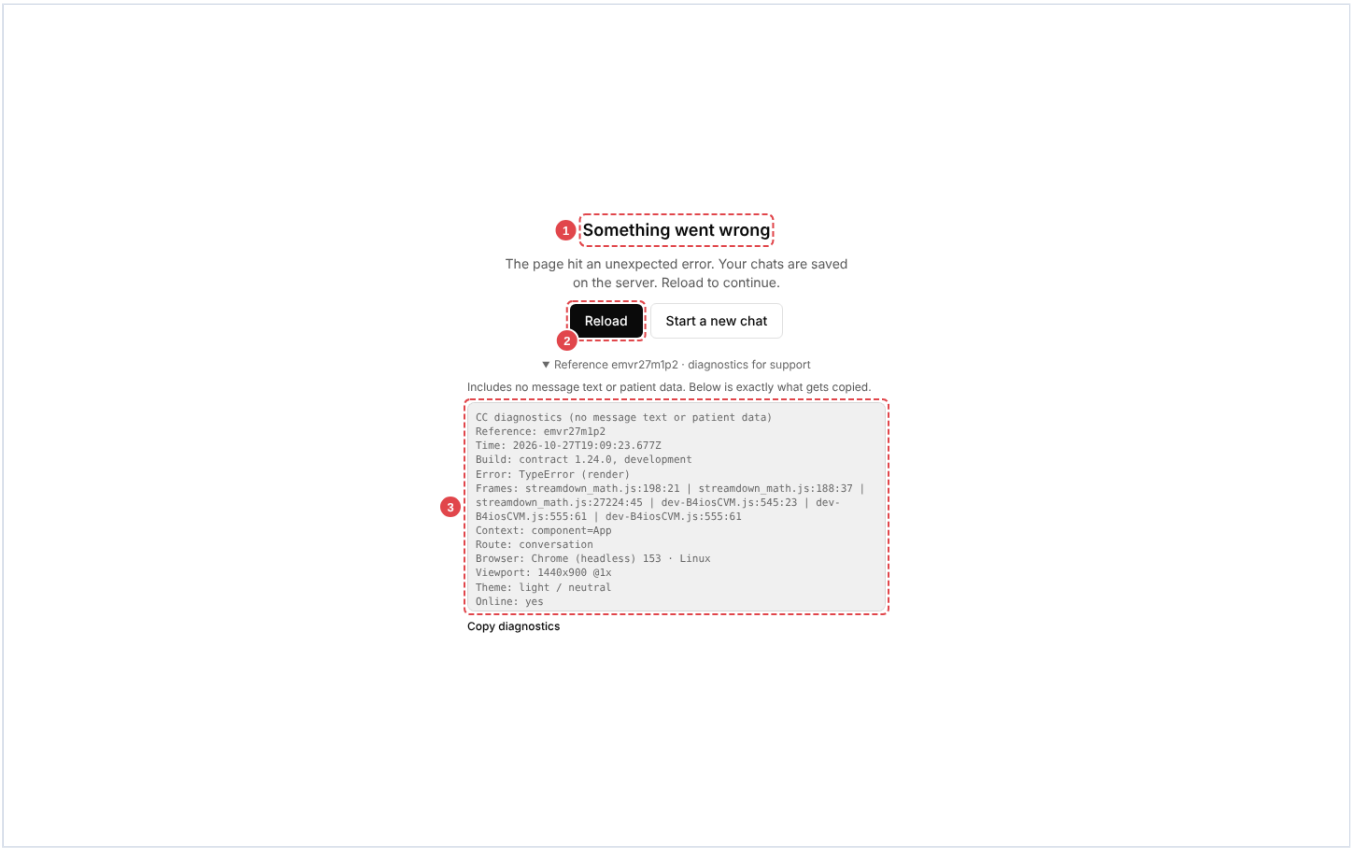
#	Element	Behaviour
1	Offline pill	After 1 s of continuous offline (`useSettledOnline`); hides as soon as the connection is back.
2	Send	Follows raw `navigator.onLine` , so nothing is sent into a blip.
3	Draft	Stays editable.

TODO screenshot: stalled stream — route POST /api/chat to send only message.start and then nothing for 45 s; before message.start the turn ends with "The analyst stopped responding" (retryable)

TODO screenshot: answer too large — serve one SSE data line over 32 MiB without a newline; the answer keeps what arrived and adds "Answer too large to show"

13.3 Render failures

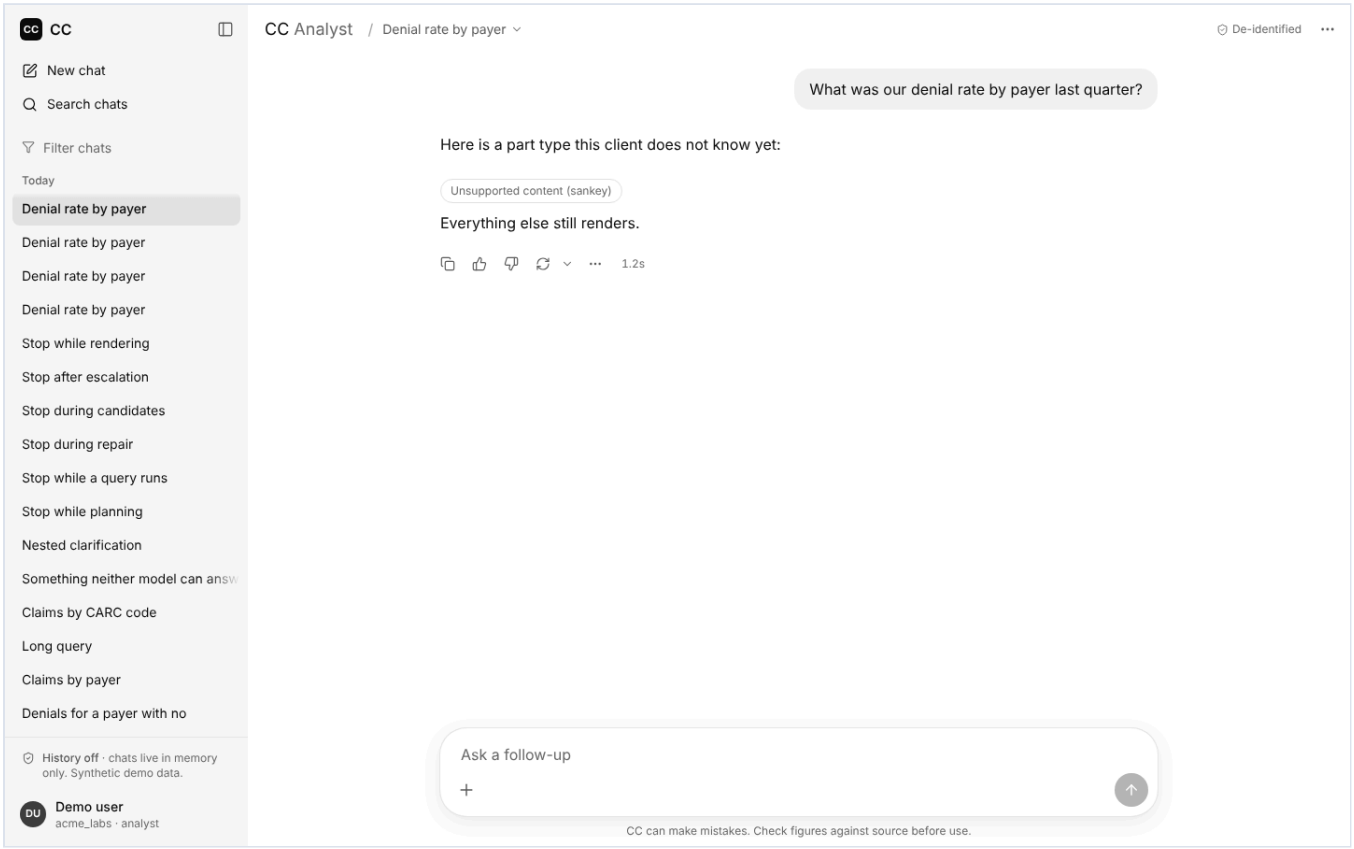




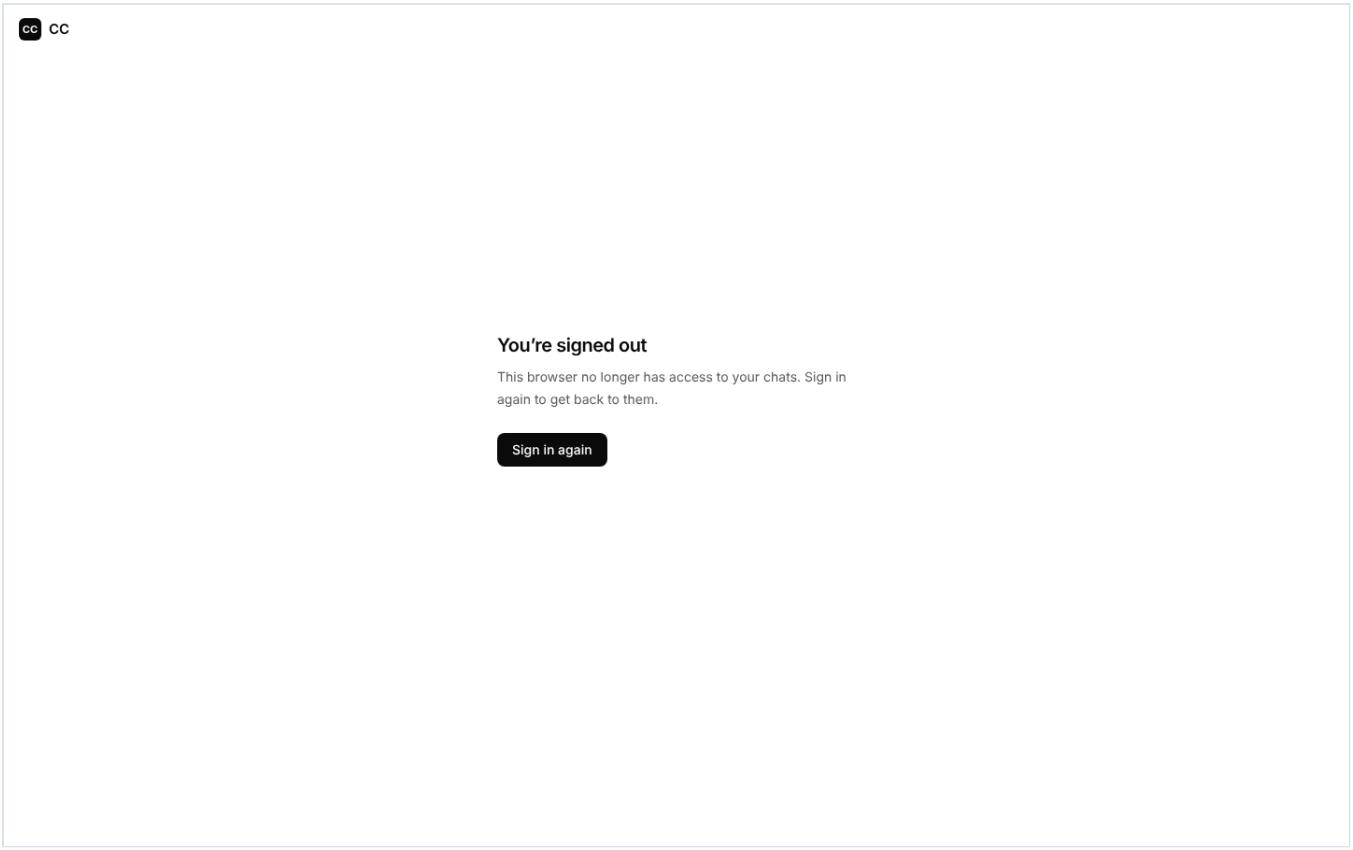
#	Element	Behaviour
1	Heading	"Something went wrong" (or "Couldn't load part of the app" for a chunk error). Never shows the error message (PHI).

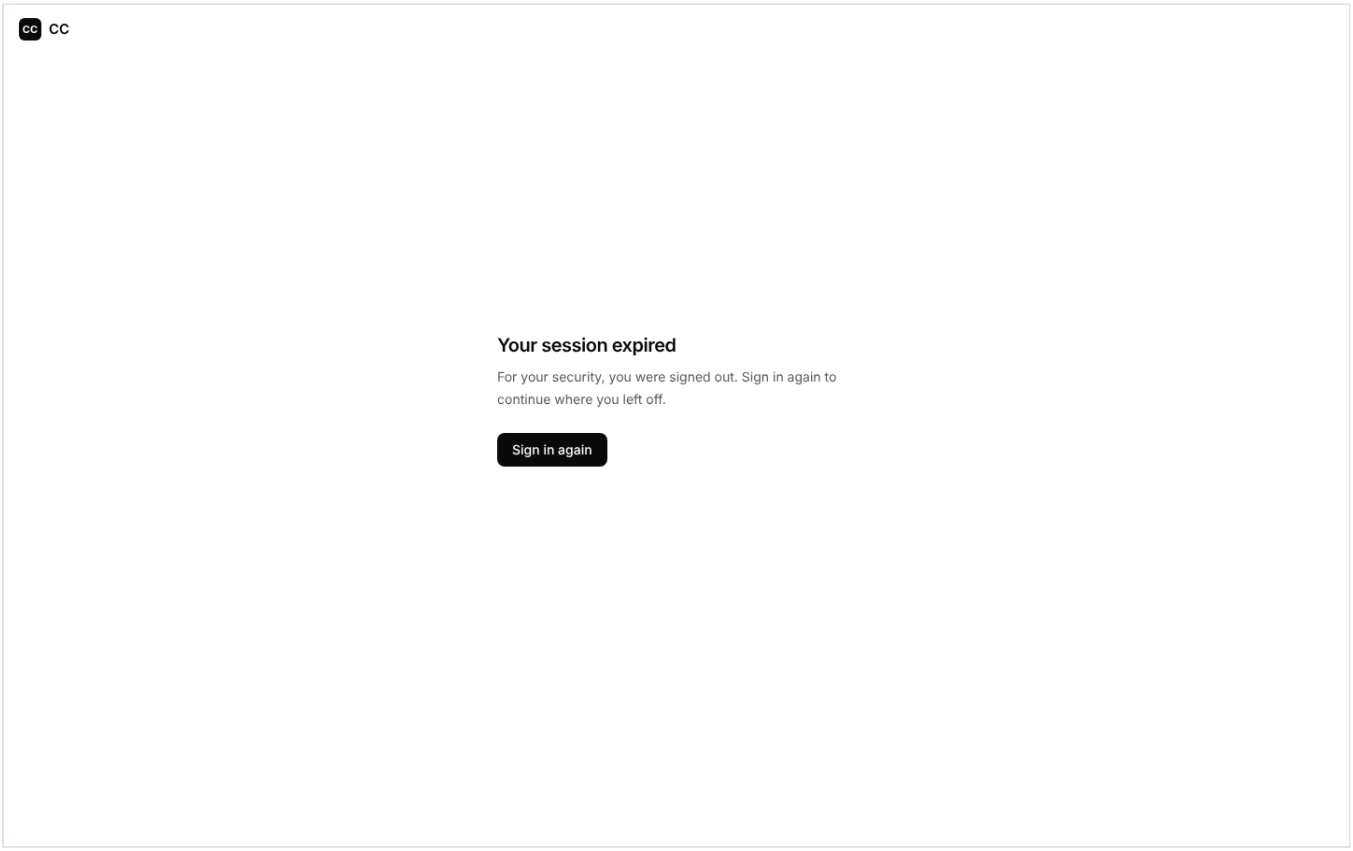
#	Element	Behaviour
2	Actions	Reload (focused) · Start a new chat (drops the `#/c/…` route so a crashing chat can't trap the user).
3	Diagnostics	Reference id + allowlisted facts (build, contract, browser, theme, online), "Copy diagnostics".

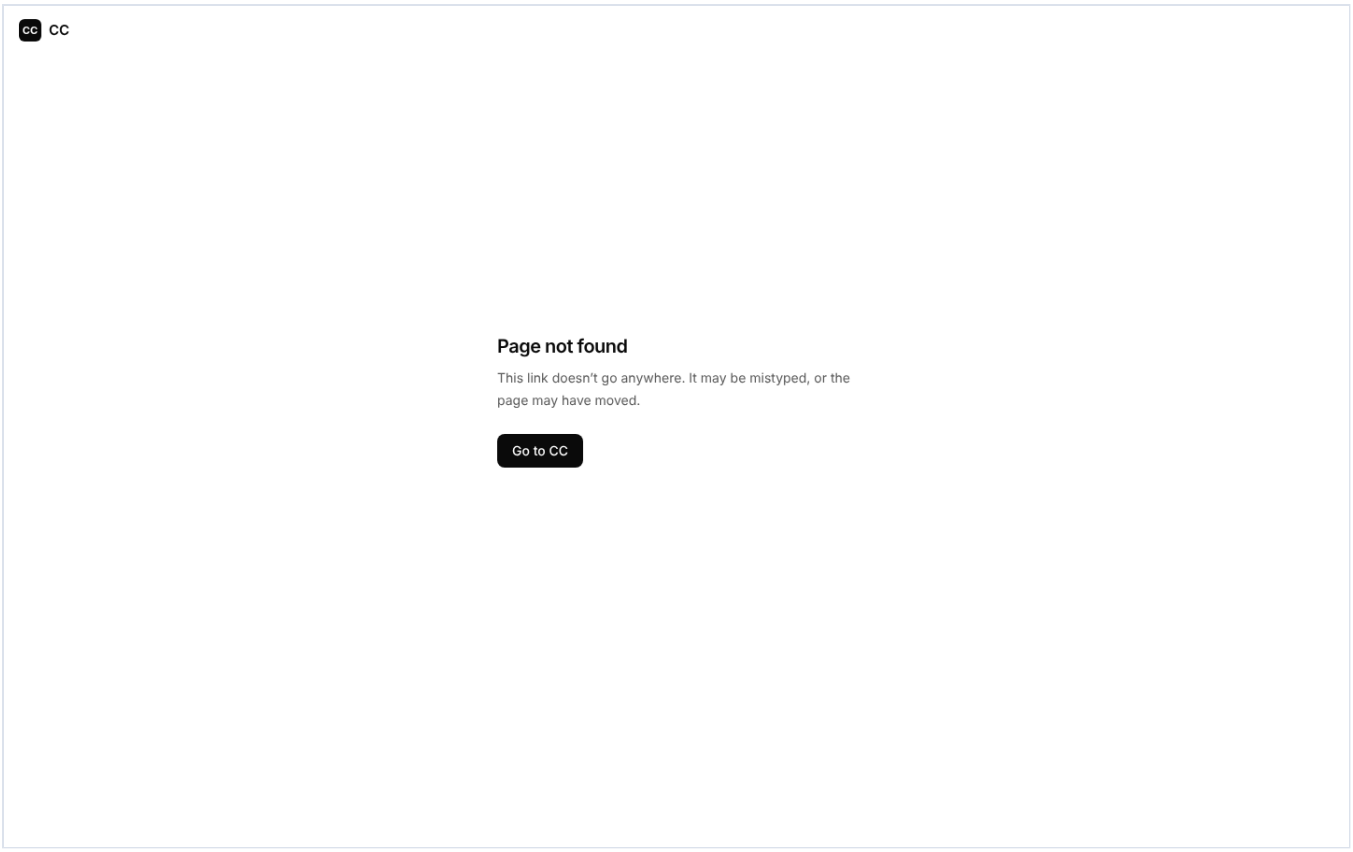
TODO screenshot: section crash — make one part throw while rendering (e.g. a malformed artifact in a test build); the part is replaced by "This part couldn't be displayed. Ref …" with Try again / Close



13.4 Status pages and sign-in





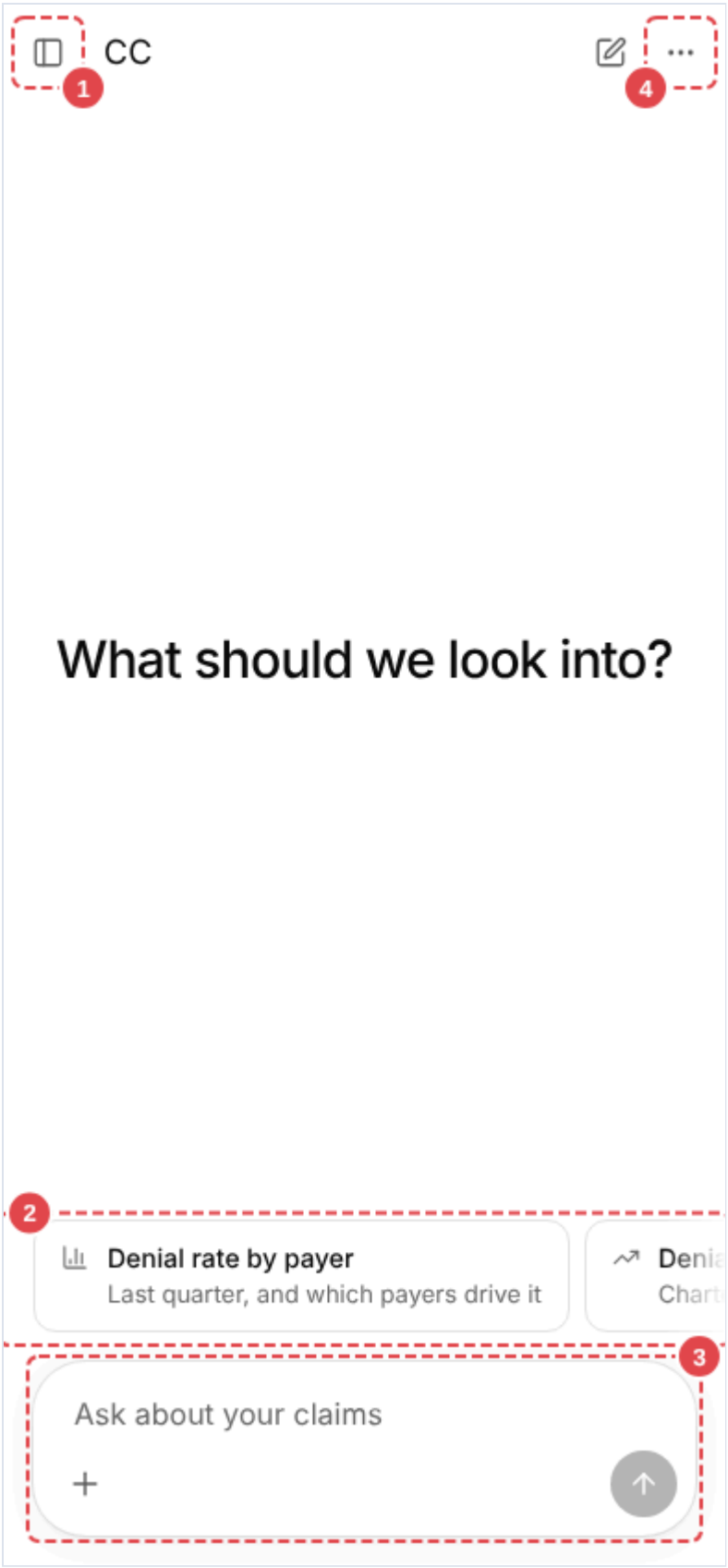


TODO screenshot: cookie sign-in — build with VITE_AUTH=cookie VITE_SIGN_IN_URL=/api/auth/mock-login against api.sim_server; capture the sign-in redirect and the first load after sign-in (ana@xifin.example)

TODO screenshot: signed out in another tab — sign out in tab A; tab B leaves to the signed-out page (BroadcastChannel)

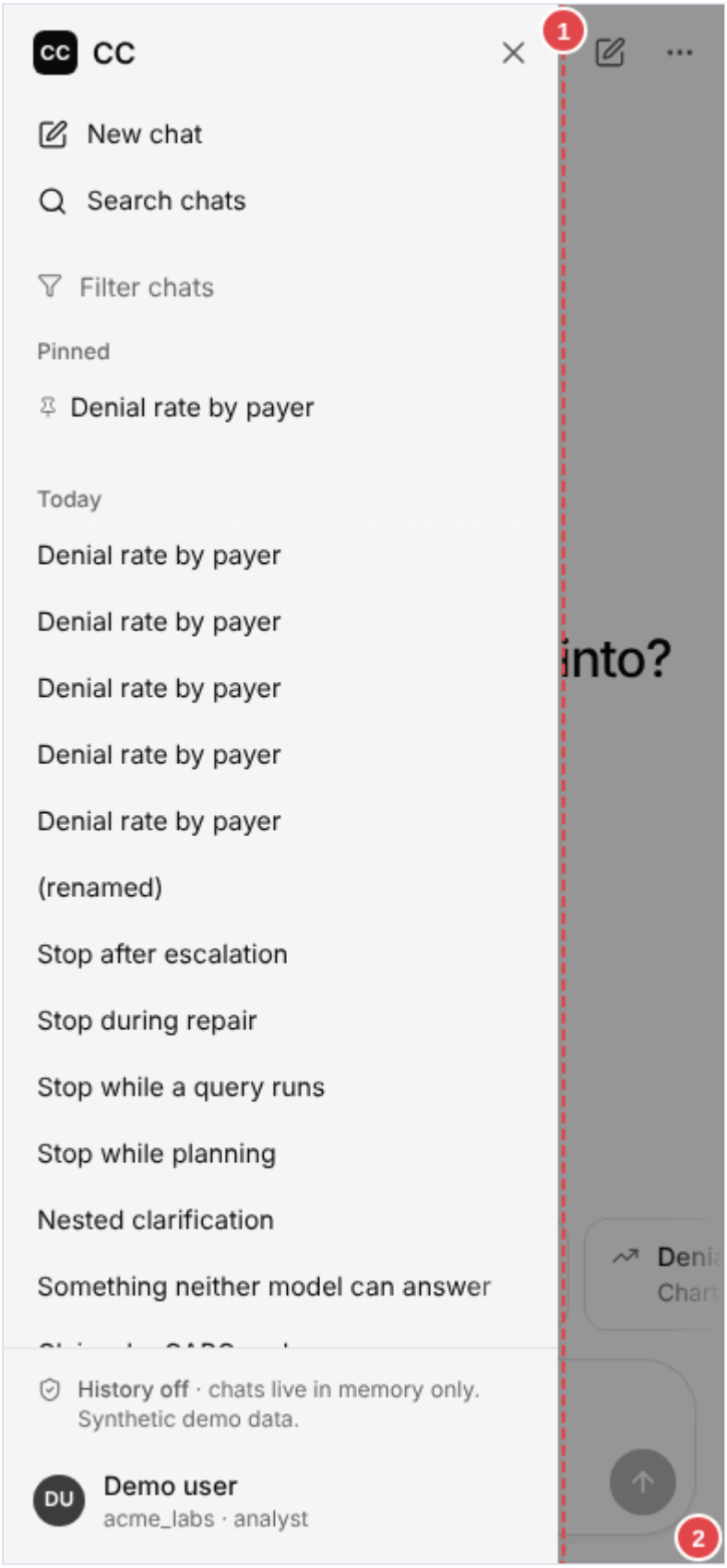
---

14. Responsive & touch



#	Element	Behaviour
1	Open sidebar	Opens the drawer (also left-edge swipe, `useEdgeSwipeToOpen`).

#	Element	Behaviour
2	Suggestion cards	One swipeable row of two-line cards above the composer; doesn't raise the keyboard (`pointer: coarse`).
3	Composer	Full width; `enterKeyHint="send"` .
4	More options	Same menu as desktop; New chat icon sits next to it.



#	Element	Behaviour
1	Drawer	`role="dialog"` `aria-modal` ; choosing a chat or New chat closes it; swipe to close.
2	Scrim	Tap to close.

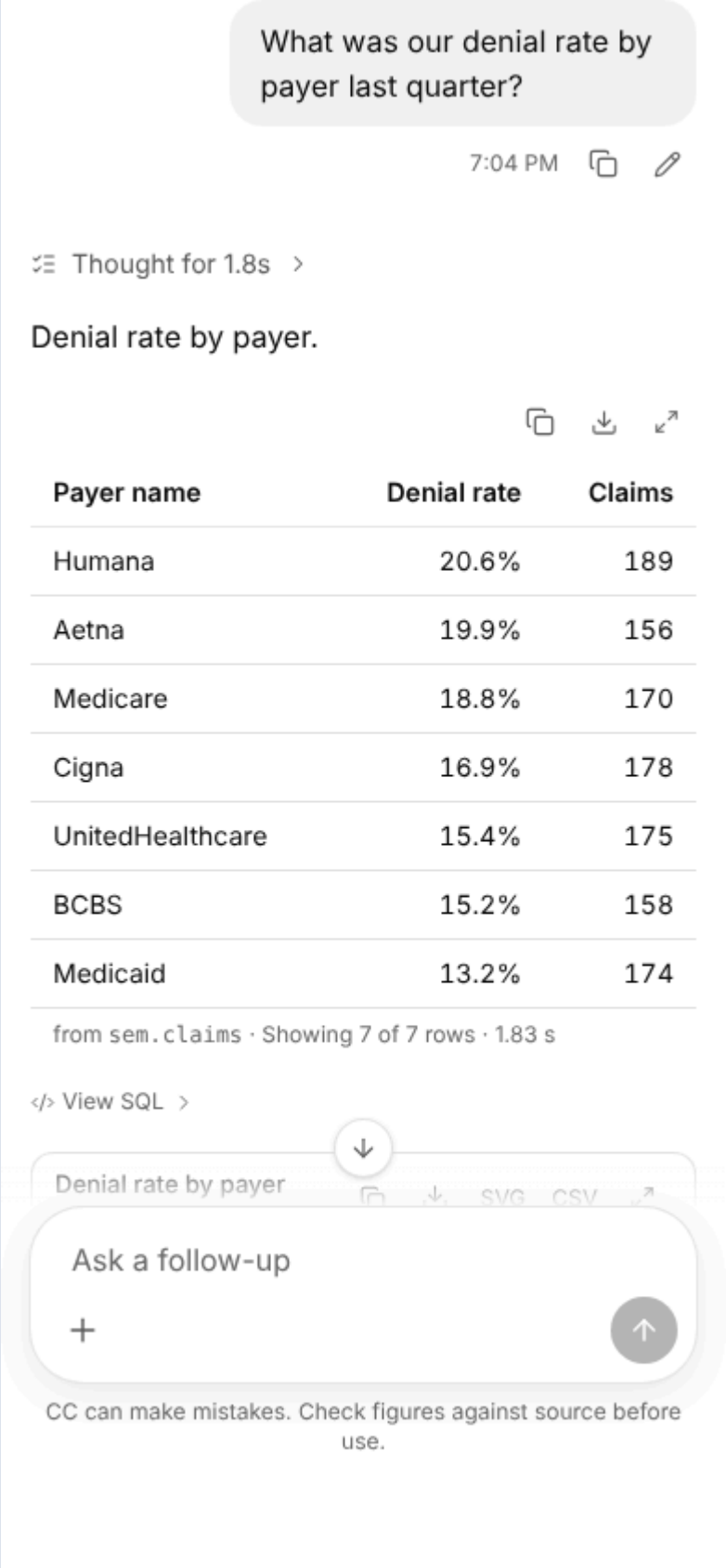


Table SQL **Result**
Table

Filter rows

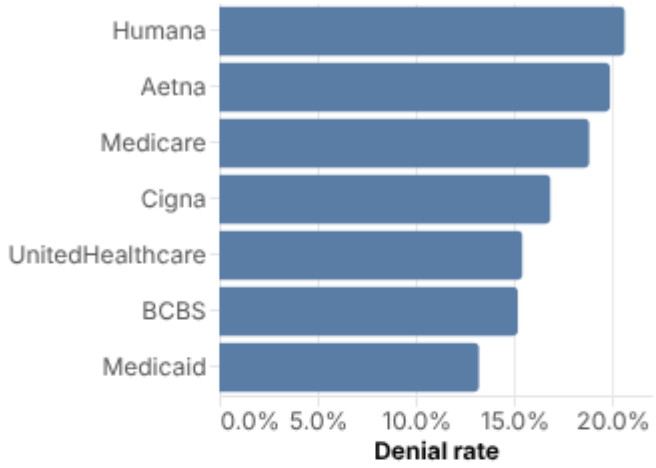
Payer name	Denial rate	Claims
Humana	20.6%	189
Aetna	19.9%	156
Medicare	18.8%	170
Cigna	16.9%	178
UnitedHealthcare	15.4%	175
BCBS	15.2%	158
Medicaid	13.2%	174

from `sem.claims` · Showing 7 of 7 rows · 1.83 s

CC

Denial rate by payer

SVG CSV



Assumptions

- Denial means the whole claim was denied; partial payments are not denials. Change

Check before using

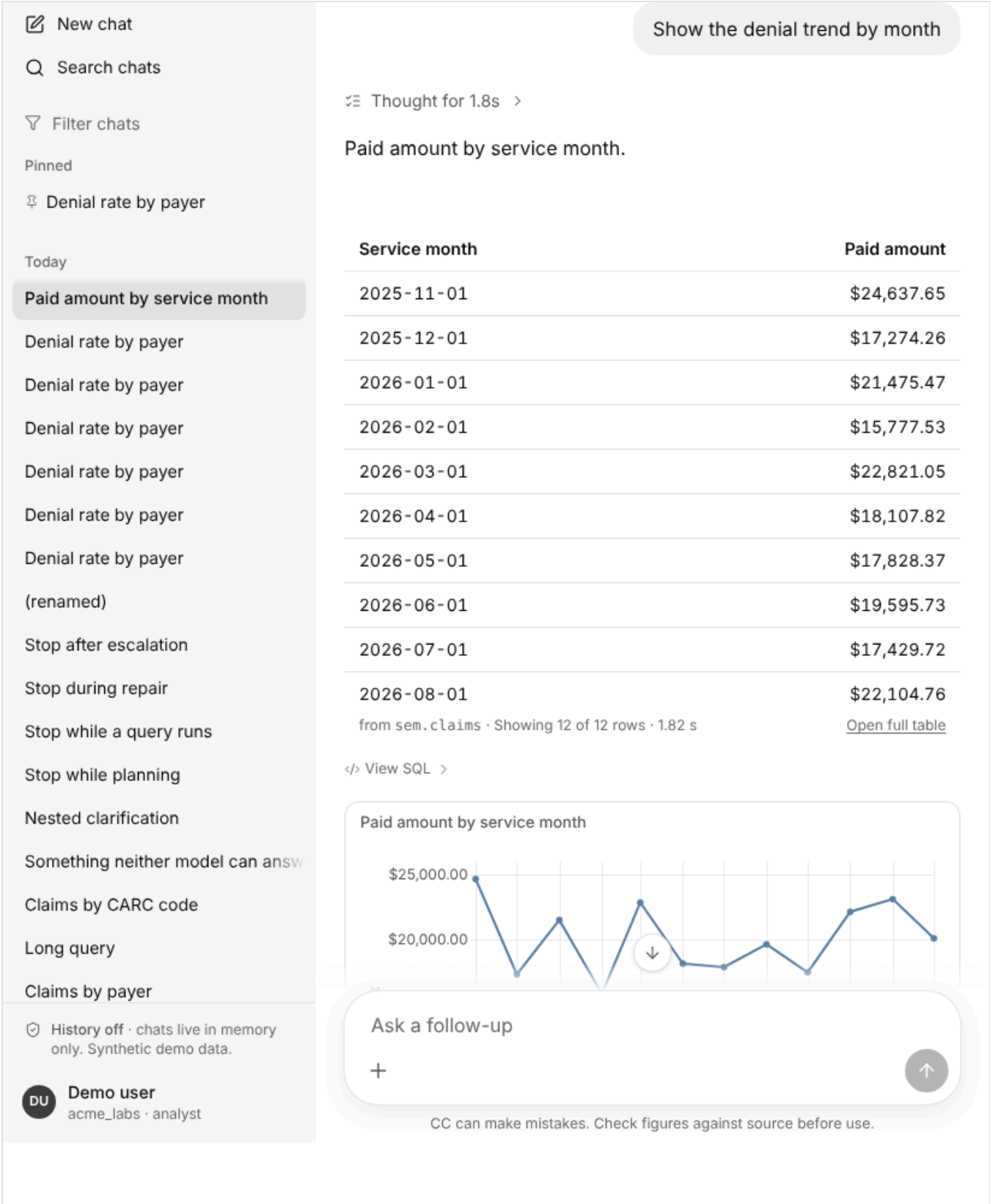
- sem.claims/claim_lines used without is_latest_submission: resubmitted claims count more than once

Looked at `sem.claims`

2 queries · 2s · <$0.01

Ask a follow-up

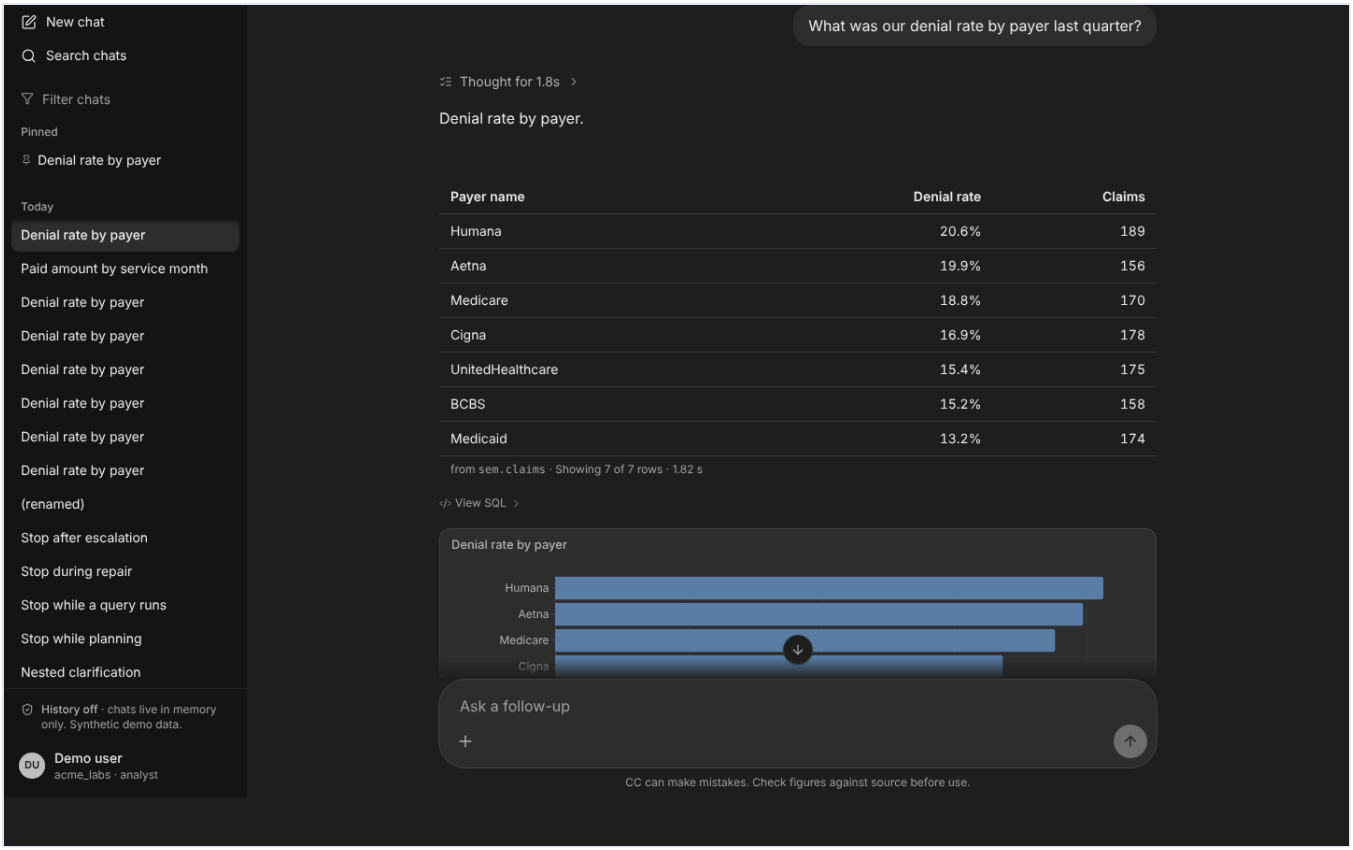
CC can make mistakes. Check figures against source before use.

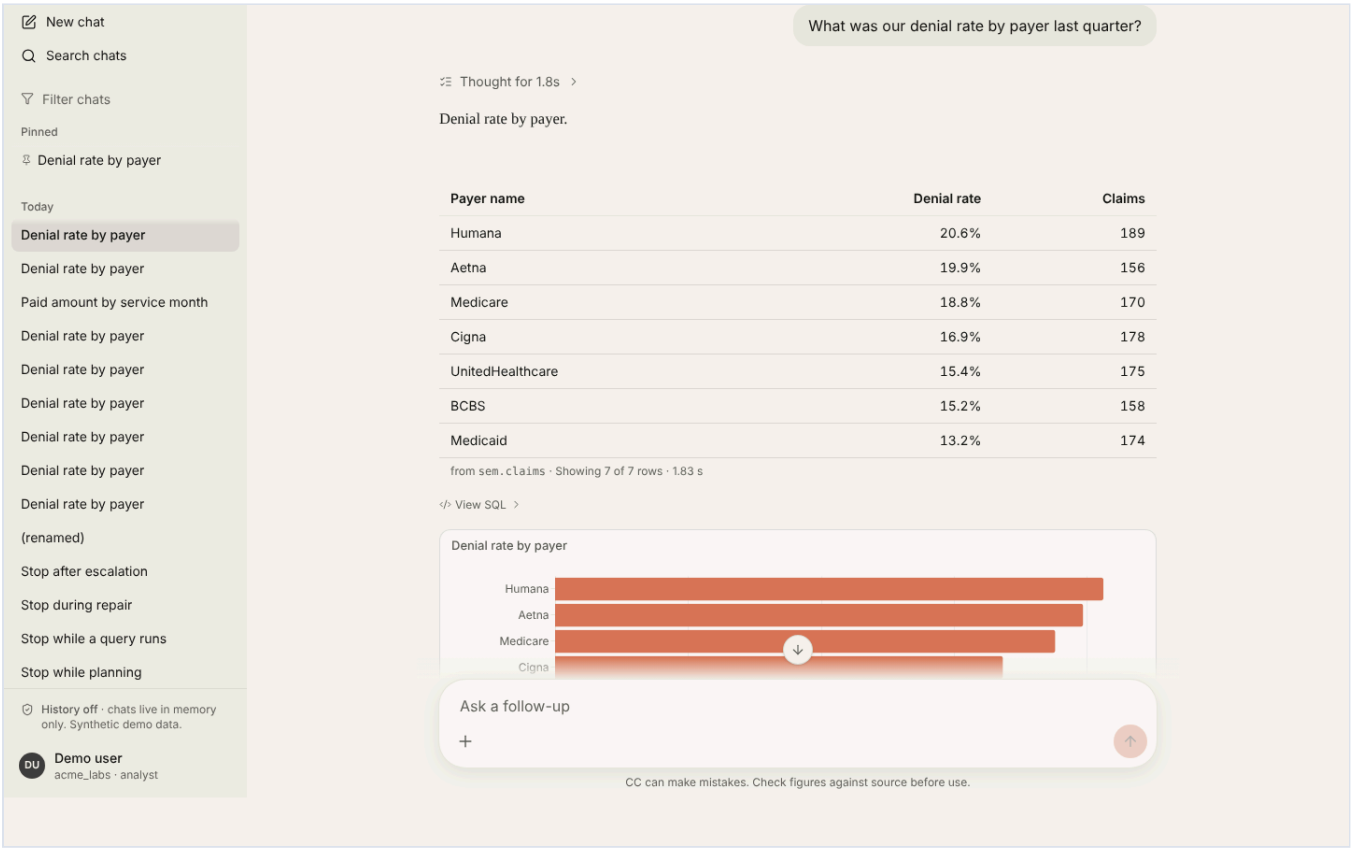


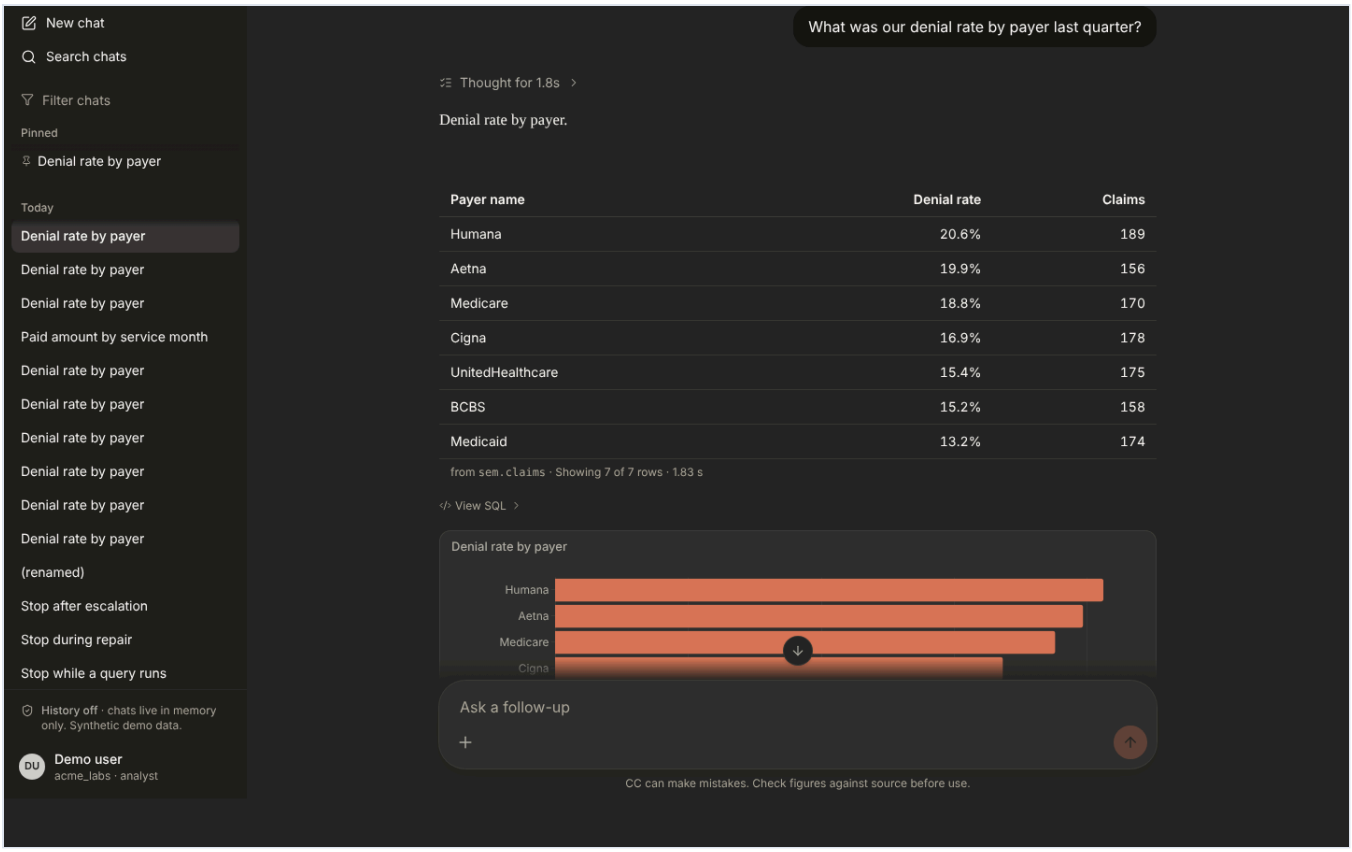
TODO screenshot: edge swipe — on a 390 px touch viewport, swipe from the left edge to open the drawer and swipe left to close (e2e/shell-swipe.spec.ts)

TODO: 200 % zoom (640 px CSS viewport): no page-level horizontal scroll (`e2e/answer-edge.spec.ts`).

15. Themes and palettes





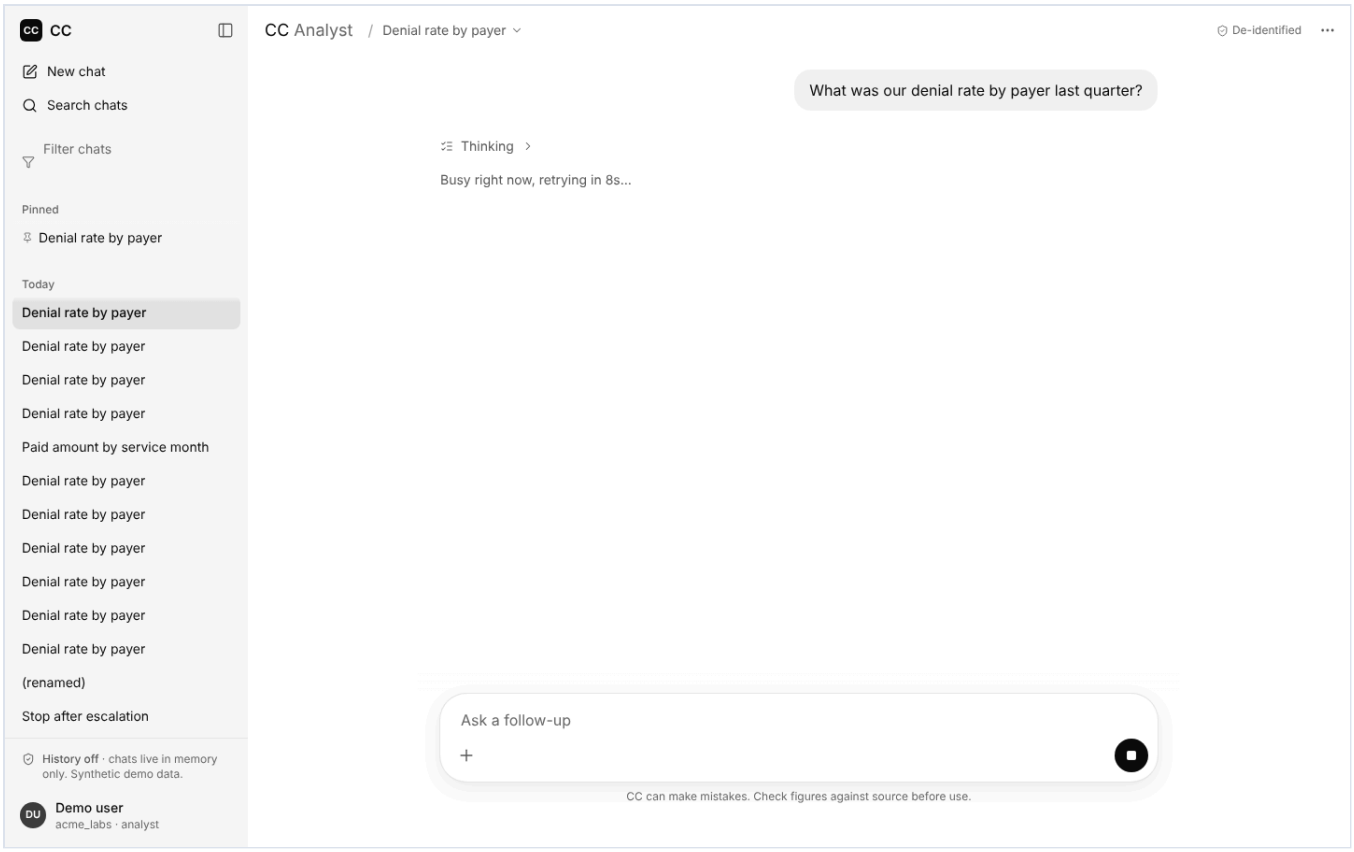


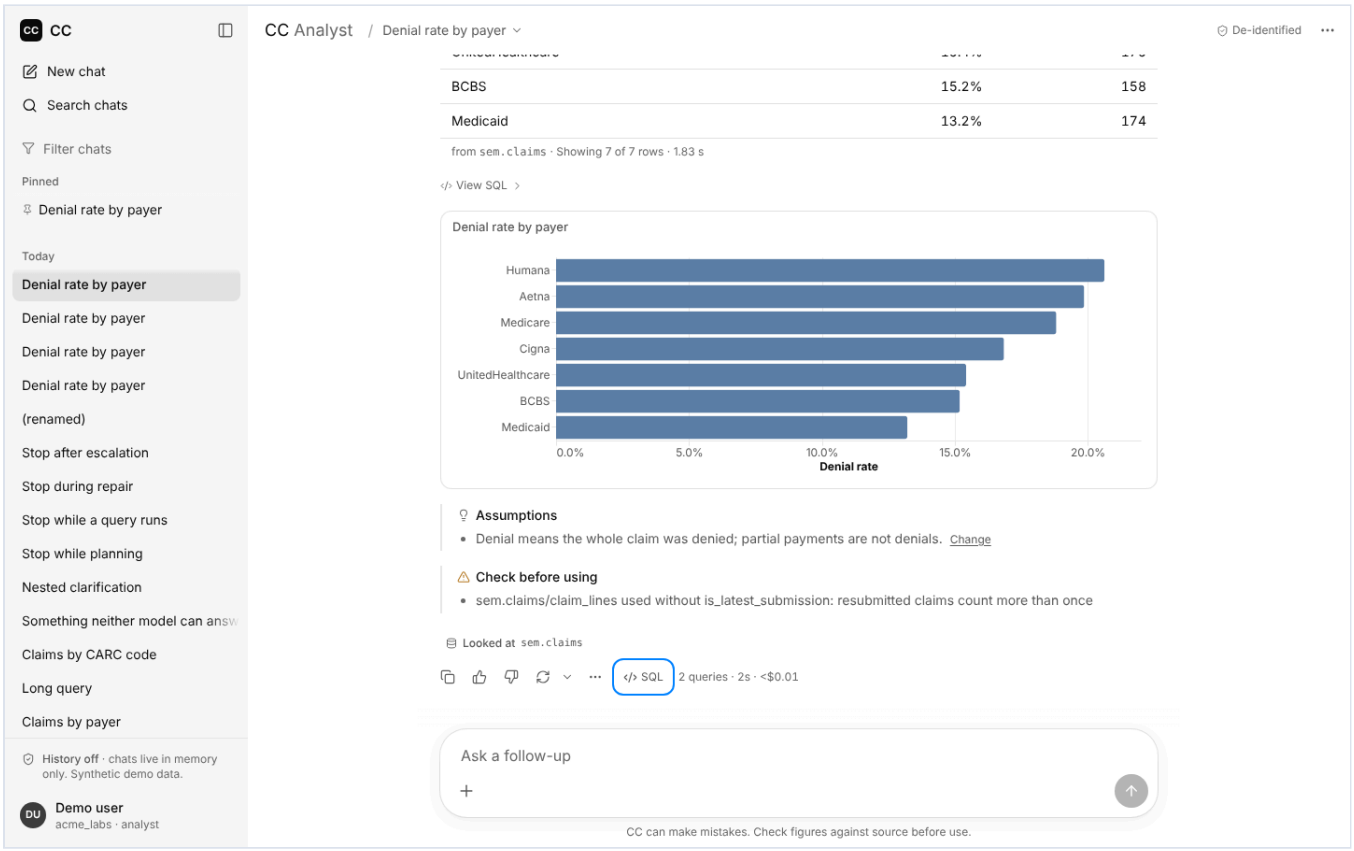
TODO: theme follows the OS on "System"; changes sync across tabs (`onPrefChange`); single-series charts follow the palette accent.

---

16. Accessibility

Area	Behaviour	Code / test
Live regions	One shell status region ("answering…" / "Answer complete"); transport announcer is polite and throttled (429 countdown spoken at start, every 5 s, and on retry)	`TransportAnnouncer`, `lib/transportAnnounce.test.ts`
Step timeline	Group named by its header; new steps spoken once; never the timer	`StepTimeline`, `answer/thinkingDisclosure.test.tsx`
Focus management	Edit/version switches put focus back on the matching control (QA-A21); dialogs trap focus; Esc closes	`shell/focus.test.tsx`, `answer/qaA21A22.test.tsx`
Keyboard	Roving focus in the sidebar list; slash menu via `aria-activedescendant`; resizable panel and columns by arrow keys	`e2e/keyboard-sr.spec.ts`
Reduced motion	Fades and paced text off	`lib/motion.ts`, `e2e/answer-edge.spec.ts`
Automated checks	axe-core in Playwright	`e2e/a11y.spec.ts`





TODO: list the ARIA names QA should use (Message, Send message, Stop generating, Chat history, Artifact: